

Narrative-Aware Thematic Trading

From Text, Topics, and Attention to Factor-Neutral Equity Baskets

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Preface

In 2024, Vistra Energy became one of the best-performing stocks in the S&P 500. It was not a chip designer, a cloud platform, or a software company. It was a competitive power producer with gas plants, solar assets, and nuclear capacity. By sector classification, it belonged with utilities. By market behavior, it belonged in the AI infrastructure trade.

That distinction is the reason this book exists.

Modern **markets** do not move only through sectors, styles, and balance sheets. They also **move through stories**: artificial intelligence, clean energy, reshoring, obesity drugs, defense technology, cybersecurity, aging populations, electrification. Some stories are empty. Some are late. Some are real but already priced. A few travel through filings, earnings calls, news, customer contracts, supply-chain bottlenecks, and capital expenditure before most investors have built the map needed to see them.

The hard problem is not noticing that a story exists. The hard problem is deciding whether the story has become an investable exposure.

A **text model** can **discover** a **cluster** of **documents** about data-center power. That is useful, but it is **not a portfolio**. A language model can summarize why **hyperscalers need electricity**. That is useful, but **it is not alpha**. An ETF can package a theme into a ticker. That is convenient, but it may arrive after the easy returns have already been earned. **Between the first appearance of a narrative and the final portfolio weight** sits the **real work**: define the theme, timestamp the evidence, map the value chain, score the companies, separate language from revenue, neutralize factors, model turnover and capacity, and test the result without leaking the future into the past.

This book is about that work.

The primary **case** is **AI infrastructure**. It is recent, visible, and cross-sectoral enough to break the standard map. Nvidia matters, but so do SK Hynix, TSMC, ASML, Broadcom, Vertiv, Eaton, Constellation, Vistra, data-center REITs, cloud platforms, and the less obvious suppliers that sit around the compute stack. The theme begins with model demand, but the investable chain quickly runs through high-bandwidth memory, advanced packaging, networking, cooling, switchgear, grid interconnection, baseload power, and software adoption. It is a good case because it is not one trade. It is a sequence of bottlenecks.

The secondary **case** is **clean energy** and the broader energy transition. The transition was real, policy support was real, and many companies had genuine exposure. The 2020–2022 public-equity trade still punished late buyers because some vehicles mixed theme exposure with long-duration growth, retail attention, crowded ownership, and a rate regime that changed underneath them. Clean energy teaches a more precise lesson than cynicism: a socially necessary and economically real transition still requires disciplined entry points, company selection, factor control, and portfolio construction.

The argument is simple enough to state and hard enough to execute. Markets are organized partly by narratives, and modern NLP can detect those narratives earlier and more systematically

than a purely manual process. But narrative detection is only the first step. A detected **topic** must **become a theme**; a **theme must become a company map**; a **company map must become an exposure score**; an **exposure score must become a risk-controlled basket**; and a **basket must survive backtesting**, transaction costs, governance, and the ordinary indignities of live markets.

The book follows that sequence.

Part I builds the market foundations: alpha, beta, benchmarks, factors, sectors, and the contamination problem. This material comes first because every text signal eventually has to answer a brutal question: did it earn something that the investor could not have bought more cheaply through beta, sector exposure, momentum, growth, value, size, or liquidity?

Part II develops the **narrative** side of the problem. It treats market stories as measurable objects without pretending that every story deserves capital. Attention, sentiment, source timing, lifecycle stage, and information lead time become part of the research process rather than color around it.

Parts III and IV turn text into topics and track those **topics through time**. They cover corpora, preprocessing, entity handling, classical and neural topic models, short-text models, labeling, dynamic topic modeling, emergence, drift, hierarchy, granularity, and evaluation. The guiding principle is practical: a topic model is valuable only if it improves an investment decision.

Part V maps **themes to companies**. This is where language starts touching balance sheets. The chapters build a theme specification, define a multi-signal exposure score, put LLMs under retrieval and governance discipline, construct value-chain maps, and add guardrails against AI-washing, greenwashing, ETF dilution, late-cycle launches, and implausible edges. The part closes with a minimum viable thematic process: the smallest end-to-end pipeline a team can defend.

Part VI turns that pipeline into portfolios. It covers basket construction, factor neutralization, rebalancing, theme decay, capacity, transaction costs, crowding, regimes, thematic ETFs, index methodologies, and prediction-market overlays. This is the practical core of the book: text signals become investable portfolios only when the weights, risks, costs, and exits are explicit.

Parts VII and VIII move from method to operation. They translate the machinery into use cases, design patterns, implementation playbooks, testing standards, attribution, governance, production controls, and open research questions. A strategy that cannot explain what it knew, when it knew it, why it expected to be paid, and who approved the decision is not ready for capital.

The intended reader is skeptical and busy. You may be a portfolio manager, quant researcher, allocator, analyst, or sophisticated investor who has watched thematic products multiply and wondered which ones contain research rather than marketing. You do not need to be an NLP specialist when you begin. You do need patience for the details that make a signal tradable: timestamps, entity resolution, factor regression, evidence trails, capacity curves, and human review.

The book is technical where it has to be, but the formulas are not decoration. Each one is here because a real portfolio mistake becomes harder to make when the idea is written down precisely. Alpha is not a vibe. A topic is not a theme. A mention is not exposure. An exposure score is not a weight. A backtest is not evidence unless every input was available at the simulated decision time.

Those distinctions sound obvious after someone names them. Most expensive mistakes do.

The hope of the book is not to make thematic investing easy. It is to make it less impressionistic. The market will keep producing stories, and companies will keep learning the language investors

reward. The task is to build a process that can listen to those stories without being captured by them: curious enough to find the next Vistra, skeptical enough to avoid the next Long Blockchain, and disciplined enough to know the difference before the trade is crowded.

Part I

Market Foundations for Thematic Alpha

Chapter 1

Why Thematic Trading Needs a Different Map

1.1 The Utility That Outperformed Nvidia

In 2024, the best-performing stock in the S&P 500 Utilities sector gained more than 250 percent.¹ The stock was Vistra Energy, a Texas electricity company. It owned natural gas plants, solar installations, and nuclear reactors. It competed for commercial and residential power contracts across the central United States. By every standard classification, it was a competitive power company.

By every market behavior, it was an AI infrastructure trade.

Vistra moved because **data centers consume extraordinary amounts of electricity**, and the companies building them needed to lock up baseload power before competitors could. Nuclear plants generate around the clock at predictable cost and without carbon emissions. When Microsoft announced a twenty-year power purchase agreement with Constellation Energy in September 2024 to restart Three Mile Island,² investors immediately began asking who else owned nuclear capacity near major load centers. Vistra had assembled a nuclear fleet spanning Texas, Pennsylvania, and Ohio. Its stock jumped sharply in the weeks that followed.

The company had not pivoted to artificial intelligence. It had not hired machine learning engineers or rebranded its investor day. Vistra simply owned the right assets at the moment a different industry needed them. The narrative attached itself to the company from outside, traveled through the financial press into broker models, and eventually reached index inclusion lists that most traditional utility analysts had no framework to anticipate.

A standard equity research process would have missed the connection entirely. A sector screen would have placed Vistra alongside Con Edison and Dominion—defensives, yield plays, utilities. A factor model would have flagged its operating leverage, debt load, and regional concentration. A geographic screen would have noted a U.S. domestic company in a regulated industry with limited growth. None of these lenses would have placed Vistra next to Nvidia, Eaton Corporation, or the operators of liquid-cooled hyperscale facilities in northern Virginia.

The market had a narrative. The standard map had a category. They were not the same thing.

¹CNBC reported Vistra's 261.3 percent return among the top S&P 500 performers for 2024 [34].

²Constellation announced the 20-year Microsoft power purchase agreement and the planned restart of Three Mile Island Unit 1, renamed the Crane Clean Energy Center, on September 20, 2024 [28].

1.2 The Map and What It Misses

Equity investors navigate by maps. The most widely used ones group companies by what they primarily sell—semiconductors, electricity, insurance, pharmaceuticals. GICS, ICB, RBICS: each taxonomy differs in granularity and philosophy, but all share the same organizing principle.³ A company belongs where its main business belongs. The map is stable by design.

Stability is the virtue and the limitation.

Markets do not care what a company primarily sells. They care about what the company's future cash flows are exposed to. And those exposures can run across many sectors simultaneously.

Consider what building the compute infrastructure for generative AI actually requires. Advanced semiconductors: Information Technology. High-bandwidth memory: Information Technology. Networking equipment and optical interconnects: Information Technology and Communication Services. Data-center real estate: Real Estate. Cloud platforms: Information Technology and Communication Services. Baseload electricity: Utilities. Power distribution equipment, transformers, and switchgear: Industrials. Cooling systems and thermal management: Industrials and Energy. Grid interconnection and fiber: Utilities and Communication Services.

A sector map places all of these in different boxes and expects them to behave separately. During 2023 and 2024 they behaved as one thing—not because the companies were similar, but because they were all exposed to the same underlying capital expenditure cycle. The hyperscaler buildout was a single story moving through multiple industries simultaneously, repricing assets in each one at different speeds.

SK Hynix is instructive. By industry classification, it is a memory chip company inside Information Technology. Its products are a commodity. Its cycle is notoriously brutal. An analyst covering memory semiconductors would have evaluated it against DRAM pricing trends and Samsung's production decisions. But inside the AI theme, SK Hynix occupied a specific position: it was one of the few companies that could manufacture the high-bandwidth memory stacks that Nvidia's H100 accelerators required. When the industry hit an HBM supply bottleneck in 2023, SK Hynix was not a memory stock caught in a commodity cycle. It was a chokepoint in the most important supply chain in technology. The company's story had changed even though the company had not.

The standard map had no language for this. The thematic map had a name for it: a value-chain bottleneck in the AI infrastructure buildout.

1.3 What a Theme Actually Is

The word “theme” appears in hundreds of ETF names, but the definitions behind those names are often circular or purely descriptive. A fund is thematic because it has a theme; the theme is whatever the fund holds. That circularity is commercially convenient and analytically useless.

In this book a theme means something specific: a transient, cross-sectoral exposure created by a shared narrative, structural shift, or value-chain relationship. It links companies that may not belong to the same sector or share the same factor characteristics, but whose cash flows, risks, or investor demand are affected by the same developing story.

Three words in that definition carry weight.

³GICS, jointly developed by MSCI and S&P Dow Jones Indices, is one of the dominant global sector classification frameworks [83].

Transient. A sector can be stable for decades. A style factor earns its premium across long historical samples. A theme has a lifecycle. It emerges, accelerates, becomes validated or rejected, attracts capital, becomes crowded or institutionalized, and eventually decays or mutates into something else. The AI infrastructure theme that mattered in 2024 was not the same narrative as the cloud buildout theme of 2015, even though both involved data centers and semiconductor demand. The beneficiaries were different. The bottlenecks were different. The lifecycle position was different.

Cross-sectoral. Themes by definition cut across the boundaries that sector classifications draw. A theme concentrated in a single sector is usually just a sector bet with extra steps. The analytical value of thematic thinking is precisely in connecting companies that conventional tools would not place together—the transformer manufacturer, the memory chip supplier, the nuclear plant operator—because the same story touches all of them.

Exposure. A theme is not merely a topic in the news. A news cluster about AI regulation is a topic. It becomes an investment theme only when it connects to firms whose economics are materially affected. The distinction between topic and investment theme is where most thematic products fail. ETF providers launch funds around topics that are easy to describe and sell. Rigorous thematic investing asks whether the companies in the basket are genuinely exposed or just loudly adjacent.

	Sector	Factor	Theme
Defined by	Primary business activity	Return characteristic	Narrative exposure
Horizon	Decades	Persistent	Months to years
Cross-sectoral	No	Yes	Yes
Changes over time	Slowly	No	Yes, by design
Example	Utilities	Momentum	AI infrastructure

The table above is more than a taxonomy. It is a warning. Because themes share the cross-sectoral, time-varying character of factors, thematic baskets often become disguised factor bets. An AI infrastructure basket may look thematic and behave like a bundle of large-cap growth, high-momentum, high-beta technology exposure. An investor paying for thematic insight may be buying a more expensive version of a growth index. Disentangling genuine theme return from ordinary factor exposure is not a technical nicety. It is the entire problem.

1.4 The Promise and the Trap

The appeal of thematic investing is genuine. It can identify economic connections that standard tools miss. A company supplying electrical switchgear to hyperscale facilities may appear unremarkable under any factor or sector lens until power infrastructure becomes the binding constraint in an AI buildout. When that constraint tightens, the switchgear company begins behaving like a key node in a pressurized supply chain. An investor who mapped the value chain before the repricing captures something that a passive sector allocation cannot.

Themes also help explain market behavior that looks anomalous under conventional maps. When a water treatment company whose largest customer is a data-center campus operator outperforms its sector peers without any apparent news, the theme is often the explanation—and the sector analyst covering water treatment stocks will not find it.

The trap is harder to see while you are inside it.

Themes are most legible after they have worked. The narrative that explains Vistra’s 2024 performance is obvious in hindsight: nuclear baseload, hyperscaler power demand, Microsoft’s announcement. Constructing that thesis in January 2024, before the announcement, from scattered data-center power demand projections and utility capacity maps, is a different exercise entirely. The skill is not in identifying which themes drove returns last year. It is in detecting which themes are accumulating early evidence now, before capital has moved.

The commercial incentives run in the wrong direction. A fund provider building a thematic product benefits from a narrative that is already understood, a track record that already exists, and a story that investors have already heard. The fund launches when the theme is visible. The investors who need the product most—the ones who could benefit from early positioning—are the ones least likely to receive it. **By the time a thematic ETF reaches the market, the founding observation has often been priced.**⁴

Systematic thematic trading is the attempt to reverse this dynamic: detect narratives earlier, validate them more rigorously, map them to companies before the obvious beneficiaries are crowded, and exit before the lifecycle peak. That is harder than it sounds and more tractable than it seems. The rest of this book is the technical argument for why.

1.5 A Running Case: **AI Infrastructure**

One example serves as the book’s running case study: artificial intelligence infrastructure.

The choice is deliberate. AI infrastructure is not a single company or a single industry. It is a **cross-sectoral** narrative about the **physical**, **electrical**, and **computational** requirements of building and operating large-scale AI systems at hyperscale. That narrative branches into distinct sub-themes, each with its own set of beneficiaries, timeline, and economic mechanism:

- advanced graphics processors and AI accelerators;
- high-bandwidth memory and advanced semiconductor packaging;
- networking equipment, switches, and optical interconnects;
- data-center real estate and colocation operators;
- baseload electricity generation, including nuclear capacity;
- power distribution, transformers, and switchgear;
- cooling systems, liquid cooling infrastructure, and thermal management;
- cloud platforms and enterprise software adopters;
- firms whose labor-intensive models face automation pressure.

Each sub-theme has a different lifecycle, different pure-play beneficiaries, different second-order connections, and different crowding dynamics. High-bandwidth memory became a bottleneck before most investors understood what HBM was. Power and cooling became the dominant sub-theme roughly eighteen months after GPU supply did. The narrative evolved, split, and recombined as each bottleneck resolved and the next one emerged.

⁴Morningstar’s thematic fund guide discusses the growth of thematic products and the investor challenge of selecting durable themes rather than late-cycle narratives [81].

This is not unique to AI. Every major theme—**clean energy**, **obesity drug supply chains**, **defense technology**, **battery materials**, **reshoring**—follows a similar pattern of emergence, branching, crowding, and decay. AI infrastructure is a useful teaching case precisely because it is recent enough to study with current data, cross-sectoral enough to illustrate every methodological challenge, and economically grounded enough to separate real exposure from marketing language.

The case study appears in this order through the book: first as a **market narrative**, then as a text signal, then as a discovered **topic cluster**, then as a value-chain map, then as a company exposure scoring problem, then as a factor-neutral basket, then as a backtesting and governance challenge. The goal is not to prescribe how to trade AI stocks. The goal is to use a concrete, well-documented example to show how a body of text becomes an investment process.

1.6 What a Better Map Must Do

The GICS framework divides the equity market into eleven sectors and twenty-five industry groups. It has changed relatively little since its introduction in 1999. In those twenty-five years, cloud computing, streaming media, electric vehicles, genomic medicine, and large-scale AI infrastructure have all emerged as major sources of return and risk. None of them fit cleanly into the existing boxes; each one stretched or straddled the boundaries the framework was designed to enforce.

Sector classification were built to be stable, comprehensive, and comparable. They do those things well. **They are poorly suited to detecting transient cross-sectoral narratives**, tracking their emergence and decay, or isolating their portfolio contribution from ordinary market exposures.

A **thematic map** must do four things a sector map cannot.

Semantic. It must understand that **data centers, GPUs, electrical transformers, high-bandwidth memory, and liquid cooling systems** belong to the **same investment story even when they appear in different filing sections**, different industry reports, and different analyst coverage universes. This requires language models and topic discovery—but the requirement precedes the technology. The map must organize by **economic meaning, not by vocabulary**.

Temporal. **Themes have lifecycles.** A classification that correctly described the AI infrastructure opportunity in early 2023 required updating by late 2024, when the power and cooling sub-themes had matured into their own crowded trades. The map must know not only what the theme is but where it sits in its lifecycle: weak signal, emerging, validated, crowded, institutionalized, decaying. Static snapshots produce static errors.

Economic. Words are not alpha. A topic model can discover a cluster of documents about AI chips. That discovery is not a portfolio. The theme must be linked to measurable economic exposure—revenue, supply-chain relationships, capital expenditure commitments, customer concentration, analyst estimate sensitivity—before it qualifies as a trading thesis. **A company that mentions AI frequently but earns no material revenue from AI demand is a marketing exercise, not a thematic investment.**

Risk-aware. The purpose of thematic investing is not to own what is popular. It is to isolate exposure to a specific narrative driver after removing the market, sector, and factor exposure that an investor could have obtained more cheaply. A thematic basket that earns 30 percent while explaining 28 percentage points through beta and growth exposure is not a triumph of thematic research. It is an expensive replica of a growth index.

At the end of this book, the map should be able to do what the sector classification could not do in the summer of 2024: place Vistra Energy next to Nvidia—not because they were similar

businesses, but because the same story was pricing both of them.

Chapter 2

Returns, Risk, and the Language of Performance

2.1 The Number That Isn't Enough

The following example is a composite drawn from multiple strategy reviews, not any single audited product. A thematic basket of semiconductor and cloud infrastructure stocks returned 23 percent over a twelve-month period. The manager presenting it to investors was confident. The number was accurate. The auditors had signed off on it.

Then the questions started. Twenty-three percent compared to what? The relevant sector benchmark had returned 31 percent over the same period. The basket's volatility was nearly double the benchmark. It had spent six weeks underwater mid-period before recovering. Monthly rebalancing had cost an estimated 1.4 percentage points in bid-ask spreads and market impact. The liquidity of the underlying names meant the strategy was practically unscalable beyond \$200 million without the basket moving prices against itself.

A single accurate number had told an almost entirely misleading story.

This is the discipline that precedes everything else in systematic investing. Before discussing topic models, sentiment signals, exposure scoring, or factor neutralization, we need a shared vocabulary for what it means to evaluate a strategy. Return is one word in that vocabulary. It is not the sentence.

2.2 Simple and Log Returns

Gains and losses are not symmetric. A strategy that rises 50 percent and then falls 50 percent does not return to its starting point. It leaves the investor with 75 cents on the dollar. This asymmetry — obvious once stated, habitually forgotten in practice — is the first reason to be precise about how returns are calculated.

Let P_t be the total-return adjusted price of an asset at time t , incorporating dividends and corporate actions. The simple return from $t - 1$ to t is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \tag{2.1}$$

Simple returns compound multiplicatively. The total return over T periods is

$$1 + R_{0,T} = \prod_{t=1}^T (1 + R_t). \quad (2.2)$$

Multiplicative compounding creates the asymmetry. A gain of x percent requires a subsequent loss of $x/(1+x)$ percent to return to the starting point, not a loss of x percent. At large moves — the kind thematic strategies routinely produce at peak and trough — the difference is substantial.

When comparing strategies with different track-record lengths, convert total return R_{total} over N years into an annualized compound rate:

$$R_{\text{ann}} = (1 + R_{\text{total}})^{1/N} - 1. \quad (2.3)$$

This is not the arithmetic average of annual returns. It is the per-year rate that, compounded for N years, produces the observed total return. A three-year strategy and a ten-year strategy reporting the same annualized return are making equivalent compound claims over any common horizon.

Log returns are defined as

$$r_t = \log(P_t) - \log(P_{t-1}) = \log(1 + R_t). \quad (2.4)$$

Their main convenience: they add through time.

$$r_{0,T} = \sum_{t=1}^T r_t. \quad (2.5)$$

For small daily moves, simple and log returns are nearly identical. For large moves, leveraged portfolios, or long evaluation periods, the gap is meaningful. Thematic strategies experience both: extended periods of unremarkable returns punctuated by sharp rallies and sudden drawdowns. A research system should be explicit about which return definition it uses and should not switch definitions mid-evaluation.

2.3 Excess Returns and the Rate Environment

In 2022, the Federal Reserve raised policy rates by approximately 425 basis points in less than a year.¹ Long-duration growth stocks — precisely the kind of company that dominates most AI and clean technology baskets — fell 40 to 70 percent in some cases. A portfolio manager who had described her strategy as a pure AI infrastructure play discovered, in real time, that it was also a significant bet on a zero-rate environment.

Raw returns had not revealed this. Excess returns would have raised the question earlier.

The excess return over a period is the return above the risk-free rate available to investors:

$$R_t^e = R_t - R_{f,t}. \quad (2.6)$$

When the risk-free rate is near zero, as it was from 2009 to 2021, simple returns and excess returns are nearly the same. The distinction seems academic. When the risk-free rate reaches 5 percent, an investor can earn that return with minimal market volatility and no thematic exposure. Any strategy must clear that bar before it deserves the additional risk.

¹The federal funds target range moved from 0–0.25% in early March 2022 to 4.25–4.50% by mid-December 2022, an increase of roughly 425 basis points across the seven FOMC meetings of that year [43].

For thematic investing, the rate environment matters at a deeper level. Themes concentrated in capital-intensive, high-growth, long-duration companies — AI infrastructure, clean energy, biotech, space — are especially sensitive to discount rates. The same narrative that drove extraordinary raw returns in a zero-rate era can produce large negative excess returns when rates normalize. Comparing a strategy's performance across rate regimes requires excess returns, not raw returns.

2.4 Portfolio Returns and Exposure

A portfolio return is the weighted average of its constituents:

$$R_{p,t} = \sum_{i=1}^N w_{i,t-1} R_{i,t}, \quad (2.7)$$

where $w_{i,t-1}$ is the weight of asset i at the start of the period. For a long-only portfolio, weights are non-negative and sum to one. For a long-short portfolio, some weights are negative.

Two summary statistics describe a long-short portfolio's structure. Gross exposure is the sum of absolute weights:

$$\text{Gross Exposure} = \sum_{i=1}^N |w_i|. \quad (2.8)$$

Net exposure is the algebraic sum:

$$\text{Net Exposure} = \sum_{i=1}^N w_i. \quad (2.9)$$

A dollar-neutral long-short portfolio — long \$1, short \$1 for every dollar of capital — has net exposure near zero and gross exposure near two. It is not risk-free. The longs can fall, the shorts can rise, borrow costs can increase, and correlations between the long and short legs can collapse at precisely the wrong moment.

This matters for thematic research because many long-short thematic strategies claim to be market-neutral by construction. Whether they are neutral in practice depends on the actual factor loadings of the long and short books, not on the net weight. An AI infrastructure long book paired with a broad market short book may look dollar-neutral and carry substantial residual sector and factor risk. Net exposure is a starting point, not a complete risk description.

2.5 Volatility: The Price of the Ride

Two portfolios that each earn 15 percent per year are not the same investment if one does so with 8 percent annual volatility and the other with 35 percent. The first delivers steady compounding. The second delivers years of stomach-churning variance around the same average. For most investors, the ride is part of the result.

If $\bar{R}$ is the average return over T periods, sample variance is

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2, \quad (2.10)$$

and volatility is $\hat{\sigma}$. Daily volatility is conventionally annualized by scaling with the square root of the number of trading days per year:

$$\sigma_{\text{ann}} = \sigma_{\text{daily}} \sqrt{252}. \quad (2.11)$$

This scaling assumes independent daily returns — an approximation that holds reasonably well for diversified portfolios but becomes less reliable for concentrated thematic positions where a single piece of news can produce several days of correlated movement.

Volatility is also incomplete as a risk measure. It does not distinguish between upside and downside dispersion. It does not capture tail events, liquidity risk, crowding risk, or model error. It treats a 10 percent gain and a 10 percent loss as equivalent contributors to risk. For thematic portfolios, which tend to produce skewed return distributions, higher-order statistics matter. Still, volatility is the entry point because it enters every standard performance ratio and risk model.

2.6 Correlation and the Illusion of Diversification

A thematic basket covering AI infrastructure may hold twenty companies across six sectors. On paper, this looks diversified. In practice, those twenty companies may all respond to the same news: a hyperscaler revising its capital expenditure guidance, a regulatory ruling on semiconductor exports, a change in wholesale electricity prices, or simply a shift in risk appetite that depresses everything connected to the AI trade simultaneously.

Diversification is not a property of how many companies a portfolio holds. It is a property of how those companies move relative to each other.

The covariance between two assets i and j is

$$\text{Cov}(R_i, R_j) = \mathbb{E}[(R_i - \mu_i)(R_j - \mu_j)]. \quad (2.12)$$

Correlation rescales covariance to a bounded measure between -1 and 1 :

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\sigma_i \sigma_j}. \quad (2.13)$$

Portfolio variance, given a weight vector w and covariance matrix Σ , is

$$\sigma_p^2 = w^\top \Sigma w. \quad (2.14)$$

This expression captures the essential fact: portfolio risk is not an average of position risks. It depends on how positions move together. When the off-diagonal entries of Σ are large — when holdings are highly correlated — portfolio variance can approach the average individual variance with no diversification benefit at all.

Themes create a specific correlation hazard. A theme's value is that it connects companies through a shared narrative. But that same connection creates shared return behavior, especially during periods of market stress. In a crowded exit — when sentiment toward AI infrastructure reverses and institutional holders reduce exposure simultaneously — semiconductor firms, data-center REITs, utilities, and industrial equipment companies can all fall together. A conventional sector-level covariance model, which would treat these as separate exposures, will underestimate the drawdown risk. The thematic correlation is real and must be modeled.

2.7 Drawdown: The Test No One Wants to Take

Between its February 2021 peak and its December 2022 trough, the ARK Innovation ETF lost more than 75 percent of its value.² The fund had produced extraordinary returns in 2020. Many investors who observed those returns bought in near the top. The subsequent experience was not a test of strategy quality. It was a test of whether investors could hold through a drawdown of a magnitude that most professional risk frameworks would classify as catastrophic.

Drawdown measures the gap between current wealth and the most recent peak. Let W_t be cumulative portfolio value, and let

$$M_t = \max_{s \leq t} W_s \quad (2.15)$$

be the running maximum. Drawdown at time t is

$$D_t = \frac{W_t - M_t}{M_t}. \quad (2.16)$$

D_t is always zero or negative; it measures how far below peak the portfolio stands at any given moment. Maximum drawdown is the most negative value of D_t over the evaluation period.

Maximum drawdown matters for thematic strategies for two reasons. First, themes are prone to crowded entries and chaotic exits. A narrative that attracts significant capital inflows tends to overshoot on the upside; when the narrative weakens, the exit can be equally disorderly. Second, investors experience returns sequentially. A 50 percent loss requires a 100 percent gain to recover. Strategies with high average returns but severe drawdowns are often impossible to hold through the stress period, which means investors typically realize the losses but not the subsequent recovery. A thematic strategy's maximum drawdown is therefore a more honest description of investor experience than its long-run average return.

2.8 Turnover, Liquidity, and Capacity

Thematic narratives evolve. Companies enter and exit the investable universe as their exposure scores change, as new beneficiaries become apparent, or as old ones become crowded. This dynamism is a feature — it is what allows a systematic thematic process to adapt to a changing narrative — and it is a cost.

Turnover measures portfolio trading intensity. A one-period turnover measure is

$$\text{Turnover}_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - w_{i,t-1}^+|, \quad (2.17)$$

where $w_{i,t-1}^+$ is the weight after price movement but before rebalancing. The factor of 1/2 avoids double-counting buys and sells in a long-only portfolio. Annual turnover above 100 percent means the portfolio is trading its entire book once per year; 200 percent means twice.

High turnover is expensive. Transaction costs include commissions, bid-ask spreads, market impact, financing costs for leveraged positions, and borrow fees for short positions. Each is small in isolation; together they compound into a material drag. A thematic strategy that earns

²Axios reported in October 2022 that ARK Innovation ETF was nearly 80 percent below its February 2021 peak [86].

4 percent of gross alpha before costs and turns over 200 percent annually may earn nothing after costs if the underlying names are illiquid.

Liquidity is the more binding constraint in thematic investing. Pure-play thematic names — the companies most directly exposed to a narrative — tend to be smaller, less liquid, and more volatile than the broad market. A basket that looks attractive in a spreadsheet may be practically unmanageable at institutional size. If the strategy's target allocation requires trading 30 percent of a name's average daily volume at each rebalance, the portfolio will move prices against itself. What looked like a paper alpha will become execution losses.

Capacity is the maximum capital a strategy can deploy before its own trading materially reduces its expected return. Capacity should be reported as a range or a curve, not a single number. A strategy may be viable at \$100 million and marginal at \$500 million. An institution that deploys \$2 billion into a strategy designed for \$200 million has not scaled the strategy. It has changed it.

For AI infrastructure specifically: the most thematically pure companies in the early buildout — specialized cooling equipment manufacturers, advanced packaging foundries, niche optical interconnect suppliers — were often small-cap and thinly traded. A thematic system that identified them early created both an opportunity and an implementation challenge that a spreadsheet return would not reveal.

2.9 A Minimal Performance Dashboard

For any thematic basket, the first evaluation report should answer at least the following questions. A strategy that cannot produce these numbers is not ready for capital allocation.

- Cumulative return, annualized compound return, and cumulative excess return over the risk-free rate.
- Annualized volatility, separately for up-market and down-market periods where sample size permits.
- Maximum drawdown, duration of the drawdown, and time to recovery.
- Turnover rate and estimated total transaction costs at the intended capital level.
- Benchmark-relative return against at least two benchmarks: a broad market index and a relevant sector or style index.
- Market beta and primary factor loadings, estimated over the full period and separately for the last twelve months.
- Sector and industry weight distribution.
- Concentration statistics: top-five and top-ten holdings as a share of total portfolio.
- Liquidity statistics: percentage of the portfolio that can be liquidated within one day, three days, and ten days at 20 percent of average daily volume.
- Capacity estimate under conservative market-impact assumptions.

This dashboard does not prove that a theme has alpha. It prevents the most common failures. It exposes whether a claimed thematic return is actually a market bet, a sector bet, a factor bet, a liquidity bet, or a small-cap premium that disappears once transaction costs are modeled.

The AI infrastructure basket introduced in Chapter 1 will reappear throughout the book with this dashboard applied at each stage. A theme that cannot survive a minimal performance dashboard has not earned a position. It has earned a question.

2.10 A Note on the AI Infrastructure Example

The questions this dashboard raises become concrete when applied to the AI infrastructure case. Suppose the basket returned 35 percent in a year when the technology sector returned 28 percent and the broad market returned 22 percent. Each comparison tells a different story. Against the broad market, the basket outperformed by 13 percentage points. Against the technology sector, the outperformance falls to 7 percentage points. Against a semiconductor index, it might disappear entirely.

Now add the risk dimension. If the basket carried annualized volatility of 28 percent during the same period — roughly twice the broad market's volatility — the Sharpe ratio shrinks considerably. If the maximum drawdown was 24 percent, and the basket recovered only because the broader technology rally continued, the manager may have generated no independent thematic alpha at all.

These are not rhetorical concerns. They are the specific questions that determine whether a systematic thematic process earns its complexity. The vocabulary here is the minimum equipment needed to ask them.

Chapter 3

Alpha, Beta, and Benchmarks

3.1 The Return That Proved Nothing

Between 2019 and 2021, a large share of actively managed equity funds beat the S&P 500. They had different investment philosophies, different research processes, and different portfolio construction rules. What many had in common was a tilt toward high-beta growth stocks: large technology platforms, fast-growing software companies, electric vehicle makers, cloud infrastructure providers. When the broad market returned 31 percent in 2019, 18 percent in 2020, and 29 percent in 2021,¹ their high-beta portfolios returned more.

Many of them called this alpha. Most of it was not.

The problem is not dishonesty. The problem is measurement. Without a reference model for what a portfolio is expected to return given its market exposure, outperformance is indistinguishable from systematic risk-taking. A portfolio with a market beta of 1.5 is structurally expected to outperform when markets rise and underperform when markets fall. That expectation is not skill. It is arithmetic.

The same problem runs through thematic investing. A basket of AI infrastructure companies will, in most environments, carry high market beta. It will tend to outperform in growth-driven bull markets and underperform in rate-driven bear markets. These outcomes are largely predictable from the beta alone. What is not predictable from beta alone — what has to be extracted as a residual — is the part of return genuinely attributable to identifying exposed companies before the market did.

Alpha is that residual. Finding it requires a model of what the portfolio was already expected to earn.

3.2 Market Beta

A stock with a beta of 1.5 is expected to fall 15 percent when the market falls 10 percent. It is also expected to rise 15 percent when the market rises 10 percent. This is not good or bad by itself. It is exposure. The investor is taking on amplified market risk; the question is whether that amplification is intentional, compensated, and separable from any additional insight the strategy claims to provide.

The market model relates an asset's excess return to the excess return of the market portfolio:

¹These figures refer to S&P 500 total returns, rounded to the nearest percentage point [103].

$$R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \epsilon_{i,t}, \quad (3.1)$$

where $R_{i,t}^e$ is the excess return of asset i , $R_{m,t}^e$ is the market excess return, β_i is market beta, α_i is the intercept, and $\epsilon_{i,t}$ is the residual return unexplained by the market.

Beta is defined as

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}. \quad (3.2)$$

An asset with $\beta > 1$ amplifies market moves. An asset with $\beta < 1$ dampens them. A portfolio with $\beta = 1.4$ will, on average, rise 1.4 percent for every 1 percent the market rises. If the market has a good year, the portfolio benefits more. If the market has a bad year, the portfolio suffers more. The investor is not getting something for nothing. The extra return in bull years is the fair price for the extra pain in bear years.

Thematic portfolios frequently have high market beta. The most thematically pure AI infrastructure companies — accelerator designers, high-bandwidth memory manufacturers, liquid cooling specialists — tend to be high-growth, high-valuation, high-volatility stocks. In the language of the market model, their excess returns are largely driven by $\beta_i R_{m,t}^e$. The α_i term, the part the fund manager wants to be large and positive, may be small or even negative once the beta is accounted for.

3.3 What the CAPM Actually Says

William Sharpe published what became the Capital Asset Pricing Model in 1964 [95]. It remains one of the most contested ideas in finance. Whether or not one accepts its assumptions, its practical value for thematic investing is clear: it provides a minimum benchmark for the return an investor should have expected, given the risk taken.

The CAPM gives expected return as a linear function of market beta:

$$\mathbb{E}[R_i] = R_f + \beta_i(\mathbb{E}[R_m] - R_f). \quad (3.3)$$

The term $\mathbb{E}[R_m] - R_f$ is the equity risk premium — the extra return the market delivers above the risk-free rate over long periods, as compensation for bearing systematic risk. The CAPM says that the expected excess return on any asset is simply its beta multiplied by that premium. An asset with beta of 1.5 should earn 1.5 times the market's excess return. Nothing more; nothing less.

Jensen's alpha is the gap between what an asset actually earned and what the CAPM predicted it should earn [68]:

$$\alpha_i = \mathbb{E}[R_i - R_f] - \beta_i \mathbb{E}[R_m - R_f]. \quad (3.4)$$

In empirical practice, alpha is estimated as the intercept from a time-series regression of excess returns on market excess returns. A positive estimated alpha says that the asset or strategy earned more than its market exposure would predict. A negative alpha says it earned less.

The interpretation requires care. A positive estimated alpha is not automatically evidence of skill. It could be sample noise — a lucky period. It could be survivorship bias — the strategy is visible precisely because it worked. It could be omitted risk factors — the strategy was exposed

to size, value, or momentum, and those factors happened to perform well. It could be illiquidity premium — the strategy held assets that were hard to trade and compensated for that difficulty. And it could be genuine insight, but distinguishing that from the other explanations requires far more than a single alpha estimate.

The CAPM is not used in this book because it is believed to be a complete theory of asset pricing. It is used because it is the simplest and most transparent version of a critical question: did the strategy earn more than its systematic exposure would predict?

3.4 Choosing the Right Benchmark

In 2022, a hypothetical AI infrastructure basket might have returned 8 percent. Against the broad market, which fell roughly 18 percent that year, the basket outperformed significantly. Against the Nasdaq-100, which fell roughly 33 percent that year, it also outperformed. Against a narrow index of pure-play semiconductor companies, which fell more than 35 percent, it outperformed considerably more.

None of those comparisons is the right one in isolation. Each tests a different claim.

A benchmark is the reference portfolio against which performance is measured. Different benchmarks test different hypotheses:

- Against the broad market: does the theme improve total equity allocation?
- Against a sector index: does the theme add value beyond a conventional sector bet?
- Against a style index: does the theme add value beyond growth, momentum, or quality exposure?
- Against a thematic ETF or peer basket: does the systematic process improve on existing industry practice?

Benchmark choice is not neutral. A strategy evaluated against the easiest benchmark looks better than one evaluated against the hardest. Credible thematic research should report performance against multiple benchmarks. A strategy that survives only the easiest comparison is fragile. One that outperforms across several comparisons is more interesting.

For thematic investing, a particularly useful benchmark is a portfolio matched by sector, size, and liquidity — one that holds similar companies in similar proportions but without the thematic selection signal. If the thematic strategy cannot beat that matched benchmark, the signal may be selecting companies that are already captured by more conventional approaches.

3.5 Tracking Error and the Information Ratio

Two strategies both earn 2 percentage points of annual alpha above their benchmark. One delivers it every year: 2.1 percent, 1.9 percent, 2.0 percent. The other achieves the same average with high variance: 18 percent in year one, negative 12 percent in year two, 2 percent in year three. Both strategies average 2 percent. Only one is useful.

Benchmark-relative return — called active return — is the difference between portfolio return and benchmark return:

$$A_t = R_{p,t} - R_{b,t}. \quad (3.5)$$

Tracking error is the volatility of active return:

$$\text{TE} = \sigma(A_t). \quad (3.6)$$

The information ratio divides average active return by tracking error:

$$\text{IR} = \frac{\mathbb{E}[A_t]}{\sigma(A_t)}. \quad (3.7)$$

The information ratio is the primary performance metric for active strategies evaluated against a benchmark [50]. It asks how reliably the manager generates active return per unit of active risk. A high average active return with low tracking error produces a high information ratio. A high average active return with erratic tracking error produces a low one.

For thematic investing, tracking error has a natural interpretation. A thematic basket departs from the benchmark in concentrated, narrative-driven ways. During the theme's active period, it should generate positive active return. During regime shifts, rate changes, or narrative reversals, it may generate negative active return. The information ratio asks whether the positive periods consistently outweigh the negative ones, controlling for the magnitude of the bets.

A thematic strategy with an information ratio above 0.5 over a meaningful sample is performing well. Above 1.0 sustained over several years would be exceptional. Most strategies that survive independent scrutiny fall somewhere between 0.3 and 0.7.

3.6 Sharpe Ratio

A long-only thematic fund and a market-neutral thematic fund can both report an information ratio of 1.0 against their respective benchmarks. They can report wildly different Sharpe ratios, because their benchmarks carry different total risk. The two metrics measure different things.

The information ratio asks: did the strategy beat its benchmark reliably? The Sharpe ratio asks: was the strategy's total excess return worth its total risk [96]?

$$\text{Sharpe} = \frac{\mathbb{E}[R_p - R_f]}{\sigma(R_p - R_f)}. \quad (3.8)$$

The Sharpe ratio is most useful for total portfolio evaluation and for comparing strategies with different benchmarks or mandates. A long-short thematic strategy benchmarked against cash is better evaluated on Sharpe ratio than information ratio. A long-only thematic overlay benchmarked against an equity index is better evaluated on information ratio.

The two metrics are related through total risk, benchmark risk, active risk, and the correlation between the strategy and its benchmark — but the relationship depends on the specific strategy structure and is not a simple closed-form shortcut. In practice, the choice of metric matters less than the discipline: always report both, always calculate them using the same return series, and never claim performance without specifying which metric is being used and against what benchmark.

3.7 Alpha Is a Residual Concept

In 2020, virtually every investor who held growth stocks looked skilled. Some of them were skilled. Most were long the right factor at the right moment. When the Federal Reserve held

rates near zero and flooded markets with liquidity, high-duration growth companies outperformed broad indices by extraordinary margins. When rates rose in 2022, those same companies gave back large portions of their gains. The factor was doing the work both times.

The central lesson is not a formula. It is a disciplined habit of thought.

Alpha is what remains after explaining returns with the right model. If the model is too simple, alpha will be overstated. The CAPM uses one factor: the market. Outperformance against the CAPM could reflect genuine insight, but it could equally reflect size, value, momentum, profitability, or investment exposure that the single-factor model fails to capture. As the factor model expands to include these risks, estimated alpha shrinks.

For thematic portfolios, the relevant factors include market, sector, growth, momentum, size, liquidity, and sometimes currency and interest-rate sensitivity. If an AI infrastructure basket earns positive CAPM alpha, the next question is whether that alpha survives after controlling for the technology sector. If it survives the sector control, does it survive after removing the momentum factor? After removing the growth tilt? After accounting for the small-cap illiquidity premium that may be embedded in the pure-play names?

Each successive control is not an attack on the strategy. It is a sharpening of the claim. If alpha survives every reasonable control, the claim is stronger. If it disappears at the first factor adjustment, the strategy was factor exposure, not thematic insight.

Before declaring any performance result, the researcher should ask:

- What exposures explain the return?
- Are those exposures intentional, measured, and disclosed?
- Would a cheaper portfolio — a growth ETF, a sector fund, a factor tilt — have delivered a similar result?
- Does the residual return align with the theme’s stated economic thesis?
- Does it survive out-of-sample after transaction costs?

If these questions have good answers, the alpha is worth claiming. If they do not, the performance is a story in search of an explanation.

3.8 The AI Infrastructure Calculation

The numerical exercise below applies these concepts to the AI infrastructure basket directly. Suppose the basket produced the following annual results:

Measure	Basket	Benchmark
Return	35%	22%
Volatility	28%	16%
Market beta	1.45	1.00
Maximum drawdown	−24%	−12%

The basket outperformed the benchmark by 13 percentage points. That number is the first thing a manager presents. It is also the least informative number in the table.

Applying the CAPM with the risk-free rate approximated at zero for simplicity, the expected return on the basket given its beta of 1.45 is

$$1.45 \times 22\% = 31.9\%. \quad (3.9)$$

The implied CAPM alpha is therefore

$$35\% - 31.9\% = 3.1\%. \quad (3.10)$$

The raw outperformance was 13 percentage points. The CAPM alpha is 3.1 percentage points. The difference — nearly 10 percentage points — was compensation for bearing amplified market risk. An investor could have gotten close to that return by leveraging a broad index fund with the same beta. They did not need a thematic portfolio to do it.

The 3.1 percent CAPM alpha may still be real. But it needs to survive further scrutiny. If the basket was overweight technology and underweight defensives, sector exposure may explain some of it. If the basket held high-momentum names, momentum exposure may explain more. If the basket held small, illiquid names, the illiquidity premium may account for the rest.

This is not pessimism. It is clarity. A thematic strategy that earns genuine residual alpha after all reasonable factor controls has found something. The process of ruling out the alternatives is how the researcher knows what they found.

Chapter 4

Factors, Sectors, and the Contamination Problem

4.1 Different Names, Same Trade

By the end of 2024, an investor browsing US-listed exchange-traded funds could find more than a dozen products with “artificial intelligence,” “robotics,” or “machine learning” in their names.¹ They had different sponsors, different selection methodologies, and different fees. Their portfolios did not differ as much as their marketing did.

Many of the largest AI-themed funds held Nvidia. Many held Microsoft, Alphabet, Meta, Broadcom, and AMD. The selection rules differed in their text; the portfolios differed less than their names suggested. An investor allocating across several such funds, in pursuit of diversified AI exposure, was usually buying overlapping tilts on the same large-capitalization technology trade.

What a standard factor model would have shown about such a basket is also predictable. The market beta is high. The growth loading is large and positive. The momentum loading is large and positive. The size loading on SMB is negative or muted, depending on the benchmark — these are mega-cap names, not small caps. The value loading is negative — they trade at high multiples. Once those exposures have been accounted for, the residual return is often modest, sometimes negative, and rarely persistent across regimes.

This is the contamination problem. A basket built around a story can earn most of its return from exposures the investor could have purchased more cheaply elsewhere. The thematic story decorates the basket. Standard factors do most of the work.

Separating the two requires a baseline tougher than a sector label and richer than one market beta. The working tools are academic factor models that thematic strategies must clear before claiming alpha, a practical inventory of the tilts thematic baskets typically carry, and two methods for isolating what remains: residualize the exposure score before construction, then hedge factor exposures after the basket exists.

¹Morningstar’s thematic fund research provides a broad taxonomy of thematic fund categories and documents the rapid expansion of thematic fund offerings [81].

4.2 Why One Beta Is Not Enough

In the early 1990s, Eugene Fama and Kenneth French published a paper showing that two simple characteristics — a company’s size and its book-to-market ratio — explained cross-sectional differences in average stock returns that the CAPM did not [41]. Cheap companies outperformed expensive ones, on average, over the sample they studied. Small companies outperformed large ones. Neither pattern was captured by market beta.

The implication was uncomfortable for active management. A manager whose portfolio was structurally tilted toward small, cheap stocks would have been expected to beat a market index, on average, just by sitting still. Calling that outperformance alpha was a measurement error.

For thematic portfolios, the problem is structural rather than incidental. A basket built around a fashionable theme is rarely neutral on these characteristics. AI infrastructure baskets tend to lean large and growth. Clean energy baskets tend to lean small and growth. Biotech baskets tend to lean small and unprofitable. Defense baskets lean toward large industrials with predictable revenue. These tilts are not accidents of construction. They are properties of the universes the themes select from. And they explain a great deal of what looks, at first, like skill.

4.3 Style Factors

By the early 2000s, dozens of return patterns had been documented in academic finance. A smaller subset proved robust enough across samples and methodologies to be priced into commercial risk models and routinely used in performance attribution.

A style factor is a systematic return pattern associated with a measurable company characteristic. The Fama–French three-factor model adds size and value to the market factor:

$$R_{p,t}^e = \alpha_p + \beta_{\text{MKT}}\text{MKT}_t + \beta_{\text{SMB}}\text{SMB}_t + \beta_{\text{HML}}\text{HML}_t + \epsilon_t. \quad (4.1)$$

Here MKT is the market excess return, SMB (“small minus big”) is the return of a portfolio long small-cap stocks and short large-cap stocks, and HML (“high minus low”) is the return of a portfolio long high book-to-market stocks and short low book-to-market stocks. The intercept α_p is the part of average return left unexplained by these three exposures.

The five-factor extension adds profitability and investment [42]:

$$R_{p,t}^e = \alpha_p + \beta_{\text{MKT}}\text{MKT}_t + \beta_{\text{SMB}}\text{SMB}_t + \beta_{\text{HML}}\text{HML}_t + \beta_{\text{RMW}}\text{RMW}_t + \beta_{\text{CMA}}\text{CMA}_t + \epsilon_t, \quad (4.2)$$

where RMW (“robust minus weak”) is the profitability factor and CMA (“conservative minus aggressive”) is the investment factor.

Many practitioners also include momentum — the tendency of stocks that have outperformed over the past six to twelve months to keep outperforming over the next few [24].

A further widely tracked premium is low volatility: the empirical pattern that lower-volatility stocks have historically delivered higher risk-adjusted returns than the CAPM would predict. Quality factors, built around profitability, leverage, and earnings stability, behave similarly. These exposures matter for thematic research because narrative-heavy themes — AI infrastructure, clean energy, biotech — tend to load *against* low volatility and quality. The companies most directly exposed to an emerging story are usually among the most volatile and least profitable names in the market.

The generalized factor regression is:

$$R_{p,t}^e = \alpha_p + \sum_{k=1}^K \beta_k F_{k,t} + \epsilon_t, \quad (4.3)$$

where $F_{k,t}$ are factor returns. For the rest of this book, the working default is the Fama–French five factors plus momentum. This is not because it is the only defensible model. It is widely available, well-documented, and consistent with both the academic literature and the most common practitioner risk frameworks. For institutional users, commercial risk models from Barra, Axioma, or MSCI provide finer-grained industry and style decompositions and would replace the academic factors in serious portfolio attribution.

The choice of factors matters less than the discipline of having a model at all. Before calling a return thematic, explain as much as possible with known, cheaper, and more general sources of return. Whatever survives that effort is what is worth defending.

4.4 The Tilts That Themes Cannot Avoid

The iShares Global Clean Energy ETF (ticker ICLN) more than doubled during 2020. It peaked in early 2021 and then lost more than half of its value from peak to trough over the following two years.² The clean energy theme had not collapsed. Solar installation volumes kept growing, electric vehicle adoption accelerated, and government incentive programs in Europe and the United States expanded. What had changed was the interest-rate environment. Long-duration cash flows had been worth more when rates were near zero; they were worth less when ten-year yields climbed past 4 percent.

The ICLN drawdown was not an argument against clean energy. It was a duration story wearing a clean energy uniform.

Thematic baskets tend to share predictable factor tilts:

- Technology and AI themes often load on growth, momentum, profitability, and market beta.
- Clean energy themes often load on small-cap growth, high investment, policy sensitivity, and interest-rate duration.
- Biotech themes often load on idiosyncratic volatility, event risk, and financing conditions.
- Defense themes may load on geopolitical risk, government spending cycles, and large-cap industrials.
- Commodity-transition themes may load on inflation, energy prices, metals, and emerging-market demand.

These tilts are not necessarily problems. An investor who wants AI infrastructure exposure has implicitly accepted some growth and technology-sector exposure. An investor who wants clean-energy exposure may accept some duration, policy, and project-finance risk. But intentional is not the same as measured. A thematic basket must be analyzed for its factor loadings before capital is committed. If the loadings explain 70 or 80 percent of the variance in basket returns, the investor is buying a factor portfolio with a thematic label.

²BlackRock’s official ICLN fund page provides fund performance history, holdings, and index objective [12].

4.5 Sectors and Industries

A clean energy ETF prospectus may list utilities, industrials, materials, and technology as the constituent sectors. In risk terms, it can behave like a single concentrated bet on solar manufacturers and lithium producers. The taxonomy of holdings is one thing. The taxonomy of risk is another.

Sector and industry classifications are a second source of contamination. A theme may appear to be cross-sectoral but still be dominated by one sector. An AI infrastructure basket may include utilities, industrials, and real estate, but if 70 percent of risk comes from semiconductors, the basket behaves like a semiconductor portfolio.

Sector neutrality is one possible control. It requires the portfolio to match benchmark sector weights or to balance long and short positions within sectors. Industry neutrality is stricter. It can help isolate within-industry stock selection from broad sector moves.

Neutrality is not always appropriate. Some themes are genuinely concentrated in certain industries. A fully sector-neutral version of a theme may remove exactly the exposure the investor wanted. The question is not whether every thematic basket must be sector neutral. The question is whether the sector exposure is intentional, measured, and priced into the evaluation.

4.6 The Contamination Problem

A thematic strategy that posts 15 percentage points of annual outperformance over a broad equity index can lose nearly all of it once a six-factor model is fit to its monthly returns. The strategy did not stop working in the data. The data was always explaining itself with cheaper variables. The manager had simply not asked it to.

Factor contamination occurs when a claimed thematic return is actually explained by known factors or sectors. The problem is not only statistical. It is economic. If a strategy's return can be replicated by buying a cheap growth ETF, a semiconductor index, or a momentum factor, then the expensive thematic research process is selling a story whose payoff was already available off the shelf.

Contamination can enter at several stages:

- **Discovery:** a text **model finds topics** that **are actually company or sector** clusters.
- **Exposure scoring:** the highest-scoring companies are those that market themselves aggressively rather than those with true economic exposure.
- **Portfolio construction:** the weighting scheme concentrates in high-beta, high-momentum, or illiquid names.
- **Evaluation:** the backtest uses a benchmark that is too easy, making ordinary factor exposure look like alpha.

The solution is not to eliminate all factor exposure mechanically. The solution is to know what is being paid for. If the goal is pure **thematic alpha**, factor exposure must be neutralized or hedged. If the goal is strategic thematic allocation, factor exposure may be accepted but should still be reported. What cannot survive is the case where factor exposure is present, unmeasured, and described as something else.

4.7 Residualizing the Exposure Score

Two approaches exist for stripping factor contamination from a thematic process. The first cleans the inputs. The second cleans the outputs. They are complementary; serious systems use both.

The input-side approach residualizes the raw theme exposure score before building the portfolio. Let $Y_{\text{raw},i}$ be the raw exposure score for company i . This score may come from revenue mapping, semantic similarity, analyst classification, or a combination of signals. Let F_i be a vector of style characteristics (size, value, momentum, profitability, investment) and S_i a vector of sector or industry indicators. A cross-sectional regression of the score on these characteristics takes the form

$$Y_{\text{raw},i} = \gamma_0 + F_i^\top \theta + S_i^\top \delta + \eta_i. \quad (4.4)$$

The residual η_i is the part of the exposure score not explained by the chosen factors and sectors. A portfolio built from η_i rather than $Y_{\text{raw},i}$ is more likely to represent unusual thematic exposure rather than ordinary style or industry membership.

This procedure must be used carefully. If the theme is inherently tied to a sector — and many are — residualization may remove too much of the signal. But it is a powerful diagnostic. If the residualized signal has no predictive value, while the raw signal looks strong, the original result was probably contamination. The thematic process was selecting the same companies that a generic growth or momentum screen would have surfaced.

4.8 Hedging the Basket's Factor Exposures

The output-side approach builds the raw basket and then strips its factor exposures after the fact. If the basket excess return is $R_{p,t}^e$, estimate factor betas through a time-series regression:

$$R_{p,t}^e = \alpha_p + \sum_{k=1}^K \beta_k F_{k,t} + \epsilon_t. \quad (4.5)$$

The factor-hedged return is the residual after subtracting the estimated factor contribution:

$$R_{p,t}^{\text{hedged}} = R_{p,t}^e - \sum_{k=1}^K \hat{\beta}_k F_{k,t}. \quad (4.6)$$

This is an estimate of what remains after removing factor exposure. If the theme is economically real, the residual should still show meaningful behavior: return persistence, response to theme-specific news flow, or correlated movement among basket constituents that the factor model fails to absorb.

In practice, factor hedging is imperfect. Betas change through time and across regimes. Factor portfolios may be costly or unavailable to short. The hedge can introduce basis risk if the theoretical factor return differs from the tradeable instrument used to replicate it. The exercise is still essential: it is the most direct test of whether the theme survives a more demanding model than the single-factor market regression.

4.9 The Role of Residual Correlation

A coherent theme should leave a footprint in what standard models cannot explain. Suppose a standard risk model describes returns through market, factor, sector, and industry exposures:

$$r = Xb + u. \quad (4.7)$$

Traditional models often treat the residuals u as largely idiosyncratic — the part of each stock’s return that moves on company-specific news rather than on broad systematic forces. A coherent theme challenges that assumption. If firms exposed to the same narrative show *correlated* residual returns during the theme’s active period, the standard model is missing a transient common driver. The augmented model is

$$r = Xb + Yg + u', \quad (4.8)$$

where Y is a matrix of thematic exposures and g is the corresponding vector of thematic returns. The theme earns its place in the model because it explains structure in what would otherwise look like residual noise. This same equation reappears in Chapter 6 as the formal definition of an investment theme.

The implication is testable. A basket is not coherent simply because its name sounds compelling. It should generate measurable residual co-movement among its constituents during the theme’s active period. If a proposed theme produces no residual correlation, no return prediction, and no improvement in portfolio risk attribution, it is a marketing category rather than an investment category.

4.10 AI Infrastructure as a Contaminated Theme

AI infrastructure offers a concrete example. A raw basket constructed from the most narrative-relevant companies in late 2024 might include Nvidia, AMD, Broadcom, TSMC, ASML, SK Hynix, Micron, Arista Networks, Vertiv, Eaton, Constellation Energy, Vistra, and Digital Realty. Each is plausibly exposed. Each carries factor baggage.

A standard six-factor regression of this kind of basket against the Fama–French five factors plus momentum will almost certainly show:

- high and statistically significant market beta;
- large positive momentum exposure;
- negative value exposure;
- concentration in mega-cap technology and semiconductors that the size factor cannot fully absorb;
- sensitivity to interest rates through the long-duration valuations of the largest constituents;
- liquidity and crowding risk reflected in correlated drawdowns when institutional ownership unwinds.

If the basket’s outperformance disappears once these exposures are removed, the investor has not found clean thematic alpha. They have bought an expensive replication of factor exposure that more conventional vehicles could have delivered.

If the basket still outperforms after careful controls, the case for thematic alpha is materially stronger. The residual return might then reflect specific economic mechanisms that factor models do not capture: supply-chain bottlenecks in advanced packaging, underestimated power demand from hyperscaler buildouts, capital-expenditure announcements that propagate slowly through analyst estimates, or second-order beneficiaries whose exposure is not yet visible to sector screens.

Buying a popular AI ETF after a large run-up is one trade. Building a factor-aware basket of underrecognized AI infrastructure beneficiaries is a different trade. The first is thematic consumption. The second is thematic research. The chapters that follow are concerned with the second.

4.11 From Foundations to Narrative Systems

The four chapters of Part I set the bar for everything that follows. A serious narrative-aware system must detect a story before it has been priced, link that story to companies whose cash flows are genuinely exposed, construct a basket whose returns can be decomposed into market, factor, sector, and residual components, and produce a residual that survives transaction costs, capacity constraints, and out-of-sample evaluation.

A text-driven system that cannot beat a Fama–French five-factor model plus a sector index has not produced investment value. It has produced a more elaborate description of risks investors already knew how to buy.

Part II

Narratives, Themes, and Signals

Chapter 5

Narrative Economics and Financial Markets

5.1 A Story That Moved for Twenty Years

In December 1996, Robert Shiller presented research to the Federal Reserve Board arguing that the **US stock market** had become **irrationally exuberant**. Two days later, Federal Reserve Chairman Alan Greenspan used the phrase in a public address [98]. The market fell briefly, then resumed climbing for more than three additional years. From Greenspan’s December 5, 1996 speech to the Nasdaq’s March 10, 2000 peak, the S&P 500 rose roughly 90 percent and the Nasdaq Composite almost quadrupled.¹ Companies with no profits, sometimes no customers, sometimes no functioning products went public at multi-billion-dollar valuations. **The story — that the internet was a new industrial revolution, that ordinary valuation methods no longer applied, that traditional analysts were rear-view-mirror thinkers** — was so coherent and so widely repeated that contradictory facts were absorbed into it rather than counted against it.

When the bubble finally broke in March 2000, the narrative had not collapsed because it was disproved. Many of its specific claims about the long-run economic impact of the internet were correct. The narrative broke because too much capital had front-loaded the future, leaving no margin for any company to disappoint. The story was substantially right. The prices were not.

This is the central problem of narrative economics: how do shared stories move markets for years, even when credible voices have identified them as fragile? Shiller’s answer, developed across decades and most fully in his 2017 American Economic Association presidential address [99], is that economic actors do not respond only to balance sheets, discount rates, and formal forecasts. **They respond to stories**. Those stories spread, mutate, attract capital, become reflexive, and eventually decay — on their own clock, not the clock of fundamentals.

For thematic investing, this is not a philosophical observation. It is the operating environment. Every tradable theme is, at some stage of its life, a story being processed by a population of investors with different beliefs, different attention budgets, and different reaction speeds. The system that wins understands where in that processing cycle the story currently sits.

¹Approximate price-index changes calculated from FRED daily levels for the S&P 500 and Nasdaq Composite [45, 44].

5.2 Markets as Story-Processing Systems

By mid-2023, the same cloud capital-expenditure line that investors once treated as a margin problem had become evidence of AI capacity scarcity. A spending increase had changed genres.

Financial markets process information, but they do not process it as a neutral machine. Investors interpret information through stories: compact causal explanations that connect facts, expectations, fears, and incentives. A revenue surprise can be read as evidence of temporary demand, durable product-market fit, management quality, or a new industry cycle. The same fact creates different price reactions depending on the narrative that surrounds it.

Some narratives compress real structural change. The AI infrastructure narrative is grounded in measurable capital expenditure, supply constraints, power demand, and shifts in software adoption. Other narratives are mostly extrapolation, marketing, or speculative enthusiasm. The task is not to dismiss stories. It is to separate economically grounded narratives from contagious but fragile ones — and to recognize that the same narrative can be both at different points in its life.

Narratives also coordinate attention. They tell investors what to watch, which companies matter, which risks are salient, and which future states feel plausible. A coordinated audience is a moving audience. When a narrative becomes dominant, prices begin to respond to new facts in compressed time. The story has done the analytical work in advance; the market only needs to confirm or reject specific data points against the existing frame.

5.3 Narratives and Expectations

On September 20, 2024, Constellation Energy announced a twenty-year power purchase agreement with Microsoft to restart Three Mile Island Unit 1 [28]. Almost immediately, investors began asking what other nuclear assets could serve the same demand. Broker research and price action shifted from conventional utility metrics toward nuclear scarcity, data-center load growth, and long-term offtake. The companies had not changed. The narrative that explained their assets had.

The price of an equity reflects expectations about future cash flows and discount rates. Narratives affect both. A story about accelerated AI adoption raises expected revenues for semiconductor firms, cloud platforms, and data-center suppliers. The same story may change discount rates by increasing uncertainty, crowding, or duration sensitivity. A story about regulatory pressure may lower expected cash flows, increase risk premia, or both.

The mechanism is expectation revision. Investors rarely observe the future directly. They infer it from signals: earnings releases, management commentary, customer demand, macro data, news flow, peer behavior, and price action. Narratives organize those signals. Once a narrative becomes dominant, new facts are interpreted through it. A data-center power shortage becomes an AI infrastructure bottleneck. A semiconductor order backlog becomes evidence of durable accelerator demand. A cloud capex increase becomes validation of the entire value chain.

The same mechanism can reverse. When a dominant story weakens, facts that previously confirmed it may be reinterpreted. A capex increase becomes overbuilding. High margins become cyclically unsustainable. Customer enthusiasm becomes pulled-forward demand. Narrative shifts can therefore move prices before fundamental numbers change in any measurable way.

5.4 Transmission and Contagion

A market story is rarely born fully formed in the financial press. It begins somewhere narrower — a research note, a podcast, a technical Twitter thread, a specialist trade publication — and travels. Each stage changes the audience and the market impact.

The AI infrastructure story is a useful trace. **The original observation was technical: training a large language model requires enormous amounts of GPU compute.** That observation moved into industry coverage as a hardware story. Then into earnings-call questions as an Nvidia bottleneck story. Then into specialized memory analysis as a high-bandwidth-memory bottleneck story. Then into utility coverage as a power-demand story. Then into nuclear coverage as a baseload story. Then into ETF inclusion lists. At each stage the narrative simplified, the audience widened, and the price effect spread to companies further from the original observation.

The language of contagion is useful. A narrative has exposure, adoption, mutation, and decay. Some market participants are susceptible because the story fits existing beliefs or portfolio needs. Others become carriers by repeating it in research notes, social media posts, fund letters, podcasts, and company presentations. The story mutates as it travels. “Large language models need GPUs” becomes “AI needs data centers,” then “AI needs power,” then “utilities are AI beneficiaries.” Each version is shorter, more memorable, and more general than the last.

Contagion is not the same as truth. A true but complex narrative may spread slowly. A simplified narrative may spread quickly even if it is incomplete. A story that is easy to repeat can attract capital before the full economics are understood. The chain matters. The same story at the GPU stage and at the utility stage prices different companies, attracts different investors, and decays on different schedules.

5.5 From Story to Price

A story does not move prices by being repeated. It moves prices by changing what investors do. A **narrative** can **affect prices** through several distinct channels:

- **Information diffusion:** investors gradually learn about a real change in cash flows.
- **Attention allocation:** investors begin to search for and buy companies connected to the story.
- **Analyst revision:** sell-side and buy-side models update assumptions, target prices, and peer groups.
- **Flow pressure:** funds, ETFs, and retail accounts allocate capital to the theme.
- **Risk repricing:** investors change the required return for exposed firms.
- **Reflexivity:** rising prices validate the story and make it easier for companies to raise capital or invest accordingly.

These channels operate at different horizons. Newswire sentiment can move prices within minutes or days. Earnings-call language affects revisions over weeks. SEC filings confirm strategic change over quarters. ETF inclusion arrives late, after a theme has become widely recognized — a pattern Morningstar’s thematic-fund research documents at the product level [81]. A systematic thematic process must know which channel it is operating in. A signal designed to capture information diffusion has nothing to say about late-stage flow pressure, and the reverse.

5.6 Fundamental Information vs. Contagious Interpretation

The iShares Global Clean Energy ETF more than doubled in 2020 [12]. The narrative that drove the run-up was substantively grounded: solar costs were falling, electric-vehicle adoption was accelerating, governments in Europe and the United States were committing to long-term emission targets. By every fundamental measure that mattered for the long-term economics of the energy transition, the story was right.

By the measures that mattered for prices over the following two years, the public-equity expression was fragile. Long-duration cash flows were worth far more when interest rates were near zero than when ten-year yields climbed past 4 percent. The clean energy narrative had attached itself to a duration trade. The story was correct about the destination. The trade was wrong about the discount rate that the market would apply to the journey.

This is the distinction between fundamental information and contagious interpretation. Fundamental information changes the expected cash flows, risks, or constraints of a company. Contagious interpretation changes how investors frame existing information. Both can move prices. They have different lifecycles, and they require different trading responses.

For a system trying to separate them, two questions help. First: would the same conclusion still hold if a sober analyst with no exposure to the narrative looked at the same facts? Second: is the price action explained by the story's specific economic mechanism, or by something the story is correlated with but not causally connected to? The clean energy trade in 2020–2022 is useful because it splits the answer. The story was about decarbonization. Much of the price action was about duration. A narrative-aware portfolio that knew the difference would have measured and stress-tested duration exposure as the rate cycle began to turn, then treated only the residual as thematic alpha.

5.7 The Economic Plausibility Test

A topic model trained on five years of financial news might discover that the phrase “supply chain visibility” co-occurs with strong subsequent returns. That is not, by itself, a reason to trade it. Statistical association is the easy part. The hard part is articulating why the association should produce returns prospectively rather than only in the rearview mirror.

Every narrative signal should therefore be forced through an economic plausibility test before it becomes a position. The test asks why the signal should earn a return. Possible answers include:

- investors are slow to process complex or cross-sectoral information;
- the affected firms are undercovered by sell-side analysts;
- the value chain is non-obvious and requires specialist mapping;
- short-sale constraints delay negative information from being reflected in prices;
- retail attention creates temporary buying pressure without changing fundamentals;
- institutional mandates delay reallocation across portfolio buckets;
- liquidity constraints prevent immediate arbitrage by larger pools of capital;
- regulation or event timing creates binary uncertainty that the market discounts unevenly.

If no plausible mechanism exists, the signal is likely a statistical artifact. The researcher must explain who is mispricing the information, why the mispricing persists long enough to trade, what the implied horizon of the edge is, and what observable evidence would falsify the thesis. A signal that survives this checklist is at least the kind of thing a thematic process can defend. A signal that does not survive it is either a coincidence or a stale, already-priced observation. This same checklist will reappear in Chapter 23 as the final guardrail before a theme is committed to capital.

5.8 Two Stories at Two Stages

Two charts can carry the same word and tell different stories. In one, attention is spreading faster than coverage; in the other, coverage is rising after price has already broken.

Within this framework, the book's two running cases occupy different positions.

AI infrastructure is, as of writing, a still-spreading narrative. The original observation about GPU demand has propagated through every stage of the transmission chain described above — from technical to industry to broker to mainstream to ETF inclusion — but the story continues to discover new beneficiaries. Power infrastructure was a serious sub-narrative in mid-2024; cooling and thermal management was a serious sub-narrative by late 2024; advanced packaging is a candidate for the next stage. The investment problem is not whether the story is real. It is which sub-stories are still underpriced and which have already become consensus.

Clean energy 2020–2022 is a mature transition narrative whose early public-equity expression crested. The story is not gone; the Inflation Reduction Act of 2022 re-injected it with policy support, but ICLN continued to struggle through 2023 [12]. The narrative no longer dominates expectations in the way it did at the peak. The story has been priced, repriced, and repriced again. A thematic process now operating in clean energy is no longer riding a broad coordinating wave; it is making selective bets within a mature transition where the easy awareness premium has already been captured. These are different investment problems even when the underlying economics overlap.

Distinguishing them is not optional. A signal that performs well during the still-spreading stage of a narrative may perform poorly during the crested stage of the next. The same statistical pattern can sit on top of two very different economic mechanisms. A theme is a story being processed. The investment problem is to know which one is being told, who is telling it, where in its life it sits, and whether the prices it is moving are still detached from the eventual destination.

Chapter 6

What Is an Investment Theme?

1. A strong long-term theme can remain fundamentally correct while becoming a poor investment once the market fully crowds into it.
2. The success of a theme's underlying economics does not guarantee continued investment returns from its public-market expression.

6.1 The Theme at Its Peak

By early January 2021, the iShares Global Clean Energy ETF had more than doubled in twelve months [12]. Capital was flowing into clean energy and broader thematic funds at record speed [69, 81]. Corporate net-zero commitments were arriving weekly. The underlying economics of the transition appeared to validate every step: solar levelized costs had continued to fall [66], electric-vehicle registrations had more than doubled in Europe during 2020 [64], and the incoming US administration had made clean-energy investment central to its economic platform.

From inside that environment, the theme felt validated. To most participants, that was the moment to put money into it.

By the end of 2022, ICLN had lost more than half of its peak value. Solar deployment had not stopped. Electric-vehicle sales had not stopped. Government incentive programs in the United States and Europe had expanded. The clean energy story had not been disproved; if anything, the underlying economics had strengthened. What had happened was that the public-equity expression had moved through several stages of its life — from validated to crowded to institutionalized to fragile — in a window that few investors had a framework to track.

A sector classification has no language for this. A factor model has no language for this. The vocabulary of returns, alpha, and benchmarks developed in Part I describes the consequences but not the mechanism. To trade themes systematically, an investor needs an explicit working definition of what a theme is, how to measure whether one exists, and where in its life the current price action sits. The definition has to be operational enough for a model and unforgiving enough for a risk committee.

6.2 A Working Definition

Vistra, Nvidia, Schneider Electric, and SK Hynix do not share a sector. They can still share a trade. That is the problem a thematic definition has to solve.

A theme is a transient, cross-sectoral exposure created by a shared narrative, structural shift, or value-chain relationship. Three words carry the weight.

Transient. A sector can remain stable for decades. A style factor earns its premium across long historical samples. A theme has a lifecycle. It emerges, spreads, becomes validated or rejected, attracts capital, becomes crowded or institutionalized, and eventually decays, mutates, or re-emerges under new conditions. The investment horizon of a theme is usually measured

in months to years, not in the multi-decade timescales of sectors or the persistent-anomaly timescales of factors.

Cross-sectoral. A theme links companies that may not belong to the same sector but whose expected cash flows, risks, or investor demand are affected by the same story. A theme concentrated in a single sector is usually just a sector bet with extra steps. The analytical value of thematic thinking is precisely in connecting companies that conventional tools would not place together.

Exposure. A theme is not merely a topic in the news. A news cluster about AI regulation is a topic. It becomes an investment theme only when it can be mapped to firms whose cash flows, valuations, or risks are plausibly affected. The distinction between topic and investment theme is where most thematic products fail. The companies in the basket must be genuinely exposed, not loudly adjacent.

These properties separate themes from three things they are routinely confused with: sectors, factors, and values-driven focused portfolios. The differences matter because they determine which analytical tools apply.

Category	Basis	Horizon	Example
Sector	Primary business activity	Long-lived	Utilities
Factor	Return characteristic	Persistent	Momentum
Focused portfolio	Preference or label	Variable	Founder-led firms
Theme	Narrative exposure	Transient	AI infrastructure

The categories overlap in practice. AI infrastructure has technology-sector concentration, growth and momentum factor exposure, and high market beta. Clean energy has small-cap growth, high investment, and policy-sensitivity exposure. This overlap does not invalidate the themes. It means the thematic component must be separated from the ordinary exposures that any factor or sector lens would already detect. The next section makes that separation formal.

6.3 The Augmented Return Model

When clean-energy stocks fell together in 2022, a simple sector map explained too little and a clean-energy slogan explained too much. The question was narrower: after market beta, size, growth, momentum, rates, and industry exposure were stripped out, did anything genuinely thematic remain?

The standard factor model decomposes returns into exposure-weighted factor returns and a residual:

$$r = Xb + u, \quad (6.1)$$

where r is a vector of asset returns, X is a matrix of factor and industry exposures, b is a vector of factor returns, and u is the residual return. Traditional risk models treat u as largely idiosyncratic — the part of each stock’s return that moves on company-specific news rather than on broad systematic drivers.

A theme challenges that treatment. If firms exposed to the same narrative show correlated residual returns during the theme’s active period, then u contains a transient common driver that the standard model has missed. The augmented model adds an explicit theme block:

$$r = Xb + Yg + u'. \quad (6.2)$$

Here Y is the matrix of thematic exposures — one row per firm, one column per theme — and g is the vector of thematic returns over the period. The residual u' is what remains after both ordinary factors and thematic exposure have been accounted for. The model is simple, but it clarifies the research problem precisely.

A claimed theme is real only if Yg explains structure in returns that Xb alone cannot. If the residual co-movement vanishes once size, value, momentum, and the technology sector are controlled for, the basket is a factor portfolio under a thematic label. If the residual co-movement persists, the basket is a candidate for genuine thematic exposure.

For AI infrastructure, X may include market beta, sector exposures, growth, momentum, size, profitability, investment, and interest-rate sensitivity. Y captures exposure to accelerators, high-bandwidth memory, data centers, power and cooling infrastructure, cloud capex, and related supply-chain roles. The research challenge — the work of Parts III, IV, and V of this book — is to estimate Y from text, revenue evidence, value-chain relationships, and analyst signals without simply rediscovering the columns of X .

6.4 Residual Coherence as the Test

At the peak of a theme, almost everything seems connected. The discipline is to ask what remains connected after the obvious exposures have been removed.

A theme should leave a footprint in residuals. After ordinary factor and sector effects are removed, the firms in a coherent theme should still show common return behavior during the theme's active period: correlated residual returns, shared sensitivity to theme-specific news, common analyst revisions, or co-movement that intensifies around theme-relevant announcements.

This is the empirical test. A marketing theme combines companies that sound related but do not move together once controls are applied. A “future technology” basket that holds software, semiconductors, electric vehicles, space, robotics, and genomics with no common residual driver is a label, not an exposure. A “circular economy” basket whose holdings move with the general industrials sector and nothing else is a label. A “cybersecurity” basket whose entire residual co-movement disappears once the software industry is controlled for is a label.

The standard is not perfection. Themes are messy. Companies have multiple business lines. Narratives mutate, and the membership of a theme changes as new beneficiaries are identified and old ones become consensus. But the basic test must clear. A theme should improve the explanation of return covariance, the prediction of return, or the attribution of portfolio risk relative to what the standard factor and sector model already delivers. If it does none of those things, the basket is a story.

6.5 The Seven-Stage Lifecycle

The same clean-energy basket could look early in 2019, obvious in 2020, dangerous in early 2021, and cheap but still uninvestable in parts of 2023. The label had not changed. The stage had.

A theme is not a static object. It moves through identifiable stages, and the right portfolio response depends on which stage the theme currently occupies. The working taxonomy used throughout this book has seven stages:

1. **Weak signal.** Scattered documents, specialist discussion, technical commentary, or early company references appear, but the market has not coordinated around the story. Coverage is concentrated in narrow channels. Prices are not yet responding systematically.
2. **Emerging theme.** Coverage accelerates. Vocabulary stabilizes. The first tradable beneficiaries become visible. A small number of forward-looking investors begin to position; broker research starts to flag the narrative.
3. **Validated theme.** Company fundamentals, analyst revisions, capital expenditure, revenue evidence, or policy action confirm the thesis. The narrative now has external support beyond the original observation.
4. **Crowded theme.** Attention, flows, valuations, and ownership concentration rise sharply. Easy beneficiaries are fully priced. Late entrants discover that the marginal trade is no longer the most thematic one but the most liquid one.
5. **Institutionalized theme.** ETFs, index products, benchmark categories, and corporate reporting normalize the theme. The story now has dedicated investment vehicles, index inclusion criteria, and standardized vocabulary. Allocation to the theme becomes a default rather than a decision.
6. **Decaying theme.** Attention fades, returns reverse, fundamentals disappoint, the macro regime turns against the theme, or the theme becomes absorbed into ordinary sector and factor exposures. The label persists; the alpha does not.
7. **Recycled or re-emerging theme.** The story returns under new conditions, often with changed beneficiaries, a different macro environment, or a fresh policy catalyst. The recycled theme is not identical to the original; it is the same narrative wearing different clothes.

This lifecycle is a map, not a law. Themes do not march through it on schedule, and some collapse before reaching institutionalization while others recycle through several full cycles. The value of the taxonomy is that it forces the question: *what stage is this theme in right now, and what does that imply for the trade?* A weak-signal theme requires discovery and patience. An emerging theme requires evidence gathering and gradual position building. A crowded theme requires valuation discipline and risk control. An institutionalized theme may still be investable, but the edge is unlikely to come from mere awareness of the story.

6.6 Clean Energy Through the Lifecycle, 2018–2024

The clean energy theme provides a complete worked example because it ran through every stage of the taxonomy in roughly six years [12].

2018 — weak signal. Solar leveled cost of energy continued its decade-long decline, moving closer to parity with new fossil-fuel generation in more markets. Electric-vehicle adoption was accelerating in Norway, the Netherlands, and parts of China. Climate-focused research notes circulated among specialist analysts, but the broad equity market had not coordinated around a clean-energy narrative. ICLN traded in a narrow range. Most generalist investors paid no attention.

2019 — emerging theme. Climate activism reached mainstream visibility. The European Commission presented the European Green Deal in December 2019 [36]. ESG inflows began

accelerating. **Earnings-call** mentions of decarbonization, net zero, and renewable transition rose meaningfully. ICLN and the Invesco Solar ETF (TAN) began to climb. The first generalist research notes appeared treating clean energy as a thematic category rather than a sector subset.

2020 — validated theme. The Biden campaign made clean-energy investment a central policy promise. European recovery spending emphasized green infrastructure. Tesla joined the S&P 500 in December and gained roughly 743 percent during the year [87]. Renewable capacity additions hit records globally, with IEA estimating almost 280 gigawatts of additions in 2020 [65]. ICLN more than doubled across the year. The narrative now had political, fundamental, and price-action support simultaneously — the textbook validated stage.

Early 2021 — crowded theme. ICLN peaked in early January. Plug Power had risen more than tenfold during 2020. Hydrogen, fuel-cell, and solar names traded at valuations implying multiple decades of high-growth cash flows. Inflows into clean-energy products reached record levels. Late entrants discovered that the most thematically pure names had already moved most.

2021 — institutionalized theme. New clean-energy ETFs launched at a rapid pace. Index providers added climate-themed benchmarks. ESG mandates became standard at large institutional allocators. Corporate sustainability reports treated clean-energy positioning as a default disclosure. The narrative now had its own infrastructure of products, indices, and reporting categories. Investing in clean energy had become a default option rather than a discretionary thematic bet.

2022 — decaying trade expression. The Federal Reserve raised policy rates by roughly 425 basis points across the year [43]. Long-duration cash flows repriced sharply downward. ICLN lost more than half of its peak value from peak to trough. Speculative hydrogen, fuel-cell, and solar names fell much more than diversified clean-energy vehicles. Solar and wind capacity additions kept rising, but parts of the public-equity basket decoupled from the underlying buildout. The narrative had not been disproved. The discount-rate environment and crowded entry point that had supported the 2020 trade were gone.

2023–2024 — recycled theme. The Inflation Reduction Act, signed in August 2022, re-injected the clean-energy theme with substantial policy support for clean electricity, electric vehicles, hydrogen, batteries, and domestic manufacturing [112]. Public clean-energy equities still struggled, which is precisely the point: the recycled theme was not a broad repeat of 2020. The investable questions shifted toward policy eligibility, domestic manufacturing, grid constraints, storage economics, and, as AI infrastructure grew, whether data-center power demand would create a new source of demand for low-carbon generation. The recycled theme is not the 2020 theme. The pure-play darlings are different. The macro setup is different. The story is connected to the original observation but no longer identical to it.

A clean energy strategy designed to ride the 2020 validated stage would have been profitable through January 2021 and painful through 2022 if it ignored valuation, rates, and crowding. A strategy designed for the 2023 recycled stage starts from very different prices, different beneficiaries, and a different macro context. The same structural transition, six years apart, is two different investment problems.

6.7 Theme **Granularity**

In 2024, “AI” meant Nvidia to some investors, Microsoft Copilot to others, data-center power to utility analysts, and memory bandwidth to semiconductor specialists. All of them were talking about AI. They were not talking about the same portfolio.

A theme can be too broad or too narrow. “Artificial intelligence” may be too broad for portfolio construction because it includes accelerators, software, automation, data infrastructure, robotics, and disrupted business models with very different lifecycles and beneficiaries. “High-bandwidth memory substrate suppliers for AI accelerators” may be economically precise but too narrow or too illiquid for most institutional portfolios. Granularity is itself an investment decision.

The right level depends on the decision problem:

- For strategic allocation, a macro theme may be sufficient.
- For alpha generation, sub-themes are usually necessary — the residual return tends to be sharper at the sub-theme level.
- For risk management, both macro and sub-theme exposures matter, because risks may concentrate at a level different from the one used for selection.
- For implementation, liquidity and capacity constrain how narrow the theme can be without becoming untradeable.

The relationship between macro and sub-themes is hierarchical, and the modelling chapters will return to it from the topic side. The investment logic, however, comes first: a topic hierarchy is useful only if it supports better portfolio decisions.

6.8 Falsifiability

Falsifiability means a theme must include clear conditions under which the thesis would be considered wrong. For example, an AI infrastructure thesis may fail if cloud companies cut capex spending, if memory shortages disappear due to oversupply, or if AI-related stocks stop moving together after adjusting for sector and market effects. A clean-energy thesis may weaken if government subsidies fail to translate into earnings growth or if adoption slows despite falling technology costs. Without predefined disconfirming signals, a theme becomes a belief rather than a testable investment hypothesis.

A theme thesis should be falsifiable. If every outcome can be interpreted as confirmation — if a price rise validates the story and a price fall is also evidence of mispricing — the theme is not a research hypothesis. It is a position the researcher has fallen in love with.

For AI infrastructure, plausible falsifiers include:

- cloud hyperscaler capex guidance weakens while the basket continues to rely on capex growth;
- power-demand projections fail to translate into utility earnings;
- memory suppliers expand high-bandwidth capacity faster than expected, removing the scarcity that drove the bottleneck premium;
- exposed firms stop showing residual co-movement after factor and sector controls;
- attention reaches saturation while forward returns deteriorate.

Each of these is observable in advance. None requires hindsight to identify. A theme thesis that does not specify what would make it wrong cannot be evaluated; it can only be advocated. Falsifiability also defines what to monitor in production. The signals that would invalidate a theme are the signals a thematic system needs to track most carefully.

6.9 A Theme Is a Hypothesis

The definition built here is operational. A theme is transient, cross-sectoral, and tied to exposure. It is separable from ordinary factors via the augmented return model. Its coherence is testable

as residual co-movement. Its position in time can be located on the seven-stage lifecycle. Its granularity is a portfolio decision. Its thesis is falsifiable.

What unifies these properties is that a theme is a hypothesis, not a position. The investor who holds AI infrastructure in 2024 is not betting that the narrative is true. They are betting that, at this stage of the theme's life, the prices of specific firms have not yet caught up with specific economic mechanisms that the narrative implies. That bet is correct or wrong on its own terms, and it is the researcher's job to specify which.

A theme without a definition is a slogan. A theme with a definition is something a system can defend.

Chapter 7

Sentiment, Attention, and News Acceleration

7.1 What the Transcripts Knew First

In July 2023, SK Hynix said demand for AI server memory had increased rapidly and that sales of premium products such as HBM3 and DDR5 had risen with it [101]. Three months later, the company attributed a sequential revenue jump partly to strong demand for HBM3, which it called a key product for AI applications [102]. The language of high-bandwidth memory bottlenecks was moving from specialist discussion into the earnings materials of a major public company before it became a mainstream market shorthand.

The lead was not magic. It was a property of where the text came from. Earnings calls and results releases sit deeper in the transmission chain described in Chapter 5 than newswires or social media, but the information they contain is more substantive and arrives well before the same content reaches SEC filings or ETF inclusion lists. Different sources release different kinds of information at different speeds, and a thematic system that treats them as interchangeable feeds will miss precisely the windows where signal is most concentrated.

The text signal has two sides. Three signal families — sentiment, attention, and acceleration — describe what a body of text contains and how loudly it is being said. A fourth dimension — the source itself — determines how that content was filtered, by whom, and with what lag. Chapter 8 is dedicated to sources; the first three families form the content-side apparatus.

7.2 Three Signal Families

The market can react to a sentence, to the number of people reading similar sentences, or to the fact that the number of readers has suddenly changed. Those are different signals.

Textual alpha signals are often described loosely as “sentiment.” That label captures only part of what matters. A useful working framework distinguishes three content-side families, all of which must be measured separately and combined:

- **Sentiment:** what the text says. The positive, negative, or neutral tone of language about a company, theme, or event.
- **Attention:** how much the market is looking. The total volume of coverage, search, discussion, or capital focused on a company or theme.

- **Acceleration:** how fast the looking is changing. The rate at which attention is rising or falling relative to a recent baseline.

These families interact. A small positive article about an undercovered firm can matter more than a positive article about a mega-cap firm that everyone already follows. A sudden spike in neutral coverage can matter because it changes who is paying attention. None of these signals is meaningful in isolation from the others, and none is interpretable independently of where it came from — which is why Chapter 8 takes up sources as a coequal dimension.

The investment system’s job is to know which family matters for which decision. A high-frequency trading desk monitoring earnings releases cares about sentiment within minutes. A thematic portfolio manager wants attention acceleration over weeks. A backtesting researcher needs to know that the same article may have produced three different sentiment readings depending on when and how it was ingested. The signal families look similar from a distance and behave very differently up close.

7.3 Sentiment

In a 10-K, “liability” is often an accounting category, not a confession. In a product review, the same word would sound ominous. Finance punishes lazy dictionaries quickly.

Sentiment converts language into a measure of tone. Early approaches used dictionaries: count positive words, negative words, uncertain words, litigious words. Modern approaches use supervised classifiers and domain-tuned language models that label sentences or documents in context.

In finance, sentiment must be domain-aware. A word that sounds negative in everyday usage may be neutral or technical in accounting disclosure. A word that is positive in a consumer review may be irrelevant in an earnings call. Loughran and McDonald demonstrated empirically that nearly three-quarters of the words flagged as negative by a standard general-purpose dictionary were not in fact negative in the language of 10-K filings [78]. The sentence “liabilities increased because of acquisition financing” is not what a generic emotional dictionary would interpret it as. Financial sentiment requires a financial dictionary, or better, a model trained on financial text.

Sentiment also has asymmetric dynamics. Tetlock’s work on media sentiment showed that pessimistic newspaper coverage can forecast downward pressure on market prices, with effects that persist beyond what a same-day reaction would explain [108]. The implication is not that sentiment is mispriced everywhere. It is that tone can interact with investor attention, trading frictions, and limits to arbitrage in ways that survive the first headline reaction.

For thematic trading, tone asymmetry matters at both the firm and theme level. A theme can have positive aggregate sentiment while specific constituents face negative second-order effects. The AI infrastructure narrative is positive for accelerator suppliers and data-center builders. It can be negative for firms whose products are commoditized by the same buildout, or whose labor-intensive business models face automation pressure. A theme-level sentiment score that averages everything together loses the differentiation the strategy actually needs.

7.4 Attention

When a stock appears in the news, posts an extreme one-day return, or shows abnormal trading volume, more investors put it on the mental menu. Barber and Odean found that individual

investors are net buyers of attention-grabbing stocks, because buying requires searching across thousands of possibilities while selling usually starts from positions already owned [8]. Da, Engelberg, and Gao later used Google search volume as a more direct proxy for investor attention and found that spikes in search interest predict short-term price pressure followed by longer-horizon reversal [29].

Attention measures the allocation of scarce investor cognition. Investors cannot analyze every stock and every document. They focus on what is salient: large price moves, media coverage, search trends, social discussion, analyst reports, and popular narratives. The questions a signal must answer are: what is currently being looked at, by whom, and is the looking unusual?

Raw volume is not enough. Large companies always receive more coverage than small companies. Some sectors are structurally more news-intensive than others. A useful attention measure compares current coverage to an appropriate baseline:

$$Z_{i,t} = \frac{V_{i,t} - \mu_{i,t}^{(L)}}{\sigma_{i,t}^{(L)}}, \quad (7.1)$$

where $V_{i,t}$ is current coverage volume for company or theme i at time t , and $\mu_{i,t}^{(L)}$ and $\sigma_{i,t}^{(L)}$ are the rolling mean and standard deviation over a lookback window L . The z-score asks whether attention is unusual for that entity, controlling for its normal coverage rate. In production, V should be deduplicated, source-weighted, and winsorized; a syndicated headline should not look like ten independent signals. Five articles about a thinly covered small-cap may be an attention shock. Five articles about a megacap may be a quiet morning.

Attention can produce two opposing effects on prices. First, it improves information incorporation. An undercovered firm may finally react when attention arrives because the market begins to process previously neglected information. Second, attention can produce overpricing. Retail investors may buy what is visible, especially when short selling is constrained and the story is easy to repeat. The same attention spike is bullish or bearish depending on context — which is why attention measures alone are not a strategy. They are a conditioning variable for other signals.

7.5 Acceleration

At the weak-signal stage from Chapter 6, almost nobody is talking. At the crowded stage, everybody is. The investable transition often sits between them, when coverage is still modest but the slope has changed.

A mature theme can carry high volume without new information. An emerging theme may have modest volume but rapidly rising coverage. The early signal is not that everyone is talking; it is that more people are talking than were doing so a few weeks earlier, using increasingly coherent language. Acceleration measures that change.

A simple ratio captures it:

$$A_{i,t} = \frac{V_{i,t}^{(S)}}{\bar{V}_{i,t}^{(L)} + \epsilon}, \quad (7.2)$$

where $V_{i,t}^{(S)}$ is short-window coverage, $\bar{V}_{i,t}^{(L)}$ is long-window average coverage, and ϵ prevents division by zero. The ratio is the speed of attention relative to its own baseline. Values close to

one indicate steady coverage. Values well above one indicate that the entity is becoming a more active topic faster than its historical rate. For sparse series, the same idea is often implemented on log counts, smoothed counts, or source-weighted counts so that one stray article does not masquerade as a regime change.

For thematic investing, acceleration is particularly diagnostic of lifecycle stage. A weak-signal or emerging theme typically shows high acceleration on low absolute volume — the conversation is small but growing fast. A crowded theme shows high absolute volume with modest acceleration — everyone is already paying attention, and the rate of new entry has slowed. An institutionalized theme shows stable volume and acceleration near one. The same numerical attention measure means different things at different stages, and the acceleration ratio is what makes the stage legible.

7.6 Signal Horizons and Decay

The first headline can be too fast to trade. The tenth annual report can be too late. A signal is valuable only if it remains tradable after the trader receives it.

Lead time without persistence is noise; persistence without lead time is consensus. The two properties together — arriving before the market has priced the content, and remaining informative for long enough to trade against — define a usable signal.

Different signals operate at different horizons:

- intraday sentiment from newswires may decay within minutes;
- daily sentiment may predict reactions over one or two days;
- weekly attention shifts may capture slower information diffusion;
- earnings-call language may inform analyst revisions over weeks;
- filing-based exposure scores may be stable for quarters;
- extreme attention may become contrarian over longer horizons;
- prediction-market changes matter only until the underlying event resolves.

A signal's horizon must match the strategy's rebalancing cadence. A monthly thematic portfolio should not depend on signals that decay in thirty minutes. A high-frequency news strategy gains nothing from quarterly filing text. The architecture of a thematic trading system reflects these horizons: fast sources belong in monitoring and event overlays, slow sources belong in exposure scoring and validation. The work of the later chapters is to connect the two — to use the language of fast-moving narratives to update slow-moving exposure scores without contaminating either with the other's timing.

7.7 Why Should the Signal Earn a Return?

The most dangerous text signal is the one that works beautifully and explains nothing. A textual signal that predicts returns in a backtest is not yet a strategy. The economic plausibility test introduced in Chapter 5 applies here in sharpened form: the researcher must articulate why the signal should be compensated, prospectively, after costs. Possible mechanisms include:

- slow analyst revision in coverage of complex or cross-sectoral stories;
- limited attention among investors who cannot monitor every signal;
- complexity of mapping theme beneficiaries to specific firms;
- short-sale constraints that delay the incorporation of negative information;
- liquidity limits that prevent immediate arbitrage in smaller stocks;
- behavioral extrapolation after salient stories receive sudden coverage;
- institutional mandate constraints that delay reallocation across portfolio buckets;
- delayed incorporation into indexes and ETF inclusion criteria.

The mechanism determines the sign, the horizon, and the universe. Slow information diffusion supports a momentum-style trade with horizons measured in weeks. Extreme retail attention supports a contrarian trade with horizons measured in months. Complex value-chain mapping supports underreaction trades in second-order beneficiaries that mainstream coverage has not yet reached. Without a mechanism, a text signal is only a pattern — and patterns without mechanisms are how a backtest becomes a story that data-mined itself into existence.

For AI infrastructure, the signal apparatus has natural applications. Sentiment can identify the tone of management commentary on hyperscaler capex commitments. Attention can flag which parts of the value chain investors are currently focused on — accelerators, memory, power, cooling, packaging — and which are still under-followed. Acceleration can show when a sub-theme is moving from specialist discussion into broader market awareness. None of these measurements means anything in isolation from where it came from. A sentiment score from a 10-K and a sentiment score from a tweet are not the same input, regardless of what the spreadsheet says.

Chapter 8

Information Sources and Lead Times

8.1 The Same Words in Two Places

In 2023 and 2024, many US-listed companies described artificial intelligence as a strategic priority in their earnings-call prepared remarks. Management talked about AI initiatives, AI capabilities, AI-enabled products, AI-driven efficiency. The language was confident and the analyst questions were attentive.

In many cases, the annual 10-K reports for the same fiscal year contained no AI segment, no AI-specific revenue line, no AI capital-expenditure disclosure, and no AI-related risk factors. The legal document and the management narrative were describing the same business, in the same quarter, to the same investors. They were not describing the same thing.

That gap is the lesson. The source through which text reaches an investor is not a delivery mechanism for content. It is part of the content. A sentence in a regulatory filing has been reviewed by lawyers and signed by an executive. The same sentence in a social-media post may have been written in two minutes by an anonymous account. The legal source is slow but binding. The social source is fast but unreliable. A thematic system that ingests both and treats them as interchangeable feeds will discover, painfully, that they are not.

Chapter 7 developed the content-side apparatus: sentiment, attention, acceleration. The source-side apparatus determines what those measurements actually mean. The same sentiment score, the same attention spike, and the same acceleration ratio can support an actionable trade or a noise position depending on where they came from.

8.2 The Transmission Chain

Chapter 5 sketched how narratives travel through markets. Sources can be arranged into the same kind of chain, ordered by speed at one end and by institutional weight at the other:

prediction markets and specialist forums → social media → newswires → broker research →
earnings calls → SEC filings → ETF and index inclusion.

The sequence is stylized rather than universal. Some corporate information starts in filings. Some macro information starts in official releases. Some event probabilities are visible in prediction markets before mainstream coverage. But the chain captures a robust pattern: narratives tend to move from narrow, fast, noisy sources toward broad, slower, institutionalized ones. The same

story arrives differently at each stop — with different lag, different audience, different verification, different incentives.

Early sources offer lead time but high noise. Late sources offer confirmation but little remaining alpha. The right source depends on the trade. A high-frequency event strategy lives at the newswire and prediction-market end. A thematic exposure-scoring system lives at the earnings-call and filing end. A monitoring system that wants to flag emerging stories lives at the social-media and specialist end. A capacity-aware portfolio that needs to anticipate institutional flows pays attention to the ETF-inclusion end.

A systematic process should not pick one source. It should know which source is informative for which question, and how to combine them without letting noise from one stage contaminate signal at another. Each stop in the chain has its own speed, audience, and failure mode.

8.3 Newswires and Premium Financial Press

When Microsoft announced the Constellation power purchase agreement, the first clean, time-stamped version of the event mattered. The second-order map from that event to power producers, nuclear suppliers, and data-center load growth mattered more.

Newswires and premium financial press are timely, structured, and widely monitored. Bloomberg, Reuters, the Dow Jones wire, the Financial Times, and the Wall Street Journal produce the first professionally verified version of most company and macro events. Their language is more disciplined than social media. Their timestamps are usually clean enough to anchor event studies. Their entity tagging is good enough to map directly to tickers.

The drawback is competition. Many professional investors receive the same feeds with similar latency. Simple headline sentiment on liquid large-cap stocks is unlikely to remain profitable when execution speeds are slower than the wire-to-fill latency of high-frequency desks. The edge in newswire data has migrated upward in the value chain: aggregation across multiple wires, cross-source confirmation, entity disambiguation, and the second-order mapping of an event onto less obvious beneficiaries. Reading the headline matters less than reading what the headline implies for companies that the headline does not name.

For thematic investing, newswires are useful for confirming events that will affect a basket — cloud capex announcements, chip supply constraints, export controls, data-center construction, utility demand projections. The first-order market reaction at the named companies is fast. The second-order reaction in suppliers, beneficiaries, and adjacent value-chain firms is often slow and tradeable.

8.4 Social Media and Specialist Forums

A ticker can become a story on social media before it becomes a business case. That is useful information, but it is not yet evidence.

Social media is fast, broad, and noisy. It reveals retail attention, emerging vocabulary, rumor propagation, and disagreement before formal sources see any of it. New product names, niche technical jargon, fringe-becoming-mainstream phrases, and early reactions to corporate events often appear on platforms before any broker analyst writes about them.

The cost of that speed is reliability. Social media is vulnerable to manipulation, coordinated posting, bot activity, selection bias, echo chambers, and the inherent unrepresentativeness of self-selected user populations. A signal extracted from social media is rarely fundamental information.

It is, most of the time, an attention signal — a measurement of how much retail or specialist focus is on a name or theme.

Two practical guidelines follow. First, social media is most useful as a leading indicator for weak signals: a theme that is gaining vocabulary on Reddit, X, or specialized Slack communities before broker research notices it is worth tracking even if no single post is itself informative. Second, social media is rarely sufficient as standalone exposure evidence. A company that is widely discussed on social media but does not appear in earnings calls, broker research, or filings as a relevant operator in the theme is more likely a meme than a thematic constituent.

Specialist forums — industry mailing lists, research preprint servers, standards bodies, procurement forums, and technical conferences — sit somewhere between social media and broker research. They are slower than mainstream social platforms but typically have higher signal-to-noise. For deep technical themes like AI accelerator architecture, advanced packaging, or grid-scale energy storage, specialist forums can lead even broker research by months. Their weakness is coverage bias: they describe what technical specialists care about, not necessarily what public equities can monetize.

8.5 Prediction Markets

An event contract is a sentence with a price attached. If a contract pays one dollar if a regulator approves a merger and zero otherwise, a price near sixty cents looks like a probability. The word “looks” is doing work.

Prediction markets price contracts on specific future events. A contract that pays out if a candidate wins an election, a drug is approved, or a regulator imposes a tariff has a current price that can be interpreted as a risk-adjusted probability of the event, subject to liquidity, fees, market structure, and participant composition.

The advantage is forward-looking aggregation. Prediction-market prices can move in response to information faster than the same information reaches newswires, because the participants are explicitly trading the probability rather than reacting to a story about the probability. Event-contract venues such as Kalshi, crypto-native markets such as Polymarket, and a small number of academic and institutional venues produce probability series for many of the events that matter for thematic baskets: election outcomes, policy actions, regulatory milestones, central-bank decisions, and earnings-related corporate events. Those series are not interchangeable. Venue access, legal status, market design, and participant composition differ materially.

The weakness is market quality. Thin liquidity can leave prices stale. Favorite-longshot bias can distort probabilities at the extremes. Strategic trading and outright manipulation are real risks in low-volume markets. Insider activity can move prices in ways that resemble information but reflect access rather than analysis. Regulatory status also matters: the CFTC settled charges against Polymarket in 2022 for offering off-exchange event-based binary options without the required registration [27], while Kalshi’s election-contract litigation in 2024 showed how contested the boundary around regulated event contracts could be [111].

Prediction-market data is therefore best used as an event overlay — a conditioning variable that adjusts the timing or sizing of a thematic position — rather than as a standalone equity portfolio engine. For AI infrastructure specifically, event contracts on energy policy, export controls, antitrust decisions, and major regulatory actions can adjust the probability mass of theme-relevant scenarios in ways that no equity-only model can replicate.

8.6 Broker Research

The first broker note that adds a data-center power-demand line to a utility model is not merely commentary. It is evidence that a narrative has entered institutional spreadsheets.

Broker research is slower than newswires but richer in interpretation. Sell-side analysts connect company fundamentals, estimates, management commentary, valuation, peer-group composition, and target prices into a coherent narrative. Their reports reveal when a theme is becoming institutionalized: when sector analysts start including AI revenue line items in their models, when utility analysts start treating data-center demand as a baseline assumption, when memory analysts upgrade HBM penetration estimates, the theme has moved past the discovery stage and into consensus.

Three practitioner notes apply. First, broker research is not a neutral corpus. It reflects client demand, banking relationships, analyst incentives, coverage constraints, and the firm's own institutional positioning. A research note is communication with a purpose, not a disinterested statement. Second, the value of broker research lies less in any single note than in the trajectory across notes: the same analyst's estimates moving sequentially in one direction tells a story that no single report does. Third, broker research is licensed and gated. Building a systematic process that depends on broker text requires real procurement, licensing, and data-engineering work, and the universe of available analysts and firms is narrower than a list of all sell-side coverage suggests.

For thematic investing, broker research is especially useful at two specific points in the lifecycle. In the emerging stage, the first analyst note to explicitly model a theme is a meaningful event. In the institutionalized stage, the consensus assumptions that broker research embeds tell you what the market now takes for granted — which is precisely the set of assumptions a contrarian strategy would interrogate.

8.7 Earnings Calls

Prepared remarks are staged. Q&A is negotiated in public. Earnings calls contain both.

Earnings calls combine two distinct kinds of text in one source. The prepared remarks show what management wants investors to hear, organized in the order management chose. The Q&A shows what analysts are trying to understand, expressed in questions management cannot fully control. The difference between the two is often more informative than either alone.

For thematic investing, earnings calls are valuable for three reasons. First, transcripts can quantify theme exposure — explicit AI revenue contribution, capex direction, supply-chain commentary, customer concentration — in a way that simpler text sources cannot. A company that names AI as a strategic priority but cannot quantify its AI revenue contribution under analyst questioning is signaling something different from one that can. Second, the language of analyst questions tracks what professional investors are currently focused on. When the same questions about AI capex sustainability or HBM supply constraints recur across multiple companies in the same quarter, the theme has reached the dominant-narrative stage. Third, the gap between prepared remarks and Q&A often reveals what management is avoiding. A theme that appears in the opening narrative but cannot survive specific analyst questioning is a marketing exercise.

Earnings-call transcripts are not as fast as newswires, but they are closer to company fundamentals than any of the faster sources. The gap between AI in earnings-call language and

AI in 10-K segment disclosure is exactly the kind of cross-source comparison earnings calls make possible. They are also a natural source for exposure scoring, where transcript-derived evidence anchors the scoring of which companies are actually materially exposed.

8.8 SEC Filings

A 10-K rarely gives the first hint of a new theme. It often gives the first legally reviewed hint that the theme has become material.

Regulatory filings are slow but legally meaningful. The 10-K, 10-Q, 8-K, and proxy filings contain business descriptions, risk factors, segment information, customer concentration, capital expenditure, executive commentary, and management discussion. The text is long, repetitive, full of boilerplate, and structured by regulatory convention. The SEC's Form 10-K instructions define the annual-report frame, and EDGAR provides public full-text access to electronic filings [116, 117]. The signal is sparse but it is verified.

Filings are usually late in the narrative chain. By the time a theme appears in many companies' filings, the market has often understood it for some time. The value of filings is not lead time. It is materiality. A theme thesis stands on stronger ground when the exposed companies disclose related capex, related customers, related risks, or related segment lines in their regulated filings than when they only mention the theme in press releases and earnings-call enthusiasm.

The 10-K is especially diagnostic because its structure — business description, risk factors, MD&A, segment disclosure — is fixed by regulation. A company that mentions AI in its business overview but not in its capex discussion, customer list, segment reporting, or risk factors has weak exposure no matter how often the CEO uses the word on calls. The 10-K cannot lie cheaply. It can be vague, stale, and lawyerly, but it cannot omit material economic facts without legal consequence. For exposure scoring, the 10-K is the source against which other sources must be checked.

8.9 ETF and Index Inclusion

ETF inclusion is the moment a story becomes packaging. That can create flows, but it also tells you the theme has become easy enough to sell.

When a theme becomes the basis for dedicated investment products, it has typically reached a broad audience. ETF launches, index methodology additions, and benchmark category creations are late-stage signals — not lead indicators. Morningstar's tracking of thematic-fund launches documents the empirical pattern at the product level: thematic ETFs tend to launch after the strongest period of the underlying narrative, and the holdings they include represent the industry's view of what is exposed rather than an independent test of exposure [81].

This makes ETF holdings useful as noisy labels but not as truth. They tell a systematic process two things. First, they show who else is positioned where — which is information about crowding, flows, and consensus rather than about fundamentals. Second, they reveal dilution: large thematic funds often include broad proxies (mega-cap technology, mega-cap utilities) because the pure-play firms are too small or illiquid to absorb the AUM. A systematic strategy can try to improve on ETF holdings by scoring materiality and value-chain roles more precisely than market-cap-weighted thematic indices do.

8.10 Comparing Sources

The summary table below collapses the discussion of the preceding sections. None of the rows tell the whole story; each summarizes a source whose detailed properties were just described.

Source	Speed	Noise	Best Use
Prediction markets	Very fast	Medium/high	Event probability overlays
Social media	Very fast	High	Attention and vocabulary drift
Newswires	Fast	Medium/low	Event detection, timestamp anchors
Broker research	Medium	Medium	Institutional interpretation
Earnings calls	Medium/slow	Medium/low	Management and analyst evidence
SEC filings	Slow	Low (boilerplate-heavy)	Materiality and validation
ETF holdings	Slow	Medium	Flow, crowding, noisy labels

The asymmetry runs along a diagonal. The fastest sources have the highest noise and the lowest verification. The slowest sources have the strictest verification and the least lead time. No row is uniformly best; each is best for a specific question. A system that draws on multiple rows simultaneously can ask better questions than a system that depends on any single one.

8.11 Lead Time and Decay

Lead time is valuable only if the signal remains tradable when it arrives. A source can be early but too noisy, or reliable but too late. Signal decay is therefore source-specific, and a thematic system's architecture should match its rebalancing cadence to the natural decay of the sources it depends on.

Intraday newswire sentiment decays within minutes. Daily news sentiment may persist for one or two days. Social-media attention spikes typically decay within days unless they are reinforced by repeated catalysts. Broker research updates persist over weeks or until the next quarter's reset. Earnings-call themes can remain informative until the next call, eight to thirteen weeks later. SEC-filing exposure scores are stable across reporting periods. ETF holdings turn over quarterly at most. Prediction-market changes matter only until the event resolves; after resolution, the contract pays out and the signal disappears.

Fast sources belong in monitoring and event overlays. Slow sources belong in exposure scoring and validation. The discipline is not to mix the two: a monthly thematic portfolio should not be built on a signal that decays in thirty minutes, and a high-frequency news strategy gains nothing from quarterly filings. Topic models, treated in Part III, are part of the bridge between fast and slow — they help identify when the language of a fast-moving narrative becomes stable enough to map to the slow universe of company exposure scores without contaminating either with the other's timing.

8.12 Sources as Diagnostic

A thematic process that draws on a single source is at the mercy of that source's specific failure modes. A process that draws on many sources can use the gaps between them as diagnostic evidence. A theme that appears in social media but not in broker research is likely in the weak-signal or emerging stage. A theme that appears in broker research but not yet in filings is likely validated but not yet institutionalized. A theme that dominates ETF inclusion lists but not the 10-K disclosure of its constituents may be a marketing label rather than a real exposure.

The opening example described one such gap: companies that talked about AI in earnings calls but did not disclose AI-related segment information in their 10-K filings. That gap was a measurement opportunity. The companies that closed it — by disclosing AI-related revenue, capex, risks, customers, or business lines in regulated filings — were a different set from the companies that did not. A systematic process that read both sources and compared them could have separated genuine exposure from narrative attachment without trusting either source on its own.

That is the entire point. A source is not a feed. It is a witness with a perspective, a speed, and a set of incentives. A thematic system that knows the perspective, the speed, and the incentives of each source can extract evidence that no single source contains.

Part III

Turning Text into Topics

Chapter 9

Financial Text as Data

9.1 The Filing Date That Lied

A research team builds a backtest on S&P 500 10-K filings. The strategy reads the risk-factor section of each annual report, scores the language, and trades at the next open after the filing date. Across ten years, the backtest looks respectable: an information ratio above 1.0, a Sharpe ratio above 0.8, and drawdowns small enough to survive an investment committee.

Then someone asks a dull question. What time did each filing actually become available?

The answer breaks the strategy. EDGAR does not treat every timestamp the way a spreadsheet treats a date. The SEC states that EDGAR operates from 6:00 a.m. to 10:00 p.m. Eastern Time on business days, but most live submissions transmitted after 5:30 p.m. receive a filing date of 6:00 a.m. the next business day and are not disseminated until that next business day [115]. EDGAR archives also expose dissemination and index mechanics that matter for automated collection [114]. A document can have an acceptance time, an official filing date, a dissemination time, a vendor ingestion time, and the time at which a particular strategy actually saw it.

The original backtest used the easiest field. It was not the right field. When the team rebuilds the simulation with the acceptance timestamp, dissemination rule, vendor delivery lag, and a one-business-day availability buffer, the information ratio falls by half. **Most of the apparent alpha was the model trading on a document before the simulated portfolio could have known the document in the form used by the research code.**

This is one of the quiet traps in financial text. The edge appears to come from language. Far more often than researchers admit, the edge comes from metadata. A topic model cannot repair a corpus that entered the lab with the wrong clock.

9.2 Sources, in the Plural

The same AI sentence carries a different meaning in a 10-K, an earnings-call answer, a Reuters headline, and a Reddit post. One is a legal disclosure. One is management under questioning. One is a verified news item. One is attention, enthusiasm, rumor, or all three at once.

A narrative-aware system rarely lives on one corpus. Each source answers a different question, and combining sources is the work that makes thematic detection possible. Headlines move fastest and show that something has happened. Filings move slowest and show whether the company has made the story material. Earnings calls combine prepared narrative with analyst pressure. Social media reveals attention and vocabulary before institutional channels catch up.

Broker research reveals how professionals translate a story into estimates. ETF holdings reveal which exposures the asset-management industry has already packaged.

The principal sources for a thematic process include:

- **Headlines:** short, timely, sparse; useful for event detection and attention measurement.
- **Full news articles:** richer context, source tone, entity relationships, and second-order implications.
- **Newsires:** professional reporting with relatively clean timestamps and broad institutional monitoring.
- **Social media and specialist forums:** fast attention and disagreement; noisy, manipulable, and selection-biased.
- **Earnings calls:** management narrative plus analyst Q&A.
- **SEC filings:** legally meaningful business, segment, risk, and financial disclosure.
- **Broker research:** institutional interpretation, estimate revisions, and target-price logic.
- **Patents and technical documents:** evidence of innovation, product direction, and engineering bottlenecks.
- **Event databases:** structured event, actor, location, and tone metadata, with GDELT as the canonical open source [109].
- **ETF holdings and index documents:** industry-defined exposure labels, useful but noisy.

Chapter 8 treated the source taxonomy from the signal-extraction side: speed, noise, audience, and lead time. Here the same sources reappear as raw material that has to be ingested, timestamped, cleaned, and made comparable before any modeling begins. The tempting shortcut is to merge every source into one stream and let a topic model find structure. That shortcut destroys the evidence. Combining sources works only when their differences survive the merge.

9.3 Time Is the Spine

A 10-K describes a fiscal year that may have ended two months earlier. The filing appears on EDGAR on a particular day. A vendor may deliver it minutes or hours later. A batch pipeline may not process it until the next morning. The portfolio decision may happen at the close.

A document is known only when it was actually available to the investor in the form the investor used. The principle sounds elementary and fails routinely. Publication timestamps get delayed, rounded, revised, stored in different time zones, or substituted for the moment of the underlying event. Filings have acceptance timestamps that differ from report periods and filing dates. Earnings-call transcripts arrive after the call ends. Broker reports may be embargoed by client channel or time zone. ETF holdings appear with regulatory and provider lag.

A robust pipeline maintains several timestamps per document, all in one canonical time zone, and never collapses them prematurely:

- **Event time:** when the underlying event occurred: the moment an executive said the sentence, a regulator issued the decision, or a price moved.

- **Publication time:** when the source emitted the document: the wire timestamp, EDGAR acceptance or dissemination time, or call transcript release.
- **Vendor ingestion time:** when the data vendor through which the strategy reads the document made it available to clients.
- **First-available-to-strategy time:** the latest relevant timestamp plus any internal processing latency. This is the only timestamp a backtest may treat as “known.”
- **Revision time:** when the document changed: corrections, restatements, amended filings, or after-the-fact edits to a story.

The chain matters because the gap between event time and first-available-to-strategy time is where leakage hides. A study that uses event time treats every event as tradable the instant it occurred. A production backtest that uses publication time but ignores vendor latency gives itself minutes or hours it did not have. The rule that subsumes the problem is the as-of invariant: every input to every simulated decision must have been available, in its recorded form, before the simulated decision time.

Revision time deserves its own line in the schema. A wire story silently rewritten three hours after publication carries different information at hour zero and hour three. A 10-K later amended by a 10-K/A changes the historical record. A robust pipeline stores the document as it appeared at each revision and resolves which version the strategy would have seen.

9.4 Document Length and Sectioning

A headline can move a stock with five words. A 10-K can bury the useful sentence on page 83. Treating those two documents as the same modeling object is how a text strategy becomes precise in code and vague in reality.

Document length shapes model choice. Classical bag-of-words topic models assume enough word co-occurrence per document to estimate topic mixtures; that assumption breaks for headlines. The specialized short-text methods introduced later in Part III exist because headlines, social posts, and search snippets do not behave like filings or transcripts.

Long documents need sectioning. A 10-K’s business description, segment disclosure, MD&A, and risk-factor section serve different purposes and follow different rhetorical conventions. Loughran and McDonald showed why finance-specific language and 10-K context matter for textual analysis; ordinary word counts can misread financial documents when vocabulary changes meaning in disclosure settings [78]. An earnings call has the same problem in miniature: prepared remarks show what management chose to emphasize, while Q&A shows what analysts forced into view.

For the running cases, the distinction is practical. A headline such as “Cloud provider raises data-center capex” flags an event without proving exposure. A 10-K segment line stating that a company derives a large share of revenue from data-center cooling validates materiality. An earnings-call exchange in which an analyst asks how much next year’s growth depends on AI customers reveals institutional concern. The same theme appears in each source, but at different lengths, with different evidential weight, and on different timelines.

9.5 Metadata Is Half the Signal

Two identical sentences do not mean the same thing when one appears in a risk factor and the other appears in a marketing blog. The words match. The evidence does not.

Document text is what investors notice first. Metadata is often where the information lives. A pipeline that treats metadata as accessory data throws away signal before the model starts.

The fields that matter most include:

- source and source type: wire, filing, transcript, social, research, ETF document;
- language and region, especially when a corpus crosses jurisdictions;
- document type and section: 10-K versus 10-Q versus 8-K; risk factors versus MD&A versus business overview;
- issuer identification: company, ticker, CUSIP, ISIN, exchange, and CIK, all with point-in-time validity;
- author or outlet, including whether the outlet originated the story or syndicated it;
- structural fields: headline, body, tags, categories, tables, exhibits, and speaker labels;
- the full timestamp chain: event, publication, ingestion, first-available, revision;
- linked entities, events, and geographies populated by upstream NLP.

Metadata supports source-specific modeling. A phrase whose appearance in a risk factor signals a serious disclosure may be boilerplate in a headline. A wire story that appears at five outlets within two minutes carries one piece of news, not five independent observations. Metadata also supports the cross-source diagnostic gap from Chapter 8: a company that names AI on its earnings call but not in any segment line of its 10-K is informative only if the pipeline tracked both sources and could compare them.

9.6 Data Quality and Vendor Reality

The 2020–2021 SPAC boom produced exactly the kind of historical corpus that flatters weak research. It had ticker changes, de-SPAC mergers, delistings, restated projections, thinly covered concept stocks, and investor presentations that read like business plans from a future that had not yet arrived. Clean-energy and EV-adjacent firms were present in that universe, but the data problem was broader than the energy transition.

Financial text data is messy in ways that have specific investment consequences. The recurring problems include:

- **Duplicate articles across syndication networks.** One wire story may appear at thirty outlets within minutes. Without near-duplicate detection, an attention model concludes that the story is thirty times more important than it is.
- **Corrected or revised stories.** The published version of a wire item at time t may differ from the version stored in the vendor archive at time $t + 24h$. A backtest that fetches the archived version reads post-correction text as if it were visible at the original moment.

- **Missing or inconsistent timestamps.** Some vendors record only dates. Others mix UTC and local time. Others store local time without the offset.
- **Stale ticker mappings.** A historical document references “XYZ Corp.” Today, that ticker belongs to a different issuer after delisting and reuse. Linking by today’s ticker map routes the old signal to the wrong security.
- **Survivorship bias in the company universe.** A corpus that includes only currently listed firms drops de-SPAC failures, acquired targets, renamed issuers, delisted experiments, and early-stage theme companies that shaped the original opportunity set.
- **Restated fundamentals.** A historical earnings number may have changed. A model that reads “missed estimates” against today’s restated figure may reverse the original market fact.
- **Corporate-action adjustment errors.** Stock splits, spinoffs, and reverse mergers can corrupt the linkage between text and price series.
- **Incomplete coverage of small firms.** The corpus may cover mega-caps exhaustively and barely cover the micro-caps where some themes are most economically pure.
- **Language and translation artifacts.** Translated text can cluster by translator or vendor process rather than by topic.
- **Licensing constraints.** Some vendors restrict storage, redistribution, or model training. Those legal constraints shape the architecture as much as the model choice.

None of these are clerical details. They can move results by more than the alpha the strategy claims to capture. A research process that ignores them will discover patterns that belong to the data vendor, the index file, or the archive date rather than the market.

9.7 Deduplication and Distribution Breadth

Reuters publishes one story. Ten outlets pick it up. Three update the headline. Two add local context. A naive article count sees sixteen signals. The market saw one story traveling through a distribution network.

Deduplication is essential for news data. Exact duplicates can be removed by hashing normalized text. Near duplicates — the same story rewritten slightly by different outlets, or updated with a fresh paragraph — require similarity methods. The system has to decide whether two near-duplicate articles count as one signal or two.

The answer depends on the use case. For sentiment, repeated syndication usually does not add new information; the canonical story should count once. For attention measurement, repetition is information because distribution breadth measures how widely the story traveled. A good pipeline preserves both pieces: a canonical story identifier for the deduplicated unit and a distribution-breadth count for the number of places it appeared, with timestamps.

The same logic applies to retransmitted corporate announcements. A press release posted on the company’s investor-relations page, picked up by three wires, and tagged by ten outlets is one piece of news with several measures of reach. Counting it correctly requires the pipeline to separate novelty from amplification.

9.8 The Corpus as a Research Object

A corpus of premium wires will find institutional information and miss most retail excitement. A corpus of social media will find excitement and miss materiality. A corpus of filings will validate exposure and discover themes late.

A corpus is not a folder of documents. It is a design choice with investment consequences. The universe of included sources determines which narratives the system can detect at all. Each restriction is also a research bet: the corpus declares which kinds of narratives the system is willing to find and which it will systematically miss.

Before fitting any topic model, the researcher should write an explicit corpus statement:

- What sources are included, and by what selection rule?
- Which timestamps are used as “known,” and what availability buffers are applied?
- What languages and regions are covered?
- Which securities can be linked, and through what point-in-time identifier system?
- What documents are excluded, and on what grounds?
- What biases does the corpus contain: survivorship, selection, syndication, translation, vendor coverage?
- What investment horizon does the corpus support, given the timestamps and source mix?

This statement is the first line of defense against overclaiming. The model can only discover what the corpus makes visible.

Chapter 10

Preprocessing, Entities, and Financial Meaning

10.1 The Topics That Were Just Companies

Run a topic model on raw financial news and the first result often looks impressive for exactly the wrong reason. One topic comes back as “Apple, iPhone, Tim Cook, Cupertino, App Store, services.” Another is “Nvidia, Jensen Huang, GPU, H100, accelerator, data center.” A third is “Tesla, Elon Musk, Model 3, Fremont, autopilot.” Each topic is coherent. None is the theme the investor wanted.

The model did its job. It found tight co-occurrence structure. In financial text, the tightest co-occurrence structure is often a single company surrounded by its own proper nouns: products, executives, facilities, brand names, tickers, and slogans. A semiconductor corpus produces a Nvidia topic, an AMD topic, and a Broadcom topic. A clean-energy corpus produces a Tesla topic, a Plug Power topic, and a First Solar topic. The model has rediscovered the company list. It has not discovered a theme.

The thematic story the model should have surfaced – the GPU bottleneck of 2023, the hydrogen enthusiasm of 2020, the data-center power constraint of 2024 – requires the model to see that Nvidia, SK Hynix, Vertiv, Eaton, and Constellation can belong to one economic story even though their names rarely appear in identical sentences. The first intervention is to remove proper nouns from the model’s input. The second is to put them back, carefully, after the topics have been discovered.

Preprocessing in financial text is not a technical prelude. It is modeling. Every decision changes what the topic model can see, and the most consequential decisions are about how the system handles named entities.

10.2 Preprocessing Is Modeling

Lowercasing “Apple” looks harmless until the corpus contains both a technology company and agricultural commodity news. Removing “of” looks harmless until the phrase is “cost of goods sold.” Stripping numbers looks harmless until the sentence contains capex, margin guidance, or an interest-rate path.

Generic NLP pipelines often treat preprocessing as a fixed sequence: lowercase, strip punctuation, remove stop words, tokenize. In financial text the choices are not mechanical, and the

order matters. Removing all named entities can reveal abstract themes but erase the link to tradable securities. Keeping all named entities can force the model to cluster by company rather than theme. Stripping stop words may help a general corpus and damage a financial one, where “The Fed” is not filler and “in line” can mean a result met estimates.

The right preprocessing pipeline depends on the investment question. If the goal is to discover macro narratives, company names should often be masked before modeling and reconstructed afterward. If the goal is company exposure scoring, company names are exactly what must be preserved, with high-precision linking. If the goal is event detection, timestamps, outlet metadata, and short source-specific phrases matter more than elegant linguistic normalization.

A pipeline that is right for one task can be wrong for another. Chapter 9 argued that a corpus is a research object. The operating rule follows naturally: preprocessing is the set of choices that decides which parts of that object the model is allowed to notice.

10.3 Cleaning and the Boilerplate Problem

A Reuters byline repeated ten thousand times can become a topic. So can a cookie notice. So can the phrase “forward-looking statements.” The model does not know that these phrases are plumbing.

Basic cleaning is the part of the pipeline most easily underestimated. The objective is not to make text pretty. The objective is to preserve meaningful language while removing artifacts that create false topics, false attention, or false similarity.

The standard operations include:

- stripping HTML, scripts, navigation, cookie banners, and CMS boilerplate from web-scraped documents;
- normalizing whitespace, punctuation, quotation marks, and character encoding;
- preserving sentence boundaries so downstream segment-level models can recover them;
- handling tables, bullet lists, exhibits, and footnotes in regulatory filings;
- removing or relocating outlet names, repeated tags, and boilerplate disclaimers into metadata;
- preserving source-specific fields, such as speakers in earnings calls and item numbers in filings.

The outlet-name problem is small enough to miss and large enough to ruin the model. If “Reuters,” “Yahoo Finance,” or “*The Wall Street Journal* reports” appears in thousands of documents, a topic model cannot distinguish the publisher’s signature from a meaningful term unless the signature is removed from the body and stored as metadata. Otherwise the system creates a “Reuters topic,” which captures everything Reuters covers and therefore explains nothing.

Tokenization sits inside cleaning but deserves special care. Classical models often work on words and short n-grams; transformer encoders work on subword tokens. Either way, finance-specific units need explicit handling: tickers, monetary values, percentages, dates, regulatory acronyms, product names, CUSIPs, and filing item labels. Tokenize “H100” badly and a GPU theme loses one of its most important terms. Split “high-bandwidth memory” into three unrelated tokens and the HBM supply bottleneck becomes harder to see.

Normalization should stay conservative. “AI” and “artificial intelligence” may be unified for some thematic tasks. “Apple” should not be normalized before the system knows whether the surrounding sentence refers to the issuer, the product, or the fruit.

10.4 Stop Words and Domain Stop Words

The word “may” is often boilerplate in a risk factor. The same word in “the FDA may approve” can be the hinge of an event trade. Frequency alone does not decide what a word means.

Stop words are frequent terms that often carry little semantic content. Standard lists – “the,” “and,” “of,” “in” – work reasonably well for general English. Financial corpora add two complications. The first is true domain boilerplate: repeated legal openings, generic reporting verbs, outlet signatures, broad category tags, and stale disclosure phrases. These can and often should be stripped or downweighted.

The second category is more delicate. Some sequences that look disposable in isolation are central financial vocabulary in context. “The Federal Reserve” or “The Fed,” stripped carelessly, can collide with generic uses of “federal.” “Of” is a stop word; “cost of goods sold” is not. “In line” can mean “met estimates,” not merely two common words adjacent to each other.

Loughran and McDonald showed that general-purpose word lists misclassify financial language in 10-Ks because words change meaning in disclosure settings [78]. The same risk applies to stop-word lists. They should be source-aware and section-aware: a filing boilerplate list should not automatically govern news headlines, earnings-call Q&A, social posts, or broker research. Every stop-word list belongs in the corpus statement from Chapter 9, because it defines what the model was prevented from seeing.

10.5 Named Entity Recognition in Finance

Bitcoin mining is not mining diamonds

“Visa beat estimates” and “visa demand recovered” are not the same sentence. A generic model can miss the difference. A trading system cannot.

Named entity recognition, or NER, finds proper nouns and assigns them types: organization, person, location, product, ticker, monetary value, percentage, date. The general task has a long NLP literature, and even broad NER systems face design choices around boundaries, label sets, and external knowledge [88]. Finance adds a harder constraint: almost every downstream action must connect the text to a security, a contract, a facility, a regulator, or a value-chain role. The need for finance-specific entity definitions is not cosmetic; recent financial NER datasets were created precisely because ordinary NER resources do not capture the semantic and lexical variation of financial entities [94].

Generic NER systems trained on ordinary news or web text routinely fail in financial settings. They confuse company names with common nouns: Apple, Visa, Target, Snowflake. They miss tickers because “AAPL” looks like an odd token rather than an issuer symbol. They misclassify products as companies and subsidiaries as parents. They fail to preserve the difference between Nvidia the issuer, H100 the product, Jensen Huang the executive, TSMC the foundry, and Taiwan the jurisdiction. They mislabel regulators such as the SEC, CFTC, FDA, or FERC as generic acronyms or fragments. The more precise the investment use case becomes, the less adequate generic labels such as organization, person, and location become.

A financial NER system should support, at minimum, the following entity types:

- company and issuer names;
- tickers, exchanges, CIKs, CUSIPs, ISINs, and internal permanent identifiers;
- products, brands, platforms, and business-line names;
- executives, board members, analysts, regulators, and agencies;
- geographies, regulatory jurisdictions, physical facilities, and operating regions;
- monetary values, percentages, share counts, units, capacity figures, and currency codes;
- corporate and regulatory events: mergers, IPOs, guidance revisions, litigation, approvals, export controls, restatements.

Each type feeds a different downstream model. A dollar value attached to a capex announcement is not the same signal as a percentage attached to gross-margin guidance. A facility location matters for power and water constraints. A product name matters for value-chain mapping. NER is not the last step in financial NLP. It is the hinge on which the rest of the pipeline turns.

10.6 Entity Linking and Point-in-Time Identifiers

“META” meant something different before June 2022. “FB” meant something different after. A point-in-time system knows both facts at once.

Extracting an entity is not enough. The extracted entity must be linked to a stable, point-in-time identifier. “Apple,” “Apple Inc.,” “AAPL,” “Cupertino-based iPhone maker,” and the analyst phrase “the largest US company by market cap” should all resolve, when appropriate, to the same issuer at the same point in time. Entity linking is the broader NLP task of assigning textual mentions to unique entities in a knowledge base [97]. In investing, the knowledge base must also be historical.

The most common failure mode is using today’s ticker map to link yesterday’s text. A ticker that identifies one issuer now may have identified a different issuer in the historical record. A company may change name, exchange, share class, parent, or listing status. SPACs merge and rename. Distressed firms delist and reappear through restructurings. The 2020–2021 SPAC and de-SPAC universe is full of this: companies that changed tickers, missed projections, merged again, delisted, or disappeared from today’s investable universe.

Point-in-time linking requires three pieces:

- a permanent issuer or security identifier that does not change when ticker and name fields change;
- a time-indexed mapping from text-side identifiers to that permanent identifier;
- the discipline never to dereference an identifier without supplying the as-of date.

The investment cost of skipping this discipline is paid twice. In research, the backtest looks better than it should because the system quietly inherits survivorship and look-ahead bias. In production, the signal can attach to the wrong ticker. That is not an abstract data-quality issue. It is an order-routing problem wearing a data-engineering costume.

10.7 Entity Masking and Reconstruction

The model cannot discover an AI infrastructure theme if every document screams one company’s name louder than the shared economics. The names have to step aside before they can become useful again.

Entity masking solves the “topics are just companies” problem from the opening section. Before topic modeling, the pipeline replaces specific entities with category tokens. Company names become `COMPANY`. Tickers become `TICKER`. Countries become `COUNTRY`. Products become `PRODUCT`. Executives become `PERSON`. The corpus no longer reflects the trivial structure “one company appears with its own proper nouns.” It reflects the shared vocabulary around the economic story.

Consider headlines across several semiconductor firms. Without masking, the model produces a Nvidia topic, an AMD topic, and a Broadcom topic. With masking, the same corpus may produce a topic whose strongest terms are `COMPANY`, `ACCELERATOR`, `DATA CENTER`, `HBM`, `EXPORT CONTROL`, and `ADVANCED PACKAGING`. That is the underlying theme the unmasked model could not see.

Masking has to be paired with reconstruction. After topics are discovered on the masked corpus, the system needs to know which masked tokens corresponded to which entities in each document. The mapping is the bridge from abstract theme to tradable universe: it tells the portfolio process which companies appeared, how often, in which source, in which section, and under which topic.

A masking step without reconstruction produces a model of language with no investable handle. Reconstruction without masking produces topics that are company lists. Both halves are required.

Two reconstruction errors deserve special attention. The first is reconstructing with the wrong identifier vintage, which injects the survivorship problem from the previous section directly into the topic-to-portfolio bridge. The second is reconstructing with the wrong granularity. Counting a parent-subsidiary structure once at the parent level may understate exposure; counting every subsidiary as if it were independently investable may overstate it. Reconstruction discipline mirrors the linking discipline that produced the entities in the first place.

10.8 Relation Extraction

“Nvidia sells GPUs” is useful. “SK Hynix supplies HBM for Nvidia accelerators” is more useful. The theme lives in the edge between the companies.

Themes often live in relationships between entities rather than in entities themselves. A company supplies another company. A utility serves a data-center region. A chip designer depends on a foundry. A drug developer waits for a regulator. Relation extraction identifies these links from unstructured text; the general task has been studied for decades as part of information extraction [126]. Recent finance-domain relation-extraction datasets make the same point in the language of filings and financial news: useful relations are directional, often numeric, and often embedded in corporate structure, revenue streams, acquisitions, and market activity rather than in generic person-organization links [70, 52].

The relations that matter most for thematic investing recur across sectors:

- supplier-of and customer-of, the value-chain backbone;

- competitor-of, for substitution and pricing-power analysis;
- acquired-by, merged-with, spun-off-from, for corporate-action handling and issuer continuity;
- exposed-to a commodity, input, regulation, or geography;
- regulated-by, licensed-by, certified-by, or approved-by;
- located-in or facility-in, which matters for power, water, geopolitics, labor, and tax;
- partner-of, licensee-of, joint-venture-with, and distributor-of.

For AI infrastructure, relation extraction identifies which firms supply networking equipment to data-center operators, which utilities serve constrained regions, which advanced-packaging foundries support accelerator designers, and which memory suppliers are tied to which customers. These extracted relations later become the value-chain graphs used in company exposure scoring. Relation extraction is a Part III implementation problem with a Part V investment payoff.

10.9 A Preprocessing Strategy for AI Infrastructure

An AI infrastructure pipeline that strips numbers, masks product names permanently, and merges earnings-call Q&A into prepared remarks has already lost the trade before the model runs.

The pipeline decisions for the running case follow from the preceding sections. A reasonable preprocessing strategy for an AI infrastructure thematic system includes:

- preserving technical multi-word phrases such as “high-bandwidth memory,” “advanced packaging,” “liquid cooling,” “hyperscale data center,” and “power purchase agreement”;
- masking company names, executive names, tickers, and product names during topic discovery, with reconstruction after;
- separating earnings-call prepared remarks from Q&A as distinct documents, since they carry different incentives;
- separating 10-K business descriptions, risk factors, segment disclosure, and MD&A as distinct sections;
- preserving capital-expenditure values, revenue values, margin guidance, capacity figures, wafer starts, megawatts, and delivery lead times as typed entities;
- extracting supplier, customer, facility, regulator, and power-market relationships into a separate relation store;
- retaining region and facility names because power, water, fiber, land, and grid constraints are local;
- maintaining a domain stop-word list specific to each source.

The same exercise for clean energy would produce a different list. Solar, wind, hydrogen, and EV sub-themes share some vocabulary and diverge on the rest. The SPAC and ticker-change history puts heavier weight on linking and reconstruction. A clean-energy pipeline would preserve different technical phrases, mask different entities, and treat lifecycle stage as part of the data problem rather than merely the investment interpretation.

No preprocessing pipeline is generic. The pipeline has to be designed around the investment question and the corpus, and the designer has to defend every choice as having a specific reason. In financial text, the tutorial shortcut is often the shortest route to topics that look meaningful and are not.

Chapter 11

Classical Topic Models

11.1 The Baseline That Became the Model

A research team builds a thematic NLP system on a corpus of earnings-call transcripts. The first attempt uses a transformer encoder, dense clustering, and a custom downstream classifier. The output looks technically impressive: low loss, tight embedding clusters, a clean UMAP projection. The problem appears at the first review meeting. The portfolio manager asks what the topics actually represent. The team cannot answer in a way that survives the next training run, because the topics shift when the model is retrained on the same data with a different random seed.

The team falls back to non-negative matrix factorization on the same corpus. The NMF topics produce top-word lists that are readable in minutes. The topics are stable across runs. They are easy to label. They are explainable to a non-technical reader who needs to understand what the strategy is actually buying. The fallback model becomes the production model. The transformer pipeline is kept upstream as a feature generator, not a topic discoverer.

This is not an unusual story. The dominant approaches to topic modeling that arrived in the late 1990s and early 2000s — Latent Semantic Indexing, Non-Negative Matrix Factorization, Probabilistic Latent Semantic Analysis, and especially Latent Dirichlet Allocation — have specific properties that the neural alternatives developed later in Part III do not match easily. They are interpretable by construction. They expose document-topic and topic-word distributions directly. They scale tractably to large corpora. They produce results that two analysts can read the same way. More than two decades after Blei, Ng, and Jordan published LDA in 2003 [16], the classical family remains the honest baseline against which every more elaborate method has to justify itself.

The classical family — LSI, NMF, pLSA, LDA, and the hierarchical and nonparametric extensions that came afterward — shares a common representation, a common set of properties, and a common set of design choices. The story begins with the representation and then walks through each algorithm and the properties that decide whether it belongs in production or only in the baseline column of a performance table.

11.2 Bag-of-Words and Document-Term Matrices

The classical family begins with a shared representation. Each document is converted into a vector over a fixed vocabulary; the corpus becomes a matrix. Let $X \in \mathbb{R}^{D \times V}$ where D is the number of documents and V is the vocabulary size. Each row represents a document; each

column represents a term. The entries may be raw counts, normalized frequencies, or TF-IDF weights that downweight terms appearing across many documents.

The representation is called bag-of-words because it ignores word order within a document. The sentences “AI demand raises power needs” and “Power needs raise AI demand” have identical bag-of-words vectors. The model sees them as the same observation even when the causal direction of the sentence matters. For financial text this is a real loss: the difference between “the company sued its supplier” and “the supplier sued the company” is exactly the kind of distinction a thematic system cares about, and bag-of-words destroys it.

The loss is partially compensated by interpretability. Because every input dimension is a vocabulary term and every output dimension can be inspected as a distribution over terms, the model never produces a result whose meaning is hidden inside a high-dimensional embedding. The structure that classical models find is structure a human can read. That property is what makes them durable.

The bag-of-words assumption also turns out to be more useful in financial text than the criticism implies. Most thematic signals are word-co-occurrence signals: the cluster of words that surrounds an emerging supply-chain bottleneck, the vocabulary that accompanies a new regulatory regime, the set of terms that appears around hyperscaler capital expenditure. These patterns survive the loss of word order. The patterns that do not survive — nuanced sentence-level causation, sarcasm, conditional reasoning — are usually not where thematic alpha lives.

11.3 Latent Semantic Indexing

Latent Semantic Indexing was the original linear-algebra approach to topic discovery [31]. It applies truncated singular value decomposition to the document-term matrix:

$$X \approx U_k \Sigma_k V_k^\top. \quad (11.1)$$

The truncated decomposition maps documents and terms into a k -dimensional latent space. Documents become low-dimensional vectors; terms become low-dimensional vectors; the inner product between any two captures their associational similarity in the corpus. LSI was a real advance over keyword matching: it could connect documents that used different vocabulary to describe the same underlying topic.

The limitation is that the components contain both positive and negative weights. A latent dimension whose top positive words are “bank,” “loan,” “deposit” and whose top negative words are “oil,” “rig,” “barrel” is conceptually saying “high on the banking axis, low on the oil axis,” which is hard to communicate to a portfolio manager. In thematic investing, interpretability is not a nice-to-have. LSI is therefore most useful as a conceptual predecessor and a dimension-reduction step inside larger systems, not as the topic model that gets shown to a non-technical reader.

11.4 Non-Negative Matrix Factorization

Non-Negative Matrix Factorization fixes the sign problem by restricting all factors to be non-negative [74]. It factorizes a non-negative document-term matrix into two non-negative components:

$$X \approx WH, \quad (11.2)$$

where $W \in \mathbb{R}_{\geq 0}^{D \times K}$ represents document-topic weights and $H \in \mathbb{R}_{\geq 0}^{K \times V}$ represents topic-word weights. Non-negativity makes the topics additive. A document is built from non-negative combinations of topics, and a topic is built from non-negative word weights. The result is that every topic can be read as a list of words, every document can be read as a list of topics, and every weight is a magnitude rather than a signed coefficient.

NMF often produces remarkably crisp and interpretable topics when applied to TF-IDF-weighted matrices. For news corpora and earnings-call transcripts of moderate size, NMF is frequently the model that survives the production round when fancier alternatives do not. Its weaknesses are real: results are sensitive to preprocessing, to initialization (which is non-convex), and to the chosen number of topics. But the weaknesses are manageable, and the strengths are exactly what a production system needs.

11.5 Probabilistic Latent Semantic Analysis

Probabilistic Latent Semantic Analysis introduced the explicit probabilistic mixture interpretation that LDA later refined [59]. It models the probability of a word in a document as a mixture over latent topics:

$$p(w \mid d) = \sum_{z=1}^K p(w \mid z) p(z \mid d). \quad (11.3)$$

Each topic z has its own word distribution $p(w \mid z)$. Each document d has its own topic distribution $p(z \mid d)$. The matrix decomposition of NMF and the mixture decomposition of pLSA produce similar low-dimensional structure under the bag-of-words assumption; pLSA's contribution was to make the latent variable a probability rather than a magnitude.

The practical limitation of pLSA is its handling of unseen documents. The topic distribution $p(z \mid d)$ is learned per-document during training; the model does not specify how to infer the topic distribution for a document it has not seen before. For a thematic system that processes new documents every day, that gap is precisely where production lives. LDA resolves it.

11.6 Latent Dirichlet Allocation

Latent Dirichlet Allocation adds a Bayesian generative structure on top of pLSA's mixture decomposition [16]. Each document has a topic mixture θ_d , and each topic has a word distribution ϕ_k . The topic mixture is drawn from a Dirichlet prior:

$$\theta_d \sim \text{Dirichlet}(\alpha), \quad (11.4)$$

and each topic-word distribution is drawn from another Dirichlet prior:

$$\phi_k \sim \text{Dirichlet}(\beta). \quad (11.5)$$

For each word position in a document, a topic is drawn from θ_d , and then a word is drawn from the selected topic's word distribution ϕ_k . The Dirichlet priors give the model two useful properties. First, they regularize the topic and word distributions, preventing the degenerate solutions that pLSA can produce on small or noisy data. Second, they define a generative process that extends naturally to unseen documents: a new document's topic mixture is inferred under the same prior used in training.

The result is a model with native soft assignment. A document can be 40 percent about supply chains, 30 percent about regulation, and 30 percent about earnings. Each topic can be inspected as a distribution over words. Each document can be inspected as a distribution over topics. The two together form the document-topic matrix that downstream models — portfolio construction, exposure scoring, time-series attribution — can consume as features.

LDA spread quickly across computational linguistics, recommender systems, bioinformatics, and political science in the years after publication. Its adoption in finance was slower and is still uneven, but the structure it provides — interpretable topics, mixed-membership documents, principled out-of-sample inference — continues to fit the thematic investing problem well.

LDA also created the bridge to the neural models that follow in Chapter 12. ProdLDA and NeuralLDA replace hand-derived inference with amortized variational inference: a neural network learns to map a document directly into an approximate topic mixture, while the topic model keeps the LDA-like generative structure [106]. That move matters historically because it shows that the boundary between classical and neural topic models is not a wall. It is a continuum. The practical question is not whether a model is old or new; it is whether the additional machinery produces better topics for the investment problem at hand.

11.7 Strengths and Limits of LDA

LDA is interpretable, mathematically elegant, and extensible. The basic model has been augmented in dozens of directions: correlated topic models that allow topics to co-occur with non-zero correlation [15], supervised topic models that condition on a target variable, dynamic topic models that allow topics to drift through time (the subject of Part IV), and hierarchical and nonparametric variants treated below.

For financial text the limitations are worth stating directly:

- LDA ignores word order and sentence-level context, which is a meaningful loss for nuanced disclosures.
- It assumes documents are long enough for bag-of-words mixtures to be estimated reliably. It struggles on headlines and other short documents, which is the central topic of the short-text chapter that follows in Part III.
- It depends heavily on preprocessing decisions made upstream. The entity-handling discipline of Chapter 10 is precisely the discipline that determines whether LDA’s topics will be thematic or company-clustered.
- It requires choosing or inferring the number of topics K , a hyperparameter to which results are sensitive.
- Topics can be redundant, overlapping, or hard to label even when the math is well-behaved.
- Generic word co-occurrence may miss domain-specific financial meaning that a financial-language model would catch.

One more limitation sits outside the model but inside the research workflow: the evaluation scores can lie. Topic coherence metrics such as NPMI are useful diagnostics, but Hoyle and coauthors show that automated coherence can diverge from human judgment, especially when comparing classical models with neural topic models [60]. A thematic system should therefore

treat coherence as a screen, not a verdict. A topic that scores well but cannot survive word-intrusion review, representative-document review, or downstream financial-utility testing is not a good topic for this book’s purposes.

For AI infrastructure, a well-tuned LDA on a clean corpus of earnings calls and broker research from 2023 can recover topics around accelerators, high-bandwidth memory, data-center power, and cloud capital expenditure. The same model can fail by collapsing two of those topics into one or by splitting one across redundant clusters. The quality of the topics depends as much on the preprocessing and corpus design of Chapters 9 and 10 as on LDA itself.

11.8 Soft Assignment and Mixed Membership

A signature property of classical topic models is soft assignment. Financial documents almost always cover multiple themes. An earnings-call transcript may discuss AI demand, currency headwinds, labor costs, regulatory exposure, and capital allocation in the space of one hour. A hard clustering model that assigns the whole transcript to one topic destroys most of that information.

In LDA, the document-topic vector θ_d directly represents mixed membership: $\theta_{d,k}$ is the probability that a randomly chosen word in document d belongs to topic k . In NMF, the document-topic weights in W provide a continuous representation that can be normalized into similar mixture-like vectors. In both cases, the per-document vector is exactly the kind of input downstream regressions, exposure models, and risk attribution systems need.

This matters for thematic investing because the operating questions are mixture questions. “How much of last quarter’s call was about AI?” “What share of this filing concerns supply-chain risk?” “Did the company’s emphasis on power constraints rise or fall between Q1 and Q3?” None of these has a one-topic answer. The classical models’ native handling of mixed membership is part of why they remain in the toolkit even as more elaborate neural alternatives become available.

11.9 Hierarchical and Nonparametric Models: hLDA and HDP

Two issues that LDA in its basic form cannot answer become important as soon as the thematic universe scales. The first is hierarchy: themes are usually nested, with broad parent narratives containing more specific child sub-narratives. AI infrastructure contains accelerators, memory, networking, power, cooling, packaging, and cloud platforms. Each of those further decomposes. Clean energy contains solar, wind, batteries, EVs, and hydrogen, with each branch further specialized. A flat topic model can represent any of these as separate topics, but it cannot represent the structural relationship between a parent topic and its children. The second issue is the number of topics. LDA requires K to be specified in advance. Hyperparameter selection on a corpus whose true thematic structure is unknown is uncomfortable.

Hierarchical Latent Dirichlet Allocation (hLDA) addresses the first issue [13]. The model assumes that each document is generated by selecting a path from the root of a topic tree down to a leaf, and then drawing words from a mixture of the topics along that path. The depth of the tree is a hyperparameter; the branching at each internal node is governed by the nested Chinese Restaurant Process, a prior that allows the model to discover the tree structure from the data rather than requiring it to be specified in advance. Topics near the root carry vocabulary shared across many documents (broad terms that show up in any AI infrastructure document); topics

near the leaves carry specialized vocabulary (the words specific to liquid-cooling beneficiaries or to nuclear-baseload power producers). The output is a hierarchy a human can read: a parent topic with its child topics, each with their own word distributions, organized by abstraction level.

For thematic investing, hLDA’s appeal is obvious. The natural questions are nested: what is happening to AI infrastructure overall, and within it, what is happening to the power sub-theme, and within the power sub-theme, what is happening to nuclear specifically. A flat model has to be queried with separate analyses for each level. A hierarchical model produces all three in one fit. The cost is computational: hLDA is more expensive to fit than LDA, and the inference is more delicate. For very large corpora the practical alternative is to fit LDA at multiple granularities separately or to use embedding-based hierarchical clustering, covered later in this part.

Hierarchical Dirichlet Processes (HDP) address the second issue: the number of topics K [107]. HDP is a nonparametric Bayesian model that lets the corpus tell the model how many topics it should have. Mechanically, HDP places a Dirichlet process prior over the topic distribution and a hierarchical structure over per-document mixtures, with the property that the number of topics actually used is determined by the data: small corpora produce few topics, large corpora produce more, and topics that the data does not support are not introduced. The model is in this sense self-pruning. HDP is also the natural starting point when the topic structure changes through time, because the right number of topics in 2018 may not be the right number in 2024.

HDP and hLDA are not interchangeable. HDP gives an arbitrary number of flat topics; hLDA gives a fixed-depth tree of topics with hierarchy. The two extensions can also be combined, though the combination becomes computationally demanding for production-scale corpora. In practice, hierarchical and nonparametric extensions are still under-used in financial thematic work — partly because of computational cost, partly because the engineering tooling around them is thinner than for plain LDA. They remain among the most natural classical models for thematic investing, however, because thematic structure is itself hierarchical and the number of relevant themes is itself unknown.

11.10 Classical Models as Production Tools

A serious thematic NLP system keeps classical models in the toolkit for reasons that have not been displaced by neural methods.

They are interpretable. The document-topic matrix and the topic-word matrix are both directly readable by a non-technical reviewer. A portfolio manager who wants to know what theme a basket is actually loaded on can be shown the topic-word distribution. A risk committee that wants to understand a model’s behavior can see exactly which words moved a classification. Neural and embedding methods can be coaxed into producing similar artifacts, but they do not produce them by construction.

They are stable. NMF and LDA, fit with reasonable initialization and adequate corpus size, produce topic structures that are recognizably the same across runs. The transformer alternative whose results shift with each retraining is a real headache for production work, and it is one of the principal reasons classical baselines stay in the pipeline even when neural models are also available.

They are scalable in a specific way: they handle corpora of millions of documents with established, well-tuned implementations that have been refined for two decades. The same is not yet uniformly true of all neural approaches.

They are the right benchmark. A neural pipeline that cannot beat a well-tuned LDA on the

relevant downstream task — exposure scoring, return prediction, portfolio risk attribution — has not justified its complexity. The benchmark role is critical: it forces every more elaborate model to demonstrate that the complexity is paying for something. In financial systems, where every additional moving part is a future maintenance cost and a future failure mode, the bar for new complexity has to be set somewhere. The classical baseline is where the bar sits.

The honest summary is that classical models are not legacy tools waiting to be replaced. They are part of the operating layer of a working thematic system, with specific properties that no current alternative dominates uniformly. Neural and embedding-based methods add capabilities classical models lack. They do not subtract the capabilities classical models still uniquely provide.

Chapter 12

Neural and Embedding-Based Topic Models

12.1 The Vocabulary Gap That LDA Could Not Cross

A research team builds a topic model on two hundred thousand technology-sector news headlines. They use Latent Dirichlet Allocation, twenty topics, careful preprocessing. The model produces respectable output: a smartphone topic, a cloud-services topic, a social-platforms topic. It also produces three separate topics about AI hardware. One is anchored to “GPU,” one to “accelerator,” one to “AI chip.” To a human reader, the three are the same story. To a bag-of-words model trained on a corpus where the three words rarely appear in the same headline, they are three different things.

The team replaces the LDA component with a BERTopic pipeline using a finance-tuned sentence encoder. The same corpus produces a single topic whose representative documents reference all three terms interchangeably, plus a fourth word — “compute” — that none of the LDA topics had picked up. The new model has not learned anything LDA did not know. It has learned to see what LDA could not: that different vocabularies for the same concept belong to the same topic regardless of whether they ever co-occurred in the same document.

This is what embedding-based topic models do. They replace word-level co-occurrence with semantic-vector proximity in a learned representation space. The result is topics defined by meaning rather than by lexical overlap. The cost is a new set of failure modes: harder interpretability, run-to-run instability, sensitivity to preprocessing and parameters, and a dependence on the quality of the underlying encoder that does not show up anywhere in the model’s API.

Classical topic models anchor any thematic system as an honest baseline. Short-text variants extend that family into the regime of headlines and social posts where bag-of-words collapses. Between the two sits the embedding-based family: BERTopic, Top2Vec, Contextualized Topic Models, and Contextualized Word Topic Models. Each makes a different bet about how to combine embeddings with topic discovery, and each is the right answer to a specific question.

12.2 From Counts to Vectors

The transition from classical to embedding-based topic modeling is a transition in representation. The document-term matrix of Chapter 11 encodes each document as a sparse vector of word

counts. An embedding-based model encodes each document as a dense vector in a learned semantic space:

$$e_d \in \mathbb{R}^m, \quad (12.1)$$

where m is the embedding dimension, typically 384, 768, or 1024 depending on the model. The vector e_d is the output of a neural network — usually a transformer trained to produce sentence-level representations [90] — that has seen enough text to know which sentences mean the same thing even when they use different words.

Similarity between documents is measured by cosine similarity in the embedding space:

$$\text{cosine}(e_i, e_j) = \frac{e_i^\top e_j}{\|e_i\| \|e_j\|}. \quad (12.2)$$

The shift from counts to vectors is consequential beyond the math. A bag-of-words model sees two documents as similar if they share words. An embedding model can see two documents as similar if they share *meaning*, including in cases where the lexical overlap is zero. A headline about “GPU shortage” and a headline about “accelerator scarcity” can have near-zero word overlap and near-identical embeddings. That is exactly the property the LDA-versus-BERTopic example at the start of the chapter demonstrates.

The other side of the trade requires equal honesty. The embedding is only as good as the encoder that produced it. A generic encoder trained on Wikipedia may not understand financial language: it can confuse ticker symbols with proper nouns, miss the meaning of “in line” as an analyst phrase, or treat “liability” as emotionally negative when it is in fact a routine accounting category. The Loughran-McDonald evidence introduced in Chapter 7 about financial dictionaries applies in sharpened form to encoders: a general-purpose encoder loses signal that a domain-tuned encoder preserves. For financial work, the practical choice is between an off-the-shelf encoder like Sentence-BERT, a financial-domain encoder like FinBERT-MPNet or FinLang, or a LoRA-tuned adaptation of a strong general encoder. The choice of encoder is a research decision, not an implementation detail.

12.3 BERTopic: The Modular Pipeline

BERTopic [51] is the embedding-era topic-model architecture that has become the de facto industry standard for English news corpora. The pipeline is modular by design and runs in four stages:

1. embed each document with a sentence or document encoder;
2. reduce embedding dimensionality, typically with UMAP, to mitigate the curse of dimensionality before clustering;
3. cluster the reduced embeddings, typically with HDBSCAN, allowing low-density points to be labeled as noise rather than forced into the nearest cluster;
4. describe each resulting cluster with a class-based TF-IDF (c-TF-IDF) score that identifies terms frequent within the cluster and infrequent across clusters.

The c-TF-IDF score for term t in cluster c is

$$\text{c-TF-IDF}_{t,c} = \text{TF}_{t,c} \cdot \log\left(1 + \frac{K}{df_t}\right), \quad (12.3)$$

where $\text{TF}_{t,c}$ is the frequency of term t in cluster c , K is the total number of clusters, and df_t is the number of clusters in which term t appears. The structure mirrors document-level TF-IDF but operates at the cluster level, producing topic descriptions that are typically more readable than LDA’s top-word lists.

Two practical details make a meaningful difference in production. First, **sum-pooling or concatenating the last four hidden layers of an SBERT** encoder, rather than relying on the default last-layer mean pooling, yields roughly 10–20% relative **improvement** in topic diversity and coherence on news corpora. Second, **stop-word removal** before embedding generation rather than only at the c-TF-IDF stage improves downstream cluster quality measurably. Neither change is large in isolation; both compound across a production pipeline. For finance specifically, a domain-tuned encoder swap (**FinBERT-MPNet**, **FinLang**, or a **LoRA-adapted** strong general encoder like bge-large) typically delivers larger gains than any of these tuning choices, because the underlying semantic space is more faithful to financial language to begin with.

BERTopic’s modularity is its second advantage. The encoder, the dimensionality reduction step, the clustering algorithm, the topic representation method, and the optional reduction of topics down to a target number can all be swapped independently. This is the practical reason BERTopic dominates exploratory thematic discovery: the same conceptual pipeline survives the substitution of any component when a domain-specific need arises.

12.4 Top2Vec: A Joint Embedding Approach

Top2Vec [5] takes a different design path. Rather than the encoder-then-cluster-then-label sequence of BERTopic, Top2Vec embeds documents, words, and topics into a single shared semantic space. Topics are identified as dense regions of document vectors. The topic’s representative words are the words whose vectors are nearest to the topic’s centroid in the same space.

The unified representation has two consequences. First, the topic labels and the topic centroids live in the same coordinate system, so the words describing a topic are guaranteed to be semantically close to the documents in that topic by construction. Second, no explicit class-based scoring step is needed; the topic-word relationship is read directly off the embedding geometry.

Top2Vec’s limitations track its design. Like BERTopic, it is fundamentally cluster-based, which produces hard assignment by default. For very large corpora, the joint embedding of documents and vocabulary becomes computationally demanding. The model also depends on the quality of the underlying word and document embeddings; a poor encoder choice does not improve through any later refinement step. In practice, Top2Vec sits alongside BERTopic as an alternative for use cases where the conceptual simplicity of a shared embedding space matters more than BERTopic’s modular flexibility.

12.5 Contextualized Topic Models

Contextualized Topic Models [10] bridge the classical and the neural by combining a variational autoencoder with two input views per document. The VAE’s inference network processes a traditional bag-of-words vector and a dense contextualized sentence embedding (typically SBERT

or a domain-tuned alternative) simultaneously. The decoder reconstructs the bag-of-words representation while conditioning on the contextual vector. The result is a model that produces probabilistic document-topic distributions in the LDA tradition while leveraging the semantic depth of transformer embeddings.

The architecture is worth the additional complexity when soft topic mixtures matter. A 10-K filing, an earnings-call transcript, or a long broker note rarely belongs to a single topic. The cluster-based output of BERTopic and Top2Vec assigns the whole document to one cluster (or to the noise bin) by default; recovering mixed membership requires post-hoc heuristics covered in the assignment section below. CTM produces a continuous topic distribution per document by construction, which is exactly the input downstream exposure-scoring and attribution models need.

CTM also extends in directions that the basic version does not capture. The CTM-Neg variant adds contrastive training: the reconstructed document is treated as an anchor whose similarity to the input is maximized while its similarity to documents reconstructed from adversarially perturbed latent vectors is minimized. The contrastive objective drives higher topic diversity and stronger coherence scores than the basic CTM. The related Embedded Topic Model (ETM) [33] represents words and topics as vectors in a shared embedding space within an LDA-like generative structure, providing yet another point on the spectrum between purely classical and purely neural. ProdLDA / NeuralLDA [106] was the first amortized variational inference for topic models and remains the historical bridge between Gibbs-sampled LDA and modern neural variants.

For financial applications, CTM is the right choice when the downstream task explicitly requires soft document-topic weights and the documents are long enough to genuinely span multiple topics. For headline-scale corpora, the additional complexity rarely pays off; BERTopic remains the more practical default.

12.6 Going to the Token Level: CWTM

Document-level autoencoders inherit one structural limitation: the bag-of-words representation that feeds the encoder discards word order and treats every term as known. Out-of-vocabulary terms and word-sense ambiguity are handled poorly, which matters in financial text where new terminology arrives faster than vocabularies can be updated and where the same word (“visa,” “apple,” “target”) can refer to entirely different entities depending on context.

The Contextualized Word Topic Model (CWTM) addresses both limitations with a Wasserstein autoencoder operating on token-level contextualized word embeddings drawn directly from BERT. By bypassing the bag-of-words assumption at the input layer, CWTM preserves word order and contextual sense disambiguation. The result is a model that is exceptionally robust to rare financial terminology and evolving domain vocabulary, two properties that classical and standard neural topic models both struggle with.

CWTM is the most recent and least battle-tested model in this family. Its appeal in finance is specifically for corpora where new vocabulary arrives faster than the encoder’s pretraining can absorb — emerging technologies, novel financial instruments, rapidly evolving regulatory language. For mainstream English news on established themes, BERTopic and CTM remain the more practical defaults.

12.7 Comparing the Families

The properties that distinguish these models for thematic investing are summarized below.

Model	Foundation	Input	Short-text	Assignment
LDA	Bayesian probabilistic	Bag-of-words	Poor	Soft (Dirichlet mixture)
NMF	Algebraic factorization	TF-IDF matrix	Moderate	Soft (matrix weights)
BERTopic	Density-based clustering	Transformer embeddings	High	Hard (with noise bin)
Top2Vec	Joint embedding	Transformer embeddings	High	Hard (cluster-based)
CTM	Variational autoencoder	BoW + transformer	High	Soft (probabilistic)
CWTM	Wasserstein autoencoder	Token-level contextual	High	Soft (distribution)

Two columns deserve special attention. The Short-text column reflects how well each model handles documents like headlines or tweets where word co-occurrence within a document is sparse. Classical LDA collapses; embedding-based models handle short text reasonably because their similarity structure is inherited from the encoder rather than estimated from the corpus. The chapter that follows develops the specialized short-text models that go further in this direction. The Assignment column reflects whether the model natively produces a per-document topic distribution. Soft assignment is the default for the classical and VAE-based families; cluster-based methods require additional handling, which is the subject of the next section.

12.8 Hard, Soft, and Sliding-Window Assignment

Cluster-based models assign each document to exactly one cluster (or to the noise bin) by design. This is reasonable for clean headline corpora where each document is naturally about one event. It breaks down for long documents that genuinely cover multiple themes — precisely the documents thematic exposure scoring most needs to handle correctly. A 10-K business overview might discuss AI capabilities, supply-chain restructuring, currency exposure, and regulatory uncertainty all in the same five pages. Forcing the whole document into a single cluster destroys most of the information the downstream system needs.

Three patterns are available for recovering soft assignment from cluster-based models. The first is the simplest: compute the cosine similarity between the document embedding and each cluster centroid, then treat the normalized similarities as a topic distribution. This works well when clusters are well-separated in the embedding space and poorly when they overlap.

The second is to operate at sub-document granularity. A long document is partitioned into sections, paragraphs, or sentences before embedding. Each piece is assigned to its nearest cluster, and the document-level distribution is recovered by aggregating the assignments weighted by piece length. For financial documents this approach has the added benefit of respecting the structural sectioning of 10-Ks and earnings-call transcripts already established in Chapter 9.

The third is the sliding window: the document is partitioned into overlapping token windows of length 4–8 with stride 1–4, and each window’s embedding is scored against all cluster centroids. The final document-topic distribution aggregates the per-window scores, typically with averaging weighted by window confidence. The sliding-window pattern allows a BERTopic-family model to produce LDA-equivalent soft assignment while keeping the semantic depth of the transformer encoder. The cost is computational: a 500-word document becomes hundreds of windows to embed and score.

The choice between these patterns depends on document length and required precision. For headlines and short alerts, the cosine-to-centroids approach suffices. For long filings and

transcripts, sub-document sectioning is usually the right default. For documents whose topical structure cannot be predicted from their sectioning (long social posts, podcast transcripts, free-form research notes), the sliding window is the more general but more expensive alternative.

12.9 Choosing a Model Family

The right model is not classical or neural in the abstract. It depends on the corpus, the document length, the soft-assignment requirement, the production stability budget, and the downstream task.

- Use NMF or LDA as interpretable baselines and as the honest comparison for any more complex model.
- Use BERTopic for exploratory discovery on long-form English text — news, transcripts, research reports — where modularity, finance-tuned encoders, and readable c-TF-IDF labels are the operating priorities.
- Use Top2Vec when the conceptual simplicity of a shared semantic space matters more than BERTopic’s modular flexibility.
- Use **CTM** or **ETM** when the downstream task explicitly requires probabilistic document-topic distributions and the documents are long enough to genuinely span multiple topics.
- Use **CWTM** when the corpus contains substantial novel terminology that pretrained encoders are unlikely to have seen.
- Use short-text specialists — BTM, GSDMM, hybrid models — for headline-scale corpora where embedding-based methods still struggle with sparsity.

The practical workflow for an AI infrastructure thematic system might use BERTopic to discover sub-themes across news and earnings calls, CTM to estimate soft exposure scores on long broker reports, and entity reconstruction from Chapter 10 to map both back to a tradable company universe. The classical baselines from Chapter 11 stay in the pipeline as a stability check. None of these methods dominates uniformly; the question is which combination matches the corpus and the decision the system has to make.

The strongest reason to keep more than one family alive in a production thematic system is that their failure modes are different. Classical models fail by collapsing distinct themes into one or by splitting one theme across redundant clusters. Embedding-based models fail by producing topics whose underlying semantic structure has shifted between training and inference, by becoming dependent on encoder choices that are hard to audit, or by inheriting biases from pretraining data that nobody on the team can interrogate. Two independent systems detecting the same theme is evidence the theme is real. One system detecting it is a hypothesis worth checking against a second.

Chapter 13

Short Texts, Headlines, and Sparse Streams

13.1 The Headline That Was Too Short for LDA

A research team builds a topic model on twelve months of business-press headlines, several hundred thousand of them, with the goal of tracking the AI infrastructure story as it develops one headline at a time. They run Latent Dirichlet Allocation, fifty topics, the same preprocessing pipeline that produced clean topics on their long-document corpus the month before.

The output is incoherent. The “AI” topic mixes earnings vocabulary, regulatory vocabulary, and merger vocabulary in the same word list. The “data center” topic contains almost no AI words and clusters around generic real-estate terms. Two headlines that should obviously belong to the same story — “Cloud capex jumps on AI demand” and “Utility shares rise on data-center load” — end up in different topics because they share almost no actual words. Repeating the run with different hyperparameters does not fix the problem. Neither does adding stop-word filtering or bigram detection.

The issue is not preprocessing or parameter choice. LDA estimates topic mixtures from word co-occurrence within each document. A ten-word headline does not provide enough word pairs to estimate anything reliably. The model is being asked to learn from data it does not have.

Short text is the part of the financial corpus that arrives fastest and matters most for emerging-theme detection. It is also the part that classical topic models handle worst. The mismatch is severe enough that an entirely separate family of models was developed for this regime — biterm models that aggregate co-occurrence globally rather than per document, mixture models that assume one topic per short document, embedding-based methods that lean on transformer-learned similarity, and hybrid pipelines that combine the strengths of each. None of them is a strict improvement on LDA. Each is the right answer to a different question about what short text actually contains.

13.2 Why Short Documents Break Bag-of-Words

The classical topic models of Chapter 11 assume that each document is a mixture over topics and that the words in the document are drawn from the topic distributions in proportion to the mixture. The mixture is estimated from the words that actually appear together within the

document. The estimate is reliable when there are many such within-document co-occurrences and unreliable when there are few.

A typical newspaper article gives the model thousands of within-document word pairs to learn from. A 10-K filing gives tens or hundreds of thousands. A headline gives the model on the order of ten words, which is roughly forty-five unordered pairs. A tweet is comparable. A push notification is shorter. At those scales, the per-document topic mixture is essentially noise. The model either produces incoherent topics, as in the example above, or collapses everything into a small number of high-frequency topics whose top words are the most common terms in the corpus.

The problem is structural, not parametric. Increasing the number of topics makes it worse, because the model has even less per-document data to allocate to each topic. Decreasing the number of topics produces broad, uninformative clusters. Hyperparameter tuning of the Dirichlet priors smooths the symptoms without addressing the cause. Stop-word filtering removes some boilerplate but does not change the fundamental data scarcity.

Short text needs a different structural assumption. Three directions are available: pool short documents together until they are long enough to behave like documents (aggregation); change the model to learn from corpus-level co-occurrence rather than per-document mixtures (BTM); or change the model to assume each short document belongs to a single topic, eliminating the mixture estimation problem entirely (GSDMM). Each gives up something the classical model assumed and gets back the ability to discover topics in regimes the classical model cannot enter.

13.3 Aggregation: Trading Timing for Density

The simplest response to short-text sparsity is to stop treating each short document as a document. Instead, pool short texts into longer pseudo-documents and run a standard topic model on the pool. Headlines from the same company in the same day become one pseudo-document. Headlines from the same source on the same theme become another. Headlines from the same week, regardless of source or company, become another. The pseudo-documents are long enough to behave like documents under bag-of-words assumptions, and the underlying topic model becomes tractable again.

Aggregation works because most short-text streams have natural grouping structure. Companies generate clusters of related headlines around earnings, events, and announcements. Newswires generate clusters around stories that get covered in waves. Days generate clusters of headlines that share the day's dominant narratives. The grouping does not have to be perfect; it has to be informative enough that the pseudo-document captures more co-occurrence structure than any individual headline did.

The cost of aggregation is timing. A model that operates on weekly pseudo-documents cannot detect topics that emerge and decay within the week. A model that pools by company cannot track narratives that span companies until aggregation happens upstream. The aggregation window has to match the rebalancing cadence of the strategy that consumes the topic output. For a monthly thematic process, weekly aggregation is fine. For a daily emerging-theme monitor, weekly aggregation is fatal. The grouping dimension also has to match the question: pooling by company is appropriate for exposure scoring, pooling by day is appropriate for attention tracking, pooling by source is appropriate for transmission-chain analysis. A single aggregation scheme rarely serves all three.

Aggregation is the right starting point precisely because it requires no new model machinery.

The classical apparatus of Chapter 11 continues to work; only the document boundaries change. When aggregation is not sufficient — when the timing the strategy needs is shorter than the window aggregation requires — the structural alternatives take over.

13.4 The Biterm Topic Model

The Biterm Topic Model [122] changes the assumption at the heart of LDA. Instead of modeling word co-occurrence within each document, BTM models word co-occurrence across the entire corpus. The unit of analysis is the biterm: an unordered pair of words appearing within a designated sliding window in some document.

Both words in a biterm are assumed to be drawn from the same latent topic, which is itself drawn from a corpus-level topic distribution. The generative story is simple: a biterm is generated by picking a topic and then drawing both words independently from that topic's word distribution. Inference aggregates biterms across all documents, learning topic-word distributions from the global pool rather than from any individual document.

The change is consequential for short text. A headline that contributes only a few biterms still contributes them to a corpus-level estimation problem that has millions of biterms total. The estimate for each topic-word distribution is based on the global pool, not on the per-document mixture LDA tries and fails to estimate. The downside is that BTM does not produce a per-document topic distribution by default; the model is about topics in the corpus, not about which topics each document covers. For thematic systems that need per-document scores, post-hoc inference can recover an approximate document-topic distribution from the biterms each document contains, but the result is less principled than LDA's native mixture vector.

BTM is the right choice when the goal is to discover topics from short text and the per-document distribution is either not needed or recoverable downstream. For headline-scale financial corpora — newswires, social posts, push alerts — BTM frequently produces more coherent and more useful topics than LDA fit to the same data. Its weakness is that it assumes both words in a biterm genuinely belong to the same topic, which holds well for short text where there is rarely room for multiple topics within a single document and holds poorly for longer documents where biterms can span topic boundaries.

13.5 GSDMM: One Topic per Document

The Gibbs Sampling Dirichlet Multinomial Mixture model [124] takes a complementary route. Rather than aggregating across documents, GSDMM assumes that each short document belongs entirely to one topic. The mixture problem disappears by assumption: a headline is not a 40%/30%/30% blend of three topics; it is a single topic plus noise.

The single-topic assumption matches the empirical reality of short text more closely than the LDA mixture model does. Most headlines are about one event, one company, or one theme. The mixture interpretation is what LDA needs to make sense of long documents; for headlines, it adds parameters the data cannot estimate. GSDMM keeps the Dirichlet priors on topic and word distributions but assigns each document to a single cluster, and the assignment is updated iteratively via Gibbs sampling that balances two principles: completeness, the preference for clusters whose existing documents resemble the new one, and consistency, the preference for clusters whose word distributions align with the new document's words.

A useful side effect of GSDMM is automatic cluster-count inference. The model starts with an over-specified number of clusters; clusters that fail to attract documents during sampling are allowed to disappear. The number of clusters that survive is informative about the true thematic structure of the corpus rather than an arbitrary choice. For exploratory work on a new short-text stream, this is a real advantage over LDA’s fixed- K requirement.

GSDMM’s limitation is the one-topic assumption itself. Headlines that genuinely cover multiple themes — “Apple buys AI chip startup” touches corporate finance, AI, and M&A — are forced into the single topic that fits best. For most short text this is acceptable because the cases where it matters are rare. For social-media streams where ambiguity is more common, the assumption is more costly.

13.6 Embedding-Based Methods for Short Text

The embedding-based topic models of Chapter 12 handle short text reasonably well by default. The semantic similarity structure they rely on comes from the encoder’s pretraining, not from corpus-level word co-occurrence. A finance-tuned sentence encoder can map “GPU shortage” and “accelerator scarcity” to nearby vectors even when neither headline ever appears in the same document. BERTopic on headlines is a usable system out of the box in a way that LDA on headlines is not.

The cost is a different set of failure modes. Embedding-based methods on short text tend to produce many noise points: short documents that do not fit any dense cluster end up in HDBSCAN’s noise bin. They tend to split topics finely, producing dozens of small clusters where a coarser model would produce a few large ones. And the cluster boundaries can shift on retraining in ways that classical models do not. The instability is acceptable for exploratory work and uncomfortable for production systems that need consistent topic IDs across days.

Two specific embedding-based interventions help on short text. First, careful entity masking before embedding — the technique introduced in Chapter 10 — prevents the model from clustering by company name rather than by topic, which is the dominant failure mode on financial headlines where company names appear in almost every document. Second, the soft-assignment patterns from Chapter 12 can be adapted: cosine similarity to cluster centroids is the natural way to recover a per-headline topic distribution from a BERTopic model trained on headlines, since the alternatives (sub-document sectioning, sliding windows) require document length that headlines do not have.

13.7 Hybrid Approaches: Gibbs-BERTopic

Hybrid pipelines combine the strengths of embedding-based and probabilistic methods. The most developed example for short text is Gibbs-BERTopic, which uses BERTopic to produce an initial high-dimensional partition of the corpus and then applies a Gibbs sampler to iteratively refine the topic-document assignments. The refined distribution is fed back into the BERTopic pipeline for final c-TF-IDF topic-word updates.

The architecture addresses two specific failure modes of standalone BERTopic on small or noisy corpora. First, BERTopic tends to over-generate outliers on datasets smaller than roughly a thousand documents, which is exactly the regime where many emerging-theme corpora live early in their lifecycle. The Gibbs refinement reclaims many of these outliers into existing clusters where they belong. Second, BERTopic’s HDBSCAN clusters can be coarse in regions

of the embedding space where document density is uneven. The Gibbs refinement smooths the boundaries by allowing documents near a cluster edge to be reassigned based on a probabilistic criterion rather than the deterministic density-based one.

On sparse and noisy datasets, Gibbs-BERTopic reports increases in silhouette clustering coefficient and topic coherence on the order of 20% compared to standard BERTopic. The improvement is meaningful for systems that need stable topic structure on small short-text corpora. The cost is computational: the Gibbs sampler adds inference time to a pipeline that was already slower than classical alternatives, and it introduces additional hyperparameters (Gibbs iterations, prior strengths) that need to be tuned.

The general lesson from hybrid pipelines is that no single short-text method dominates. Embedding-based models handle semantic similarity well; probabilistic models handle uncertainty and refinement well; pooling them in well-designed sequences is often the best result a thematic system can achieve on real short-text data.

13.8 Noise, Outliers, and First-Story Detection

Short-text streams contain noise. Repeated headlines from syndication, irrelevant alerts, ticker-only messages, market memes, spam, and boilerplate disclosures all appear in a typical financial headline feed. Some of this noise is genuinely uninformative and should be filtered upstream. Some looks like noise but is the early form of a signal that has not yet attracted enough documents to form a cluster.

HDBSCAN-style methods explicitly label low-density points as noise, which is normally a useful filtering step. For emerging-theme detection it is also where the most valuable signal sometimes lives. A headline that appears in the noise bin today, alongside a handful of semantically similar headlines that also appear in the noise bin, is a candidate first story for a theme that has not yet crystallized. If a similar headline appears again next week, and the cluster of noise points grows, the cluster is on its way to becoming a topic. A monitoring system that discards noise outright will not see this. A monitoring system that maintains an outlier watchlist — semantically clustered noise points tracked over time — will.

This is the connection between short-text modeling and the temporal-emergence work in Part IV. Most emerging themes do not arrive in a recognizable form; they arrive as a small, semantically related cluster of noise points that grows over the next few weeks. The short-text pipeline is the layer that produces those noise points in a usable form. The temporal layer is what spots them coalescing into a coherent topic before mainstream coverage catches up.

13.9 Evaluating Short-Text Topic Models

Evaluation is harder for short text than for long-document topic modeling. The top-words lists are sparser. Coherence metrics that work reasonably on long documents (NPMI, UMass, C_v) become noisier when the underlying topic-word distributions are estimated from less data. The standard caveat about coherence metrics — that they often diverge from human judgment, as Hoyle and coauthors documented [60] — applies in sharpened form to short-text models, where the coherence numbers can fluctuate substantially across runs that produce similar topics.

The practical checks for short-text topic quality include representative-headline review (does a human analyst recognize the topic from the top documents?), cluster purity by analyst judgment, stability across adjacent rolling windows, entity distribution per cluster (does a cluster have a

balanced mix of companies or is it dominated by one?), and downstream relationship to volume, volatility, or returns. For a headline-scale thematic system the most reliable validation is extrinsic: does the cluster’s appearance and disappearance line up with known event timelines, and does it carry investment information that a simpler signal does not?

A topic that looks semantically coherent on inspection but has no relationship to subsequent market behavior is still useful as a monitoring signal — it tells the system that some narrative is gaining attention — but it should not be treated as alpha. The discipline of distinguishing semantically coherent topics from investment-coherent topics is essential at the short-text scale, where the model is producing many candidate topics and most of them will not survive the test.

13.10 A Short-Text Pipeline for AI Infrastructure

For AI infrastructure specifically, a usable short-text pipeline combines the methods from the preceding sections. Headlines about chip orders, power shortages, data-center expansion, memory pricing, export controls, and cloud capex are the fast-moving end of the corpus. They use different vocabulary from earnings calls and filings, they arrive in bursts, and they require source-aware deduplication before any topic model sees them.

A reasonable design starts with entity masking and outlet-tag removal from Chapter 10, runs BERTopic with a finance-tuned encoder on the cleaned headlines, recovers a soft per-headline topic distribution via cosine similarity to cluster centroids, and maintains an HDBSCAN noise watchlist for emerging-theme candidates. GSDMM or BTM serves as a probabilistic baseline that catches cases where BERTopic over-fragments or treats a coherent theme as outliers. Aggregation by day or by company-day provides the pseudo-documents that a classical baseline can consume for stability checks. Each component contributes a piece of the answer; none of them is sufficient alone.

The same pipeline applied to clean-energy headlines from 2020–2022 would have detected the early signs of the duration unwind through a different path: not in any single headline, but in the pattern of headlines about rates, inflation, and growth-stock valuations that started clustering with clean-energy coverage in late 2021. A short-text system that tracks how its topic boundaries shift over time is doing the work the long-document corpus is too slow to do — and short text is precisely where most narratives first become visible to a system that knows how to read them.

Chapter 14

Labeling, Interpretation, and Analyst Review

14.1 The Label That Sold a Thesis

A research team runs BERTopic on six months of semiconductor-sector news. The model produces a tight cluster: roughly forty headlines about export controls, equipment shipments to China, advanced-node restrictions, and customs filings. The cluster is internally coherent. The top c-TF-IDF terms include “chip,” “export,” “China,” “ASML,” “semiconductor,” and “advanced.” An LLM is asked to generate a human-readable label from the top terms and three representative headlines.

The label comes back as “AI Chip Boom.”

The label is not wrong about the headlines being related to each other. It is wrong about what the headlines say. The cluster is not about an AI chip boom; it is about regulatory headwinds restricting the equipment that produces advanced chips. A portfolio constructed from the label would buy the firms most exposed to the restrictions. A portfolio constructed from the underlying evidence would short them, or at minimum flag them for hedging. The label and the evidence point in opposite trading directions.

This kind of failure happens because labeling is not neutral. A label inherits the priors of the model that produced it. “AI” is a more frequent collocation in the LLM’s pretraining data than “export controls,” so the LLM produces the more familiar phrase even when the evidence supports the less familiar one. The model is doing what it was designed to do. It is producing readable, plausible labels. It is not producing labels that survive analyst review.

Topics are mathematical objects: clusters, word distributions, document-topic weights, embedding neighborhoods. Themes are investment objects: economic exposures with directions, horizons, and named beneficiaries. The translation between them is a labeling process whose failure modes are different from the failure modes of the topic model itself — and whose discipline determines whether the topic-model output is investment-grade or merely technically impressive.

14.2 Topics Are Not Yet Themes

A topic model can be perfectly calibrated and still produce labels that mislead the portfolio that consumes them. The mismatch is structural: the model evaluates topics on word-co-occurrence properties, while the investor evaluates labels on economic plausibility. A topic that scores well

on coherence metrics like NPMI can carry a label that an analyst would reject in seconds. A topic that scores poorly on the same metrics can carry a label that an analyst would defend with a paragraph of evidence.

The labeling discipline sits between the topic model and the theme library. Its job is to turn each topic into an artifact that an analyst can review, that a portfolio manager can understand, that a risk committee can audit, and that does not encode an investment direction the evidence has not earned.

A poor label can create false confidence. If a cluster of words is presented as “AI winners,” the label has already implied an investment conclusion. A better label might be “data-center power-demand coverage” or “advanced-chip export-control restrictions.” Good labels describe the evidence before they sell a thesis. The opening example failed at exactly this step: the label sold a thesis the evidence did not support.

14.3 Word Lists Are a Starting Point

Classical topic models represent topics with ranked word lists. Embedding-based methods produce c-TF-IDF top words at the cluster level. Either way, the analyst’s first input is a list of terms ranked by importance for the topic. The list is useful and incomplete.

A topic with top words “power, data, center, demand, grid, utility” may suggest a theme, but the words alone do not say whether the topic concerns electricity demand growth, transmission constraints, renewable integration, or rate-case regulation. Each of those would imply different beneficiaries. Without further evidence, the word list is consistent with all of them.

Word lists should be read with care. Three failure modes are common. First, high-frequency common terms can dominate the ranking even when domain-specific stop-word handling has been applied; “company,” “market,” and “year” are not informative regardless of the topic they appear in. Second, entity names can obscure the underlying theme; if “Nvidia” is the top word, the topic is partly a Nvidia topic regardless of whether the cluster also captures something general about accelerator demand. Third, broad words can make a topic look more coherent than it is; a topic whose top words are all generic technology vocabulary may appear unified while actually mixing several distinct sub-stories. The top words are a hypothesis to verify, not the verification itself.

14.4 Representative Documents as Evidence Packages

The most reliable input to a labeling decision is not the word list but the documents themselves. A small set of representative documents shows how the topic actually appears in context: what words surround the top terms, which entities are mentioned alongside which, what kinds of events are being reported, and whether the cluster is a single coherent story or two stories that happen to share vocabulary.

For each topic awaiting a label, a useful review package contains:

- the top c-TF-IDF or weighted terms;
- five to ten representative documents near the cluster center;
- three to five documents near the cluster boundary, which often reveal where the topic might fragment;

- a small set of outlier documents that did not fit any cluster but are semantically close to this one;
- the source distribution across newswires, transcripts, filings, social, and broker research;
- a time series of topic document volume;
- the top linked entities reconstructed via the entity-handling pipeline of Chapter 10;
- a stability indicator showing how the cluster boundaries shifted across recent rolling-window refits.

The package is the unit on which a labeling decision is made. A label assigned without the package is a label assigned to the word list, which is precisely the move that produced the opening section’s “AI Chip Boom” failure. The package costs more to produce. It is worth the cost.

14.5 Entity-Aware Labels

Financial topics often require entity context that pure word frequencies hide. A topic about “chips, export, controls, China” could involve semiconductor designers, equipment makers, end-customer firms, regulators, customs agencies, or several countries. The label should reflect the economic relationship the cluster actually documents, not just the most frequent words.

Entity-aware labeling uses the reconstructed entity layer from Chapter 10. After entity masking during topic discovery prevents the cluster from collapsing into a company list, the reconstruction step restores which entities appeared in which documents. The labeling step can then ask:

- Which companies appear most often, and across how many distinct sources?
- Are the entities suppliers, customers, regulators, or competitors to one another?
- Are entity mentions evidence of material exposure or merely examples used to illustrate a broader phenomenon?
- Does the entity mix suggest a value-chain story (suppliers and customers appearing together) or a sector story (competitors appearing together)?

For AI infrastructure, entity-aware labeling can distinguish “AI chip export controls” from “AI memory supply shortage” even if both topics share semiconductor and data-center vocabulary. The difference is in which entities recur: regulators and equipment makers for the first, memory suppliers and accelerator customers for the second. Entity-aware labels make this difference visible.

14.6 Phrase Labels

Single words are usually too ambiguous to serve as topic labels. “Power” could mean electricity, market dominance, or political authority. “Cooling” could mean heat management or market deceleration. “Memory” could mean DRAM or a corporate-memory metaphor. Phrase extraction produces labels that preserve domain meaning: “high-bandwidth memory,” “data-center power

demand,” “liquid cooling,” “advanced packaging,” “export controls,” “cloud capex,” “hyperscaler buildout.”

Phrase labels can be extracted using n -gram statistics, part-of-speech patterns over candidate noun phrases, embedding similarity to known domain-vocabulary anchors, or language models constrained to selecting from the cluster’s documents. Whatever method generates them, phrase labels need validation against representative documents. A phrase that sounds compelling but appears rarely in the underlying cluster is more likely a hallucinated summary than a faithful label, and the discipline of checking phrase labels against document frequency is what separates label generation that supports the analyst from label generation that performs for the analyst.

The financial-domain language problem documented by Loughran and McDonald [78] applies here in a sharpened form. A general-purpose phrase extractor will preferentially surface phrases that are common in its training corpus, not phrases that are common in the cluster being labeled. For finance, the preferred phrases are often domain-specific terminology that a generic phrase extractor underweights. A finance-tuned phrase extractor, or an explicit constraint that labels must be drawn from the cluster’s actual documents, prevents the model from substituting plausible-sounding generic vocabulary for the technical vocabulary the cluster actually contains.

14.7 LLM-Assisted Labeling

Large language models can convert topic evidence into readable names and summaries faster than any analyst can. They can also generate the kind of failure illustrated in the opening section, where the label is plausible and wrong in a direction the model has no way to detect. The discipline of LLM-assisted labeling is the discipline of constraining the LLM to summarize what the evidence says rather than infer what the evidence might imply.

A well-constructed labeling prompt provides:

- the top c -TF-IDF or weighted terms;
- a sample of representative documents (typically 5–10);
- the source-type and date-range distribution;
- the top linked entities;
- explicit instructions to base the label on the supplied evidence and to avoid extrapolating to investment recommendations;
- a required output schema with separate fields for label, summary, supporting evidence, and uncertainty.

The model should not be asked to infer investment direction from topic words. It should be asked to describe what kind of story the cluster contains, with citations to specific representative documents. Human review then approves, revises, or rejects the proposed label before it enters a production theme library.

The chapter on LLM workflows in Part V develops the broader discipline — prompt design, output stability, retrieval-augmented evidence packages, hallucination detection, evaluation under model cutoff and leakage — in detail. The labeling step is one specific application of that broader discipline, with the same operating constraint: the LLM is a summarization tool, not an analytical tool. Anything that requires judgment about what the evidence *means* for an investment decision belongs to the analyst and the portfolio manager, not to the model.

14.8 Analyst Review as a Control

Analyst review is sometimes treated as a weakness in an otherwise automated pipeline. It is not a weakness. It is a control. Analysts catch the categories of error that automated systems generate by construction: incoherent clusters that look unified on metrics, misleading labels generated from biased priors, boilerplate topics that should not have entered the corpus, stale narratives that the recency-weighted clustering preserved by accident, and false company links from imperfect entity reconstruction.

A structured review process produces auditable decisions. For each topic, the analyst:

1. inspects the evidence package;
2. proposes or approves a label, and revises the LLM-generated label if needed;
3. classifies the topic into one of a fixed set of states — monitoring-only, candidate theme, active theme, deprecated, rejected;
4. records the reason and a confidence indicator;
5. specifies the falsifiability criteria from Chapter 6: what observable evidence would change the classification.

Every analyst decision is logged with the timestamp it was made, the model version it was made against, and the identity of the analyst. The logs serve two purposes. First, they make the historical labeling decisions auditable when a portfolio question arises about why a particular theme was active at a particular date. Second, they prevent the most common contamination of historical research: revising old labels with knowledge that did not exist at the time. A backtest that uses today’s analyst-approved label for a 2022 topic is using future information unless the historical label is recoverable from the logs.

14.9 Label Stability Across Time

Labels need to be stable enough to support monitoring and flexible enough to evolve with the underlying topic. A label that changes every week breaks any time-series analysis that depends on consistent topic identification. A label that never changes hides important evolution in what the topic actually contains.

The resolution is to separate the topic identifier from the topic description. A stable internal ID persists as long as the aligned topic remains semantically coherent. The human-readable description — which is what the analyst sees and what the report uses — can update as vocabulary shifts. An AI-infrastructure cluster that began life in early 2023 as “GPU supply constraints” can become “AI accelerator demand” by mid-2023, “high-bandwidth memory bottleneck” by late 2023, and “advanced-packaging capacity” by 2024, all under the same stable ID, as long as the topic-alignment machinery of Part IV can establish that the cluster’s center has moved continuously rather than discontinuously.

Discontinuous changes — when a topic actually splits into two or merges with another — need explicit handling. A split that is silently treated as label evolution destroys the continuity of any historical analysis that depended on the original topic. The conservative discipline is to retire the original ID, issue new IDs for the descendants, and record the split event in the same logs that hold the analyst decisions. The cost is a small amount of bookkeeping. The benefit is

that historical attribution of returns to themes does not become ambiguous when the underlying topic structure shifts.

The same logic applies to label instability across model runs, which Hoyle and coauthors documented as a general weakness of automated topic-model evaluation [60]. Two retrainings of the same model on the same data can produce topics that an automated coherence metric scores as equivalent but that a label-stability check reveals as substantially different. The label review process should include a stability indicator: how much does the topic’s word distribution shift between adjacent runs, and how much does the recommended label change as a consequence? A label that changes substantially across runs of the same model is a label that should be reviewed by a human before it is published anywhere downstream.

14.10 From Topic Label to Theme Candidate

A labeled topic is not yet a theme. The label is necessary; it is not sufficient. The promotion from labeled topic to theme candidate requires the additional checks that any investment process applies to a hypothesis:

- Is the topic economically meaningful, or is it a vocabulary cluster with no underlying mechanism?
- Does it connect to tradable companies in a verifiable way, or does it describe a phenomenon (regulatory uncertainty, geopolitical risk, a macro shift) without specific equity exposure?
- Is there evidence of material exposure — revenue, capex, supply-chain relationships, customer concentration — or only mentions in press releases and analyst questions?
- Where does the topic sit on the seven-stage lifecycle from Chapter 6: weak signal, emerging, validated, crowded, institutionalized, decaying, recycled?
- Does the topic duplicate exposure that already exists in a known sector or factor, or does it survive the residualization tests introduced in Chapter 4?
- Is the expected horizon of the implied trade compatible with the source mix and document type that produced the topic?

For AI infrastructure, a topic labeled “data-center power-demand coverage” becomes a candidate theme if it links to utilities with new long-term offtake agreements, grid-equipment suppliers with order-book acceleration, cooling-system providers with named hyperscaler customers, and operators with material data-center exposure — and if the residual returns of a basket built from those firms survive controls for the utility sector and for interest-rate sensitivity. A topic labeled “AI mentioned in press releases” fails the same test for the obvious reason: mentions are not exposures.

The opening section described a label that sold a thesis the evidence did not support. The closing discipline is the same in the other direction. A label that survives the full set of checks above — evidence-based, entity-grounded, lifecycle-classified, factor-controlled — is the kind of label that can defend itself when a portfolio manager asks what the theme actually is. The labels that cannot defend themselves at that question are the ones the review process exists to catch.

Part IV

Tracking Themes Through Time

Chapter 15

Dynamic Topic Modeling

15.1 The Theme That Changed Its Vocabulary

In early 2023, the AI infrastructure trade sounded like a semiconductor trade. Nvidia was the name everyone knew. The market learned to say GPU, accelerator, training cluster, and inference chip. The vocabulary was technical, but the portfolio map was simple: find the companies that sold the machines.

Then the story moved.

By late 2023 and 2024, the same theme had acquired a different set of bottlenecks. High-bandwidth memory entered the conversation. So did advanced packaging, optical networking, hyperscale capex, liquid cooling, grid interconnection, and electricity supply. A static topic model would be tempted to call all of this “AI infrastructure” and move on. That label is not wrong. It is just too blunt.

The investor’s problem is not only to recognize the theme. The investor needs to know which part of the theme is becoming marginally more important now.

Static topic models, the focus of Part III, give us a vocabulary map. They help separate earnings-call language about AI chips from language about clean energy, drug discovery, or consumer software. But a theme is not a fixed bag of words. It is a living object. It attracts new companies, drops old buzzwords, splits into sub-themes, and sometimes keeps the same label while its economic center of gravity has moved.

Dynamic topic modeling turns time from a timestamp into part of the model. The question changes from “What topics exist in this corpus?” to “How did this topic get here, what did it use to mean, and what is it becoming?”

15.2 What Time Adds to a Topic

An analyst who reads every Nvidia, SK Hynix, TSMC, Microsoft, and utility transcript from 2022 through 2024 does not store those documents as one undifferentiated pile. She remembers sequence. First came the sudden demand for accelerators. Then came the supply squeeze in memory and packaging. Then hyperscalers discussed data-center capex with a new intensity. Then electricity providers, grid operators, and cooling specialists entered a conversation that had looked, at first, like a pure semiconductor story.

A dynamic topic model tries to preserve that sequence at scale.

Four pieces matter.

1. **Identity.** A topic needs a recognizable thread across time. If “AI infrastructure” appears in March, June, and September, the system should not treat those as three unrelated discoveries merely because the top words changed.
2. **Vocabulary drift.** The thread must be allowed to change. A model that freezes the early GPU vocabulary will miss the later memory, power, and cooling vocabulary.
3. **Birth and death.** Some topics do not drift. They arrive. A headline cluster around export controls, grid interconnection, or a surprise product cycle may have no clean predecessor.
4. **Point-in-time discipline.** A model used for backtesting cannot learn a topic’s full 2024 vocabulary and then pretend it knew that vocabulary in 2022. Dynamic modeling is useful only if time flows in the same direction as the portfolio.

The last point is the one that separates research graphics from investable signals. A beautiful retrospective topic timeline can be made with the full corpus. A trading system needs an as-of timeline: what the model knew on each date, using only documents available by that date.

15.3 The Classical Dynamic Topic Model

The classical Dynamic Topic Model, introduced by Blei and Lafferty, begins with the same intuition as LDA: each document mixes several latent topics, and each topic is a probability distribution over words [16, 14]. The difference is that the topic-word distributions evolve through ordered time slices.

Imagine one topic called “AI infrastructure.” In one quarter, the topic assigns high probability to words such as GPU, accelerator, training, CUDA, and cluster. In the next quarter, it should not forget those words completely. But it should be allowed to raise the probability of memory, HBM, packaging, networking, and power if those words begin appearing in the same thematic neighborhood.

The model handles this by evolving the natural parameters of the topic-word distribution. For topic k at time t , write those parameters as $\beta_{k,t}$. The dynamic prior says:

$$\beta_{k,t} \mid \beta_{k,t-1} \sim \mathcal{N}(\beta_{k,t-1}, \sigma^2 I). \quad (15.1)$$

The equation says something simple. A topic this month should look like its version last month, plus noise. The parameter σ^2 controls how much movement the model permits. A small value makes topics conservative and smooth. A large value lets them mutate faster.

The actual word probabilities come after a softmax transform:

$$\phi_{k,t,w} = \frac{\exp(\beta_{k,t,w})}{\sum_{v=1}^V \exp(\beta_{k,t,v})}, \quad (15.2)$$

where $\phi_{k,t,w}$ is the probability of word w in topic k at time t , and V is the vocabulary size. The softmax turns unconstrained real-valued parameters into probabilities that sum to one.

The mechanics are elegant because they encode a useful investment prior. Most durable themes do not reinvent themselves every week. They rotate. The AI infrastructure story can move from GPUs to memory to power, but a model should demand evidence before declaring that a completely new topic has replaced the old one.

The cost is rigidity. Classical DTM assumes an ordered set of time slices and smooth movement between adjacent slices. That is a reasonable assumption for a century of scientific

abstracts, the original style of application, but financial narratives often move in bursts. A regulator announces an export rule. A hyperscaler changes capex guidance. A company reports an unexpected bottleneck. The vocabulary can jump before the smooth chain has time to adapt.

15.4 LDASequence and the Practical Chain

Production systems often need the spirit of DTM without the full state-space machinery. LDASequence is the pragmatic version. Fit an LDA model on one time slice. Use the converged parameters to initialize the next slice. Repeat.

The method does not jointly infer the entire temporal process. It creates a chain of dependence by handing yesterday’s topic map to today’s model. That makes it easier to run, easier to explain, and easier to debug. It also makes it path-dependent.

Path dependence cuts both ways. It stabilizes topic identities, which helps the analyst follow a theme through time. It can also preserve an early mistake. If the initial “AI infrastructure” topic is really just a noisy cluster of mega-cap technology names, the chain can keep dragging that error forward even after the corpus contains enough evidence to separate chips, cloud platforms, memory, and power.

This is why every chained method needs diagnostic pressure. The analyst should inspect not only the topic label, but the movement of top words, representative documents, entity composition, and downstream portfolio exposure. A topic that keeps its name while its holdings become a disguised sector basket has not been tracked. It has been embalmed.

15.5 Irregular Time and Sparse Vocabulary

Calendar time is a poor clock for markets. Some weeks contain one routine earnings call. Others contain a shock that changes every analyst model before lunch.

Continuous-time dynamic topic models address this by replacing fixed discrete steps with a process that can move over irregular intervals. Instead of pretending that every month is equally informative, the model lets the distance between observations matter. Gaussian-process variants extend the idea further by choosing kernels that encode different temporal behaviors: mean reversion, smooth drift, bursts, or long memory.

In the technical literature, the simple continuous-time version often uses Brownian motion. More flexible variants use Gaussian-process kernels. An Ornstein-Uhlenbeck kernel pulls a topic back toward a long-run mean, which fits narratives that flare and normalize. A squared-exponential kernel favors smooth drift, which fits slow changes in management language. Dynamic embedded topic models, or **D-ETM-style systems, add embedding structure so semantically related words can move together rather than forcing every rare word to earn its own history.**

The intuition matters more than the notation. A theme about utility demand from data centers may accumulate slowly as executives repeat capex plans over several quarters. A theme about export controls may arrive abruptly in a burst of news and then fade into legal and compliance language. A single monthly clock treats those dynamics too similarly.

Specialized financial vocabularies add another problem: rarity. The words that matter most can be infrequent. HBM, EUV, chiplet, interposer, colocation, and grid queue are not common across a general business corpus. They are exactly the words a thematic investor cares about.

Multi-output Gaussian-process dynamic topic models, such as the MIST family discussed in the research literature, attack this by sharing information across related words and topics.

Rather than evolving each topic-word probability in isolation, they learn correlation structure among topics and among word groups. That helps when a rare word belongs to a small but economically important sub-field.

In finance, that rare-word problem appears constantly. The first documents that reveal a new bottleneck may not dominate document volume. They may be a handful of technical paragraphs in supplier transcripts. A model that waits for the vocabulary to become common will discover the theme after the market already has a ticker list.

15.6 Rolling Windows and Online Learning

The live news feed does not wait for a quarterly modeling run. Headlines arrive one at a time, vendors timestamp them in milliseconds, and investors want to know whether a cluster is forming before the weekly meeting.

Rolling-window methods fit or update a topic model on a recent slice of documents: the last thirty days of headlines, the last quarter of transcripts, the last year of filings. As the window moves, old documents fall out and new documents enter. The model stays responsive without pretending that a document from five years ago should carry the same weight as yesterday's earnings call.

Window length is an investment choice. A short window sees bursts and breaks. It also hallucinates structure from noise. A long window stabilizes topics, but reacts slowly when the market narrative rotates. For AI infrastructure, a short headline window may catch a sudden burst in power-grid language. A longer transcript window may better track whether management teams actually change capex plans, customer disclosures, or capacity commitments.

Online LDA gives the probabilistic version of this impulse. Hoffman, Bach, and Blei developed a stochastic variational inference approach that can update LDA on massive or streaming corpora [58]. The point is not only speed. The point is institutional memory. A system that updates continuously can carry yesterday's topic structure forward while incorporating today's documents without retraining from scratch.

Streaming cluster methods push further. They assign each incoming document to an active centroid if the document is close enough. If it is too far from every known topic, they spawn a new cluster. TopicStream-style systems are appealing for monitoring because they mimic how a human desk behaves: most headlines go into known folders; a few make the analyst create a new folder.

The danger is folder sprawl. Without decay rules, inactive topics clutter the system. Without merge rules, one theme becomes a dozen aliases. Without birth thresholds, every noisy headline becomes a discovery. Streaming models need governance as much as algorithms: expiration rules, minimum volume, novelty thresholds, and human review of new clusters that could affect capital.

15.7 BERTopic Over Time

Embedding models changed the practical frontier because they made semantic similarity cheap. BERTopic, discussed in Chapter 12, embeds documents, clusters them, and labels clusters with class-based TF-IDF [51]. Its temporal workflow starts from a global set of topic identities, then recomputes topic representations inside time bins.

That is useful when the analyst wants a stable map. Suppose the global model has a topic we call "AI infrastructure." For each month, the system can compute the topic's local c-TF-IDF

representation and show which words define it at that point. **The topic identity stays fixed.** The **vocabulary moves.**

The smoothing step is:

$$c_{k,t}^{\text{smoothed}} = \lambda c_{k,t} + (1 - \lambda) c_{k,\text{ref}}, \quad (15.3)$$

where $c_{k,t}$ is the time-specific c-TF-IDF vector for topic k , $c_{k,\text{ref}}$ is a reference vector, and λ controls how much the current time bin can deviate from the reference.

Two references are common. Global tuning uses the full global topic representation as the anchor. Evolutionary tuning uses the previous time bin as the anchor. Global tuning answers, “How did this month’s version differ from the overall topic?” Evolutionary tuning answers, “How did this topic move from the last period?”

The first is excellent for retrospective explanation. It is also dangerous in a backtest if the global model used future documents. A model fit on the whole 2024 corpus will know that power and cooling belong to AI infrastructure, then project that knowledge backward into 2022. The chart may be informative. The signal is contaminated.

Point-in-time BERTopic is possible, but it is more demanding. **The global map must be refit or extended only with documents known as of the decision date.** Labels must be generated from as-of representative documents. Topic IDs may need alignment because the cluster geometry changes as the corpus grows. Those inconveniences are the price of an honest backtest.

15.8 Chain-Free Tracking

Strict chains can become prisoners of their own continuity. A Markovian model asks topic t to follow topic $t - 1$. That works for gradual drift. It struggles when a theme splits, merges, disappears, or returns under new language.

Clean energy shows the problem. Solar, wind, hydrogen, EVs, and batteries can trade as one macro narrative when rates fall and capital floods into growth. Later, they can separate sharply. A chained model may force them to remain too close because they once shared a parent story. An independent-window model may do the opposite, treating each window’s topics as unrelated discoveries.

Chain-free dynamic topic models try to escape that binary. Instead of forcing one topic to inherit directly from the previous time slice, they place topics in a shared representation space and learn temporal relationships through similarity, contrastive learning, or matching. The CFDTM family is the explicit version of this idea: it decouples the Markov chain entirely and uses an Evolution-Tracking Contrastive (ETC) objective to keep semantically related topics close across time without forcing one slice to inherit from the previous. The framing reduces two failure modes chained models often produce: repetitive topics, where the chain accumulates stale vocabulary, and unassociated topics, where the chain drifts so far that the historical link no longer means much.

For thematic trading, the appeal is practical. A chain-free tracker can say: this new “data-center power” cluster is related to AI infrastructure, but it is not identical to the earlier GPU cluster. It can preserve ancestry without pretending nothing changed.

15.9 Choosing the Temporal Machinery

The best temporal method depends on the decision the portfolio must make. A monthly factor-neutral basket does not need the same machinery as an intraday alert system. A research note explaining a theme's history can use information a live strategy could not use. A production model cannot.

Approach	How it preserves time	Best use	Main risk
Classical DTM	Smooth Gaussian evolution of topic-word parameters across slices	Interpretable histories of durable themes	Too smooth for shocks; heavier inference
LDASequences	Initializes each slice from the previous fitted model	Stable topic IDs with simpler operations	Error propagation from early slices
Rolling window	Refits or updates on a recent document window	Responsive weekly or monthly monitoring	Window length drives results
Online LDA	Updates variational parameters as documents arrive	Large streaming corpora with probabilistic structure	Needs careful decay and diagnostics
BERTopic over time	Fixes or extends semantic topic IDs; recomputes time-bin c-TF-IDF	Analyst timelines and explainable vocabulary drift	Look-ahead if global model uses future data
Streaming clusters	Updates centroids and spawns clusters for distant documents	Live alerting and first-story detection	Topic proliferation and stale clusters
Chain-free tracking	Matches topics in a shared representation space rather than forcing one-step inheritance	Split, merge, and re-emergence behavior	More engineering and validation burden

The table hides a useful rule of thumb. If the problem is explanation, use the cleanest map. If the problem is live capital allocation, use the map that can be rebuilt honestly at every historical date.

15.10 The AI Infrastructure Thread

AI infrastructure is not one topic. It is a thread of related bottlenecks.

The first public version was easy to name: GPUs and accelerators. The next version was more industrial: high-bandwidth memory, foundry capacity, packaging, substrates, networking, power equipment, land, cooling, and grid access. Later versions may include software efficiency, inference economics, sovereign AI procurement, or regulation around data-center energy use.

A static topic model can discover the broad cluster. A dynamic model tells the investor whether the cluster is rotating toward a new scarce resource. That difference matters. A portfolio built around yesterday's bottleneck may own the right theme and the wrong companies.

Dynamic topic modeling does not solve thematic investing by itself. It supplies the time axis. The next problem is alignment: deciding when two topic snapshots are the same thread, when one thread has split, and when a new topic deserves to be treated as a real emerging theme rather than noise.

Chapter 16

Alignment, Drift, and Emerging Theme Detection

16.1 The Week One Topic Became Three

In the first wave of generative AI coverage, a lot of market language collapsed into one crowded bucket. Nvidia. ChatGPT. GPUs. Cloud capex. Model training. Every article seemed to point back to the same obvious trade.

Then the bucket cracked.

One branch became AI infrastructure: accelerators, memory, advanced packaging, networking, power, and cooling. Another became AI software: copilots, enterprise adoption, workflow automation, and subscription uplift. A third became AI regulation: copyright, data privacy, export controls, model safety, and national-security restrictions.

A static topic model can find those clusters after the fact. A dynamic model can show that the old AI topic changed. But an investment system needs one more layer: it must decide whether today's topic is the continuation of yesterday's topic, a split from it, a merger of several older threads, or a genuinely new candidate theme.

That layer is alignment. Without it, emergence detection is mostly theater. The system can announce that a new cluster has appeared, but it cannot say whether the cluster is actually new, merely renamed, seasonally recurring, or an old theme wearing a new set of words.

Alignment is the bookkeeping that makes temporal themes tradable.

16.2 The Topic Identity Problem

The phrase “data-center power” meant something different before AI capex became the market's favorite infrastructure story. It could refer to ordinary utility demand, regional grid constraints, or a commercial real estate issue. After the AI buildout accelerated, the same words started to sit near hyperscalers, GPUs, interconnection queues, cooling systems, and nuclear power.

The vocabulary overlap remained. The economic meaning changed.

Topic identity is hard because language moves in two directions at once. A theme can change its words while keeping its identity. It can also keep its words while changing its identity. The first case is drift. The second is semantic contamination.

For a portfolio, both are dangerous. If the model treats every vocabulary change as a new theme, the signal fragments into noise. If it treats every recurring word cluster as the same

theme, it misses the moment when a broad story becomes a different economic exposure.

An operational topic thread therefore needs more than a label. It needs a history:

- the topic IDs linked across windows;
- the labels used at each point in time;
- top words, representative documents, and linked entities;
- similarity scores between adjacent and historical windows;
- volume, acceleration, source breadth, and entity breadth;
- split, merge, re-emergence, and analyst-review flags.

The thread becomes the unit of monitoring. The cluster is what the model finds. The thread is what the investment process follows.

16.3 Representing a Topic for Alignment

Two clean-looking topic labels can hide very different fingerprints. “AI power demand” could be a utility transcript topic, a semiconductor-supply topic, a data-center construction topic, or a political topic about electricity prices. The label alone cannot align them.

A topic can be represented in several complementary ways:

- a sparse vector of top words or c-TF-IDF weights;
- a dense embedding centroid built from representative documents;
- a probability distribution over the vocabulary;
- a set of top entities, tickers, industries, countries, or value-chain roles;
- a source profile, such as newswires versus earnings calls versus filings;
- a return profile, if the topic has already been mapped to securities.

Each view catches a different failure. Word overlap catches obvious continuity. Embeddings catch semantic continuity when vocabulary changes. Entity overlap catches the finance-specific distinction between the same abstract topic and the same tradable exposure. Source profiles help distinguish a specialist rumor from a management-adopted theme.

For finance, entity continuity deserves special respect. If a topic’s words drift from “GPU” to “HBM” while its entities move from Nvidia to SK Hynix, Micron, and TSMC suppliers, the drift may be a legitimate value-chain rotation. If the words stay the same but the entities move from utilities to software companies, the model may be mixing two narratives that share surface language.

16.4 Similarity Measures and Matching

The first version of alignment is simple. Fit topics in adjacent windows. Compare every old topic with every new topic. Link the pairs that look sufficiently similar.

Cosine similarity works well for dense embeddings and TF-IDF-like vectors:

$$\text{cosine}(a, b) = \frac{a^\top b}{\|a\| \|b\|}. \quad (16.1)$$

Jaccard overlap works well for sets, such as the top ten entities or top twenty keywords:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (16.2)$$

Jensen-Shannon divergence works when topics are represented as word distributions. With $M = (P + Q)/2$,

$$\text{JSD}(P, Q) = \frac{1}{2} D_{\text{KL}}(P \| M) + \frac{1}{2} D_{\text{KL}}(Q \| M). \quad (16.3)$$

Because JSD is a divergence rather than a similarity, systems often convert it into a score such as $1 - \text{JSD}(P, Q)$ after normalizing the scale.

In practice, a composite alignment score is more robust than any single metric:

$$A(i, j) = w_x C_{ij} + w_e E_{ij} + w_v V_{ij} + w_s S_{ij}, \quad (16.4)$$

where C_{ij} is content similarity, E_{ij} is entity overlap, V_{ij} is vocabulary-distribution similarity, S_{ij} is source-profile similarity, and the weights reflect the use case. A headline-alert system may emphasize content and volume. A portfolio system should usually give more weight to entities and value-chain evidence.

Once the score matrix is built, the Hungarian algorithm can produce an optimal one-to-one matching between topics in window $t - 1$ and window t . This is a useful baseline. It creates clean topic threads and prevents every old topic from matching every new topic.

The one-to-one assumption is also where the trouble begins.

16.5 When One-to-One Matching Breaks

The clean-energy trade did not remain one thing after 2020. Solar, wind, batteries, EVs, hydrogen, grid equipment, and green hydrogen each carried different economics. They shared a political and rate-sensitive umbrella for a while, but they did not have the same cash-flow drivers.

A one-to-one topic matcher tends to prefer tidiness. Markets prefer mess.

A *split* occurs when one old topic matches several new topics. The old AI topic can split into AI infrastructure, AI software, and AI regulation. Within infrastructure, a broad cluster can split again into GPUs, memory, networking, power equipment, and cooling. Splits often mark the transition from broad narrative awareness to value-chain differentiation. The theme has matured enough that investors no longer buy the obvious label; they hunt for the scarce input.

A *merge* occurs when several old topics become one new topic. Power shortages, grid interconnection delays, nuclear restarts, and data-center construction may merge into a single “AI electricity bottleneck” narrative. Merges can create attention because they simplify complexity. They can also become dangerous because the new umbrella pulls in weakly exposed companies.

A *re-emergence* occurs when a topic disappears and returns. Regulation reappears around elections and court decisions. Retail demand reappears around product cycles. Energy themes reappear around commodity shocks. Comparing only with the previous window will misclassify many re-emergences as new themes.

Seasonality adds a quieter trap. Holiday retail demand, annual cloud capex guidance, regulatory filing cycles, and election-year policy stories can recur with enough new documents to look like fresh emergence. Treating every recurrence as a new theme overstates novelty and clutters the review queue.

The alignment layer should therefore compare a candidate topic with both the recent past and a longer historical library. The recent past finds continuity. The library catches seasonal and cyclical echoes.

16.6 Drift: Healthy Evolution or Identity Loss

Drift is not a bug. A theme that never changes may be a stale label.

The investor cares about the rate and direction of drift. Slow drift from “GPU” to “accelerator supply” to “HBM” may describe the same AI infrastructure chain becoming more precise. Fast drift from “GPU” to “electricity bills” may describe a genuine sub-theme birth. Drift from AI vocabulary into generic mega-cap technology language may indicate dilution rather than evolution.

A useful drift report separates four dimensions:

- **Vocabulary drift:** top words and c-TF-IDF weights change.
- **Entity drift:** the companies, suppliers, customers, or countries change.
- **Source drift:** the topic moves from specialist outlets to newswires, transcripts, filings, or broker research.
- **Return drift:** the mapped basket starts behaving like a different sector, factor, or value-chain exposure.

The last dimension is easy to ignore and expensive to miss. A text thread can remain coherent while the portfolio exposure becomes contaminated. If an AI infrastructure topic slowly becomes a mega-cap growth basket, the theme may still sound right in the documents while the return stream loses its residual character.

16.7 Emergence as Novelty Plus Support

An emerging theme is not just a topic that appears in the latest window. A one-off headline can be new. A marketing campaign can be loud. A vendor taxonomy change can look like a burst. None of those deserves capital by itself.

Emergence combines novelty and support. The topic must differ from historical topics, and it must gather enough evidence to matter.

If P_t is the word distribution of a candidate topic and P_0 is a historical background distribution, a simple novelty score is KL divergence:

$$D_{\text{KL}}(P_t \| P_0) = \sum_w P_t(w) \log \frac{P_t(w)}{P_0(w)}. \quad (16.5)$$

KL divergence is useful because it punishes topics whose vocabulary looks like the background. It is dangerous because rare words can dominate. A single obscure token, duplicate headline,

or vendor artifact can create fake novelty. In production, novelty should combine divergence, embedding distance, entity checks, and human review.

Support can be measured through volume and acceleration. Let V_t be topic volume in window t and $\bar{V}_{t-L}$ a historical baseline:

$$L_t = \frac{V_t}{\bar{V}_{t-L} + \epsilon}. \quad (16.6)$$

The ratio L_t is a crude lift score. It catches bursts. It misses slow emergence. Tu and Seng’s novelty-index and published-volume-index framing is useful here because it treats emergence as the joint movement of newness and publication volume, rather than frequency alone [110]. In market text, the same idea becomes a two-axis screen: how different is the topic from the past, and how much evidence is accumulating now?

Attention acceleration adds a third check. A theme with modest current volume but fast acceleration may matter more than a mature theme with high but flat volume. The system should track short-window growth, long-window growth, source breadth, entity breadth, and whether the topic has moved from specialist sources into newswires, transcripts, filings, or broker research. A story that appears only in one outlet may be a local story. A story that crosses sources is moving through the market’s transmission chain.

Unmatched clusters deserve the same discipline. A new cluster that fails to align with the historical library is an alert, not a theme. The review queue should ask whether the cluster has enough documents, whether representative documents are coherent, whether the vocabulary is genuinely new rather than renamed, whether linked entities are tradable, and whether the cluster reappears in the next window.

The best candidates often sit in the uncomfortable middle. They are not yet obvious enough to dominate news volume. They are not so isolated that the model sees only one strange document. They are coherent, growing, and mappable.

Slow emergence and sudden bursts carry different investment meanings. A regulatory shock may support an event trade before fundamentals are clear. A slow build in filings and earnings calls may support portfolio construction because it reflects repeated evidence rather than one news event. The architecture should be able to catch both without pretending they are the same signal.

16.8 Outliers as First Stories

Density-based models such as HDBSCAN often label documents as noise. In a static topic model, noise is usually swept away. In an emerging-theme system, noise can be where the first useful document lives.

The first article about data-center power constraints may not fit the early GPU cluster. The first transcript answer about liquid cooling may not fit the broader cloud capex topic. The first filing language about export restrictions may sit alone before it becomes a regulatory cluster.

A practical system should **keep an outlier buffer**. It embeds unmatched documents, watches whether nearby outliers accumulate, and promotes a candidate when the sparse region becomes a dense one. The signal is not that one document is strange. The signal is that strangeness repeats.

BERTrend is a useful recent reference point because it turns this monitoring problem into an **online topic-tracking workflow**: time-sliced BERTopic models, topic merging across periods,

popularity measures that combine document counts and update frequency, adaptive thresholds, and weak-signal classification [19]. The lesson for investing is not that one framework should be imported unchanged. It is that emergence detection needs memory, decay, and a thresholding rule that changes with the background level of attention.

16.9 False Positives That Look Like Themes

Most new clusters are not themes. This is not a defect. It is the base rate.

Common false positives include:

- **duplicate** syndication across news vendors;
- a **new source added to the corpus**;
- **ticker**, entity, or company-name mapping **errors**;
- **seasonal** topics around earnings, holidays, elections, or product launches;
- **coordinated** marketing **language**;
- one-off legal or regulatory events;
- social-media **manipulation**;
- **changes** in embedding **model**, preprocessing, or clustering parameters.

The nasty cases are pipeline changes. If the embedding model is updated, if a vendor changes ingestion rules, or if a new source appears in the corpus, the topic map may shift even though the market did not. That is model drift, not concept drift.

Production systems should version corpora, preprocessing, embeddings, clustering parameters, and alignment thresholds. The system should be able to answer a boring but essential question: did the world move, or did the machinery move?

16.10 From Candidate Cluster to Tradable Theme

Early detection has no investment value unless it can survive the next steps. A topic that cannot be mapped to liquid securities is useful for monitoring, not portfolio construction. A topic whose mapped basket is just sector beta is a story, not residual alpha.

An emerging-topic review should therefore ask:

1. Is the cluster coherent in representative documents, not just in top words?
2. Is it novel relative to both the recent window and the historical library?
3. Is volume or source breadth increasing?
4. Are linked entities tradable, material, and correctly mapped?
5. Does the topic identify a value-chain role, not merely a buzzword?
6. Does the candidate basket retain residual behavior after factor and sector controls?
7. Is the signal still early enough for a holding-period thesis?

The answer can be no. It often should be no. The purpose of emergence detection is not to turn every new phrase into a portfolio. It is to build a review queue where the right few candidates receive analyst attention before they become common language.

16.11 The Thread Library

By the time Part IV is finished, the system should have more than a collection of topics. It should have a library of threads.

For AI infrastructure, the library might show a parent thread with children for accelerators, memory, packaging, networking, data-center construction, power equipment, cooling, and grid access. Some children emerge slowly through transcripts. Some burst through news. Some merge during periods of broad attention. Some decay.

For clean energy, the library should preserve a different lesson. A broad structural story can split into solar, wind, batteries, EVs, hydrogen, storage, and grid equipment, then reveal that some sub-themes are policy-sensitive, some are rate-sensitive, some are manufacturing-cycle trades, and some are only weakly connected marketing labels.

Alignment gives the library continuity. Drift analysis keeps continuity honest. Emergence detection decides when the library needs a new shelf.

Chapter 17

Hierarchies and Theme Granularity

17.1 The Theme That Was Too Big to Trade

“AI” was an investable word before it was an investable map. In 2023, almost any company could say it. Chipmakers said it. Cloud platforms said it. Consulting firms said it. Software vendors said it. Utilities eventually said it too, because data centers need electricity before they need a pitch deck.

The word was everywhere. That was the problem.

A topic model that discovers one large AI topic has found attention, not structure. A portfolio built from that topic may become a disguised mega-cap technology basket. A model that discovers fifty tiny AI topics has the opposite problem: it may find every linguistic nuance while producing clusters too narrow, unstable, or illiquid to trade.

The hard question is granularity. At what level should the book of themes be read?

For a human analyst, the answer changes by task. A chief investment officer may want the macro theme: AI infrastructure. A portfolio manager may want the active sub-theme: high-bandwidth memory, grid equipment, or liquid cooling. A risk manager may want to know whether all of those sub-themes load on the same semiconductor factor. A client may need a label simple enough to explain without lying.

Hierarchy is the machinery for moving between those levels. It lets the same theme library serve discovery, monitoring, portfolio construction, and communication without pretending that one level is right for every decision.

17.2 Granularity Is an Investment Decision

The phrase “data-center power constraints” is more specific than “AI infrastructure.” It is also more fragile. The phrase “high-voltage transformer suppliers for Northern Virginia data centers” is even more precise, and may be economically revealing, but it is probably too narrow for a diversified equity basket.

Every granularity choice trades specificity against robustness.

Broad themes offer liquidity, communication, and stability. They also dilute exposure. Narrow themes offer cleaner economic logic and earlier second-order beneficiaries. They also raise turnover, concentration, and false-positive risk.

A theme library should usually carry at least three levels:

- **Macro themes:** broad narratives such as AI infrastructure, energy transition, defense modernization, obesity therapeutics, or supply-chain reshoring.
- **Sub-themes:** specific economic drivers such as HBM, grid equipment, GLP-1 manufacturing capacity, satellite communications, or export-control compliance.
- **Events and catalysts:** policy decisions, earnings revisions, clinical-trial readouts, product launches, regulation, or customer announcements.

The same document can belong to several levels. A Microsoft transcript paragraph about data-center capex may belong to AI infrastructure, cloud capex, power demand, and enterprise AI adoption. A flat topic model forces one view. A hierarchy preserves several.

17.3 Tree Models, DAGs, and Theme Maps

Classical hierarchical LDA gives the cleanest mathematical metaphor. It arranges topics in a tree. Higher nodes capture broad language. Lower nodes capture more specific language. Each document follows a path through the tree, drawing words from topics along that path [13].

This matches the way investors talk. “AI infrastructure” can sit above “accelerators,” “memory,” “networking,” “power,” and “cooling.” A document about high-bandwidth memory can share broad AI infrastructure words while also using the specialized vocabulary of memory bandwidth, substrates, packaging, and supply agreements.

The tree is also too tidy. Financial narratives overlap. A nuclear-utility topic can belong to AI infrastructure, power scarcity, energy security, and clean-energy politics. A semiconductor equipment company can sit inside AI infrastructure, national security, reshoring, and capex-cycle risk. A pure tree makes one parent the truth.

Pachinko Allocation addresses this by organizing topics as a directed acyclic graph rather than a strict tree [75]. The point for this book is not that every trading desk should implement Pachinko Allocation. The point is conceptual: theme structure often needs multiple ancestry. A narrative map should allow overlap without turning every overlap into a new theme.

Hierarchical Dirichlet Processes offer another useful idea: the number of topics does not need to be fixed in advance [107]. That matters in emerging-theme discovery, where the researcher does not know how many economically relevant clusters will appear. In production, though, fully nonparametric Bayesian models can be harder to operate than modular embedding systems with explicit review and reduction steps.

17.4 Hierarchies in Embedding-Based Systems

BERTopic-style workflows often start flat and build hierarchy afterward. Fit the model. Represent each topic with a c-TF-IDF vector. Compute distances between topics. Use hierarchical agglomerative clustering to merge close topics into broader groups [51].

This post-hoc hierarchy has a practical advantage: the analyst can inspect the dendrogram. If “AI accelerators” and “HBM” merge before either merges with generic software language, the hierarchy has economic plausibility. If “AI regulation” merges with “cloud capex” merely because both contain the word AI, the model is telling the analyst to intervene.

A useful production pattern is two-level rather than infinitely deep:

- Level 1 contains roughly 20 to 40 macro themes, depending on corpus size and mandate.

- Level 2 contains roughly 100 to 300 sub-themes, each mapped to representative documents, entities, and candidate exposures.

The numbers are not sacred. They are operating ranges. A global multi-asset research platform may need more. A focused equity strategy may need fewer. The point is to separate the vocabulary map used for discovery from the theme library used for capital allocation.

17.5 Granularity Controls

Granularity knobs look technical, but each one changes the kind of portfolio the system can build.

The main controls include:

- the number of topics in LDA, NMF, or CTM;
- HDBSCAN minimum cluster size and minimum samples;
- BERTopic topic-reduction thresholds;
- agglomerative-clustering cut levels;
- sub-clustering rules for large topics;
- analyst-defined taxonomy anchors;
- liquidity, capacity, and concentration constraints.

Portfolio constraints belong on this list. A semantically pure theme with three illiquid stocks may be a good research note and a bad institutional basket. A broad theme with 200 stocks may trade easily while diluting away the exposure that made the theme interesting.

This is where model selection and portfolio construction begin to touch. Topic granularity cannot be optimized in the abstract. It has to be optimized for the action the system will take.

17.6 Granularity by Use Case

A research platform and a portfolio engine do not need the same topic map. The first can tolerate nuance. The second needs stable exposures, tradable names, and risk controls.

Granularity should be judged by use case:

- **Monitoring:** topics should be interpretable and stable enough for analysts to follow.
- **Emergence detection:** topics should be sensitive enough to catch new language before it becomes a consensus label.
- **Portfolio construction:** topics should map to enough liquid securities without diluting the economic exposure.
- **Risk management:** topics should explain residual co-movement that sectors and factors do not explain.
- **Communication:** labels should be understandable without flattening the economics.

This checklist prevents a common mistake: choosing the topic count that makes the dashboard look best. A beautiful hierarchy that cannot support any decision is taxonomy theater.

17.7 When Geometry Matters

Theme maps do not live naturally on a spreadsheet. They look more like networks. A topic can be near its parent, near its siblings, and also near a different branch through a shared company, product, or regulatory exposure.

Hyperbolic embeddings, including Poincaré-style representations, are useful because tree-like structures distort in ordinary Euclidean space [85]. Hyperbolic space expands as one moves away from the origin, which lets broad parent concepts sit near the center while many fine-grained descendants spread outward.

The practical lesson is modest. A theme library should support graph views, not only lists. Nodes can represent themes, sub-themes, companies, products, events, and sources. Edges can represent semantic similarity, supplier links, customer links, shared documents, or shared return behavior. Later chapters use this graph logic for company exposure scoring and value-chain mapping.

For now, the hierarchy must answer one question: does this level of detail improve the decision?

17.8 The AI Infrastructure Hierarchy

A useful AI infrastructure hierarchy should do more than draw a neat tree. It should help investors see where the scarce input has moved.

One practical map might start with AI infrastructure as the macro theme. Below it sit accelerators, high-bandwidth memory, advanced packaging, optical networking, data-center construction, power equipment, cooling systems, cloud capex, and sovereign AI infrastructure. Some of those sub-themes are suppliers. Some are customers. Some are bottlenecks. Some are policy-sensitive. Lumping them together hides the rotation.

The hierarchy also clarifies what clean energy taught in reverse. A broad clean-energy topic may be coherent and popular while hiding separate solar, wind, hydrogen, EV, and battery economics. Those sub-themes can share a label during a liquidity boom and then separate under rate pressure, policy disappointment, or commodity stress.

A hierarchy is not a decoration. It is how the book of themes becomes readable at multiple scales.

17.9 Where a Clean Hierarchy Still Goes Wrong

A hierarchy can look elegant and still fail. The labels may be unsupported by the documents underneath them. The topics may shift under reseeding while the dendrogram looks unchanged. Coherence scores may reward stale, familiar language at the expense of the new vocabulary that an emerging sub-theme actually uses. Two clusters can sit close on a c-TF-IDF distance plot and map to wildly different baskets — one liquid, one not; one factor-neutral, one a disguised growth tilt.

Taxonomy is necessary. It is not sufficient. A theme library that has been organized but not evaluated is still an opinion.

Chapter 18

Evaluating Topic Quality for Investment Use

18.1 The Beautiful Topic That Should Not Be Traded

In January 2021, the iShares Global Clean Energy ETF was not an obscure vehicle. ICLN had more than doubled during 2020, retail money had discovered the energy-transition story, and the topic words were easy to love: solar, wind, hydrogen, batteries, decarbonization, policy support.

A topic dashboard would have looked beautiful.

The portfolio did not. From its 2021 peak into the rate-shock years, the clean-energy trade gave back a large share of the boom. The narrative had not vanished. The topic was still coherent. The problem was that coherence did not tell investors whether the basket was crowded, rate-sensitive, diluted, expensive, or exposed to companies whose economics did not match the label.

This is where bad thematic processes become expensive.

Topic modeling is unsupervised. There is usually no single correct answer. Two models can produce different topic maps that are both defensible. A coherent topic may have no investment value. A noisy-looking topic may be the first trace of an economically important bottleneck. A high-scoring model may learn the corpus beautifully and teach the portfolio nothing.

For thematic trading, topic evaluation must be more severe than model evaluation in a notebook. It must answer four questions:

1. Does the topic make semantic sense?
2. Does the topic remain stable enough to monitor?
3. Does the label survive human inspection under point-in-time evidence?
4. Does the topic improve an investment decision?

The fourth question is the hardest. It is also the one that matters.

18.2 Coherence Is a Starting Point

The most familiar proxy for topic quality is coherence. It asks whether the top words in a topic tend to make sense together.

Normalized Pointwise Mutual Information for two words w_i and w_j can be written as

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{P(w_i, w_j) + \epsilon}{P(w_i)P(w_j)}}{-\log(P(w_i, w_j) + \epsilon)}. \quad (18.1)$$

The topic-level score averages this quantity across pairs of top words. A high score means the words tend to co-occur in the reference corpus. For a topic about high-bandwidth memory, that is reassuring if words such as HBM, memory, bandwidth, packaging, and accelerator appear together repeatedly.

NPMI is one member of a broader coherence family. UMass coherence uses document co-occurrence in the training corpus. UCI and NPMI-style measures use sliding-window co-occurrence in a reference corpus. C_v combines several design choices into a score that often works well for classical topic inspection. These names matter because two papers can both report “coherence” while computing meaningfully different things.

But coherence is not truth. It is a proxy built on a reference corpus. Change the corpus, window size, preprocessing, or vocabulary, and the score can move. In finance, entity-heavy topics and short headline clusters often behave differently from long-form academic or news corpora.

Hoyle and coauthors show why this matters: automated topic-model evaluation can diverge sharply from human judgment, especially when researchers compare different model families with incompatible coherence setups [60]. Coherence metrics were built for an older generation of topic models. Neural topics and embedding clusters can defeat their assumptions.

The investment implication is blunt. Do not rank BERTopic, LDA, CTM, and short-text models on NPMI alone. A score that rewards familiar co-occurrence may prefer old language over emerging language. Alpha often begins where the vocabulary is not yet comfortable.

18.3 Diversity and Redundancy

A model can produce coherent topics that all say the same thing. In an AI corpus, ten variations of “mega-cap AI software” may each look reasonable. Together they create hidden concentration.

Topic diversity measures whether topics are distinct from one another. A simple score is the fraction of unique words among the top N words of all K topics:

$$\text{Diversity} = \frac{|\bigcup_{k=1}^K W_k^{(N)}|}{KN}, \quad (18.2)$$

where $W_k^{(N)}$ is the set of top N words for topic k . Low diversity means the model may be producing many versions of the same topic.

For a portfolio, redundancy is not an aesthetic flaw. It can lead the construction engine to believe it owns several independent signals when it really owns one crowded exposure. If three topics all map to the same handful of mega-cap stocks, the topic system has multiplied the label, not the information.

18.4 Perplexity and the Fit Trap

Probabilistic topic models often report held-out likelihood or perplexity:

$$\text{Perplexity} = \exp \left(-\frac{1}{N} \sum_{d=1}^D \log p(w_d) \right). \quad (18.3)$$

Lower perplexity means the model predicts held-out text better. That sounds like progress. Sometimes it is.

But a model can predict word patterns while producing topics that humans cannot label, companies cannot be mapped to, and portfolios cannot use. Perplexity measures fit to text, not fit to a decision. In this book, likelihood is a diagnostic for some model families. It is never the final criterion.

18.5 Stability and Temporal Quality

A topic model that changes completely when the random seed changes is not a research engine. It is a slot machine with embeddings.

Stability measures whether topics persist across seeds, samples, parameter choices, and adjacent time windows. Jaccard overlap can compare top-word sets. Cosine similarity can compare topic embeddings. Rank-biased overlap can compare ordered word lists while giving more weight to the highest-ranked terms.

The evaluation should separate three kinds of stability:

- **Run stability:** does the same corpus produce similar topics under small modeling changes?
- **Sample stability:** does the topic survive reasonable document resampling?
- **Temporal stability:** does the topic evolve plausibly through time?

Temporal stability is not the same as stillness. A theme should move when the world moves. The AI infrastructure thread should drift from GPUs toward memory, power, and cooling as the evidence changes. A clean-energy topic should split when solar, wind, hydrogen, batteries, and EVs stop behaving like one trade. A model that freezes those relationships may look stable while becoming wrong.

The right temporal evaluation asks whether the model moves for the right reasons.

18.6 Human Evaluation and Analyst Review

The cleanest topic model in the dashboard can still fail the analyst's five-minute test. Representative documents may not support the label. The top words may mix suppliers, customers, and regulators. The entity list may be dominated by one large company whose filings mention everything.

Human evaluation gives structure to that judgment. Chang and coauthors introduced word intrusion and topic intrusion tasks as direct tests of whether humans perceive the same topic structure the model claims to find [25]. In word intrusion, an evaluator sees top words plus one intruder and tries to identify the out-of-place word. In topic intrusion, the evaluator sees a document and several topic labels, one of which should not belong.

For this book's use case, analyst review should score four things:

- **Coherence:** do representative documents support the label?

- **Novelty:** is the theme genuinely different from the existing library?
- **Materiality:** does the topic connect to revenues, costs, capex, regulation, or competitive advantage?
- **Investability:** can it be mapped to liquid securities with defensible exposure?

The review must be point-in-time. An analyst reviewing a 2022 topic should not use 2024 hindsight to call it “data-center power.” Labels, representative documents, company links, and judgments must be recorded with the information available at the time.

18.7 Extrinsic Evaluation: The Market Gets a Vote

The final test is not whether a topic sounds good. It is whether the topic improves a decision. For thematic trading, useful extrinsic tests include:

- out-of-sample information coefficient for firm or basket returns;
- incremental R^2 in return, volatility, flow, or spread models;
- improvement in factor and sector risk attribution;
- cleaner exposure mapping to companies and value-chain roles;
- earlier detection of known historical theme transitions;
- overlap with thematic ETF holdings, treated as a noisy benchmark rather than truth;
- reduction in portfolio turnover, concentration, or unexplained drawdown.

The ETF-overlap test deserves a closer look, because the temptation to use thematic-ETF holdings as ground truth is strong and the trap is well disguised. A topic model that reproduces ICLN’s clean-energy roster looks validated. It may instead have learned the index provider’s eligibility rules, the launch-date marketing rationale, and the holdings dilution that comes with rebalancing for liquidity. Index methodologies (MSCI relevance scoring, S&P thematic indices, RBICS revenue mapping, Solactive-style NLP scoring) and analyst-curated baskets carry the same risk: each is an opinionated label, useful as a check but circular as a target. The cleanest use of these external maps is as noisy labels in a multi-source agreement test. A theme is more credible if four independent maps — model topic, ETF holdings, index methodology, and a revenue-taxonomy slice such as RBICS — overlap on the same companies. It is less credible if the model only reproduces one of them.

This is where granularity returns, but as one piece of a larger test. A macro AI topic may score well on stability and liquidity but poorly on residual return prediction. A fine-grained HBM topic may score well on materiality but fail capacity. A power-equipment sub-theme may have lower document volume but better exposure mapping and cleaner factor-adjusted behavior.

There is no universal best topic count. There is only a best topic map for a decision under constraints.

18.8 Leakage in Topic Evaluation

Topic evaluation can leak future information in quiet ways.

A full-sample topic map can project 2024 vocabulary into 2022. An LLM label can use knowledge outside the historical window. ETF holdings can be used before the holdings were public. Entity mappings can reflect future corporate actions. Analyst review can be colored by a return outcome that had not yet happened.

The rule is the same as in backtesting: if the topic is used at time t , its representation, label, company links, hierarchy position, and evaluation evidence must have existed at time t .

This requirement makes evaluation slower. It also makes it honest.

18.9 A Minimum Evaluation Stack

A practical thematic topic system should not graduate a model because one coherence score improved. The minimum stack is heavier:

1. intrinsic coherence and diversity;
2. stability across seeds, samples, and time windows;
3. representative-document inspection;
4. analyst scoring for coherence, novelty, materiality, and investability;
5. hierarchy review: too broad, too narrow, or at a useful decision level;
6. downstream validation against returns, volatility, exposure, flows, or risk attribution;
7. leakage audit for labels, corpus construction, entity mapping, and historical benchmarks.

This stack is heavier than a single dashboard metric, but the stakes justify it. A topic model that looks impressive and feeds bad exposures into a portfolio is worse than no model at all. It gives false precision a UI.

18.10 Evaluating the AI Infrastructure Map

A useful AI infrastructure topic system should answer questions a portfolio process actually has.

Does the map separate accelerators, memory, advanced packaging, networking, cloud capex, power equipment, cooling, and grid access? Are the labels supported by point-in-time documents? Do topic threads move around known events and disclosure changes? Do company links match economic exposure rather than press-release vocabulary? Do sub-themes explain residual return or risk after sector and factor controls?

The same discipline should catch the clean-energy granularity problem. A broad clean-energy topic may be coherent and popular while hiding separate solar, wind, hydrogen, EV, and battery economics. If the model does not reveal those distinctions, it will not explain why a policy narrative, a duration trade, and a commodity-cycle exposure should not be treated as one clean portfolio.

Evaluation ends where exposure begins. A theme that cannot be mapped to companies remains a beautiful abstraction. The portfolio cannot buy an abstraction.

Part V

Mapping Themes to Companies

Chapter 19

From Topic to Investable Theme

19.1 The Topic That Had No Companies

On June 30, 2021, Roundhill launched the first United States metaverse ETF under the ticker META; the fund later changed its ticker to METV after Facebook became Meta [91, 92]. The launch landed on a topic that had just become impossible to miss. Documents about virtual worlds, digital identity, augmented-reality hardware, immersive commerce, gaming engines, and the future of remote work were clustering in news, broker research, social media, and corporate slide decks. A topic model run on almost any contemporary corpus would have found the cluster on the first pass.

In October of the same year, Facebook renamed itself Meta.

The topic was real. The investable theme was harder to find.

The fund rose with the late-2021 enthusiasm and then fell sharply through the 2022 growth-stock unwind. The holdings list explained the gap between the topic and the basket. Many of the largest weights were mega-cap technology names whose metaverse-specific economics were hard to isolate at the time of inclusion: Apple, Microsoft, Nvidia. The nearer pure plays — Roblox, Unity, Snap’s augmented-reality unit — were small, expensive, and concentrated in ways an index could not absorb without diluting the exposure. A dashboard could discover the topic. It could not, by itself, build a portfolio that expressed the topic cleanly.

This is the conversion problem. A coherent topic is not yet a decision object. It is a model output: a cluster, a label, a thread of representative documents, an embedding neighborhood. A theme that an institution can underwrite is something else. It has a definition with edges, a timestamp, a hypothesis about why the exposure should be paid for, a candidate universe, a falsification condition, and a place in a portfolio that someone is willing to sign for. The machinery between the two is not optional. The metaverse trade did not fail because the topic was imaginary. It failed because the market packaged the topic before most investors had done the conversion work.

19.2 A Topic Is Not a Decision

A coherent cluster of documents about “data-center power” is a model output. It is not yet a thesis.

The model has shown that a group of articles and transcripts use similar vocabulary, name overlapping entities, and behave like a single thread through time. That is a useful finding. It is

the beginning of the work, not the end. None of the next questions a portfolio process needs to answer have been asked yet. What is the economic mechanism that should turn data-center power into a return? Which companies are exposed in a way that matters to their cash flows? How early is the signal relative to the price already paying for it? What evidence would tell the team that the thesis has broken?

The gap between a topic and a theme is filled by an artifact. Institutional desks do not approve themes from a dashboard. They approve them from a written specification that survives review by someone who was not in the room when the topic was discovered. The artifact is the bridge.

19.3 The Theme Specification

Every investable theme should carry a point-in-time specification. The specification has at least the following fields:

- theme name and a stable identifier;
- a one-paragraph narrative description with the proposed economic mechanism;
- the date of first detection and the date of approval;
- source evidence and a small set of representative documents;
- inclusion and exclusion criteria written in mappable terms;
- the lifecycle stage assigned at approval;
- the expected investment horizon;
- the hypothesized beneficiary chain;
- a preliminary company universe with the evidence that placed each name;
- known factor, sector, and liquidity risks;
- the validation tests that would confirm the thesis and the conditions that would falsify it.

The specification is both documentation and a control against drift. If the theme changes meaning over time, the spec should be versioned rather than silently rewritten. A library of specifications becomes the record of how the theme library learned — which themes graduated from monitor-only to live capital, which were retired, which were renamed, and on what evidence.

A spec is not the same as a marketing one-pager. It is a structured object that downstream systems can read. The exposure-scoring layer reads the inclusion criteria and the narrative description to decide what evidence matters per firm. The backtesting layer reads the date of first detection to know what the model could have known when. The governance layer reads the falsification conditions to decide when to stop trading the theme. A theme with no spec is a theme that cannot be operated.

This is why the spec should store examples as well as rules. For each theme, the team should keep at least one positive archetype, one negative archetype, and one borderline case. The positive archetype shows what obviously belongs. The negative archetype shows what vocabulary alone would have admitted but economics should reject. The borderline case forces the team to state whether ambiguity is resolved by revenue share, strategic importance, customer exposure,

or analyst judgment. Those three examples make later scoring less theatrical. They turn a name-by-name argument into an audit of whether the evidence still matches the rule.

19.4 The Narrative Description That Survives Mapping

“AI” is a topic. It is not a usable narrative description for a theme. The word is on too many cash-flow statements and too few in a way that matters.

A narrative description earns its place by being precise enough to drive company mapping. “Data-center power demand caused by AI workloads” is better. “Utilities, grid-equipment suppliers, and backup-power providers with exposure to incremental electricity demand from AI data centers” is better still, because it names the mechanism, the value-chain role, and the kind of revenue that should move.

A good description reads like this:

AI model training and inference are driving sustained growth in high-density data-center demand. This raises electricity load, grid-interconnection queues, backup-power requirements, cooling intensity, and capital expenditure across the data-center supply chain. Firms with material exposure to these bottlenecks may experience revenue growth or pricing power that standard sector classifications do not capture.

The description does three things. It names the cause (AI workloads). It names the consequence path (electricity, interconnection, cooling, capex). It names the kind of firm that should benefit (bottleneck suppliers with measurable exposure). A description that does none of these turns inclusion into a debate. A description that does all three turns inclusion into a checklist.

It also names the non-theme. This matters more than it sounds. If the theme is AI-driven data-center power, then a generic enterprise-software vendor using AI internally is not a beneficiary of this theme, even if its earnings call contains the word “AI” twenty times. If the theme is grid equipment for data centers, then a utility with broad regulated load growth may belong only if the incremental data-center exposure is large enough to matter. A narrative description that contains no exclusions is only half a definition.

19.5 Inclusion and Exclusion as Defenses

A theme without inclusion criteria is a marketing label that other people will populate. The METV experience is the canonical version of that failure: a vague theme attracts the megacap holdings that pass liquidity filters, and the basket becomes the average of what was already there.

For the AI data-center power theme, inclusion criteria might require:

- disclosed revenue from data-center power equipment, grid infrastructure, backup generation, cooling systems, or electrical components;
- disclosed customer exposure to hyperscalers or large data-center operators;
- confirmed capital expenditure or order backlog tied to AI data-center demand;
- physical operation in regions where data-center load growth is straining the grid.

Exclusion criteria are at least as important:

- generic AI mentions without measurable business exposure;
- diversified firms where the theme is immaterial to revenue or earnings;
- firms exposed only through speculative investor attention rather than fundamentals;
- firms whose exposure is negative unless the strategy explicitly trades disrupted incumbents.

The discipline is procedural. Criteria are written before the backtest is evaluated. Otherwise the researcher can tune the theme definition to the names that happened to work, and the criteria become a story told after the fact. The point-in-time spec exists to make that drift visible.

Inclusion and exclusion criteria are also where cleaner narrative-excess cases do useful work. The metaverse trade, blockchain rebrands, and parts of the SPAC boom all showed the same failure: a company could borrow the vocabulary of a theme long before the economics were material. A theme definition that admits any firm with the right words in a presentation pulls in shell companies, early-stage concept stocks, and diversified incumbents with token exposure. The narrative may be coherent. The portfolio that follows may not be.

The criteria therefore need two thresholds. The first is semantic: does the company plausibly belong to the theme? The second is economic: is the exposure large enough, timely enough, and directional enough to matter to future returns? Semantic inclusion gets a company onto the research list. Economic inclusion gets it into the investable universe. Confusing those two steps is how a watchlist becomes a portfolio without anyone noticing.

19.6 Lifecycle Stage as a Position-Sizing Input

A theme's lifecycle stage — weak signal, emerging, validated, crowded, institutionalized, decaying, recycled, as developed in Chapter 6 — is not an academic label. It is a position-sizing input.

An emerging theme justifies exploration and a small probe. The evidence is thin, the names are illiquid, and the path to consensus is uncertain. A validated theme justifies a larger basket if valuation and crowding are acceptable. A crowded theme requires exposure caps, hedges, or contrarian timing. A decaying theme is still useful, but mostly as a short candidate, a risk reducer, or a continued monitoring target.

The stage assigned in the spec should be evidence-based, not aspirational. Candidate evidence includes attention acceleration, source breadth (specialist media to newswires to earnings calls to filings to broker research), analyst adoption, ETF creation activity, valuation expansion in the candidate universe, fund flows into adjacent vehicles, and the residual return behavior of preliminary baskets. The clean-energy trade was an institutionalized theme by early 2021 by almost every one of these measures, and a position size built for an emerging theme would have looked wildly wrong against the actual stage.

This assignment should be made before the basket is optimized. Otherwise stage becomes another parameter that can be fit to a desired result. If the team calls a theme emerging, the evidence package should look like an emerging theme: few sources, weak institutional adoption, high uncertainty, and a small investable universe. If it calls the theme crowded, the evidence package should show price response, fund flows, valuation pressure, or ETF/index adoption. The label is not a mood. It is a constraint on how much capital the theme is allowed to carry.

19.7 Horizon and Source Match

Themes operate at different horizons. A policy event may matter for days. A supply-chain bottleneck may matter for quarters. A demographic shift may matter for years.

The expected horizon belongs in the spec because it has to match the data source and the rebalancing rule. A signal that comes from intraday news acceleration loses most of its information by a monthly rebalance. A signal that comes from SEC filings does not benefit from daily turnover — it benefits from the patience to wait for the filings to arrive. Horizon mismatch is one of the quietest sources of disappointing backtests, because the model can be right about the direction and wrong about the holding period that captures it.

The horizon is also where source choice becomes a portfolio choice. Themes whose primary evidence lives in 10-K segments belong in slower books. Themes whose primary evidence lives in transcript Q&A or specialist trade press belong in books that can move at that speed. A spec that names the horizon and names the source in the same paragraph forces the question to be answered before it costs money.

There is a second horizon as well: the thesis horizon. A source may update quickly while the economics unfold slowly. Hyperscaler capex commentary can change in an afternoon, but transformer backlogs, grid interconnection queues, and cooling-system capacity can take quarters to clear. The spec should separate signal refresh frequency from expected return horizon. A monthly rebalance may be too slow for a headline shock and too fast for a capital-cycle thesis. The theme definition should say which problem it is solving.

19.8 The Beneficiary Chain

A theme without a beneficiary chain treats every related company as if it were the same kind of bet. The AI infrastructure trade makes the cost of that mistake clear.

The chain for AI infrastructure runs roughly:

- model demand creates compute demand;
- compute demand creates accelerator and memory demand;
- compute clusters require networking, power, cooling, and physical real estate;
- infrastructure constraints create pricing power for bottleneck suppliers;
- cloud platforms monetize the capacity, and software adopters capture the productivity later;
- incumbents whose products become obsolete sit on the other side of the chain.

A pure-play bottleneck supplier, a diversified adopter, and a potentially disrupted incumbent have different exposures, different signs, and different horizons. A theme that puts all three in the same basket with equal weights can cancel the exposure it is trying to buy. The spec should name the roles the basket will hold and the roles it will avoid. The full machinery for mapping firms to roles — supply-chain graphs, relation extraction, centrality measures — belongs in the value-chain chapter later in this part. The spec only needs to identify which roles the theme is buying.

19.9 Approval, Monitoring, and the Versioned Library

The best research desks reject most candidate themes. That is not a failure of the discovery system. It is the discovery system working.

Theme approval should be a structured gate rather than a yes-or-no question. The questions to answer at the gate are familiar but worth writing down:

- Is the topic coherent and stable enough to monitor?
- Is the proposed economic mechanism plausible?
- Can companies be mapped with evidence, not just vocabulary?
- Is the exposure material at the firm level?
- Is the theme early enough that the price has not already discounted it?
- Can the basket be implemented with acceptable liquidity, capacity, and factor risk?
- What evidence would falsify the thesis?

The answer can be “monitor only.” Monitor-only themes are not consolation prizes. They preserve awareness without forcing every narrative into the portfolio, and they create the option to upgrade the theme later if the evidence improves. A theme library that contains only approved themes has lost its early-warning value.

Point-in-time discipline ties the library together. A theme as defined on March 1 may differ from the same theme as defined on June 1. If an analyst later improves a label or finds a better company mapping, the improvement should not be retroactively applied to a historical backtest unless the simulation explicitly reflects the information available then. The library therefore stores versions: each theme’s specification at each point in time it changed, with the evidence that motivated the change. This is not bookkeeping for its own sake. It is how the system can be audited, replayed, and trusted by capital allocators who were not in the room when the topic first appeared.

The version history should also record rejected themes. Rejections are often more informative than approvals. They reveal which topics were coherent but not investable, which themes lacked material company exposure, which baskets failed factor or liquidity tests, and which narratives were already too crowded when discovered. In later research, that archive becomes a training set for skepticism. It teaches the system what a seductive but non-investable topic looks like.

19.10 From Specification to Company

A specification is necessary. It is not sufficient. A theme that has been named, scoped, and approved is still an abstraction until specific firms can be tied to it with evidence that survives skepticism. The next question is which companies are exposed and how exposure should be measured. The portfolio cannot hold a description; it can only hold tickers.

Chapter 20

Company Exposure Scoring

20.1 The Companies That Said “AI” the Most Were Not the AI Companies

In 2023 and 2024, almost every large US-listed firm that wanted to talk about AI did. AI mentions in earnings calls hit records. AI mentions in 10-K risk factors hit records. Risk-factor language from consumer-staples conglomerates explained, with surprising care, how artificial intelligence might affect everything from supply-chain logistics to advertising spend. The top of the AI-mention league table read like a list of the largest companies in the index.

The top of the AI-exposure league table was a different list.

It included a Korean memory company whose HBM line ran flat-out and whose customers had to wait in a queue. It included a Texas competitive power generator whose nuclear and gas fleet sat next to data-center load growth that utilities had not modeled. It included a Cleveland-rooted electrical-equipment maker whose switchgear backlog grew with hyperscaler capex. None of these firms led the mention-count rankings. They earned the theme from the part of the income statement that already moved.

This is the exposure problem. Mention frequency is cheap. Exposure — the part of a firm’s cash flows, risks, valuation, or investor demand that a theme actually moves — is harder to detect, harder to measure, and the only thing a portfolio should pay for. An exposure-scoring system that ranks companies by how often they say the word will systematically over-weight diversified mega-caps and under-weight the firms whose business has already changed.

The job of an exposure score is to put the second list on top.

20.2 The Multi-Signal Score

No single piece of evidence is enough to score exposure. Keywords are easy to game. Segment disclosure is partial. Embeddings drift toward the average mega-cap. LLMs hallucinate. Supply-chain graphs are noisy. Each signal is right about something the others miss, and wrong in ways the others can sometimes catch.

Let $S_{i,\theta,t}$ be the exposure score of company i to theme θ at time t . A multi-signal score is a function of several evidence streams:

$$S_{i,\theta,t} = f(K_{i,\theta,t}, E_{i,\theta,t}, R_{i,\theta,t}, L_{i,\theta,t}, G_{i,\theta,t}), \quad (20.1)$$

where K is keyword or phrase evidence, E is semantic embedding evidence, R is revenue or segment evidence, L is language-model classification evidence, and G is graph or value-chain evidence.

The temptation is to set f to a weighted average and move on. That is almost always wrong. The signals are not commensurable. Revenue evidence is usually stronger than a press-release mention. A direct customer-supplier relationship is usually stronger than generic semantic similarity. A confident LLM classification with three filing citations is stronger than a confident LLM classification with none. An honest scoring function preserves the type and confidence of each piece of evidence rather than averaging them into a number that hides what it is made of.

In practice, the scoring object should be richer than a scalar. The scalar $S_{i,\theta,t}$ is what the portfolio optimizer eventually consumes, but the research system should also store the role, sign, confidence, source trail, and as-of timestamp:

$$\mathcal{E}_{i,\theta,t} = (S_{i,\theta,t}, \rho_{i,\theta,t}, \sigma_{i,\theta,t}, C_{i,\theta,t}, A_{i,\theta,t}). \quad (20.2)$$

Here ρ is the value-chain role, σ is the sign of exposure, C is confidence, and A is the evidence set available as of time t . A memory supplier, a cloud adopter, and a disrupted incumbent can all be close to the same theme. They should not collapse into the same number.

The signals are introduced one at a time below, with a closing section on how to combine them without erasing what each one knows.

20.3 Keyword and Phrase Evidence

Keyword matching is the cheapest and most transparent signal. For each theme, define a vocabulary of terms and phrases. For AI infrastructure, the list includes “AI accelerator,” “high-bandwidth memory,” “advanced packaging,” “liquid cooling,” “data-center power,” “grid interconnection,” and “cloud capex.” A useful dictionary also includes negatives that should not count: “machine learning” in a hiring posting, “AI” in a risk-factor boilerplate paragraph, or “data center” in a corporate-real-estate disclosure unrelated to capacity build.

The strength of keyword evidence is precision when the vocabulary is curated. The weakness is brittleness. Keywords miss synonyms (“GPU cluster” versus “training fabric”), miss implicit exposure (a switchgear vendor that never says the word AI), and reward keyword stuffing (a 10-K that mentions AI in twelve places without revenue attribution). The history of greenwashing in sustainability disclosures is the canonical warning: companies learn what investors are scanning for, and the scanned vocabulary appears in the next filing whether or not the underlying business changed.

Keyword evidence is best used as one layer in a stack. It contributes high-precision positive hits and high-precision negative filters. It does not, on its own, decide what a basket looks like.

The keyword layer should also be versioned. Theme vocabularies mutate quickly: “AI accelerator” gives way to “inference ASIC,” “HBM3” gives way to “HBM3E,” and yesterday’s precise phrase becomes tomorrow’s boilerplate. A backtest that applies a 2025 vocabulary to a 2021 filing has already leaked the future. The keyword dictionary is therefore part of the as-of research record, not a convenience file sitting beside the code.

20.4 Revenue and Segment Evidence

Revenue evidence is the strongest signal that exposure scoring routinely has access to. If a company discloses revenue by business segment, product line, or end market, those disclosures can be mapped to theme-relevant categories with a real economic anchor underneath.

The standard institutional version of this mapping is FactSet’s RBICS Revenue. RBICS decomposes a firm’s operations into a multi-sector participation tree up to six levels deep, covering more than 1,800 sector groups, and quantifies the share of revenue that comes from each [40]. The granularity matters. Mapping a hyperscaler’s revenue from cloud database hosting separately from its e-commerce revenue, or separating a chemical company’s lithium-precursor business from its broader specialty-chemicals line, produces theme exposures that headline GICS sectors cannot represent. A standard revenue-based score takes the form

$$R_{i,\theta,t} = \sum_{s=1}^{S_i} \text{RevenueShare}_{i,s,t} \times \text{ThemeRelevance}_{s,\theta,t}, \quad (20.3)$$

where s indexes the firm’s segments, RevenueShare is the share of revenue from segment s , and ThemeRelevance is the score assigned to that segment for the theme at the time the score is computed.

The limitations are familiar but worth naming. Revenue disclosure lags — the most recent 10-K can be six months old, and segment breakdowns change without warning when a company reorganizes its reporting. Segment definitions are inconsistent across firms; one firm’s “cloud and AI” line includes services another firm reports inside “software.” Many segments are too coarse to isolate the theme-relevant business. Capex disclosure, order backlog, and end-market commentary in transcripts often add the granularity that the segment table misses, and a serious revenue layer reads all three.

When revenue evidence is available and material, it should usually dominate. A firm that earns half its revenue from data-center power equipment does not need an embedding to confirm what its income statement already says. But revenue evidence should not be allowed to pretend it is fresher than it is. The score should store both fiscal-period date and availability date, because a December segment table that became available in February is not usable for a January decision.

20.5 Embedding Similarity

Embedding similarity is the signal that catches what keywords miss. It compares a vector representation of a company — built from business descriptions, filings, transcripts, or product copy — with a vector representation of the theme. With e_i as the company embedding and e_θ as the theme embedding, cosine similarity is

$$\text{Similarity}(i, \theta) = \frac{e_i^\top e_\theta}{\|e_i\| \|e_\theta\|}. \quad (20.4)$$

The academic precedent for using text similarity to classify firms is Hoberg and Phillips’s Text-based Network Industry Classification [56]. TNIC computes pairwise cosine similarity over 10-K business descriptions and treats firms as competitors when their descriptions cross a threshold. The framework was updated more recently with doc2vec-based embeddings (ETNIC), which capture more semantic structure than the original TF-IDF version [57]. Thematic exposure

scoring is the same idea pointed at a different question: rather than asking which firms describe similar businesses, the system asks which firms describe businesses similar to a theme.

The strength of embedding similarity is that it surfaces firms whose exposure is real but unnamed. A switchgear vendor that supplies hyperscaler data-center buildouts may never use the phrase “AI infrastructure,” but its product copy and customer descriptions can sit close to a theme embedding built from data-center capex commentary. Keywords would miss it. Embeddings catch it.

The weakness is false proximity. A diversified industrial whose 10-K mentions data centers in one paragraph can score close to the theme even when the underlying revenue share is immaterial. The embedding does not distinguish “we operate data centers as part of our business” from “we are exposed to data-center demand growth” from “we mention data centers because our risk-factor disclosure is comprehensive.” Embedding evidence should not be acted on alone; it should be combined with revenue and materiality filters that test whether the proximity reflects an exposure or a vocabulary coincidence.

There is also a scale problem. Large firms have longer filings, broader vocabularies, and more reasons to mention every emerging technology. Without controls, embedding similarity can become a disguised size and complexity factor. A practical system therefore computes embeddings on comparable text units: business-description sections, theme-relevant excerpts, segment descriptions, or retrieved passages rather than the entire 10-K indiscriminately. The closer the text unit is to the claimed mechanism, the more the similarity score means.

20.6 LLM Classification with Retrieved Evidence

Large language models can classify exposure in a way that earlier methods could not, because they read context. A prompt can ask the model to assign categories such as direct pure-play exposure, material partial exposure, indirect beneficiary, potential adopter, disrupted incumbent, or no material exposure, and to justify each label.

The standard pattern is retrieval-constrained: the model is given a small set of point-in-time evidence (10-K segments, transcript excerpts, broker research, news headlines) and asked to classify the firm using that evidence and to cite the passages that support the classification. The model is forbidden from scoring from memory, which limits both hallucination and look-ahead leakage from training data. The output is structured (JSON, a fixed-schema CSV) so a downstream system can audit which passages drove which labels.

The rubric should be more important than the prose. A useful LLM output records role, materiality, sign, evidence quotes, uncertainty, and the reason a higher exposure label was rejected. That last field is not bureaucratic decoration. It is what keeps a model from upgrading a company simply because the passage sounds exciting. The model should have to say, for example, “partial exposure, not pure-play, because the disclosed revenue line remains diversified” or “monitor only because the customer relationship is inferred, not stated.”

Done well, LLM classification handles the cases the other signals struggle with: multi-clause strategy descriptions, indirect exposure that depends on context, disclosures whose meaning depends on what the firm did not say. Done badly, it confidently invents customer relationships and revenue lines that do not exist. The discipline that separates the two is structural — retrieval, schema, citation, audit — not prompt-engineering luck. LLMs therefore belong in the pipeline as an operating layer in their own right, with evaluation under model cutoff and retrieval-constrained workflows.

20.7 Graph and Value-Chain Evidence

Some exposures are not in a firm’s own filings. They are in its customers’ filings.

A graph layer treats the company universe as nodes and customer-supplier, partnership, and disclosure-co-mention relationships as edges. A switchgear vendor’s exposure to AI data-center demand is visible in the hyperscalers’ capex commentary even if the vendor never publishes it. A memory company’s exposure to a specific accelerator vendor is visible in that vendor’s supply-chain disclosures. The graph signal $G_{i,\theta,t}$ captures evidence that lives in the neighborhood of the firm rather than at the firm itself.

Light versions of this signal use named-entity co-occurrence across filings and transcripts. Heavier versions use relation-extraction models trained for financial text to populate a directed supply-chain graph and compute centrality measures that quantify whether the firm sits at a bottleneck, a redundancy point, or a periphery. Graph evidence is one of the few signals that can distinguish a critical supplier from an interchangeable one. The full machinery for value-chain mapping — relation extraction, network construction, centrality, beneficiary roles — is developed in its own chapter later in this part. For scoring, what matters is that graph evidence belongs in the stack and that it carries information neither text nor segment data contains.

Graph evidence also needs direction. A link from a cloud platform to a power-equipment supplier is not the same as a link from a software vendor to a cloud platform. One may identify a bottleneck supplier; the other may identify a customer whose margins are pressured by the bottleneck. The score should preserve edge direction and relationship type until the portfolio rule decides whether the exposure is long, short, hedge, or monitor-only.

20.8 Materiality: When the Theme Touches Cash Flow

Two firms can be semantically close to the same theme and economically far apart. The difference is materiality.

A supplier may be genuinely exposed to data-center power, but if the business line accounts for two percent of revenue, the stock will not respond to data-center demand surprises in a way that supports a thematic basket. Materiality converts signal magnitude into investment magnitude. Useful indicators include:

- revenue share of the theme-relevant business;
- earnings contribution after accounting for line-level margins;
- capital expenditure share directed at the theme;
- order backlog or contracted commitments tied to theme demand;
- customer concentration in theme-exposed buyers;
- explicit management guidance on the theme’s revenue impact;
- analyst-estimate sensitivity to theme news;
- price reaction to theme-specific events, controlling for sector moves.

Materiality is not a one-time check. It moves. A firm whose data-center business is two percent of revenue today can become a different bet at twenty percent. The exposure score should reflect that trajectory rather than freezing the firm at its last filing.

For this reason, materiality should be scored on both level and change. A small business line growing from two percent to six percent of revenue may matter more to expected returns than a mature thirty-percent line growing with the rest of the company. The former is a revision story; the latter is a classification fact. A good exposure score separates the two, because portfolio construction may want the acceleration even when the current revenue share is still modest.

20.9 Purity Versus Diversification

Two firms can both be exposed to AI infrastructure and represent very different bets.

SK Hynix is a memory pure-play. Its HBM line is at the heart of the accelerator supply chain; almost every dollar of its revenue depends on the same end demand. The exposure score should be near the top of any AI infrastructure list. The same exposure makes the stock volatile, capacity-constrained, and crowded with thematic capital. A small position is concentrated; a large position is expensive and hard to scale.

Samsung is also exposed — it makes memory, foundry services, and components used in AI hardware. Its consumer-electronics, display, and appliance businesses dilute the exposure. The thematic share of revenue is smaller, the volatility is lower, the liquidity is higher, and the basket can hold more of it without paying a premium. The score should be lower than SK Hynix's, but the position can sometimes be larger.

The same trade-off appears in the power case. Vistra and Constellation are competitive generators with concentrated exposure to AI-driven data-center demand. NextEra is a diversified utility with renewables and a regulated business; the thematic share is real but smaller, and the rate-regulated cash flows damp the upside. A pure-play basket buys Vistra and Constellation; a scalable basket also holds NextEra.

Neither approach is correct in the abstract. Purity offers cleaner thematic exposure and higher beta to the narrative; diversification offers liquidity, capacity, and lower idiosyncratic risk. The exposure score should distinguish them. The portfolio rule should decide which it wants.

The clean separation is exposure score versus portfolio weight. Exposure asks how much of the firm the theme plausibly moves. Weight asks how much of the portfolio should be allowed to depend on that firm. A pure-play can receive a high exposure score and a modest weight because liquidity, valuation, or single-name risk is unattractive. A diversified giant can receive a lower exposure score and a larger weight because it is liquid and useful for capacity. Mixing those two decisions inside the exposure score makes the score impossible to interpret.

20.10 Calibration, Confidence, and a Worked Scoring Pass

A raw score is not yet usable. Two further questions decide whether the number means what it appears to mean.

Calibration asks whether a score of 0.8 on one theme means the same thing as 0.8 on another, and whether last year's 0.8 means the same thing as this year's. Without calibration, the construction engine treats different themes as if their scores were comparable, which they usually are not. Useful calibration anchors include analyst-confirmed labels, RBICS revenue thresholds, prior ETF inclusion (treated as a noisy label rather than ground truth), reactions to theme-specific events, and held-out confirmations from future filings.

Confidence is separate from exposure. A company may have high possible exposure with low confidence because evidence is sparse: one transcript mention, no segment disclosure, no

graph link. Another may have moderate exposure with high confidence: a clear segment line, a public customer, a named backlog. Treating these as the same number throws away information the construction engine needs. A scoring system that emits both an expected exposure and a confidence interval — or at minimum a confidence band — gives the portfolio a way to size positions by certainty as well as magnitude.

A worked AI infrastructure scoring pass combines the evidence streams in roughly this order. Revenue evidence first: RBICS and segment disclosure place firms on a coarse exposure ladder anchored to cash flows. Keyword and phrase evidence sharpens the list, adding precision hits and removing keyword-stuffing false positives. Embedding similarity surfaces firms whose exposure is real but not named — switchgear, optical interconnect, specialty cooling — and pulls them into the candidate set. LLM classification with retrieval reads transcripts and filings for the remaining cases and assigns role labels (pure-play, partial, indirect, adopter, disrupted) with cited passages. Graph evidence adds the customer-side information that confirms or contradicts firm-side disclosure. Materiality and confidence then qualify the score for portfolio use.

The final combination should be monotonic with respect to hard evidence. A company with explicit theme revenue should not be pushed below a company with only semantic similarity unless the materiality, sign, or confidence fields explain why. Likewise, an LLM should be allowed to downgrade a keyword hit when retrieved evidence shows boilerplate, but it should not upgrade a company to pure-play without revenue, customer, or graph support. The combination layer is less a blender than a rule-governed court of appeal: each signal can challenge the others, but it must bring evidence.

The highest-scoring firms in the resulting pass are not the ones that say “AI” the most. They are the ones whose income statements, customer lists, and capex disclosures have already started to reflect it. That is the only ranking the basket should care about.

A score is still an input, not a portfolio. Before it becomes a basket, the system needs two more layers: the LLM operating discipline that governs some of the evidence used above, and the value-chain map that explains where exposure sits, what sign it carries, and how it propagates across firms.

Chapter 21

LLMs in the Thematic Pipeline

21.1 The 2019 Filing the Model Classified With 2022 Knowledge

An analyst opens a Q3 2019 earnings transcript and pastes it into a research assistant built on a large language model. The prompt is straightforward: read this transcript and assess the company’s AI infrastructure exposure as of the date of the call.

The model responds with confidence. The company, it explains, is materially exposed to AI infrastructure: it pivoted into AI hardware in late 2022, won major data-center customer contracts in 2023, and reorganized a segment around accelerator memory in 2024. The classification is detailed, well-written, and useful.

The analyst rereads the transcript. The 2019 call discussed PC and mobile demand, inventory destocking, and a cautious outlook for consumer end markets. The word “AI” does not appear. The model did not read the transcript. It recalled the company’s later history.

The same problem appears at scale. Lopez-Lira and Tang reported in 2023 that GPT-4 prompted to classify financial headlines as good or bad news achieved roughly ninety percent portfolio-day hit rates on the *initial market reaction* to strictly post-training-cutoff headlines, with significant additional drift in the days that followed [76]. The number is striking. It is also a useful warning. Even when cutoffs are enforced honestly, a model trained on the entire prior history of financial language brings strong priors into its predictions; on shorter horizons or older windows, those priors blur into outright recall.

LLMs are the most powerful tool added to the text-driven investing toolkit in a generation. They can classify exposure, extract structured data from unstructured filings, summarize complex disclosures, and reason about value-chain relationships in ways that earlier systems could not. They are also the easiest tool to misuse. The right stance is operational rather than magical: an LLM is a capability to run under controls, not a feature to turn on.

21.2 What an LLM Actually Adds

Earlier chapters of the book put LLM-style tooling to work in narrow ways: as label generators for topic clusters in Chapter 14, as one signal among many in the exposure score of Chapter 20. The role expands here because LLMs are the only tool in the stack that operates fluently on the level of meaning rather than the level of vocabulary.

Four roles recur across thematic pipelines.

- **Classification.** Given a document and a rubric, assign a label. Most common: theme exposure type (pure-play, partial, indirect, adopter, disrupted, none), sentiment, materiality, novelty.
- **Structured extraction.** Given a document, return a structured object: revenue lines, segment shares, capex commitments, customer names, supplier relationships, contract terms, geographic exposure.
- **Summarization and evidence packaging.** Given a corpus of documents about a theme or company, produce a compact, citable summary that an analyst can review faster than the source material.
- **Reasoning over relationships.** Given a value chain or a regulatory environment, answer questions whose answers are not in any single document — inferred customer exposure, second-order beneficiaries, regulatory cascade effects.

Each role replaces work that previously required either manual analyst review or a brittle pipeline of bespoke models. Each role also fails in a different way: classification hallucinates labels under prompt ambiguity, extraction invents fields under schema pressure, summarization smooths over contradictions, and reasoning fabricates relationships that sound plausible. Architecture is what separates the productive uses from the impressive demos.

The useful mental model is a contract. Every LLM call should declare five things before it runs: the task, the allowed evidence, the output schema, the abstention condition, and the downstream authority of the result. A call that cannot name those five fields is not yet part of a research pipeline. It is an analyst asking a clever model a question in prose.

21.3 Encoder, LLM, or Hybrid

Not every task in the pipeline needs a large general-purpose model. Many do not. The decision is a triage rather than a competition.

Domain-tuned encoder models — FinBERT and its successors, finance-tuned MPNet variants, LoRA-tuned bge-large embeddings — are the right tool when the question is a similarity question or a vector-space question. Theme embeddings, company-document embeddings for cosine similarity, retrieval indexes for filings and transcripts, and clustering pipelines all live here. Encoders are fast, cheap, and reproducible. They do not hallucinate because they do not generate.

Large general-purpose LLMs are the right tool when the question requires reading and reasoning. Classifying a multi-paragraph strategy disclosure against a six-category rubric, extracting structured supplier relationships from a transcript Q&A, deciding whether a disclosure paragraph names a beneficiary or a disruptee — these require the model to read in context. An encoder can flag that two documents are similar; only a generative model can explain why and decide which category they share.

Domain-tuned LLMs sit in between. BloombergGPT showed the appeal of a large model trained directly on financial and general-domain data [120]. FinGPT showed the opposite operating philosophy: start from open models, curate financial data, and adapt cheaply with lightweight fine-tuning [123]. In-house fine-tunes are a third version of the same trade-off. They earn their cost only when finance-specific language, proprietary documents, or deployment constraints reduce error enough to justify the maintenance burden.

The default architecture in a production thematic pipeline is a hybrid. Encoders carry the high-volume similarity and retrieval work. A general LLM handles the case-by-case reasoning that retrieval surfaces. The two communicate through retrieved evidence: the encoder finds the passages, the LLM reads them, and the system records what each layer did.

The hybrid choice also protects against a common category error. Encoders are retrieval engines, not judges. LLMs are judges and extractors, not databases. A system that asks an LLM to remember the filing archive is giving the model the job of the retriever and then wondering why provenance disappears. A system that asks an encoder to decide whether a supplier is strategically material is asking geometry to do judgment. The clean architecture lets each tool fail where another layer can inspect it.

21.4 Structured Extraction from Filings and Transcripts

The job is mechanical-sounding and consequential. Read a 10-K and return a structured object with the firm's segments, the revenue share of each, the geographic split, the named customers, the capex categories, and the disclosed risks. Read a transcript and return the list of supplier and customer names mentioned, the questions analysts asked, the topics management dodged, the forward guidance, and any numeric commitments. The output is a database row that downstream systems can join, filter, and rank.

Two architectural decisions decide whether the output is trustworthy.

First, the schema. An LLM that returns free-form prose forces the next layer to parse it. An LLM that returns a fixed JSON schema — with field names, types, allowed values, and per-field source-passage citations — can be validated programmatically. Schema enforcement is not a stylistic preference. It is the difference between an extractor that can be audited and a chatbot that produces plausible text.

Second, the citation requirement. Every extracted field should carry a pointer to the passage that justifies it. A revenue-share number with no cited paragraph is a guess. A customer name with no cited mention is a hallucination waiting to be discovered. A pipeline that refuses to record un-cited fields catches almost all the failures that quietly accumulate in less disciplined systems.

The downstream value is multiplicative. A scored extraction database lets exposure scoring (Chapter 20) read revenue share and customer relationships directly; it lets value-chain mapping (developed in the following chapter) construct supplier graphs from real disclosed edges rather than from co-mention noise; it lets governance audit the model by asking which passages drove which decisions.

Extraction also needs an abstention rule. If the relevant field is absent, the correct output is not an elegant guess; it is `null`, with a reason code such as “not disclosed,” “ambiguous,” or “conflicting evidence.” In financial text, missingness is information. A company that refuses to quantify a supposedly strategic segment is telling the investor something. The extractor should preserve that silence rather than decorate it.

21.5 Retrieval-Augmented Evidence Packages

Retrieval-augmented generation is the operating pattern that makes LLM outputs auditable rather than oracular. The shape is simple. A query is issued. A retriever pulls a small set of relevant documents from an indexed corpus. The retrieved documents are passed to the LLM

along with the question. The model is instructed to answer using only the retrieved context and to cite the passages it used.

For thematic work the retriever is typically built on a finance-tuned encoder over a corpus of filings, transcripts, broker research, and curated news. Retrieval is constrained to a time window: when the system is being run as of date t , only documents available at t are eligible. Each retrieved passage is tagged with its document identifier, source, date, and licensing terms.

The output is an evidence package. For an exposure-scoring decision, the package contains the LLM's classification (pure-play, partial, indirect, no exposure), a one-paragraph justification, and a short list of cited passages with document identifiers and dates. For a theme-monitoring decision, the package contains a summary of recent developments, the cited sources, and a confidence band. An analyst can review the package in minutes rather than the underlying corpus in hours. A governance reviewer can replay the package by re-running the same query against the same as-of corpus.

The discipline matters because the alternative is the demo failure mode: an impressive answer with no provenance, generated from a mixture of retrieved context, training-data recall, and prompt-induced confabulation. The user cannot tell which part came from where. Retrieval-augmented generation, done strictly, makes that question answerable.

Strict retrieval also means retrieval failure is a first-class outcome. If the retriever returns stale passages, low-similarity passages, or documents from the wrong source class, the LLM should not rescue the system by improvising. It should return insufficient evidence. The worst exposure mistakes often begin one layer earlier than the LLM: a bad retriever hands the model the wrong context, and the model does what it was trained to do — make the wrong context sound coherent.

21.6 The Cutoff Problem

Every LLM has a training cutoff. The model has been fit on text up to a date and not beyond. In principle, this lets a researcher test the model on strictly post-cutoff data and treat the result as honest out-of-sample evidence.

In practice, cutoffs are softer than they look. Three leaks recur.

The first is retrieval-side leakage. A retrieval index built from a current corpus can contain documents whose contents reflect post-cutoff knowledge. A 2024 broker note about a 2019 corporate event imports 2024 hindsight into the model's context window even if the model itself was trained only through 2023.

The second is training-side absorption. Models trained on broad internet text have been exposed to financial reporting, market commentary, historical analyses, and after-the-fact narratives that summarize what happened. A model asked about a 2019 filing may have, somewhere in its weights, a 2021 retrospective that already named the answer.

The third is fine-tuning and instruction tuning. A model whose base weights end at 2023 may have been instruction-tuned on 2024 data, or fine-tuned by a vendor with proprietary post-cutoff inputs. Cutoff dates published in model documentation often refer to the base pre-training corpus only.

These leaks are why the Lopez-Lira and Tang ninety percent hit rate is informative but not directly investable [76]. The study is careful about cutoff enforcement, which is what makes the result publishable evidence of an emergent capability. Whether the same number reflects what an investor would have earned in real time, with realistic latency and without the benefit of any

retrospective narrative the model may have absorbed, is a separate question. The number is a measurement of a model's behavior. It is not a backtest of a strategy.

Three adjacent results help triangulate what is plausibly real signal versus possible contamination. Bybee, Kelly, Manela, and Xiu fit a 180-topic LDA to roughly 800,000 Wall Street Journal articles spanning 1984 to 2017 and showed that topic-attention vectors track macro indicators and forecast aggregate market returns [22]. Calomiris and Mamaysky extended topic, sentiment, and word-flow entropy measures across Reuters coverage of fifty-one countries and showed predictability of returns, volatility, and drawdowns at country level, with stronger year-ahead than monthly predictions [23]. Ke, Kelly, and Xiu developed a supervised text-mining framework that adapts textual features directly to return prediction rather than relying on generic sentiment dictionaries [72]. These studies use methods whose training did not depend on knowing the outcome in the same way a modern LLM might. They corroborate that text carries signal at horizons relevant to thematic investing. They also bound the LLM-only results: text-based forecasting was substantial before LLMs, and a careful LLM result should improve on the pre-LLM baseline rather than replace its plausibility.

21.7 Retrieval-Constrained Workflows

The discipline that contains the cutoff problem is workflow design, not model choice. Two patterns dominate.

The first is the as-of corpus. Every retrieval index used in research or simulation is rebuilt at each historical decision date t , using only documents available as of t in their as-published form. A document revised in 2023 about a 2019 event appears in the 2019 index in its 2019 form, not its 2023 form. A vendor data update is replayed in the order the vendor delivered it, not in its current cleaned state. The discipline is expensive. It is also the only honest way to test a system that uses retrieved evidence at decision time.

The second is the cutoff-respecting prompt. The LLM is told explicitly that it is operating as of date t , that any knowledge it has of events after t should be excluded, that retrieved passages have been filtered to that date, and that it must classify or extract using only the provided context. This is a soft constraint; the model can still leak through training-data absorption. It is, however, the minimum standard for honest reporting. A research report that does not record what the prompt told the model is a report that cannot be replayed.

Two further guardrails reduce residual contamination. Periodic out-of-time evaluation: hold out a strictly post-cutoff window and measure whether performance degrades to plausible levels when no recall is possible. Counterfactual prompting: ask the model questions whose true answers it could only know via recall (named outcomes of historical events) and measure how often it produces them despite instruction; this calibrates how strong the recall is at the task in question.

A third guardrail is model freezing. Historical research should record the exact provider, model identifier, system prompt, retrieval index, embedding model, sampling parameters, and schema version used for each run. If a vendor silently updates a hosted model, the same prompt can become a different experiment. For production that may be acceptable; for research it is poison. A reproducible LLM experiment pins the model or stores the raw inputs and outputs so the decision can be replayed even when the model cannot.

The full backtest-evaluation framework for LLM workflows — including how to design experiments whose hit rates are interpretable rather than impressive — belongs to the chapter

on backtesting in Part VIII. The workflow side is the standing requirement: every LLM call that touches a research or production decision should be designed so that someone can ask, later, what the model knew when, and the answer should be specific.

21.8 Prompt Stability, Schema, and Hallucination Control

Two LLM calls with the same input can produce different outputs. Two calls with slightly different prompts can produce sharply different outputs. This is a property, not a bug. It is also a problem for a system that has to be auditable.

Stability is enforced through three habits. Temperature and sampling parameters are pinned and recorded. Prompts are versioned the same way code is versioned, with diffs and dates. Output schemas are strict — the model is required to return a JSON object that validates against a declared structure, and a non-validating output is retried rather than silently passed downstream.

Hallucination control runs through the same machinery already named. Retrieval grounding limits the model to material it can cite. Schema enforcement prevents free-form prose from leaking through as data. Citation requirements force every claim to point at a passage. Periodic audits sample outputs and check whether the cited passages actually support the claims; an output that cites a passage that does not contain the cited fact is a hallucination, however confidently delivered.

The absolute volume of hallucination cannot be reduced to zero. The relevant question is whether the rate is low enough, and the detection mechanism fast enough, that the operating risk is acceptable for the use case. Theme labeling at the topic level can tolerate some failure — analysts catch most errors in review. Exposure classification at the portfolio level cannot — one confidently hallucinated pure-play assignment can push a basket into a position no one approved. The use case sets the tolerance.

Evaluation should therefore be layer-specific. Extraction is evaluated against field-level precision, recall, and citation support. Classification is evaluated against analyst labels, disagreement rates, and calibration: when the model says high confidence, is it actually right more often? Summarization is evaluated by omission and contradiction audits, not by whether the prose reads well. Reasoning over value chains is evaluated by whether the inferred edges later appear in filings, contracts, customer disclosures, or analyst-confirmed maps. A single “LLM accuracy” number hides exactly the differences the pipeline needs to manage.

21.9 Governance: What the Model Is Allowed to Decide

The last decision is the easiest to skip and the most consequential.

Every place an LLM touches the pipeline, the system has to record what level of authority the model has at that step. Three levels are useful.

- **Automatic.** The model’s output is used directly. Suitable only for low-stakes, high-volume decisions where errors are caught downstream (initial topic labeling, document deduplication, language detection, simple field extraction with schema validation).
- **Analyst-reviewed.** The model produces a recommendation; an analyst approves, rejects, or modifies it before it enters the portfolio path. Suitable for exposure classifications, value-chain assignments, lifecycle-stage updates, and theme approvals.

- **Committee-reviewed.** The model’s recommendation enters a structured review that involves more than one person and produces an auditable approval record. Suitable for new-theme onboarding, large position changes, and any decision whose reversal would be operationally expensive.

Override logs are part of governance. When an analyst rejects or modifies a model recommendation, the system records the original recommendation, the override, the analyst’s reasoning, and the evidence cited. Over time, override logs become a training signal in their own right — they show where the model is systematically wrong, where the analysts are systematically over-correcting, and where the rubric should be rewritten. Without override logs, the human-in-the-loop layer is invisible to the system that produced the recommendations.

A pipeline that does not record what authority the model had at each step cannot answer the only questions that matter when something goes wrong: what did the model decide, what did the human decide, and on what evidence. That accounting is the operating definition of LLM governance for thematic investing. The tools that earn their place are the ones that make that accounting possible.

The final governance rule is uncomfortable but useful: the model should be allowed to say no. No relevant evidence. No material exposure. No reliable classification. No basis for a score change. LLMs are persuasive machines; investment systems need them to become disciplined machines. The easiest way to tell the difference is to count how often the model abstains when the evidence is thin.

A pipeline that has named its themes, scored its companies, retrieved its evidence, and governed its models has done most of the upstream work a portfolio process needs. The remaining text-side question is structural: how the companies relate to each other along the value chain, and which positions in that chain carry the exposure the basket is supposed to capture.

Chapter 22

Value Chains, Networks, and Beneficiary Types

22.1 The Edge That Took One Press Release to See

On September 20, 2024, Constellation Energy announced a 20-year power purchase agreement with Microsoft to restart Unit 1 at Three Mile Island, rebranded as the Crane Clean Energy Center [28]. A nuclear reactor that had been retired in 2019 for economic reasons was now scheduled to produce electricity for a hyperscaler’s data centers.

The press release was treated as a surprise, and for most investors who held only sector-classified portfolios, it deserved to be one. Constellation sat in the Utilities sector, Microsoft sat in software and cloud, and the PPA did not fit any standard map.

For anyone who had been reading the right transcripts, the edge had been visible for a year. Hyperscaler capex commentary had been pointing at electricity as a binding constraint. Grid-interconnection queues in the mid-Atlantic and Texas had been lengthening. Vistra’s nuclear and gas fleet had been outperforming. The vocabulary the market was using — “data-center power,” “baseload generation,” “behind-the-meter,” “baseload PPA” — described a relationship between firms that were not in the same GICS sector and would never appear together in a sector-classified basket. The Microsoft–Constellation deal made the edge visible to the press. The edge itself was older.

This is what value-chain mapping is for. A theme that touches a portfolio touches it through specific firms in specific roles, connected by specific economic relationships. A list of theme-exposed companies that does not encode those relationships is throwing away the structure that decides which firms have pricing power, which firms are bottlenecks, which firms are interchangeable, and which firms are about to be disrupted. The sector map says Constellation is a utility. The value-chain map says Constellation is an AI infrastructure node connected to Microsoft through a long-dated baseload contract. Those are not the same statement.

22.2 Themes Are Edges, Not Just Nodes

A list of companies is not a theme; a list of companies plus the edges between them is much closer to one.

The point sounds slight until the portfolio has to explain why it owns what it owns. The same firm can sit in a basket as a marketing label or as a position in a chain that explains

why the firm earns its return, and the difference is operational. A basket that holds Nvidia, SK Hynix, Vertiv, Eaton, and Constellation as five names with five exposure scores has thrown away the information that links them: GPUs require HBM, HBM ships from Korea, racks of accelerators dissipate heat that requires liquid cooling, data-center buildouts require switchgear and transformers, and the whole stack runs on power that is suddenly scarce in specific grids. Each firm's exposure and risk depend on its position in that chain.

The augmented return model introduced in Chapter 6 and used since — $r = Xb + Yg + u'$ — assumes that thematic exposure Y can be measured at the firm level. The value-chain layer is what makes that measurement structural rather than nominal. A firm's Y score is shaped by who its customers are, who its competitors are, who its substitutes are, and where in the chain the binding constraint sits. Without those edges, Y collapses to mention frequency in a more decorated form.

The most important edge in a theme is often not the most visible company. In 2023, the market learned to price Nvidia quickly. It took longer to price the firms that supplied the memory, power, cooling, and grid equipment around the GPU. A value-chain map is the tool that keeps the obvious node from swallowing the whole thesis.

22.3 Beneficiary Types

A practical taxonomy of roles in a theme's value chain runs to six categories. Each has a different economic profile, a different sensitivity to the theme, and a different sign and horizon for the position.

- **Upstream suppliers and enablers.** Provide inputs, components, raw materials, equipment, or specialized services that the theme requires. Example for AI infrastructure: ASML for lithography, SK Hynix and Micron for HBM, Applied Materials and Lam Research for fab equipment.
- **Core pure-plays.** Derive a large share of revenue directly from the theme. Example for AI infrastructure: Nvidia for accelerators, Arista for AI networking, Vertiv for data-center thermal and power management.
- **Infrastructure and platform enablers.** Provide the substrate on which the theme operates. Example for AI infrastructure: hyperscaler cloud platforms (Microsoft Azure, Google Cloud, AWS), data-center REITs (Digital Realty, Equinix), competitive power generators (Vistra, Constellation).
- **Adopters.** Use the theme to improve operations, products, or margins. Example for AI infrastructure: software firms embedding copilots, life-sciences firms running AI-accelerated discovery, financial firms automating workflows.
- **Downstream beneficiaries.** Capture second-order demand or cost changes. Example for AI infrastructure: industrial electrical-equipment makers (Eaton, Schneider Electric, nVent) selling switchgear and enclosures into data-center buildouts.
- **Disrupted incumbents.** Face revenue, margin, or competitive pressure from the theme. Example for AI infrastructure: legacy software vendors whose products are commoditized by generative tools, IT-services firms whose labor-intensive offerings face automation.

The same company often occupies more than one role. Microsoft is an adopter through Copilot, an enabler through Azure, and a buyer of accelerators and power. Alphabet is an adopter, an enabler, and a designer of its own TPUs. Role classification should be explicit rather than buried inside a single exposure number; the augmented model accommodates several roles with several signs at the same firm, and the value-chain map is how that gets recorded.

Role is not the same as exposure score, and neither is the same as portfolio weight. Role says where the company sits in the chain; exposure says how much the theme plausibly moves its fundamentals; weight says how much capital the strategy is willing to allocate after liquidity, valuation, factor, and concentration constraints. Keeping these three objects separate prevents a common mistake: giving every upstream supplier a high portfolio weight simply because the graph says it is upstream.

The clean-energy 2020–2022 case gives the same taxonomy with different residents: polysilicon refiners, lithium and cobalt miners, and wind-turbine component suppliers upstream; solar-panel manufacturers, residential solar installers, and hydrogen-electrolyzer firms as core pure-plays; utility-scale developers, grid operators, and financiers as enablers; corporate purchasers of clean-energy PPAs as adopters; rare-earth processors selling into wind generators as second-order beneficiaries; coal-asset owners and refining-heavy oil majors with little energy-transition capex on the disrupted side. The lesson is not that the taxonomy fails. It is that during the 2020–2021 rally, many ETFs treated all of these roles as equivalent “clean-energy beneficiaries” and ended up with baskets whose constituents had little in common except the label.

22.4 The Directed Graph and Its Edges

A value chain is a directed graph:

$$G = (V, E), \quad (22.1)$$

where V is the node set and E is the edge set. Nodes represent companies, but they can also represent products (HBM, lithium-iron-phosphate batteries), facilities (a specific data-center campus, a single fab), geographies (Northern Virginia, the Permian Basin), technologies (liquid cooling), or themes themselves. Edges represent directional economic relationships: supplier-of, customer-of, partner-of, competitor-of, exposed-to-input, regulated-by, beneficiary-of-policy, located-in.

Some edges are obvious in disclosure. A 10-K that names “Microsoft, Amazon, and Google” as customers makes a customer edge explicit. Other edges are implicit and require evidence to construct. A switchgear vendor whose revenue is growing alongside hyperscaler capex may have no public customer-name disclosure; the edge has to be inferred from product mix, end-market commentary, and order-backlog disclosures.

Three properties make the graph useful for portfolio work. Direction separates a supplier edge from a customer edge, and a firm with many customers is positioned differently than a firm with many suppliers. Weight separates a customer who accounts for thirty percent of revenue from one who accounts for one percent. Time separates a live relationship from a stale one: contracts expire, suppliers get swapped, and geographies shift production. A graph that records a firm’s customers as of 2019 may be wrong about 2024, and a system that uses it as if it were current is using a stale map.

The graph therefore needs an edge record, not just an edge. A usable edge stores at least:

- source node, target node, and relationship type;
- direction convention, for example supplier $\rightarrow$ customer;
- edge weight, such as revenue share, disclosed contract size, capacity share, or confidence-weighted relevance;
- sign of the economic exposure: positive, negative, hedge, or ambiguous;
- evidence passages, document identifiers, source type, and availability date;
- validity window and confidence score.

This looks like bookkeeping until a historical backtest asks whether a supplier relationship was knowable in 2021, whether it was still active in 2023, and whether the edge was explicit or inferred. Then it becomes the difference between a map and a story.

22.5 Relation Extraction at Scale

The graph does not build itself; its edges have to be extracted from filings, transcripts, news, broker research, and supply-chain databases.

The general apparatus for entity and relation extraction was developed in Chapter 10, including the finance-specific relation datasets that name the kinds of links thematic systems most need: supplier-of, acquired-by, partner-of, competitor-of, exposed-to [70, 52]. For the value-chain layer, the operating discipline is to store evidence with every edge. A graph without per-edge source passages, document identifiers, and dates is a graph that cannot be audited and cannot be replayed at a historical decision date.

The retrieval-augmented patterns from Chapter 21 apply directly. An LLM asked “does this company supply HBM to Nvidia” should answer using retrieved passages from the company’s own filings, named-supplier disclosures, and transcript Q&A, and should cite the passages that justify the answer. An LLM asked the same question from general knowledge produces an answer that sounds right and cannot be defended. The graph that gets used in production is built from cited extractions, not from unsourced inferences.

Two practical refinements make the relation layer survive contact with messy data. Edges should carry confidence scores because an explicitly named customer relationship is stronger evidence than a transcript co-mention, and they should carry validity windows because a supplier relationship disclosed in 2020 and unmentioned in 2024 should not be assumed to persist forever. Both refinements are simple to add and routinely skipped.

The relation layer should also preserve disagreement. One source may name a customer directly, another may describe the same exposure generically, and a third may imply that the relationship has ended. Collapsing that disagreement into a single binary edge loses information the analyst needs, whereas a good graph can say “edge exists, high confidence, stale evidence” or “edge plausible, low confidence, no named counterparty.” Those states should lead to different exposure scores.

22.6 Centrality and What It Catches

Network centrality measures convert graph structure into firm-level scores. Three measures recur in thematic work, and each catches a different property of a firm’s position.

- **Out-degree centrality.** Under the supplier $\rightarrow$ customer convention, this is the number (or weighted volume) of direct customer edges. High out-degree indicates breadth of demand and reduced single-customer concentration risk. A switchgear vendor selling to all major hyperscalers ranks higher than one tied to a single buyer.
- **Betweenness centrality.** Frequency with which the firm sits on shortest paths between other firms in the graph. High betweenness identifies bottlenecks — firms whose disruption propagates through the network. SK Hynix’s HBM line during the 2023–2024 accelerator shortage was a textbook high-betweenness node: many leading-edge AI clusters depended on scarce memory supply.
- **Closeness centrality.** Inverse of average distance to other nodes. High closeness suggests organizational reach and adaptive capacity — the firm can interact quickly with the rest of the network when conditions change. Useful for identifying enablers and integrators rather than specialists.

Other measures (eigenvector centrality, PageRank-style scores, k -core decomposition) add nuance and are worth running when the graph is rich enough to support them. None of them is investable on its own. A high-centrality firm without pricing power, capacity, or financial discipline does not earn the theme; a low-centrality bottleneck with one constrained input and patient customers may earn it disproportionately. Centrality is an input to a thesis, not a substitute for one.

The link back to exposure scoring (Chapter 20) is direct. The G component of the multi-signal score $S(i, \theta, t) = f(K, E, R, L, G)$ draws from centrality and edge-evidence directly; without the graph layer, the G component degenerates into co-mention statistics. The score should never treat centrality as a free lunch. High centrality raises the question “why might this firm matter?” but it does not answer “is this firm mispriced?”

22.7 Propagation of Exposure

Themes propagate along chains. If hyperscalers raise data-center capex, demand flows downstream to accelerator vendors, then to memory and packaging firms, then to fab equipment, then to substrate and chemical suppliers. Demand also flows sideways to power, cooling, real estate, and grid equipment. A propagation model spreads theme intensity across graph edges with decay:

$$S^{(k+1)} = \lambda A S^{(k)} + S^{(0)}, \quad (22.2)$$

where $S^{(k)}$ is the exposure vector at iteration k , $S^{(0)}$ is the initial exposure assigned by the upstream signal, A is a normalized adjacency matrix, and $\lambda \in [0, 1]$ is the propagation parameter. The recursion converges when propagated exposure equals decayed-back-incoming-exposure. The fixed point distributes theme intensity across the network in a way that reflects the graph’s structure.

The formula is useful as a sanity check and dangerous as a portfolio rule because it assumes uniform pass-through along edges, which the real economy never delivers. A doubling of GPU demand does not double cooling demand at the same speed; it does not double switchgear demand at all if the existing power infrastructure can absorb the marginal load. Margins, capacity, contracts, substitution, and competitive structure decide whether demand becomes earnings.

The first practical correction is edge-type decay. A disclosed revenue edge should transmit more exposure than a loose co-mention edge. A signed customer contract should transmit more than a trade-press inference. A bottleneck input should transmit more than an interchangeable service. In notation, the single λA term usually becomes an edge-weighted operator whose weights depend on relationship type, evidence quality, and economic pass-through.

The right use of propagation is qualitative more than numeric: it identifies candidate beneficiaries that direct disclosure would have missed, ranks them as a hypothesis, and hands the ranking to the materiality and confidence filters from Chapter 20 before any of it enters a basket. A propagation model used as the final exposure score for portfolio construction will reliably over-include weakly exposed peripheral firms.

22.8 Negative Exposure

Themes create losers as well as winners, and the value-chain map gives the cleanest way to find both.

Automation reduces demand for some labor-intensive services. Domain-specific generative tools commoditize parts of the IT-services and legal-research markets. Cheaper renewable generation pressures the economics of high-cost legacy assets. Stricter regulation of one technology benefits compliant incumbents and harms exposed ones. Each of these is an edge in the graph with a negative sign attached.

A complete theme map allows negative exposure as a first-class concept. Long-only baskets can use it to avoid disrupted firms whose factor profiles would otherwise pass eligibility filters. Long-short baskets can use it to size the short side from the same evidence base that sizes the long side, rather than treating shorts as a separate research question. Risk management can use it to identify firms whose negative theme exposure would amplify rather than diversify a basket's existing position.

The cost of ignoring the negative side shows up most clearly in baskets that hold both a beneficiary and a disrupted firm at the same time. The two positions cancel some of the theme exposure the manager believes the basket has, and the basket's residual return ends up smaller and noisier than the manager expects.

Negative exposure should not be confused with a short recommendation. A firm can be negatively exposed to a theme and still be a bad short because valuation, balance-sheet strength, buybacks, or unrelated business lines dominate the stock. The graph only says where the pressure enters; the portfolio chapter decides whether that pressure is worth trading.

22.9 The AI Infrastructure Chain in Practice

A worked AI infrastructure value chain, organized by edge:

- **Model demand to compute demand.** Hyperscalers, large-model developers, and sovereign AI initiatives are the customers; their training and inference loads create the upstream pull.
- **Compute demand to hardware demand.** Accelerator designers (Nvidia, AMD, custom in-house silicon at Google, Amazon, Meta) sit at the first hardware layer; their orders pull on memory (SK Hynix, Micron, Samsung), advanced packaging (TSMC CoWoS, Amkor, ASE), substrates, and high-end networking silicon (Broadcom, Marvell, Astera Labs).

- **Hardware to fab and equipment.** TSMC and a small number of leading-edge fabs sit between accelerator demand and process-equipment vendors (ASML for EUV, Applied Materials, Lam Research, KLA, Tokyo Electron); chemical and materials suppliers feed the same chain.
- **Hardware to data-center construction.** Accelerator deployments require campuses; data-center REITs (Digital Realty, Equinix) and specialized developers run the buildouts.
- **Data centers to power.** Buildouts require electricity; competitive generators (Vistra, Constellation, Talen) and grid-interconnection providers sit at this edge, with the Microsoft–Constellation PPA as the canonical visible-edge event.
- **Data centers to thermal and electrical infrastructure.** Cooling vendors (Vertiv on the air side, specialized liquid-cooling firms on the dense-rack side), electrical-equipment makers (Eaton, Schneider Electric, nVent) supply the racks and the room around the racks.
- **Cloud platforms to enterprise adopters.** Cloud-platform monetization feeds back into software vendors integrating AI features, IT-services firms helping enterprises deploy, and downstream productivity gains in adjacent industries.
- **Disrupted side.** Some labor-intensive IT-services models, some legacy on-premise software vendors, and parts of the customer-support and basic-research markets face structural compression.

A basket built from this chain looks different from a basket built from a sector list. It can include industrial firms (Eaton, Vertiv) whose GICS classification puts them outside technology, utilities (Vistra, Constellation) whose regulated peers do not earn the theme, and foreign-listed firms (SK Hynix, TSMC, ASML) whose value-chain position makes them more exposed to AI than many US-listed software names. The map decides what is in; the sector classification does not.

The clean-energy 2020–2022 chain shows why transition investing needs richer edge labels than “beneficiary” and “pure play.” The narrative chain was real — policy support to manufacturer demand to installer growth to grid integration — but the rate-driven discount-rate reset cut across many nodes at once, and the baskets that had not separated rate-sensitive duration positions from theme-sensitive operating positions absorbed the reset uniformly. A value-chain map that records the rate sensitivity of each edge would have caught it. A basket that records only the theme label could not.

The difference between the two cases is the lesson. AI infrastructure in 2023–2024 had visible bottlenecks in HBM supply, advanced packaging capacity, data-center power, cooling, and switchgear. Clean energy in 2020 had many legitimate nodes, but not every node had protected pricing power once rates moved. A value-chain map should make that distinction explicit because bottlenecks tend to create pricing power in specific nodes, while broad enthusiasm can overfund parts of a transition before the cash-flow evidence has caught up.

Once the score is set and the chain is mapped, the remaining question is whether the structure that the system has built is robust to the failure modes a thematic process is most likely to hit: dilution, illusion, vocabulary capture, and timing. That is the work the guardrails chapter takes up.

Chapter 23

Guardrails Against Thematic Illusions and Implausible Edges

23.1 The Iced Tea Company That Became a Blockchain Company

On December 21, 2017, a small beverage distributor called Long Island Iced Tea Corp. issued a press release announcing that it would change its name to Long Blockchain Corp. and refocus toward blockchain technology. The stock rose roughly two hundred percent intraday, even though the company had no blockchain products, no blockchain customers, no blockchain engineers, and no blockchain revenue. It had a new ticker and a press release. Within months the Nasdaq moved to delist the shares; the SEC eventually brought insider-trading charges against individuals tied to the trade [113].

The episode is comic because the gap between the theme and the company was so large, but the structural problem is not comic at all. The same mechanism that produced the Long Blockchain trade — a coherent investor narrative, a vocabulary that signals exposure, and a company that adapts its disclosure to capture flows — operates more subtly in every thematic cycle. The 2018–2019 cannabis ETF wave, the 2020–2021 SPAC boom, the 2021 metaverse repositioning, the 2022 “Web3” rebrand, and the 2023–2024 AI-pivot announcements all run the same script with different vocabularies: the market reacts to the language, and the basket inherits the company.

Thematic investing is vulnerable to this pattern because thematic systems are good at detecting coherent vocabulary and only sometimes good at detecting whether the vocabulary corresponds to a business. A plausible story is not yet an investable edge. Guardrails are the difference between a research process that survives a cycle and one that becomes its anecdote.

23.2 A Plausible Story Is Not an Investable Edge

The chapters that built the thematic apparatus all assumed something the apparatus cannot enforce on its own. Topic discovery (Part III) assumed the discovered topic was coherent. The investable-theme specification (Chapter 19) assumed the team would actually fill in the falsification fields. Exposure scoring (Chapter 20) assumed the evidence streams would not be manipulated by the firms being scored. Value-chain mapping (Chapter 22) assumed the edges

were structural rather than aspirational. None of those assumptions holds without enforcement.

The required posture is unusual in the order it puts things. Discovery rewards enthusiasm, while validation rewards skepticism. A process that is enthusiastic in both phases is selling; a process that is skeptical in both phases misses themes. The discipline is to be enthusiastic about discovery and ruthless about validation, and to keep the two operating modes from contaminating each other. The analyst who found the theme should not also be the analyst who certifies that it has survived the kill checks. Guardrails are how the institution makes that separation operational.

The value-chain map asks, “could this relationship transmit the theme?” The guardrail layer asks, “what would make us wrong?” A company can sit in the correct place in the chain and still be a poor constituent because the exposure is too small, the valuation already too rich, the factor contamination too large, the liquidity too thin, or the evidence too dependent on management’s preferred vocabulary. Guardrails do not replace mapping; they interrogate it.

23.3 Mention Is Not Exposure

The first guardrail is the simplest and the most often skipped: mentioning a theme is not the same as benefiting from it. Companies know which narratives investors care about. Management teams emphasize AI, sustainability, reshoring, or digital transformation because those words affect the cost of capital. A 10-K risk-factor paragraph about artificial intelligence carries weight in a keyword-scoring system and carries no weight in the income statement.

A scoring model that has internalized this distinction requires evidence beyond mentions before it admits a firm to a basket:

- revenue or segment exposure tied to the theme;
- product or service linkage that can be named;
- disclosed customer or supplier relationship;
- order backlog or contracted commitments;
- capex commitment matched to the theme;
- pricing power consistent with bottleneck position;
- analyst-estimate sensitivity to theme-specific news;
- observed stock reaction to theme-specific events, controlling for sector moves.

Language is one of the weakest evidence types, and a firm whose only evidence is language deserves a score that reflects that weakness, regardless of how often the language repeats. The audit-side check is straightforward: take the highest-scoring firms in each theme, strip the keyword and embedding signals, and see whether any other evidence remains. If nothing remains, the firm earned the score from rhetoric.

The evidence ladder should be explicit. Generic mentions and boilerplate risk factors sit at the bottom; theme-specific product descriptions and management commentary sit above them; customer names, backlog, capex, segment revenue, pricing, margin, and realized estimate revisions sit higher still. At the top are observable cash-flow changes and residual stock reactions to theme-specific events. A company can move up the ladder, but it should not be allowed to jump from the bottom to the basket because its language is fashionable.

23.4 The Anatomy of Name Capture: Greenwashing and AI-Washing

Greenwashing and AI-washing are the same mechanism with different vocabularies: a firm whose business has not changed adopts the language of a popular theme, and a system that scores firms on language admits the firm to the basket. The vocabulary changes between cycles; the mechanism does not.

Detection requires reading rhetoric against fundamentals. If AI mentions rise but the corresponding revenue lines, capex categories, hiring patterns, product disclosures, and customer relationships do not move, the exposure score should be discounted. If sustainability language increases but the underlying business mix, capital allocation, or operating metrics do not shift, the same principle applies. The discipline is comparative rather than absolute: a firm that mentions AI more than its peers but discloses less material exposure than its peers is sending a clear signal, and the scoring system should be configured to read it.

Two anti-patterns recur in systems that try to catch this without enough structure. The first is treating any keyword stuffing as automatic disqualification, which throws out firms whose disclosure happens to be comprehensive. The second is treating any rhetoric-fundamentals mismatch as a temporary lag rather than a signal, which keeps firms in baskets long after the mismatch has stabilized into a pattern. The middle path is what the multi-signal score is supposed to deliver: rhetoric without backing reduces confidence, even when it does not reduce raw exposure magnitude.

One useful test is adversarial: ask what the disclosure would look like if management wanted to appear exposed without changing the business. More mentions would appear in prepared remarks, risk-factor language would broaden, product pages would add the new vocabulary, and investor presentations would move the theme closer to the front. Segment reporting, revenue attribution, customer disclosures, and capex would remain stubbornly vague. A guardrail process should search for exactly that pattern, not because it proves deception, but because it proves that language has outrun evidence.

23.5 ETF Dilution and the Liquidity Tax

Thematic ETFs face a structural trade-off between purity and capacity. To run at scale, they include large liquid proxies whose theme exposure is limited but whose float can absorb the basket weight. The label on the wrapper says “AI” or “clean energy,” while the holdings drift toward the mega-caps the broader index already contains.

The empirical evidence on the cost of this drift is one of the most important findings in the recent thematic literature. Ben-David, Franzoni, Kim, and Moussawi showed that specialized ETFs — the academic name for the thematic category — lose roughly three percent per year in risk-adjusted performance after fees over their first five years, with newly launched specialized vehicles giving up closer to six percent per year; cumulative risk-adjusted underperformance is on the order of thirty percent over the five-year window [9]. The mechanism the paper documents is launches that cluster at peaks of investor attention, holdings whose valuations mean-revert after the launch, and dilution that prevents the wrapper from delivering the theme’s pure-play return profile. Specialized ETFs collected roughly eighteen percent of industry AUM and approximately thirty-six percent of industry fee revenue at average expense ratios near fifty-five basis points; the segmentation persists because the same retail flows are insensitive to fees and aggressive in

response to past performance.

The practical implication runs in both directions. ETF holdings are useful as one signal because they show what other systems classify as theme-exposed, but they cannot be ground truth. An exposure model that learns to imitate ETF holdings inherits the dilution that the academic literature has measured. The model has to be able to disagree with the wrapper, and to do so on the basis of evidence that the wrapper's eligibility rules did not weigh.

ETF overlap is therefore a diagnostic, not a target. High overlap with a thematic ETF may mean the model has rediscovered the market's consensus basket, which can be useful for benchmarking; it may also mean the model has inherited the same timing, valuation, and liquidity compromises. The question is not whether the model agrees with the ETF, but where it disagrees, why it disagrees, and whether those disagreements are supported by fresher or more material evidence.

23.6 Late-Cycle Theme Launches

Themes become products after performance, not before. The launch calendar for thematic ETFs concentrates near peaks of attention, when the easy returns have already been earned and the valuation premium of the named constituents has already expanded. A fund opened at peak attention gives late investors clean access to the story, but often at prices set by earlier believers.

A monitoring layer that catches the late-cycle pattern reads several signals together:

- attention relative to a historical baseline (Chapter 7's acceleration framework provides the apparatus);
- valuation expansion in the candidate universe relative to the broader market;
- ETF launch and flow activity in the theme and its adjacencies;
- media saturation and source-breadth saturation;
- ownership concentration in the named pure-plays;
- forward-return deterioration despite continued coverage growth.

High attention is not, by itself, a sell signal; themes that genuinely change the economic structure can support extended periods of high attention. The pattern that matters is the conjunction: extreme attention plus valuation expansion plus launch activity plus declining forward returns. When all four conditions are present, the prior on the theme being early should be replaced by a prior on the theme being late.

Late-cycle detection should also be run at the constituent level. A theme can still be valid while its most obvious names have already become too expensive. In that case the right response is not always to abandon the theme; it may be to move along the value chain toward less crowded bottlenecks, reduce pure-play concentration, or require stronger valuation support before adding exposure. The theme may remain alive while the first basket is no longer attractive.

23.7 The Economic Plausibility Checklist

Before a theme enters a basket, the team should be able to answer eight questions explicitly. The questions are the ones the book has been building toward since Chapter 5:

- Who is underreacting or overreacting, and why?
- What constraint prevents immediate arbitrage — short-sale costs, capacity, information complexity, regulatory uncertainty, liquidity?
- Is the mechanism information diffusion, attention, complexity, behavioral extrapolation, or structural change?
- On what horizon should the edge operate, and does the data source match it?
- What observable evidence would falsify the thesis?
- Does the signal survive after factor, sector, and liquidity controls?
- Does the basket’s residual return retain residual character after Fama–French and momentum hedging?
- Is the strategy implementable at scale after realistic transaction costs?

Weak answers do not mean the theme is wrong; they mean it is not yet ready to be capitalized. The right destination for a theme with weak plausibility answers is the monitor-only queue, not the basket. A monitor-only theme that later develops strong plausibility evidence can be upgraded with a clean audit trail.

The checklist works best as three gates. The first is economic: why should this theme affect cash flows or discount rates? The second is temporal: why should the effect be available before the market fully prices it? The third is portfolio-level: why should the effect survive after factors, sectors, liquidity, and implementation costs? A theme that passes only the first gate is a good story; a theme that passes all three is at least eligible to be tested as an investment.

23.8 Backtest Overfitting and the As-Of Discipline

Thematic backtests are unusually easy to overfit because the researcher knows which themes worked. A theme defined after the winners are known is a backtest of analyst hindsight, not of the system that supposedly discovered it; an exposure rule frozen after the basket has been observed is a curve fit; a universe that excludes delisted firms turns survivorship bias into performance.

The defenses are not novel and they are not optional. Theme specifications must be point-in-time, with first-detection and approval dates that the system can prove rather than assert. Exposure rules must be frozen at the time of theme approval, with any subsequent changes recorded and re-evaluated rather than silently absorbed. The corpus, the entity mapping, and the embedding model used in any historical experiment must be the as-of versions, not the current state. The factor model used for residualization must be the version that was published as of the simulated date, not the current revision.

Walk-forward evaluation is the operating discipline that wraps these requirements. The full machinery — including how to design experiments whose hit rates are interpretable rather than impressive, and how to handle LLM workflows whose models may already know the answers — is developed in the backtesting chapter of Part VIII. The guardrail at this stage is procedural: a theme that cannot pass the as-of discipline at the research stage will not pass it in production either, and approving it on the basis of in-sample performance is approving a story.

Placebo tests are a useful addition. Run the same exposure machinery on themes that should not plausibly predict the target universe, on randomized approval dates, and on shuffled company

labels; then run it on stale versions of the theme after the vocabulary has already diffused. If the model still produces attractive backtests, the strategy is probably harvesting a factor, a sector, a liquidity effect, or a data artifact rather than the theme. A thematic signal that cannot beat its placebos has not earned capital.

23.9 Falsification and Kill Rules

Every theme that enters a basket should leave the basket through a documented rule. Kill rules are not pessimism; they are the operational form of the falsification fields the theme specification was required to carry.

A useful kill-rule library includes:

- exposure no longer predicts residual returns over the chosen horizon;
- attention reaches extreme levels and forward returns deteriorate across the candidate universe;
- material evidence (revenue, capex, customer commitments) fails to appear at the firms that scored highest;
- the basket becomes statistically indistinguishable from a sector or factor proxy under standard regression decomposition;
- liquidity, crowding, or capacity risk exceeds the residual alpha the basket can earn;
- the economic mechanism is invalidated by an event the theme specification named as falsifying.

A kill rule firing does not mean the theme was a mistake. It usually means the lifecycle has changed (Chapter 6’s seven-stage framework is the right map): a validated theme has crowded, a crowded theme has decayed, or a decaying theme has stopped earning the residual that justified the position. Retiring the theme cleanly preserves the option to bring it back if it recycles. Keeping it in the basket on the strength of past performance is how thematic strategies turn into legacy positions that nobody can defend.

Kill rules should have owners because a rule without an owner is only a reminder. “Review if crowding rises” may prompt a conversation; “reduce exposure by half if ETF-overlap, valuation, and attention thresholds are all breached for two rebalance dates” is an operating control. The exact thresholds will differ by strategy, but the principle is stable: the exit should be written before the position is emotionally expensive to close.

23.10 Discipline That Separates Discovery from Story

A thematic system that has named its themes, scored its companies, mapped its value chains, and run its evidence through the guardrails is doing recognizable work. A system that does the first three and skips the fourth is doing storytelling that uses the same vocabulary.

For AI infrastructure, the guardrail layer prevents the basket from becoming a generic mega-cap technology proxy, from over-weighting weakly exposed firms whose disclosures only mention the theme, from concentrating in one or two chip designers whose valuation has already expanded, and from inheriting the dilution that thematic ETFs systematically carry. Useful concrete checks

include the share of basket weight in firms whose revenue exposure exceeds a stated materiality threshold, the basket's residual return after FF5 + momentum hedging, the overlap with the largest AI ETFs and the broader growth indexes, the basket's sensitivity to interest rates and to momentum factors, and explicit evidence that the named second-order beneficiaries (electrical equipment, cooling, power) actually retain pricing power in the period under test.

For clean energy 2020–2022, the same guardrails would have helped distinguish a real structural transition from fragile public-market expressions of it. The narrative was coherent, the vocabulary was correct, and the materiality of the exposure was real for some firms and aspirational for others. The ETFs that launched into high attention diluted the exposure further, while the factor decomposition would have shown that much of the basket return was duration disguised as theme. Any one of the guardrails enumerated above would have produced a signal; the combination would have been hard to miss. For pure narrative excess, cleaner examples are Long Blockchain, metaverse repositioning, and concept-stock SPACs, where the gap between the label and the cash-flow exposure was much wider.

The discipline that separates discovery from story is not a single check. It is the operating habit of running every theme through the same eight or nine filters before any capital is committed, and of running them again on a schedule afterward. The output is not only a yes or no; it is an evidence log showing which guardrails passed, which failed, which were overridden, who approved the override, and what would force the decision to be revisited.

With the guardrails in place, the upstream apparatus the book has built — text into topics, topics into themes, themes into companies, companies into value chains — becomes a basket the institution can hold. The portfolio-construction and risk-control questions that follow take that basket as their input.

Chapter 24

The Minimum Viable Thematic Process

24.1 The Smallest Pipeline That Can Be Defended

A new analyst joins a research team in early 2024 with a specific assignment: build a thematic basket for AI infrastructure. The temptation is to reach for the most sophisticated tool available — the newest neural topic model, an LLM at every stage, a multi-output Gaussian-process dynamic tracker, a hybrid BERTopic–CTM ensemble. None of those tools is wrong. None of them is the right place to start.

The minimum viable thematic process is a working version of the whole pipeline, built with the simplest defensible choice at each step, and run end-to-end before any single step is improved. The MVP is not a research demo. It is the smallest object the team can defend in front of an investment committee, the smallest object the system can run on a schedule without supervision, and the smallest object the rest of the book can sharpen, one piece at a time.

The discipline matters because thematic pipelines fail in characteristic ways when they are built in pieces without ever assembling. A topic discovery that has never been mapped to companies optimizes for coherence the basket does not care about. An exposure score that has never been put through factor regression rewards firms whose loadings are doing the work the team thinks the score is doing. A backtest that has never been run point-in-time looks like a strategy and is a curve fit. The MVP catches these gaps because it forces every layer to communicate with the next, on real data, before anything is polished.

The eight stages that follow are not a research paper. They are a checklist that a small team can execute in a few weeks, using the Part V machinery just developed and handing the resulting basket to the Part VI portfolio chapters.

24.2 Eight Stages, One Sentence

The simplest end-to-end version of the pipeline runs through eight stages, each connected to chapters that develop the stage in depth.

1. **Assemble the corpus.** A point-in-time collection of filings, transcripts, and headlines (Chapter 9).

2. **Preprocess and reconstruct entities.** Tokenize, normalize, link company mentions to identifiers (Chapter 10).
3. **Discover candidate topics.** A single BERTopic-style pass on the cleaned corpus, with manual review of the AI-infrastructure-adjacent clusters (Chapter 12, Chapter 14).
4. **Write the theme specification.** Apply the Chapter 19 template to one chosen topic.
5. **Score the universe.** A minimum multi-signal exposure score using keyword evidence, segment revenue, and embedding similarity (Chapter 20).
6. **Build the basket.** Apply eligibility filters, exposure caps, and a sector neutralization to the top-scoring firms.
7. **Run the factor regression.** Regress basket returns on FF5 + momentum and inspect residual alpha, using the baseline introduced in Chapter 4.
8. **Evaluate walk-forward.** Roll the entire pipeline forward through time with strict as-of discipline before drawing any conclusion about edge.

Every stage uses the simplest tool that works. The point is not to do any one stage well. The point is to do all eight well enough to have a system worth improving.

24.3 Building the Corpus

The MVP corpus for AI infrastructure has three components and one rule.

The three components are SEC EDGAR filings (10-K, 10-Q, and 8-K for a candidate universe of roughly three thousand US-listed firms), a transcript feed covering the same firms, and a headline feed from a major news vendor or GDELT. Five years of history is enough to cover a thematic cycle without becoming unwieldy — for AI infrastructure, a corpus that starts in 2019 captures the pre-ChatGPT baseline, the late-2022 inflection, the 2023 enthusiasm, and the 2024 broadening into power and cooling.

The rule is point-in-time. Every document is stored with its event time, its publication time, and the time at which the vendor delivered it. Backtest queries return only documents available as of the simulated decision date in the form they had then. The discipline is unglamorous and worth the cost: a system that does not enforce it cannot tell whether its results are a signal or a memoir. The full apparatus for handling timestamps, restatements, and vendor lag was developed earlier in the book; the MVP version is the smallest implementation that gets the rule right rather than the most thorough one.

Two early-stage failures recur and both can be caught cheaply. Duplicate syndication across vendors inflates document counts and biases topic discovery toward popular stories; a deduplication pass on near-identical text catches most of it. Stale ticker mappings link old document IDs to current company IDs even when the underlying firm reorganized, was acquired, or relisted; a single check that compares as-of CUSIPs with current CUSIPs catches the worst cases.

24.4 Preprocessing, Entities, and the First Topic Pass

The preprocessing layer for the MVP is deliberately minimal. Lowercase, strip boilerplate (legal disclaimers, outlet boilerplate, EDGAR header text), tokenize, remove standard stopwords plus

a small finance-specific stopword list, and lemmatize. The first time the topic model is run, the analyst inspects the top words in each topic and adds three or four obvious vocabulary-pollution items to the stopword list. Preprocessing decisions made before any topics are visible are usually too cautious or too aggressive; preprocessing decisions made with one round of inspection feedback are usually fine.

Entity reconstruction is where the MVP earns its first lesson. Without it, the top topics in any financial corpus turn into company clusters: a “topic” that is actually thirty Nvidia documents, another that is twenty Microsoft documents. The minimum version masks company mentions with stable placeholders during the topic pass (`<ORG_NVDA>`), then unmask for downstream display. The result is topics about *things companies do* (accelerators, memory, power) rather than topics that are companies.

Topic discovery in the MVP is one pass of BERTopic on the masked, cleaned corpus, using a finance-tuned sentence-transformer encoder for embeddings, UMAP for dimensionality reduction, and HDBSCAN for clustering. Set the minimum cluster size large enough that the result is a few dozen interpretable topics rather than several hundred fragments. Review every topic by hand against its top words, its representative documents, and its top entities. Most will be uninteresting — guidance language, capital-structure topics, generic risk-factor boilerplate. A few will look like investable themes. For the AI infrastructure assignment, expect at least one topic that bundles GPUs, accelerators, training clusters, and hyperscaler capex, and possibly a second that bundles data-center power, grid interconnection, and cooling. These are the candidates the rest of the pipeline operates on.

The labeling of each candidate topic at this stage is also minimal: c-TF-IDF top phrases, a manually written one-line description, and one to three representative documents pinned for downstream display. LLM-assisted label generation, which a later stage of the system will use, is not part of the MVP.

24.5 From Topic to Specification

A topic that has survived inspection is still not a theme. The MVP turns it into one by applying the specification template from Chapter 19 to the candidate topic before any company is scored.

The discipline that the template enforces is partly forensic: writing down the inclusion criteria, the falsification conditions, and the expected horizon *before* the universe is scored prevents the team from quietly tuning the theme definition to the names that come out on top. For the AI infrastructure example, the specification at this stage might read:

- **Theme:** AI data-center buildout.
- **Mechanism:** model demand drives accelerator and memory demand; hardware demand drives data-center construction; data-center capacity is bottlenecked by power, cooling, and grid interconnection.
- **Inclusion:** disclosed revenue, capex, or backlog tied to accelerator-grade hardware, advanced packaging, HBM, data-center construction, electrical equipment for data centers, cooling, or grid interconnection.
- **Exclusion:** generic AI mentions without measurable business exposure; diversified megacaps where the theme is immaterial; firms exposed only through speculative attention.
- **Horizon:** months to quarters.

- **Lifecycle stage at approval:** validated, possibly entering crowded.
- **Falsification:** top-scoring firms fail to show segment-revenue or capex evidence within two reporting cycles; or the basket's residual return after FF5 + momentum hedging collapses for two consecutive rebalances.

The spec is short by design. A long spec drafted before any of it has been tested is mostly aspirational. The MVP spec is the seed of the version that the team will revise after the first walk-forward.

24.6 Scoring the Universe

The MVP exposure score for each firm is the simplest defensible version of the multi-signal score from Chapter 20:

$$S_{i,\theta,t} = w_K K_{i,\theta,t} + w_R R_{i,\theta,t} + w_E E_{i,\theta,t}, \quad (24.1)$$

with K a normalized keyword-evidence score, R a segment-revenue exposure derived from disclosed segment lines (or RBICS Revenue where available), and E a cosine similarity between the firm's 10-K business description and the theme description. The LLM component L and graph component G from the full multi-signal formulation are deliberately switched off in the first run. Chapters 21 and 22 explain how to add them; the MVP can be defended without them.

The weights are set deliberately rather than optimized. A reasonable starting point assigns more weight to revenue evidence than to embedding or keyword evidence, because revenue exposure has the strongest economic anchor and the weakest temptation to game. A typical first pass uses something like $w_R = 0.5, w_E = 0.3, w_K = 0.2$. The weights themselves are part of the spec; the team commits to them before looking at the resulting ranking, and any later change has to be justified in writing.

Two filters live alongside the score. A confidence filter discards firms whose evidence base is thin (fewer than two distinct evidence types contributing nonzero signal); these are the firms most likely to be keyword-stuffed. A materiality filter discards firms whose strongest evidence is real but small (for instance, revenue exposure below a stated threshold of three to five percent). Both filters are blunt; both prevent the MVP basket from inheriting the most common failure modes of thematic scoring.

For AI infrastructure, the MVP score applied to a US-listed candidate universe should put the recognizable names near the top: Nvidia, SK Hynix's US-listed ADR if included, Vertiv, Eaton, Vistra, Constellation. It should also surface a handful of less-obvious names whose disclosure evidence is strong even though their market reputation is not AI-first: specialty thermal-management firms, electrical-equipment vendors with hyperscaler customer concentration, fabless designers focused on networking silicon. If the top twenty by score is dominated by mega-cap technology names with no segment-revenue evidence, the score is wrong and the team rebuilds it before going to a basket.

24.7 Building the Basket

The basket is the simplest construction object that can be evaluated. Take the top thirty firms by score from the eligible universe. Apply an eight-percent single-name cap and a twenty-five-percent

single-sector cap. Weight the rest in proportion to score, then renormalize. Rebalance monthly using only the score and evidence available as of the rebalance date.

Two construction choices that the full value-chain and Part VI apparatus will refine are made simply in the MVP. The first is the long-only assumption: the MVP basket holds positive exposures, not negative. The second is no factor pre-orthogonalization: the score is used as-is, with factor exposure handled at evaluation rather than at construction. Both are defensible for an MVP. Both will be replaced.

Eligibility is also deliberately narrow. A firm enters the universe only if it has an active US listing, average daily traded volume above a stated threshold (typically a few tens of millions of dollars), at least four quarters of as-of segment disclosure, and no pending corporate action that would compromise as-of pricing. The eligibility rule is part of the spec and is frozen at construction time; new firms become eligible only when the rule is reapplied at the next rebalance.

A representative MVP basket constructed under these rules and these weights, run as of mid-2024 with point-in-time data, would hold roughly thirty names spanning accelerators, memory, data-center REITs and operators, electrical equipment, cooling, and competitive power generation. The exact composition is less important than the fact that the basket is reproducible: another analyst, given the same corpus, the same score, and the same eligibility rule, produces a basket that looks the same.

24.8 Factor Regression and Walk-Forward Evaluation

A basket is not a result. The MVP turns the basket into evidence by running it through two evaluation steps.

The first is the factor regression. Regress the basket's daily or monthly excess returns on the Fama-French five factors plus momentum, in the form

$$\begin{aligned} r_{p,t} - r_{f,t} = & \alpha_p + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t \\ & + \beta_{\text{RMW}} \text{RMW}_t + \beta_{\text{CMA}} \text{CMA}_t + \beta_{\text{MOM}} \text{MOM}_t + \epsilon_t. \end{aligned} \quad (24.2)$$

The team reads three numbers off the result. The alpha point estimate and its t -statistic say how much of the basket's return is left after factor exposure is accounted for. The set of factor betas says what the basket was implicitly buying — a large positive market beta and momentum beta with a small alpha means the basket was a leveraged momentum trade with a thematic label. The residual return series itself is then examined for stability: did the residual produce its alpha smoothly, or in one narrow window?

The interpretation is more careful than the regression. A small but positive alpha with reasonable factor exposures is the MVP succeeding. A large alpha driven by one quarter is the MVP behaving like a luck draw. A negative residual after stripping out momentum is the MVP confessing that the theme exposure was a factor exposure in disguise.

The second step is walk-forward evaluation. Rerun the entire pipeline at each historical rebalance date using only data available then. The corpus index is rebuilt as of that date. The score is recomputed using the as-of segment disclosures and the as-of embedding model. The basket is constructed using the as-of eligibility rule. The factor regression covers the realized period only. Walk-forward turns the result from “a basket that worked” into “a process that would have worked under the same discipline at each date.”

Walk-forward is also where the leakage failures from the topic-modeling and LLM chapters reappear in the most expensive form. A topic model fit on the full corpus and then projected

backwards through time imports 2024 vocabulary into 2021 decisions. An LLM asked to classify a 2020 filing imports its memory of what happened next. A factor model used at all dates with its current revision smuggles in coefficients estimated after the simulated decisions were made. The MVP closes all three doors at this stage by reaffirming the point-in-time discipline at every layer.

24.9 What the MVP Defers and Where It Goes Next

A working MVP is a baseline, not a finish line. Each major layer of the book sharpens a specific stage.

The LLM layer the MVP refused to use was developed in Chapter 21: structured extraction from filings and transcripts, retrieval-augmented evidence packages, classification with cited passages, and the cutoff-respecting workflow that makes the layer auditable. Adding LLM evidence to the score replaces some manual review of borderline names and improves the discovery of firms whose exposure is real but unnamed.

The value-chain layer that the MVP also deferred was developed in Chapter 22: relation extraction, supply-chain graphs, centrality measures, and the propagation of exposure along edges. Adding graph evidence allows the score to distinguish a critical bottleneck supplier from an interchangeable peer, and to find second-order beneficiaries the disclosure layer alone misses.

The guardrails chapter, Chapter 23, formalizes the checks the MVP applied informally: mention-is-not-exposure, name capture, ETF dilution, late-cycle launches, plausibility, overfitting, falsification, and kill rules. Each guardrail becomes a programmatic check that runs alongside the score and the basket.

Portfolio construction and factor neutralization in Part VI replace the MVP's at-evaluation factor handling with at-construction pre-orthogonalization and explicit factor hedging. Rebalancing, theme decay, capacity, and transaction-cost modeling sharpen the basket from monthly score-weighted to a process that respects turnover, slippage, and the dual-gate activation logic. The crowding and regime chapter introduces stress tests for the late-cycle and rate-shock conditions the MVP cannot see in a five-year backtest.

The backtesting and governance chapters of Part VIII turn the walk-forward step into a full evaluation framework, including the LLM-leakage formalism that the MVP can only approximate.

The MVP, finally, is the object the rest of the book is allowed to talk about. Without it, every refinement is theoretical. With it, every refinement has a target: the basket gets a better score, the score gets a better evidence base, the evidence base gets a tighter audit trail, the audit trail makes the strategy defensible to a committee that did not write it. A theme that has survived the MVP is ready for the construction and risk-control discipline that the portfolio chapters apply next.

Part VI

Portfolio Construction and Risk Control

Chapter 25

Building and Neutralizing the Thematic Basket

25.1 The Portfolio That Was Not What It Claimed to Be

In 2022, Acadian Asset Management revisited one of the oldest trades in quantitative equity investing: buy cheap stocks, short expensive stocks, and call the spread value. The headline story sounded familiar. After years of lagging growth stocks, value was back. The more uncomfortable finding sat underneath the headline. For a simple book-to-price value-minus-growth portfolio from 2010 to 2021, Acadian attributed 92 percent of returns to industry allocation effects and only 8 percent to stock selection [1].

That result should make every thematic investor uneasy. The value label is older, better studied, and more disciplined than most thematic labels, yet a crude implementation still became mostly a sector-positioning machine. If that can happen to value, it can happen more easily to AI infrastructure, clean energy, cybersecurity, reshoring, obesity drugs, or any other theme whose company list arrives through language, index-provider rules, or analyst enthusiasm.

The failure is not that the portfolio owned the wrong stocks. The failure is that the portfolio did not know what it owned. A basket can say “AI infrastructure” on the cover and still be a levered bet on mega-cap growth, momentum, semiconductors, and market beta. A clean-energy basket can contain real energy-transition companies and still behave like a long-duration small-cap growth trade when rates rise. The score that enters the portfolio may be thematic; the return that leaves the portfolio may be something else.

Construction and neutralization are where the pipeline becomes capital. Naming themes, scoring companies, mapping value chains, and guarding against illusion does not yet say how much to buy, what to short, what to cap, what benchmark to use, or how to decide whether the residual return is still thematic after known factors have been removed. The two tasks are not separate chores. They are one decision: how to convert exposure evidence into a tradable portfolio whose risks are legible before the market teaches the lesson more expensively.

25.2 Scores Are Inputs, Not Weights

The exposure score from Chapter 20 answers a narrow question: how strongly does company i appear exposed to theme θ at time t , given keyword evidence, embeddings, revenue information, LLM classification, and graph position? A portfolio weight answers a different question: how

much capital should the strategy allocate to that company after considering expected return, risk, liquidity, valuation, concentration, factor exposure, confidence, and mandate constraints?

Confusing the two is one of the fastest ways to turn a good research signal into a poor portfolio. Nvidia may have high AI-infrastructure exposure, but a portfolio that gives it a weight equal to its exposure score may become a one-stock momentum trade. A small cooling-equipment supplier may be more thematically pure than a diversified mega-cap, but the position may be impossible to size without moving the price. A utility with a data-center power contract may add the right value-chain edge and the wrong interest-rate exposure. The score tells the portfolio where to look. It does not tell it how hard to press.

A complete construction rule has seven objects:

- an eligible universe defined before scoring;
- materiality and evidence thresholds for inclusion;
- liquidity and capacity filters;
- a weighting rule;
- concentration, turnover, sector, country, and factor constraints;
- a benchmark or comparison stack;
- a neutralization and attribution method.

These objects should be written before the backtest. If the universe, filters, and caps change after the winners are visible, the portfolio is no longer testing the theme. It is testing the researcher's ability to recognize the past.

25.3 The Universe Is the First Risk Model

The universe should be defined before the exposure model starts ranking companies. That sounds administrative, but it is the first portfolio decision. A universe of the 500 largest U.S. equities will produce a liquid, scalable basket and miss many foreign suppliers, small industrial bottlenecks, and early-stage pure plays. A global developed-market universe can include SK Hynix, ASML, TSMC, Schneider Electric, and other non-U.S. nodes, but it introduces currency, market-hour, withholding-tax, accounting, and data-coverage complications. A small-cap universe may contain stronger inefficiencies and weaker capacity. A benchmark-constituent universe may be implementable and institutionally familiar, while also being late to themes that form outside benchmark boundaries.

The same theme therefore has different investable shapes under different universes. AI infrastructure inside the S&P 500 will lean toward Nvidia, Microsoft, Broadcom, data-center REITs, electrical-equipment companies, and power producers. A global universe brings in memory, lithography, foundry, and advanced-packaging nodes that may be closer to the bottleneck. A liquidity-constrained long-short universe may exclude exactly the small firms whose language moves earliest.

The rule is simple: define the universe as part of the thesis. If the strategy claims to capture cross-sector AI infrastructure but excludes non-U.S. semiconductors and all stocks below a large-cap threshold, the thesis has already accepted a particular kind of dilution. That may be

the right choice for a large allocator. It should not be an accidental choice disguised as data hygiene.

25.4 Eligibility, Liquidity, and Materiality

Eligibility filters translate implementation reality into research. They usually include minimum market capitalization, minimum average daily value traded, minimum price, listing venue, currency, borrow availability for shorts, minimum text coverage, maximum missing data, and mandate exclusions for sanctions, compliance, or client restrictions. Each filter should be point-in-time. Today's liquidity, borrow availability, or index membership cannot be imported into a historical test without creating the kind of look-ahead bias Part VIII will formalize.

Materiality filters do a different job. They prevent the basket from owning firms whose language is thematically rich and whose economics are irrelevant. A minimum exposure score may be one filter, but it should rarely be the only one. For an AI-infrastructure basket, the rule might require at least one of the following: disclosed revenue or segment exposure, named product linkage, customer or supplier evidence, capex sensitivity, value-chain bottleneck status, or repeated analyst-estimate sensitivity to theme-specific news. A firm that clears the semantic threshold but fails every materiality test belongs in the monitoring file, not in the portfolio.

Liquidity filters should also shape position size, not merely admission. A common practitioner rule caps any single-name weight at 8 percent, a constraint widely used in narrative-portfolio implementations and simple enough to audit. A better implementation also ties the maximum dollar position to average daily value traded. A 5 percent weight in a mega-cap and a 5 percent weight in a thinly traded supplier are not the same portfolio act.

25.5 A First Weighting Rule

The cleanest starting point is a score-weighted long-only basket. Let $S_{i,\theta,t}$ be the exposure score for firm i to theme θ at time t , and let τ be the minimum inclusion threshold. Raw weights can be written as

$$\tilde{w}_{i,t} = \max(S_{i,\theta,t} - \tau, 0). \quad (25.1)$$

The portfolio then normalizes those raw weights:

$$w_{i,t} = \frac{\tilde{w}_{i,t}}{\sum_j \tilde{w}_{j,t}}. \quad (25.2)$$

The intuition is straightforward. A company just above the threshold enters with a small weight; a company with much stronger evidence receives more capital; firms below the threshold receive none. This rule is transparent, easy to reproduce, and useful as a baseline.

It is also incomplete. If the highest scores all belong to semiconductor firms, the basket becomes a semiconductor basket. If the highest scores belong to illiquid pure plays, the basket becomes a capacity problem. If the highest scores belong to the firms every thematic ETF already owns, the basket may be late and crowded. The raw weighting rule therefore needs caps and overlays before it becomes investable.

Common overlays include a maximum single-name weight, maximum sub-theme weight, maximum sector or industry exposure, maximum country or currency exposure, minimum and

maximum number of holdings, liquidity-adjusted position limits, turnover limits, and factor exposure bounds. For AI infrastructure, a useful set of caps might limit single names to 5–8 percent, prevent the semiconductor sleeve from swallowing the entire basket, and reserve room for power, cooling, electrical-equipment, data-center, and networking beneficiaries when their evidence is strong enough.

25.6 Long-Only Expression and Long-Short Testing

Long-only portfolios are the natural format for many allocators. They express positive exposure to a theme, avoid the operational complexity of shorting, and can be compared with broad-market or thematic ETF benchmarks. Their weakness is that they inherit market beta and usually inherit sector and style tilts. A long-only AI-infrastructure basket will probably be long equities, long technology, long momentum, and long growth unless the construction rule actively prevents it.

Long-short portfolios ask a sharper research question. If the strategy goes long high-exposure firms and short low-exposure firms within the same sector, industry, region, or risk bucket, it tests whether thematic exposure matters beyond the ordinary characteristics of the companies being compared. The long-short version is often the better scientific instrument even when the eventual product is long-only.

The costs are real. Short books require borrow, financing, locate management, short-sale compliance, and tighter risk controls. They can also create false comfort. A badly designed short leg can introduce its own factor bet: short low-exposure firms may also mean short value stocks, short defensives, or short low-volatility companies. The long-short design should therefore be paired with the neutralization tools below, not treated as a substitute for them.

25.7 Neutralize Before Construction

The most direct way to control contamination is to neutralize the exposure score before it becomes a weight. Let $Y_{\text{raw},i,t}$ be the raw thematic exposure score for company i . At each rebalance date, run a cross-sectional regression of that score on style characteristics and sector indicators:

$$Y_{\text{raw},i,t} = \gamma_t + F_{i,t}^\top \theta_t + S_{i,t}^\top \delta_t + \eta_{i,t}. \quad (25.3)$$

Here $F_{i,t}$ contains style exposures such as beta, size, value, momentum, profitability, investment, leverage, quality, and liquidity. $S_{i,t}$ contains sector or industry dummies. The residual $\eta_{i,t}$ is the portion of the thematic score not explained by those known characteristics.

This is the construction analogue of asking a more disciplined question. A raw AI score asks, “which firms look most exposed to AI infrastructure?” A residualized AI score asks, “which firms look unusually exposed to AI infrastructure after accounting for the fact that some sectors and styles are naturally closer to the theme?” The second question is better for alpha testing because it reduces the chance that the basket merely rediscovered mega-cap growth or semiconductors.

Residualization can also go too far. If the theme is genuinely concentrated in semiconductors, full industry neutrality may remove a real part of the mechanism. If data-center power is naturally concentrated in competitive generators and electrical equipment, forcing perfect sector neutrality may turn the theme into a statistical artifact. The right choice depends on the mandate. A pure alpha test should residualize aggressively. A strategic thematic allocation should report the residualized version but may not want to hold it as the primary portfolio.

25.8 Neutralize After Construction

The second defense happens after the portfolio is built. Estimate the portfolio's factor loadings through a time-series regression. The book's working baseline, established in Chapter 4, is the Fama–French five-factor model plus momentum [42, 24]:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{\text{MKT}}\text{MKT}_t + \beta_{\text{SMB}}\text{SMB}_t + \beta_{\text{HML}}\text{HML}_t + \beta_{\text{RMW}}\text{RMW}_t + \beta_{\text{CMA}}\text{CMA}_t + \beta_{\text{MOM}}\text{MOM}_t + \epsilon_t. \quad (25.4)$$

$R_{p,t}$ is the portfolio return, $R_{f,t}$ is the risk-free rate, and the factor returns represent market, size, value, profitability, investment, and momentum. The intercept α_p is the part of the portfolio's average excess return not explained by those factors, subject to all the usual caveats about estimation error, changing betas, omitted variables, and implementation costs.

If the factor portfolios are tradable, the hedged return can be written as

$$R_{p,t}^{\text{neutral}} = R_{p,t} - R_{f,t} - \sum_k \hat{\beta}_{k,t} F_{k,t}. \quad (25.5)$$

This expression is not magic. The betas are estimated with noise, the hedge may be costly, and real portfolios may not have clean access to every factor portfolio. But the decomposition is still essential. If the AI basket's attractive return disappears after market, growth, and momentum exposure are removed, the strategy has not found thematic alpha. It has found a familiar factor package with a better story.

25.9 Optimization Is a Tool, Not a Verdict

Portfolio optimization can combine expected alpha, risk, liquidity, and constraints in one problem. A standard form is

$$\max_w \quad w^\top \hat{\alpha} - \lambda w^\top \Sigma w \quad (25.6)$$

$$\text{subject to} \quad \mathbf{1}^\top w = 1, \quad (25.7)$$

$$B^\top w = b^*, \quad (25.8)$$

$$l_i \leq w_i \leq u_i. \quad (25.9)$$

The vector $\hat{\alpha}$ contains expected alpha estimates, Σ is the covariance matrix, B is a matrix of factor exposures, b^* is the target factor exposure, and l_i and u_i are lower and upper bounds for each position. In a factor-neutral long-short portfolio, b^* may be close to zero. In a long-only strategic basket, b^* may allow some market and sector exposure while constraining the worst contaminants.

Optimization is powerful precisely because it is dangerous. Small changes in expected alpha or covariance estimates can produce large changes in weights. A noisy thematic score can become an apparently precise allocation because the optimizer is forced to choose. The remedy is not to avoid optimization; it is to keep the objective simple, constrain the output heavily, and compare it with transparent baselines such as equal weight, score weight, market-cap weight, and score-times-liquidity weight [50].

Every optimized portfolio should answer a practical question after the math is done: which constraints are binding? If the optimizer wants to put 35 percent of the basket into one chip designer and the cap stops it, the cap is not a technical nuisance. It is the risk control doing its job. If the optimizer can produce alpha only by loading on a sector or factor the mandate was supposed to avoid, the theme is not yet carrying the portfolio.

25.10 Residual Coherence and the Benchmark Stack

Neutralization is not only about reported alpha. It also tests whether the theme is economically coherent after ordinary risks are removed. Let $u_{i,t}$ be each firm's return residual after factor and sector controls. For a candidate basket B , compute the average pairwise residual correlation:

$$\bar{\rho}_B = \frac{2}{|B|(|B| - 1)} \sum_{i < j, i, j \in B} \text{Corr}(u_i, u_j). \quad (25.10)$$

If the basket's residual correlation is no higher than that of comparable random baskets, the companies may share a label without sharing a return driver. If residual correlation rises during theme-specific events and fades afterward, the theme may be real but episodic. If residual correlation remains high after controls, the portfolio has evidence that the theme is doing something sectors and factors did not already explain.

Performance should be reported as a stack rather than a single number:

- raw thematic return;
- benchmark-relative return against the eligible universe;
- sector-adjusted return;
- style-factor-adjusted return;
- FF5 + momentum residual return;
- implementation-adjusted return after turnover, slippage, borrow, and financing costs.

The pattern across the stack tells the truth. A theme that looks brilliant raw and ordinary after factor adjustment is probably a factor trade. A theme that survives sector and factor controls but disappears after costs may be real and untradeable. A theme that survives all layers deserves the next question: whether it can be held at scale through time.

25.11 The AI Infrastructure Basket in Practice

An AI-infrastructure basket built under this discipline would not begin with the question, "which AI stocks should we buy?" It would begin with an eligible universe and an exposure definition. The universe might include global developed-market equities above a liquidity threshold. The exposure model would score accelerators, HBM, networking, advanced packaging, data centers, power, cooling, switchgear, and cloud capex. The value-chain layer would distinguish core pure plays from suppliers, platform enablers, adopters, second-order beneficiaries, and disrupted firms.

The first construction pass might include companies above a materiality threshold, cap single names at 5–8 percent, cap sub-themes so semiconductors cannot consume the entire basket, and require that each position has at least one non-language evidence source. A long-only version

would express the investable theme for allocators who want exposure. A long-short research version would rank companies by residualized exposure within sectors or industries and test whether unusual exposure predicts residual returns.

The neutralization pass would then ask what the basket really is. How much of its return comes from market beta? How much from momentum? How much from growth? How much from semiconductors? How much from rates through power and utility-like names? How much remains after FF5 + momentum hedging? The answer may be uncomfortable, but discomfort is the point. A thematic strategy should find out in research what a careless strategy finds out in a drawdown.

The clean-energy 2020–2022 case shows the same machinery from the other side. A raw clean-energy basket in late 2020 looked like exposure to policy support, decarbonization, solar adoption, EVs, and hydrogen. A risk-aware construction process would also have shown small-cap growth exposure, valuation expansion, duration sensitivity, ETF-flow pressure, and weak residual separation for many constituents. The transition was real. The portfolio was overloaded with risks the theme label did not name.

The output is therefore not a single preferred portfolio. It is a construction discipline: define the universe before scoring, require material exposure before inclusion, translate scores into weights with caps and liquidity constraints, test long-only expression and long-short alpha separately, neutralize before and after construction, and report the whole benchmark stack. Only then does the thematic basket become something that can be rebalanced, sized, stress-tested, and governed without surprises.

Chapter 26

Rebalancing, Theme Decay, and Capacity

26.1 The Trade That Was Right Until It Was Too Large

Clean energy did not stop being important in February 2021. Solar panels did not become irrelevant, EV adoption did not reverse, and the policy story did not disappear. Yet many clean-energy portfolios that looked brilliant in 2020 became painful shortly afterward because the portfolio problem had changed faster than the theme narrative. The story was no longer early. The obvious names were crowded. Valuations had expanded. The rate regime was turning hostile. The basket still had the same label, but it no longer had the same expected return.

That is the uncomfortable truth about thematic portfolios: the theme can remain valid while the trade decays. A portfolio is not a monument to an idea; it is a dated expression of that idea under current attention, valuation, liquidity, factor, and capacity conditions. The same AI-infrastructure basket that worked when the market was learning about HBM scarcity may need different weights when power, cooling, switchgear, and data-center constraints become the marginal bottlenecks. The same clean-energy basket that captured early policy enthusiasm may become a duration trade when rates rise and ETF flows have already done the easy buying.

Rebalancing is the mechanism that lets the portfolio admit this. Timing rules decide when a theme is active, decay models decide how long old evidence remains useful, turnover controls decide how quickly weights can change, and capacity rules decide how much capital the strategy can absorb before its own trading consumes the edge. A factor-aware basket has been built. How that basket survives time is what the discipline below answers.

26.2 Lifecycle Becomes Position Size

Chapter 6 described a seven-stage lifecycle: weak signal, emerging theme, validated theme, crowded theme, institutionalized theme, decaying theme, and recycled or re-emerging theme. In portfolio construction, that lifecycle should not sit in a memo. It should change position size.

A weak signal belongs in a watchlist or research sleeve. An emerging theme may deserve a small allocation, especially when attention is rising from a low base and evidence is spreading across independent sources. A validated theme can carry larger weight when exposure scores, value-chain evidence, residual returns, and analyst review point in the same direction. A crowded or institutionalized theme may still be economically real, but it should face tighter caps, stronger

valuation requirements, and more frequent review. A decaying theme should lose capital even if the vocabulary remains coherent.

The lifecycle rule prevents a common thematic error: treating a theme as binary. The portfolio does not need to choose between “AI matters” and “AI is over.” It can say that GPU designers are mature and crowded, HBM supply is still constrained, power is becoming the binding variable, and some software-adopter claims remain unproven. Each sub-theme can sit at a different lifecycle stage, and the rebalance should reflect that granularity.

26.3 The Rebalancing Clock

The right rebalancing frequency depends on the half-life of the signal. Intraday sentiment belongs to a trading desk. Annual 10-K segment changes belong to a slower research process. Most thematic equity baskets sit between those extremes because narratives diffuse through earnings calls, news, analyst revisions, ETF flows, and price action over weeks and months.

The research notes point toward monthly rebalancing as the practical default for narrative and exposure signals. Quarterly schedules can miss momentum decay; weekly schedules often create turnover that looks responsive in a backtest and expensive in production. Monthly rebalancing is not sacred. It is a useful starting point because it gives the system enough time to distinguish a story from a headline without letting a stale theme sit untouched for a quarter.

A good implementation separates the observation clock from the trading clock. Attention, source breadth, price reaction, and evidence updates can be monitored daily. The portfolio may still trade monthly unless a hard risk rule fires. This distinction matters because it lets the system notice change without paying the spread every time the language shifts.

26.4 Decay-Weighted Evidence

A document from yesterday usually tells the portfolio more than a document from last year. That does not mean old evidence is worthless. A named customer relationship, a long-term power contract, or a revenue segment can remain relevant for years, while a news headline may decay in days. The portfolio needs a memory that fades at the right speed.

A simple decay-weighted signal can be written as

$$A_{i,t} = \sum_{\tau \leq t} \exp[-\lambda(t - \tau)] x_{i,\tau}, \quad (26.1)$$

where $x_{i,\tau}$ is the theme-related evidence for company i observed at time τ , and λ controls how quickly the signal forgets. A larger λ makes the signal more responsive and less stable. A smaller λ makes it smoother and more vulnerable to stale evidence.

The intuition is easier than the formula. If a hyperscaler mentions power constraints on three consecutive earnings calls, the signal should build persistence. If one executive drops a fashionable phrase once and never returns to it, the signal should fade. The empirical case for the decay-weighted form over a binary shock indicator is concrete. In a firm-level narrative-forecasting study of Microsoft monthly returns from February 2021 through March 2025, the decay-weighted Media and Public Perception narrative carried a coefficient of +0.0128 ($p = 0.002$), while the corresponding binary MPP shock came in at $\beta = -0.0758$ ($p = 0.043$) at $t - 1$ and $\beta = -0.1048$ ($p = 0.007$) at $t - 2$. The decay-weighted form captured persistent positive attention; the binary

form caught short-horizon overreaction. Combining the two added information that neither carried alone, with a likelihood-ratio test versus the baseline ARIMA at $p < 0.01$.

Decay rates should vary by source. Filings, segment revenue, and signed contracts decay slowly. News and social attention decay quickly. Analyst revisions and management commentary sit in the middle. A single global half-life is convenient and usually wrong.

26.5 Activation Gates

Many bad thematic trades begin with one impressive spike. A theme appears in headlines, the model detects acceleration, and the portfolio buys a burst that never becomes a sustained narrative. Activation gates exist to prevent that failure.

A two-gate activation test from practitioner narrative-portfolio implementations addresses this directly. A narrative becomes active only when both medium-term persistence and short-term elevation are present:

$$A_{\theta,t}^{90} \geq 0.80 \cdot \max_{s \in [t-126,t]} A_{\theta,s}^{90}, \quad (26.2)$$

$$A_{\theta,t}^{30} \geq 0.50 \cdot \max_{s \in [t-126,t]} A_{\theta,s}^{30}. \quad (26.3)$$

Here $A_{\theta,t}^{90}$ is a rolling 90-day attention or narrative level for theme θ , $A_{\theta,t}^{30}$ is the corresponding 30-day level, and the trailing window approximates six months of trading days. The first gate asks whether the theme remains near its medium-term high. The second asks whether recent attention is still alive. The AND structure matters: a one-day spike fails, and so does an old story whose recent attention has faded.

Other gates can sit alongside the attention gates: source breadth, residual return confirmation, material company-level evidence, analyst approval, valuation discipline, and liquidity feasibility. The right question is not whether the system has heard the story. It is whether the story has enough persistence, breadth, and economic support to justify trading.

26.6 Momentum First, Contrarian Later

Thematic timing has to allow two facts that sound contradictory. Emerging themes can trend because information diffuses slowly, and late-cycle themes can reverse because attention, valuation, and ownership overshoot. A timing system that is always momentum-driven buys peaks. A timing system that is always contrarian shorts every real structural change too early.

Momentum logic is strongest when the theme is emerging or validated, attention is rising from a modest base, source breadth is increasing, fundamentals or analyst estimates are improving, and residual returns are positive without extreme valuation expansion. That was the attractive phase for parts of AI infrastructure before every investor presentation had a slide on accelerators, power, and cooling.

Contrarian logic becomes stronger when attention reaches historical extremes, ETF launches and inflows accelerate, valuation expansion dominates earnings revisions, ownership concentrates in the obvious pure plays, and forward residual returns deteriorate. This does not automatically mean the theme is false. It means the portfolio should require more evidence for each additional dollar of exposure.

The clean-energy case is the transition-investing version. By late 2020 and early 2021, the narrative had broad policy support and real corporate activity, but the trade had also absorbed enormous attention, valuation expansion, and product flows. The right response was not to declare decarbonization dead. It was to reduce exposure to the crowded expression and demand stronger evidence from each constituent.

The case for this discipline at the strategy level is documented in the practitioner literature. A long top-quintile / short bottom-quintile thematic-ETF rotation, rebalanced monthly with a 12-month trailing-return lookback, has produced approximately a 25 percent CAGR over recent years, against roughly 11 percent for traditional sector rotation under the same construction rules. Monthly cadence, exposure capping at 8 percent per name, and decay-weighted activation are what kept the long-short separation alive between rebalances; quarterly or annual schedules let the momentum signal decay before the trade could capture it.

26.7 Theme Decay

Theme decay occurs when the narrative loses investment usefulness. Sometimes the underlying story becomes priced. Sometimes it gets absorbed into ordinary sector classifications. Sometimes broad adoption dilutes the advantage of early beneficiaries. Sometimes the fundamental mechanism disappoints. Sometimes the basket keeps the theme label while the residual return disappears.

Useful decay signals include falling topic volume, shrinking source breadth, weaker residual co-movement, weaker return prediction, declining estimate revisions, increasing ETF dilution, poor response to positive news, and rising dependence on factor returns. No single signal proves decay. The pattern matters.

Decay should reduce exposure even when the theme remains semantically coherent. That distinction is central. A theme can still appear in documents, still have a plausible long-term story, and still be a poor current trade. The portfolio should ask whether exposure still predicts residual returns over the intended horizon, not whether the phrase still appears in filings.

Recycling is the exception that keeps the process honest. The 2020 clean-energy public-equity expression weakened after the 2021 peak, but policy, energy security, and grid-investment catalysts later revived parts of the opportunity set. A mature or repriced theme should not be deleted from institutional memory. It should move back to monitoring with a clear record of what would justify re-entry.

26.8 Turnover Is a Cost, Not a Virtue

Dynamic themes invite dynamic portfolios, and dynamic portfolios can quietly donate their alpha to the market. A model that changes its mind at every rebalance may look intelligent in a research notebook and look expensive on an execution blotter.

Turnover controls include buffer zones around inclusion thresholds, gradual weight changes, maximum turnover per rebalance, minimum holding periods, trade prioritization by expected alpha net of cost, and liquidity-aware execution schedules. A company whose score slips from 0.701 to 0.699 should not force a full exit if the threshold is 0.700. A new entrant whose evidence is promising but not urgent can be phased in over several trading days or rebalances.

The turnover rule should distinguish information from noise. If a company loses its named customer, misses a theme-critical product milestone, or fails a kill rule from Chapter 23, the

portfolio may need to move quickly. If its embedding score drifts slightly because a batch of headlines changed the topic centroid, patience may be the better trade.

26.9 Transaction Costs and Slippage

The implementation problem becomes concrete when a target weight becomes an order. Let portfolio assets be A , stock price be $P_{i,t}$, and the desired weight change be $\Delta w_{i,t}$. The dollar trade is

$$V_{i,t} = A|\Delta w_{i,t}|, \quad (26.4)$$

and the share quantity is $q_{i,t} = V_{i,t}/P_{i,t}$. If $\text{ADV}_{i,t}^{\$}$ is average daily dollar volume, the participation rate is

$$\pi_{i,t} = \frac{V_{i,t}}{\text{ADV}_{i,t}^{\$}}. \quad (26.5)$$

A simple implementation-cost model can then write expected cost in basis points as

$$c_{i,t} = \frac{1}{2}\text{Spread}_{i,t} + \kappa_i \sigma_{i,t} \pi_{i,t}^{\phi} + c_{i,t}^{\text{borrow}} + c_{i,t}^{\text{finance}}, \quad (26.6)$$

where the first term is half the bid–ask spread, $\sigma_{i,t}$ is volatility, κ_i is an impact coefficient, ϕ is typically between zero and one in practitioner impact models, and the last two terms capture borrow and financing costs when relevant. The exact calibration belongs to transaction-cost analysis, but the structure reflects the central lesson of optimal-execution models: trading faster and larger relative to available liquidity increases expected cost and risk [3].

The portfolio-level cost for a rebalance is approximately

$$\text{TC}_t = \sum_i V_{i,t} \cdot c_{i,t}. \quad (26.7)$$

This formula is deliberately plain. It forces the research process to convert an elegant signal into a trade value, compare that trade with liquidity, and subtract realistic costs before reporting performance. It also makes clear why small-cap thematic purity can be expensive. The purest name in the value chain may be the first name whose alpha disappears after impact.

26.10 Capacity Is a Curve

Capacity is often discussed as a single number: this strategy can run \$500 million, or \$2 billion, or \$10 billion. That framing is too crude for thematic portfolios. Capacity is a curve because the marginal dollar usually goes into less liquid names, larger participation rates, wider impact, or more diluted substitutes.

Let gross expected alpha at asset level A be $\alpha^{\text{gross}}(A)$ and implementation cost be $\text{TC}(A)$. Net alpha is

$$\alpha^{\text{net}}(A) = \alpha^{\text{gross}}(A) - \text{TC}(A) - \text{Borrow}(A) - \text{Finance}(A). \quad (26.8)$$

As A rises, $\text{TC}(A)$ usually rises faster than assets because participation rates increase and the portfolio must either take larger positions in the same names or broaden into lower-conviction

names. The relevant capacity is not the largest amount the portfolio can physically trade. It is the amount at which net alpha, net information ratio, drawdown tolerance, or client liquidity terms stop meeting the mandate.

An AI-infrastructure strategy illustrates the curve. A small portfolio can own specialized cooling, power-equipment, and regional power beneficiaries without dominating their volume. A larger portfolio may need more Nvidia, Microsoft, Broadcom, or large industrials simply to absorb capital. Capacity rises, but purity falls. The manager should show the curve: expected alpha, tracking error, turnover, implementation cost, and theme purity at different asset levels.

26.11 AI Infrastructure Through Time

The AI-infrastructure portfolio should rebalance as the bottleneck moves. In 2023, accelerator supply and HBM scarcity may have carried the strongest signal. As the market priced those nodes, attention shifted toward advanced packaging, data-center construction, power availability, thermal management, and electrical equipment. A static basket would keep owning the first winners in the first weights. A dynamic basket asks which part of the chain remains underpriced now.

The rebalance should not chase every sub-theme rotation. It should require activation gates, materiality evidence, valuation support, and cost-aware implementation. If cooling mentions accelerate but the investable cooling names are illiquid and already expensive, the correct response may be a smaller position, a slower build, or no trade. If power constraints appear across hyperscaler transcripts, grid queues, PPA announcements, and estimate revisions, the theme has more than a headline.

Clean energy supplies the portfolio-expression lesson. A disciplined process in late 2020 would not have needed to know the exact path of rates in 2021. It would have seen a theme moving from validated toward crowded, obvious pure plays absorbing flows, valuations outrunning revisions, and implementation risk rising. The rebalance rule would have reduced exposure, rotated toward less crowded nodes, or demanded stronger valuation support before adding capital.

26.12 A Basket That Can Change Its Mind

The output of a rebalancing process is not activity. It is controlled adaptation. The portfolio should know when old evidence decays, when a theme is active, when momentum deserves capital, when contrarian caution should dominate, when turnover is too expensive, and when asset growth has pushed the strategy beyond its useful capacity.

That discipline leads directly into the next layer of risk. Rebalancing handles the portfolio's ordinary passage through time. Regime shifts and crowding handle what happens when the environment changes abruptly and many investors discover that they own the same trade. A basket that can change its mind is better prepared for that stress, but it still has to be tested against it.

Chapter 27

Regime Dependence, Crowding, and Scenario Risk

27.1 The Theme Was Right. The Regime Was Wrong.

The clean-energy trade of 2020 did not unwind because investors suddenly stopped believing in decarbonization. It unwound because the market changed the discount rate on the cash flows investors were willing to imagine. Many constituents were long-duration equities: expensive, growth-oriented, policy-sensitive, and dependent on future earnings arriving far enough ahead that a rise in rates mattered immediately. When the rate regime shifted in 2021 and 2022, the basket behaved less like a pure decarbonization thesis and more like a financing-sensitive growth portfolio.

That is the lesson regime analysis forces into the thematic process. A theme can be economically correct and still lose money in the wrong macro environment. A theme can have real company exposure and still be overowned. A theme can have high residual uniqueness and still be vulnerable to a crowded exit if many investors hold the same expression through ETFs, model portfolios, hedge-fund books, and retail accounts.

Rebalancing addressed how a thematic basket changes through ordinary time. Stress raises a different question: how the basket behaves when the environment changes abruptly, when many investors discover they own the same trade, or when the specific mechanism behind a thesis breaks. Regime dependence asks whether performance belongs to the theme or to the surrounding market environment. Crowding asks whether too many investors are trying to own the same expression of the theme. Scenario risk asks what happens when the specific mechanism behind the thesis breaks, accelerates, or collides with a macro shock.

27.2 Unique Risk Is Not Free Alpha

In a 2026 analysis of thematic investments in portfolios, BlackRock reported that a selected group of thematic iShares ETFs carried 27 percent average unique risk, compared with 21 percent for global sector ETFs, 18 percent for growth-style ETFs, and 14 percent for value-style ETFs [67]. BlackRock defines unique risk in that discussion as equity-specific risk not explained by broader market movements or traditional risk factors, with the analysis measured relative to the MSCI ACWI IMI benchmark and using data as of December 31, 2025.

The statistic is useful because it cuts both ways. It supports the book's premise that themes can carry information not captured by ordinary sectors, countries, styles, or foreign-exchange exposures. It also warns against treating unexplained risk as automatically desirable. A portfolio can have high unique risk because it has found a structural narrative the market has not fully priced. It can also have high unique risk because it owns a handful of volatile concept stocks, crowded pure plays, or illiquid companies whose idiosyncratic losses will not be diversified away.

The discipline that follows distinguishes those cases. Residual risk becomes attractive only when the process can explain why it should be rewarded, when the exposure survives factor and sector controls, and when the basket remains robust under regimes that have historically damaged similar trades. A theme that is unique but fragile is not an alpha source. It is a concentrated bet with a better label.

27.3 The Regimes That Matter

Regimes are simplified descriptions of the market environment. They are crude by design. A portfolio does not need a perfect macro taxonomy; it needs a set of states that explain when the theme's return driver is supported, when it is impaired, and when its risks become correlated with everything else the investor owns.

The most useful regimes for thematic equities include:

- **Low-rate growth markets.** Long-duration themes, expensive growth stocks, venture-like public equities, and concept-heavy baskets often perform well because distant cash flows receive generous valuations.
- **Rising-rate inflation shocks.** Financing-sensitive, high-multiple, and capital-intensive themes can reprice even when operating fundamentals remain intact.
- **Liquidity contractions.** Small-cap pure plays, crowded ETF constituents, and high-turnover strategies suffer when spreads widen and market depth disappears.
- **Value rotations.** Themes whose constituents overlap with growth and momentum can underperform even if the narrative remains alive.
- **Risk-off volatility spikes.** Correlations rise, hedges become imperfect, and residual thematic differences can be overwhelmed by de-risking.
- **Commodity shocks.** Energy, metals, batteries, power, infrastructure, and industrial themes can be helped or harmed depending on whether the basket owns producers, consumers, or bottlenecks.
- **Geopolitical and regulatory shocks.** Export controls, sanctions, procurement rules, antitrust actions, and industrial policy can change the value chain directly.
- **Recession and earnings-downgrade cycles.** Adoption stories that depend on capex budgets or consumer demand can slow when customers retrench.

The same AI-infrastructure basket can look different in each state. In a low-rate growth regime, accelerator designers and cloud platforms may dominate. In a power-scarcity regime, competitive generators and electrical-equipment suppliers may carry the edge. In an export-control regime, the same semiconductor exposure may split into harmed firms, domestic substitutes, and non-U.S. competitors. The theme is one phrase. The regime turns it into several trades.

27.4 Classifying Regimes Without Cheating

Regime labels must be available at the time the portfolio uses them. A label such as “the 2022 inflation shock” is obvious after the fact and useless if the backtest applies it before investors could have known it. The as-of invariant from Part VIII applies here as much as it applies to documents and model outputs.

Practical regime variables include rate levels and rate changes, inflation surprises, credit spreads, volatility indexes, market trend, value-growth spreads, liquidity measures, commodity price shocks, currency moves, and policy-event calendars. A simple regime classifier might define a rising-rate inflation regime when realized inflation surprises are positive, rate changes exceed a rolling threshold, and growth stocks underperform value stocks. A liquidity-contraction regime might combine widening bid–ask spreads, falling market depth, higher volatility, and rising credit spreads.

The exact classifier matters less than the discipline around it. The rule should be written before the performance comparison. It should use data available as of the rebalance date. It should be simple enough that an investment committee can understand why the portfolio changed risk. A regime model that requires a dozen hidden statistical states may fit history beautifully and give the portfolio no usable instruction.

27.5 Regime-Conditional Alpha

A thematic strategy can have positive average alpha and still depend on one friendly environment. The test is to estimate performance conditional on regime:

$$\alpha^{(r)} = \mathbb{E} \left[R_{p,t}^{\text{neutral}} \mid \text{Regime}_t = r \right], \quad (27.1)$$

where $R_{p,t}^{\text{neutral}}$ is the factor-neutral portfolio return and r indexes the regime. The comparison should report not only average alpha, but hit rate, drawdown, turnover, factor drift, liquidity cost, and residual correlation inside each regime.

If clean-energy alpha appears only in low-rate growth markets, the strategy may be a duration trade with green vocabulary. If an AI-infrastructure basket works only when momentum is strong, it may be a momentum strategy with better company mapping. If residual alpha persists through rate changes, value rotations, and risk-off episodes, the case for a distinct theme is stronger.

Regime-conditional tests should also separate theme validity from expression validity. A regime may harm one part of the value chain and help another. Higher rates can hurt long-duration growth beneficiaries while helping cash-generative infrastructure owners. Higher power prices can hurt data-center operators and help generators. The regime question is not merely “should we own the theme?” It is “which expression of the theme survives this environment?”

27.6 Crowding Is Regime Risk With More Owners

Crowding occurs when many investors hold similar exposures, especially when those investors have similar reasons to exit. It can be measured through ETF overlap, mutual-fund holdings, hedge-fund filings where available, short interest, analyst coverage, retail-flow proxies, news saturation, valuation dispersion, borrow costs, and factor exposure. None of these measures is

sufficient alone. Together they reveal whether the portfolio owns a differentiated expression or the same trade everyone else bought last month.

An overlap measure gives the first diagnostic. For portfolio weights w and thematic ETF or peer-basket weights v , define

$$\text{Overlap}(w, v) = \sum_i \min(w_i, v_i). \quad (27.2)$$

High overlap is not automatically bad. If the theme is mature and the objective is benchmark-aware exposure, overlap may be intentional. For an alpha strategy, high overlap is a warning that the basket may inherit the timing, valuation, and liquidity compromises of the public wrapper. It also means the exit door is shared.

Crowding becomes dangerous when it meets a hostile regime. In calm markets, crowded ownership can push prices higher and make the theme look better than it is. In a volatility spike or rate shock, the same ownership can create forced selling, widening spreads, and declining market depth. The liquidity spiral literature describes why market liquidity and funding liquidity reinforce one another under stress: traders' ability to provide liquidity depends on their funding, and deteriorating liquidity can damage funding conditions further [20]. A thematic basket does not need leverage to feel the effect. It only needs many owners trying to reduce exposure at once.

27.7 Crowding Diagnostics

The practical crowding dashboard should combine ownership, attention, valuation, and liquidity:

- **ETF and index overlap:** percentage of basket weight also held by large thematic ETFs, sector ETFs, and benchmark indexes.
- **Flow pressure:** recent inflows into ETFs and funds that hold the same names.
- **Ownership concentration:** institutional ownership, top-holder concentration, hedge-fund ownership where available, and retail-flow proxies.
- **Valuation stretch:** multiples relative to history, sector, expected growth, and estimate revisions.
- **Liquidity fragility:** ADV, spreads, market depth, volatility, and expected liquidation days at a stated participation rate.
- **Borrow and short constraints:** borrow cost, utilization, recall risk, and short-interest changes.
- **Narrative saturation:** attention z-scores, source breadth, repeated media framing, and analyst coverage density.
- **Factor convergence:** rising exposure to momentum, growth, beta, rates, or another common factor shared by the rest of the crowd.

The dashboard should produce actions, not decoration. A basket with high residual alpha, low overlap, modest valuations, and good liquidity can carry more risk. A basket with high attention, high overlap, expanding valuations, and deteriorating liquidity should face lower caps, slower trade schedules, and stronger evidence requirements. When crowding and implementation risk grow faster than expected alpha, position size should fall even if the theme remains compelling.

27.8 Scenario Analysis

Scenario analysis asks a more concrete question than regime classification. Instead of asking how the basket behaves in a broad inflation regime, it asks what happens if rates rise 100 basis points, cloud capex guidance falls, semiconductor export controls tighten, power prices spike, credit spreads widen, or market volatility doubles.

A scenario can be written as a vector of shocks, Δz , mapped into portfolio loss through factor, sector, and idiosyncratic sensitivities:

$$\Delta R_p \approx w^\top B \Delta z + w^\top \Delta u. \quad (27.3)$$

Here w is the portfolio weight vector, B maps holdings to scenario exposures, and Δu captures company-specific residual moves. The formula is intentionally approximate. Its value lies in forcing the team to name the channels: rates, credit, commodities, policy, capex, valuation, liquidity, and value-chain disruption.

For AI infrastructure, useful scenarios include:

- cloud capex guidance is cut by 15 percent;
- advanced GPU export controls tighten;
- HBM supply normalizes faster than expected;
- power prices rise in constrained data-center regions;
- a major hyperscaler delays data-center buildouts;
- rates rise 100 basis points while growth stocks derate;
- a key supplier misses capacity-expansion targets;
- market volatility doubles and ETF outflows force selling in crowded names.

The clean-energy analogue is equally direct: rates rise, subsidy timing slips, commodity inputs spike, residential demand weakens, grid interconnection delays worsen, or ETF flows reverse. The scenario should connect to the economic mechanism, not merely shock the whole basket by an arbitrary percentage.

27.9 Stress Testing the Exit

Most backtests test entry better than exit. They ask whether the signal would have found the opportunity, then assume the portfolio could rebalance at reported prices. Crowding reverses that assumption. The relevant stress question is whether the portfolio can reduce exposure when liquidity is worse, volatility is higher, and other investors are selling the same names.

A useful liquidation stress starts with desired trade value V_i , average daily dollar volume $\text{ADV}_i^\$$, and a maximum participation rate π^* . The minimum number of trading days required to liquidate is approximately

$$D_i = \frac{V_i}{\pi^* \cdot \text{ADV}_i^\$}. \quad (27.4)$$

Under stress, both parts of the denominator can deteriorate: ADV may fall for smaller names, and the acceptable participation rate may need to drop to avoid excessive impact. The portfolio should report liquidation days under normal and stressed assumptions. If a theme looks liquid only because recent volume was inflated by attention, the stress test should haircut that volume.

Stress testing should include historical shocks and hypothetical shocks. Historical tests reveal how similar baskets behaved during rate spikes, volatility selloffs, sector crashes, and liquidity contractions. Hypothetical tests ask about events the historical sample has not yet seen: a specific export-control rule, a hyperscaler capex pause, a nuclear-policy reversal, or a power-market intervention. The goal is not prediction. It is vulnerability mapping.

27.10 Adaptive Risk Budgets

Regime and crowding diagnostics should feed risk budgets. A theme may receive higher risk when it is validated, underowned, liquid, and supported by the current regime. It may receive lower risk when the same theme becomes crowded, expensive, factor-contaminated, or exposed to a hostile macro state.

A simple adaptive rule can scale target exposure:

$$W_{\theta,t} = W_{\theta,t}^0 \cdot g(R_t, C_t, L_t), \quad (27.5)$$

where $W_{\theta,t}^0$ is the baseline theme weight, R_t summarizes regime support, C_t summarizes crowding, and L_t summarizes liquidity. The function g should be monotonic in the directions the investment committee actually believes: lower support, higher crowding, or weaker liquidity should not increase exposure unless the strategy explicitly trades contrarian reversals and has evidence that it can survive the path.

The rule should be boring. A complex regime-switching model can overfit quickly, especially when thematic histories are short. The better version is a small number of transparent adjustments: reduce gross exposure when rates and valuation sensitivity move against the theme; tighten single-name caps when ETF overlap and flow pressure rise; require stronger materiality evidence when attention reaches extremes; slow trade schedules when liquidity worsens.

27.11 AI Infrastructure Under Stress

AI infrastructure has several regime sensitivities that should be visible before the drawdown. Chip designers and advanced-packaging suppliers carry semiconductor-cycle risk. Cloud platforms depend on capex budgets and monetization of AI workloads. Power and electrical-equipment beneficiaries face commodity, rate, and regulatory exposures. Utilities and generators can behave like rate-sensitive infrastructure even when the AI link is real. Export controls can split the value chain by geography. Crowded ownership in the obvious winners can turn good news into a valuation problem.

The scenario table should therefore separate sub-themes. A cloud capex cut hurts accelerators, networking, memory, data-center REITs, and some power-equipment demand. A power-price spike may hurt data-center operators and help generators. Export controls may hurt U.S. chip firms selling into restricted markets while helping domestic substitutes elsewhere. A rate shock may hit high-multiple semiconductor and software names harder than cash-generative industrials.

This is where the value-chain map earns its keep. Sector labels cannot show that one AI-exposed company is hurt by a power shock while another benefits. A thematic graph can. The

portfolio's stress test should report not only total loss, but which value-chain roles caused it: core pure plays, upstream suppliers, infrastructure enablers, adopters, downstream beneficiaries, or disrupted incumbents.

27.12 The Clean-Energy Stress Test Revisited

Clean energy in 2020–2022 is the chapter's transition stress test because it combined almost every portfolio stress channel. The theme had real policy support and real corporate activity, but many baskets carried high valuation sensitivity, small-cap growth exposure, ETF-flow pressure, limited liquidity in pure plays, and weak separation between theme return and rate-sensitive factor return. When rates rose and growth stocks derated, the basket did not need the decarbonization story to collapse. It only needed the regime to stop subsidizing its expression.

A disciplined stress process would not have required perfect foresight. It would have shown that a 100-basis-point rate shock damaged the basket, that ETF overlap was high, that recent flows had pushed ownership into the same obvious names, that forward returns were deteriorating despite narrative strength, and that liquidation would be costly if attention reversed. Any one signal might have been debated. Together they would have demanded a smaller position.

The lesson is not that thematic investing should avoid popular themes. Popularity is part of how themes transmit. The lesson is that popularity changes the risk. A theme that begins as underreaction can become overreaction, and the portfolio has to notice when the investor base has become part of the thesis.

27.13 Risk Controls That Preserve the Theme

The purpose of regime, crowding, and scenario analysis is not to sand every edge off the portfolio. A thematic strategy that hedges away every source of discomfort may also hedge away the reason it exists. The objective is more precise: keep the exposure that the thesis can defend, reduce the exposures that the label is hiding, and make the exit path explicit before stress arrives.

The operating checklist is short. Classify regimes point-in-time. Estimate alpha by regime. Track ETF overlap, flows, valuation, ownership, borrow, and liquidity. Run scenario tests tied to the economic mechanism. Stress the exit, not only the entry. Scale risk budgets transparently. Report which value-chain roles drive losses under each shock.

After this work, a theme is no longer just a basket that worked in a backtest. It is a portfolio whose behavior has been interrogated under the conditions most likely to expose it. Industry wrappers give the same diagnostic from a different angle: thematic ETFs and index methodologies show what consensus has been willing to package, and where that packaging has already created its own timing, dilution, and crowding problems.

Chapter 28

Thematic ETFs, Index Methodologies, and Industry Practice

28.1 The Product Is Also Evidence

When a theme becomes an ETF, something important has happened. The story has crossed from research into packaging. A sponsor has decided that the phrase is recognizable enough to sell, an index provider has translated the phrase into rules, market makers can quote the vehicle, platforms can display it, and investors can buy it with one ticker.

That makes thematic ETFs useful evidence. They show how the industry has chosen to define a theme, which companies are considered representative, how much liquidity the product can absorb, and where the theme has already become visible to ordinary investors. For an AI infrastructure strategy, the ETF shelf can reveal whether the market still treats the theme as a chip-and-cloud trade or has begun to include power, cooling, grid equipment, data-center real estate, networking, and construction bottlenecks.

The same evidence is dangerous. By the time a theme has a shelf of products, the market may already have converted curiosity into flows. The ETF is not only a benchmark; it is also a trace of demand. It may tell the researcher what the market has discovered, what it has overpaid for, and what it has simplified to make the product easy to hold.

The right treatment is neither suspicion by default nor deference by default. Thematic ETFs and index methodologies are instruments. They are useful for comparison, labeling, monitoring, and capacity analysis. They are weak as final definitions of a theme. The index methodology tells us how a commercial product solved a commercial problem. The research question is what that solution preserves, what it dilutes, and what it hides.

28.2 The Index Provider's Problem

An index provider starts with a sentence and must end with a portfolio. The sentence may be “artificial intelligence,” “clean energy,” “cybersecurity,” “aging population,” or “future mobility.” The portfolio must have eligible securities, weights, caps, rebalance dates, corporate-action rules, liquidity filters, and a published methodology that can be applied repeatedly.

The typical thematic index pipeline has six steps:

1. define the theme and its economic activities;

2. choose a parent universe, usually filtered by listing venue, market capitalization, liquidity, and investability;
3. map companies to the theme using revenues, segment descriptions, industry taxonomies, filings, business descriptions, news, or proprietary text scores;
4. compute a relevance or exposure score;
5. select and weight constituents subject to concentration, liquidity, and turnover constraints;
6. rebalance and reconstitute on a fixed schedule.

This is not so different from the machinery we built in Chapters 24 through 27. The difference is objective. A systematic trading strategy can say no. It can hold cash, shrink exposure, wait for decay, or remove a name whose factor-adjusted alpha has disappeared. An index product usually has to remain invested. It must offer continuous exposure to the advertised theme even when the theme is crowded, stale, or temporarily dominated by macro forces.

That distinction matters because product design pushes every methodology toward compromise. A pure-play basket may be too small, illiquid, and volatile. A liquid basket may be full of diversified giants. A revenue-pure basket may miss emerging firms whose theme exposure is real but not yet disclosed in segment reporting. A text-driven basket may find those firms, but with more model risk and less auditability. Index construction is therefore a negotiation between purity, investability, transparency, and commercial usability.

28.3 Revenue Maps, Text Maps, and Hybrid Maps

The cleanest evidence of thematic exposure is revenue. If a company earns a measurable share of sales from solar inverters, data-center cooling equipment, defense drones, diabetes devices, or semiconductor capital equipment, the exposure has economic substance. Revenue-based methodologies use taxonomies such as FactSet RBICS with Revenue, which maps reported business segments into granular industry categories and revenue shares [40]. This makes the exposure easier to audit than a keyword match.

Revenue evidence has two weaknesses. First, segment reporting is delayed and coarse. A company may disclose “industrial automation” or “data center solutions” without enough detail to isolate the theme. Second, the most important part of a theme is often the part that is not yet large enough to appear cleanly in revenue. Early in the AI infrastructure cycle, a power equipment supplier might win orders because of data-center grid demand, while its formal revenue segments still look like ordinary electrification.

Text-based methodologies attack the opposite side of the problem. They scan filings, business descriptions, company publications, and news for evidence that a company is connected to a theme. MSCI’s Thematic Exposure Standard, for example, is presented as a security-level framework for assessing exposure to structural trends across roughly 40,000 global securities as of September 30, 2024 [82]. Solactive’s ARTIS platform is described as an AI-based stock selection tool that searches large text corpora, evaluates document relevance, extracts company mentions, and assigns thematic relevance scores for index construction [105].

Text evidence is earlier and broader than revenue evidence, but it is easier to fool and harder to audit. Management can change language faster than it can change sales. A diversified company can mention “AI” because it uses AI internally, sells AI-adjacent software, buys AI chips, partners

with an AI firm, or merely wants investors to hear the word. The sentence alone does not tell us whether the exposure is monetizable.

The practical answer is hybrid mapping. Revenue is treated as high-precision, low-recall evidence. Text is treated as higher-recall, lower-precision evidence. The strongest candidates have both: economic materiality in disclosed activities and fresh textual evidence that the theme is becoming more important. A weaker but still tradeable candidate may have only one, provided the missing side is explained. A company with strong text but no revenue evidence needs retrieval-backed justification. A company with revenue evidence but no narrative traction may belong in the theme, but not necessarily in a timely trading basket.

28.4 The Dilution Problem

The central problem in thematic ETF construction is dilution. A theme may be economically sharp, but the product wrapper can make it blunt.

Suppose an index wants exposure to AI infrastructure. The purest names may include small or mid-cap suppliers of power equipment, thermal management, networking components, specialized semiconductors, data-center construction, and grid services. Some are illiquid. Some are foreign-listed. Some are private. Some have only partial exposure. To make a daily ETF, the methodology may add mega-cap cloud platforms, diversified chipmakers, software firms, and broad technology companies whose theme revenue is large in dollars but small as a share of enterprise value.

The result is a familiar paradox. The product looks thematic to the buyer but behaves partly like a technology, growth, momentum, or quality portfolio. Chapter 25 described this as the contamination problem at the portfolio level. Here it appears in product form. The ETF may be an honest implementation of its methodology and still be a weak expression of the theme.

Weighting rules determine how severe the dilution becomes. Market-cap weighting increases liquidity and capacity, but it often turns the basket into a small variation on the largest stocks. Equal weighting gives smaller companies more voice, but it can overload the portfolio with illiquidity. Relevance weighting improves thematic purity, but it can concentrate the portfolio in expensive pure plays. Caps limit single-name risk, but they also force capital into less relevant substitutes.

There is no perfect weighting rule. There is only a rule whose failure mode is understood. For a benchmark ETF, capacity and transparency may dominate. For a trading strategy, the better answer may be to separate the research basket from the execution basket: score the pure thematic universe, then allocate only where liquidity, valuation, factor neutrality, and decay controls permit.

28.5 The Attention-Catering Trap

The strongest warning about thematic ETFs comes from the evidence on specialized ETF launches. Ben-David, Franzoni, Kim, and Moussawi study competition for investor attention in the ETF market and show that specialized ETFs are structurally different from broad index ETFs [9]. Broad products compete mainly on price. Specialized products compete by making an investment idea salient.

The paper's empirical result is uncomfortable for anyone who wants to treat ETF creation as validation. Over the first five years after launch, specialized ETFs lose about 30% in risk-adjusted

terms. The authors attribute the underperformance not mainly to fees or hedging demand, but to the overvaluation of the underlying stocks at launch. Providers create products when investor attention is high, and attention is often high after the constituents have already performed well.

In annualized language, the book-plan notes the same result as an average loss of roughly 3.1% per year in risk-adjusted performance after fees for specialized ETFs, with newly launched thematic ETFs closer to 6% per year. The exact estimate is less important than the mechanism. The ETF does not create the overvaluation by itself. It arrives when the theme has become easy to sell, and easy to sell often means late.

This is why ETF launches and flows should enter a thematic trading system as state variables rather than applause. A new product can mark institutionalization, improved access, or broader recognition. It can also mark crowded attention. Large inflows can support prices in the short run, but they may increase the probability that future returns are pulled forward. The theme may be real and the trade may still be late.

28.6 Fees, Flows, and the Price of a Story

The attention evidence also explains why thematic products can charge more than broad beta. Broad ETFs live in a price-salient market: investors compare fees, spreads, tracking error, and index exposure. Thematic ETFs live closer to a story-salient market: investors compare narratives, recent returns, and whether the ticker gives them access to the thing everyone is discussing.

The segmentation in the underlying ETF data is sharp. As of late 2019, specialized ETFs represented about 18% of ETF industry assets but generated roughly 36% of fee revenue, with average expense ratios around 0.55%. Flows into specialized products were much less sensitive to fees than flows into broad products and more responsive to past performance and media exposure [9].

That is not simply a consumer-protection point. It is an alpha point. If flows respond to recent performance and media attention, then ETF demand is partly an endogenous variable in the thematic lifecycle. The same attention signal that helps identify a theme in Chapters 7 and 8 can become a reversal signal once it is extreme, commercialized, and accompanied by expensive product creation.

The clean-energy episode is the structural-transition case running through this book. The energy transition was real, but the vehicle-level trade in 2020 and early 2021 mixed narrative excitement, low-rate duration exposure, crowded growth positioning, and retail-accessible products. When the rate regime changed, the theme did not need to die for the ETF trade to suffer. It only needed the product's embedded exposures to be repriced. For pure narrative excess, metaverse funds, concept-stock SPACs, and blockchain rebrands provide cleaner warnings.

28.7 ETF Holdings as Noisy Labels

ETF holdings are tempting ground truth. They are easy to download, easy to compare, and already curated by a methodology. If several credible AI-themed ETFs hold the same company, it feels natural to call that company an AI stock.

For model development, this is useful but hazardous. ETF holdings are labels produced by another model, not observations of nature. They reflect a parent universe, eligibility screens, liquidity filters, index reconstitution dates, issuer preferences, licensing constraints, and product

capacity. They may include a company because it is the best available public proxy, not because it is the cleanest thematic exposure.

The right use is comparative. Given a model score s_i for company i and an ETF membership indicator h_i , the researcher can examine:

$$\text{Overlap@}K = \frac{1}{K} \sum_{i \in \text{TopK}(s)} h_i.$$

This asks a modest question: how many of the model's top-ranked names appear in a relevant ETF basket? High overlap is not proof of correctness. Low overlap is not proof of failure. It is a diagnostic that should trigger review. If the model finds many names outside ETF holdings, those names require evidence retrieval: revenue links, supply-chain links, earnings-call language, patents, customer announcements, or price co-movement. If the model mostly reproduces ETF holdings, the researcher must ask whether the system has discovered a theme or merely learned the industry's packaging.

This distinction matters especially for machine-learning pipelines. ETF holdings can seed a training set, benchmark a classifier, or evaluate recall against industry practice. They should not be the only target. A model trained solely to imitate existing products will inherit their dilution, timing, and attention bias. The better target is a triangulated label: ETF membership, revenue mapping, text evidence, analyst baskets, residual return co-movement, and human review.

28.8 Industry Practice as a Risk Monitor

Thematic ETFs can hurt as alpha signals and still help as risk signals. Their holdings reveal overlap among products. Their flows reveal demand pressure. Their rebalances can create predictable trading needs. Their sector and factor exposures show how a public theme is being packaged. Their launches show which narratives have become commercially legible.

A systematic desk should therefore maintain an ETF and index monitor around each tradable theme. For every relevant product, the monitor should track:

- constituent overlap with the internal basket;
- active share versus broad sector and style benchmarks;
- factor loadings to the book's FF5 plus momentum baseline;
- concentration in the top ten names;
- estimated revenue purity where data are available;
- flows, assets under management, and abnormal volume;
- rebalance dates and index-review calendars;
- valuation spreads between ETF constituents and near-miss candidates.

The monitor does not decide the trade. It changes the burden of proof. When ETF overlap is low, the internal model must explain its differentiated names. When overlap is high and flows are extreme, the model must explain why the trade is not simply crowded beta in thematic clothing. When an index rebalance adds a name the model already liked, the desk must decide whether that is confirmation, a liquidity catalyst, or the beginning of crowding.

28.9 Active and Passive Are Not Moral Categories

It is easy to make passive thematic products the villain and active thematic research the hero. That would be too simple. Passive products are transparent, scalable, tax-efficient in many jurisdictions, and useful for allocators who want broad exposure without building internal infrastructure. Active strategies can be adaptive, but they can also overfit, overtrade, tell better stories than their data support, and charge for judgment they do not actually possess.

The real distinction is not passive versus active. It is whether the process can update when the evidence changes. A rules-based index can be excellent if the rule captures a durable exposure, controls dilution, and is honest about timing. An active strategy can be poor if it chases the same attention cycle with more discretion and less transparency.

The systematic approach in this book borrows from both sides. From index practice it takes reproducibility, eligibility discipline, published rules, concentration limits, and operational clarity. From active research it takes materiality judgment, narrative timing, valuation overlays, regime awareness, and the right to reduce exposure when the theme becomes overowned.

28.10 Using Industry Products Without Being Used by Them

The final lesson is practical. Do not ignore thematic ETFs, and do not outsource the definition of the theme to them.

Use them to learn the market's vocabulary. Use them to identify benchmark contamination. Use them to audit whether your basket is merely a sector fund with a newer name. Use them to find crowded ownership and rebalance pressure. Use their holdings as noisy labels, not as truth. Use their launches as evidence that the theme has become legible, but ask whether legibility arrived before or after the alpha.

For AI infrastructure, this means the internal basket should be compared with AI, robotics, semiconductor, cloud, power, data-center, and automation products. The point is not to maximize overlap. The point is to understand overlap. A basket that owns only the same mega-cap stocks as every public AI ETF may be liquid and easy to defend, but it may not deserve the word infrastructure. A basket that owns only obscure second-order beneficiaries may be pure, but it may fail capacity, liquidity, and valuation tests. The craft is to know which compromise has been made.

The product wrapper is one of the market's best summaries of consensus. A trading strategy should read that summary closely. Then it should decide, with evidence, where consensus is useful, where it is late, and where it is wrong.

Chapter 29

Prediction Markets and Event Overlays

29.1 The Market Before the News

Some thematic risks do not arrive gradually. A government tightens export controls. A court blocks a merger. A regulator approves a drug. An election changes the probability of subsidies, tariffs, defense spending, or antitrust pressure. A private company delays an IPO and changes the implied value of every public proxy around it.

Text systems are good at measuring the spread of a story. Topic models, earnings-call embeddings, news classifiers, and value-chain maps tell us which companies are exposed to a theme and how the language around that theme is changing. Prediction markets answer a narrower question: what price is the market placing on a discrete event?

That narrowness is their virtue. A prediction-market price is not a full investment thesis. It does not know whether a semiconductor equipment supplier has revenue exposure to China, whether a utility can pass through grid investment costs, or whether a biotech company has enough cash to survive a failed trial. But it can tell us that the probability of the triggering event has moved before the event resolves and sometimes before conventional news has fully organized the story.

For thematic trading, prediction markets belong as event overlays. They sit on top of the equity system built in the previous chapters. The equity system maps companies to themes, constructs baskets, neutralizes factors, controls capacity, and monitors crowding. The event overlay changes the conditional probabilities around that system. It says: if this event becomes more likely, which exposures should be reviewed, scaled, hedged, or removed?

29.2 Binary Contracts and Probability

A simple binary event contract pays one unit if an event occurs and zero otherwise. If a contract pays \$1.00 on success and trades at price p_t , the first-order reading is:

$$\Pr_t(\text{event}) \approx p_t.$$

This is why prediction markets are attractive. They compress dispersed beliefs, incentives, and information into a number that updates continuously. Wolfers and Zitzewitz's canonical

survey describes prediction markets as mechanisms for aggregating information and producing forecasts that often compare well with more conventional benchmarks [119]. Later work uses prediction-market prices in event studies, including political events whose expected policy effects can be linked to financial markets [104].

The approximation is not literal truth. The price is closer to a market-implied, risk-neutral probability after fees, liquidity, risk preferences, position limits, and microstructure frictions. A thin market at 72 cents is not the same evidence as a deep market at 72 cents. A contract with a vague resolution rule is not the same evidence as a contract with a mechanical resolution rule. A market dominated by one large trader is not the same evidence as one with broad, competitive order flow.

Still, changes in p_t can be valuable. The level says what the market currently believes. The update says how beliefs are changing. In an equity overlay, the update is often more useful than the level because it marks the arrival of new information or new attention.

29.3 Why Logit Updates Help

Probabilities are bounded between zero and one. A move from 50% to 60% is not the same kind of move as a move from 90% to 99%, even though both are ten percentage points. Near the boundaries, ordinary probability differences can understate how much the odds have changed.

A useful transformation is the logit:

$$\ell_t = \log \left(\frac{p_t}{1 - p_t} \right).$$

The logit maps a probability into log-odds. The update is:

$$\Delta \ell_t = \ell_t - \ell_{t-1}.$$

If an export-control contract moves from 50% to 60%, the logit update is about 0.41. If it moves from 90% to 99%, the update is about 2.40. The second move is much larger in odds space because the market has moved from “very likely” to “almost certain.” In a portfolio system, this matters because event-sensitive exposures often become nonlinear near resolution. A company may not care much whether a policy is 52% or 58% likely, but it may care a great deal whether the policy is 90% or 99% likely before a rebalance or earnings date.

For implementation, logit changes should be clipped when prices are extremely close to zero or one. A one-tick move in an illiquid contract should not create an infinite signal. The practical rule is to transform a bounded price such as $\tilde{p}_t = \min(\max(p_t, \epsilon), 1 - \epsilon)$, where ϵ is a small floor such as 0.01 or 0.02, and then ask whether the update is persistent and liquid enough to matter.

29.4 From Event Probability to Equity Exposure

The overlay begins after the company map is already built. Let $a_{i,e}$ be the exposure of company i to event e . Positive values mean the company benefits if the event occurs. Negative values mean it is hurt. Zero means the event is not expected to matter. The exposure can come from analyst judgment, value-chain mapping, revenue geography, regulatory filings, supply-chain data, or scenario analysis.

A simple event-adjustment rule is:

$$\Delta s_{i,t}^{(e)} = \lambda_e a_{i,e} \Delta \ell_{e,t},$$

where $\Delta s_{i,t}^{(e)}$ is the event-driven adjustment to the company score, $\Delta \ell_{e,t}$ is the logit probability update for the event, and λ_e controls how much the portfolio trusts that event market. The formula is deliberately modest. It does not say that prediction markets determine the portfolio. It says that, once the system knows which companies are exposed, a credible change in event probability can adjust their scores.

The intuition is straightforward. If the probability of stricter semiconductor export controls rises, negatively exposed exporters receive a negative adjustment and domestic substitutes or non-restricted competitors may receive a positive one. If the probability of a clean-energy tax-credit extension rises, companies with direct subsidy sensitivity receive a positive adjustment, while companies whose exposure is only narrative-level receive little or none. The same event probability is translated differently for each company because the value-chain map is different.

This is the key separation. Prediction markets supply event probabilities. The thematic system supplies company meaning. Confusing the two is how an event signal becomes a vague macro opinion.

29.5 Signal Credibility

Prediction markets are markets, not oracles. They can be shallow, noisy, manipulated, jurisdictionally constrained, and dominated by specialized traders. A usable overlay therefore needs a credibility layer before any probability update touches the portfolio.

A credibility review should include at least seven checks:

- **Liquidity:** volume, open interest, depth at the best bid and offer, and executable size.
- **Spread:** whether the bid–ask spread is tight enough for the displayed midpoint to be meaningful.
- **Persistence:** whether the probability move survives beyond a few prints or one visible trader.
- **Resolution clarity:** whether the contract definition and settlement source are objective.
- **Cross-venue confirmation:** whether related markets, news, options, rates, or equities tell a compatible story.
- **Trader concentration:** whether the move appears to be broad information aggregation or one-sided pressure.
- **Timestamp discipline:** whether the prediction-market update genuinely precedes the news or merely reacts to it.

These checks turn a price into a research object. A persistent move on deep volume with a clear resolution rule deserves attention. A jump in a thin contract with a wide spread and ambiguous settlement language belongs in the monitoring queue, not the optimizer.

One practical implementation is a signal credibility index:

$$SCI_{e,t} = \omega_L L_{e,t} + \omega_P P_{e,t} + \omega_C C_{e,t} + \omega_R R_e - \omega_S S_{e,t},$$

where L measures liquidity, P persistence, C cross-source confirmation, R resolution clarity, and S spread or slippage. The weights should not be overfit. Their purpose is operational triage: which event moves deserve automatic scaling, which deserve analyst review, and which should be ignored.

29.6 Venue Microstructure and Regulation

The venue matters because the same economic question can trade differently across platforms. Kalshi is a CFTC-regulated event-contract exchange in the United States, while Polymarket developed as a crypto-native prediction market. Polymarket was the subject of a 2022 CFTC settlement concerning off-exchange event-based binary options offered without required registration [27]. Kalshi's election-event contracts became a prominent legal test of how far regulated event markets could extend into political outcomes [111].

The venues are not equally rich as forecasting instruments either. Polymarket's contract design — weighted toward forward-start binary outcomes on long-dated political, geopolitical, and policy events — produces a variance risk premium roughly an order of magnitude larger than Kalshi's, where contracts skew shorter-dated and more macro-mechanical. The structural implication for an equity overlay is that Polymarket prices often carry more probability information at longer horizons, while the same contracts carry more microstructure risk per unit of information. A desk that wants forecasting depth pays for it in execution friction.

By 2025 and 2026, prediction markets had also moved toward institutional data infrastructure. In October 2025, Intercontinental Exchange announced a strategic investment of up to \$2 billion in Polymarket and said it would become a global distributor of Polymarket's event-driven data [62]. In February 2026, ICE announced Polymarket Signals and Sentiment, a service that aggregates and normalizes Polymarket data for institutional users through ICE data infrastructure [63]. For a thematic desk, this is a meaningful change. It reduces the operational gap between a crypto-native probability market and the data feeds used by conventional quantitative workflows.

Institutional access does not remove microstructure risk. Research on Kalshi transaction data finds informative prices that improve as markets approach closing, but also a favorite-longshot bias: low-price contracts win less often than break-even pricing would imply, while high-price contracts perform better [21]. Research on Polymarket finds a striking concentration of trading gains, with the top 1% of users capturing 76.5% of all trading gains in a large 2022–2026 sample [2]. These findings do not mean the prices are useless. They mean the overlay should treat prediction markets as information sources before treating them as instruments to trade directly.

29.7 Direct Trading Versus Informational Use

The book's bias is clear: most equity investors should use prediction markets primarily for information, not as standalone return engines. That is not because direct trading is impossible. Specialized traders can have domain knowledge, speed, cross-venue access, and execution discipline. But for a thematic equity process, the scalable opportunity is usually elsewhere.

The informational use is cleaner. An event market can update scenario probabilities. Those probabilities can alter basket weights, hedge ratios, review priorities, stop levels, or exposure caps. The equity portfolio remains the main investment instrument, where the manager has better tools for factor neutrality, liquidity control, tax treatment, mandate compliance, and position sizing.

Consider the difference between two questions. The first is: should we buy a 17-cent contract on a longshot policy outcome? That question requires prediction-market execution skill. The second is: if the probability of that policy outcome moves from 17% to 35% on credible volume, which companies in our basket are mis-sized? That question belongs naturally inside thematic portfolio management.

29.8 AI Infrastructure Events

AI infrastructure is full of event-sensitive branches. Export controls affect advanced semiconductor supply chains. Energy policy affects grid equipment, nuclear restarts, renewable interconnection, and data-center power procurement. Elections change the probability of industrial policy, antitrust enforcement, defense procurement, immigration rules for technical labor, and restrictions on cross-border technology transfer. Private company milestones can affect public proxies when investors use listed suppliers, cloud partners, or comparable firms as ways to express exposure to companies they cannot own directly.

The overlay should never say simply “buy AI because the event probability rose.” It should ask a narrower sequence of questions:

1. What event moved?
2. Which sub-theme does it affect?
3. Which companies have measurable exposure to that sub-theme?
4. Is the sign of exposure positive or negative?
5. Is the probability move credible by liquidity, persistence, and cross-source checks?
6. Does the resulting score adjustment survive factor, sector, and liquidity constraints?

Suppose a market-implied probability of tighter export controls rises sharply. The immediate losers may be suppliers with high revenue exposure to restricted jurisdictions or customers. The possible beneficiaries may include domestic substitutes, non-restricted suppliers, regional competitors, or firms whose backlog is insulated from the controlled technology. The same event can hurt one part of the theme and help another. Only the value-chain map can tell the portfolio which is which.

Energy policy has a different shape. A rising probability of accelerated permitting reform may benefit grid equipment, transmission construction, power-generation developers, and data-center power suppliers. But if the same event raises the probability of ratepayer pushback, project delays, or regulatory intervention, the benefit may be smaller for regulated utilities than for equipment vendors. The event is common; the company translation is specific.

29.9 Clean Energy and the Lesson of Conditional Exposure

Clean energy shows why event overlays must be conditional rather than thematic slogans. A tax credit extension, a permitting decision, a rate shock, and a commodity-price move can all matter to the same clean-energy basket, but they do not push every constituent in the same direction.

Solar installers, turbine manufacturers, battery suppliers, utilities, hydrogen firms, and transmission equipment companies have different sensitivities. Some benefit from subsidies.

Some are hurt by higher discount rates. Some are exposed to commodity inputs. Some need low financing costs more than they need policy support. A prediction-market update about an election or energy bill is therefore not a clean-energy signal by itself. It is a scenario update that must be multiplied by company-specific exposure.

This distinction would have helped investors during the 2020–2022 clean-energy cycle. The narrative remained powerful, but the portfolio expression was contaminated by duration, growth, crowding, and product flows. An event overlay could not have solved those problems alone. It could, however, have separated policy probability from rate sensitivity and prevented the investor from treating every supportive headline as a reason to increase the same basket.

29.10 The Overlay Workflow

A production event-overlay system should be boring in the best sense. The workflow should be explicit enough that a probability spike does not turn into discretionary excitement.

1. **Define the event library.** List events relevant to each theme: export rules, subsidy decisions, elections, central-bank meetings, clinical readouts, litigation dates, M&A approvals, tariff actions, and private-market milestones.
2. **Map events to companies.** Maintain event-exposure scores $a_{i,e}$ with sign, magnitude, evidence, and review date.
3. **Ingest market probabilities.** Store bid, ask, last price, midpoint, volume, open interest, timestamps, and resolution metadata.
4. **Transform and smooth.** Convert probabilities to logit space, clip extremes, and compute persistent updates rather than one-print shocks.
5. **Score credibility.** Apply liquidity, spread, persistence, cross-source, and resolution-rule checks.
6. **Translate to portfolio actions.** Adjust scores, scale exposures, update hedges, or trigger human review according to pre-specified thresholds.
7. **Audit outcomes.** After resolution, compare predicted exposure signs, portfolio actions, and realized company-level returns.

The audit step matters. Prediction-market prices can be useful even when they are imperfect, but the overlay should learn where they help. Political markets may be valuable for defense, energy, and regulation. Clinical-trial markets may be useful only when liquidity is deep and resolution rules are precise. Macro contracts may be informative for volatility but redundant with rates markets. The system should discover those domain differences through post-event review.

29.11 Ending the Portfolio Core

Parts V and VI have moved from language to portfolios. We began with company exposure, value chains, retrieval, and guardrails against thematic illusion. We then built baskets, neutralized factors, controlled turnover and capacity, stress-tested regimes and crowding, compared our work with industry products, and finally added event probabilities as conditional overlays.

The result is not a single model. It is an operating discipline. Text finds the story. Topic models organize it. Retrieval grounds it. Exposure maps translate it into companies. Portfolio construction turns it into weights. Risk control keeps it from becoming disguised beta. Event overlays update it when the world changes discretely.

That discipline is the bridge to practice. Once the portfolio core is explicit, use cases, design patterns, anti-patterns, and implementation playbooks can turn the machinery into repeatable work.

Part VII

Use Cases, Design Patterns, and Architectural Guidelines

Chapter 30

Methodological Guidelines for Narrative-Aware Research

30.1 The Backtest That Knew Too Much

In 2023, Alejandro Lopez-Lira and Yuehua Tang reported a striking result: GPT-4, prompted on post-cutoff financial headlines, could classify the initial market reaction with unusually high hit rates [76]. The result was exciting because it suggested that large language models had learned something about the economic meaning of news. It was also uncomfortable because it exposed the methodological problem that follows every text-driven investment study: how do we know the model is forecasting rather than remembering, classifying rather than explaining, or describing rather than earning a return?

That question is not a technical footnote. It is the difference between research and theater.

Narrative-aware investing attracts fragile evidence. The charts are clean, the labels are fluent, and the stories sound plausible. A topic model discovers an elegant cluster. An LLM assigns a company to a theme with beautiful prose. A basket earns a strong raw return. Then the first serious audit arrives and asks the only questions that matter: what was known at the time, why should the signal earn a return, what benchmark did it beat, which factor exposures explain it, and what would have falsified it?

Research discipline comes before architecture because every system inherits the standards of the research that created it. A weak question built on a strong model remains a weak question. The practical machinery only matters after the methodology has made the claim testable, dated, benchmarked, and falsifiable.

30.2 Start With the Investment Question

The most common mistake in text-driven investing begins with a model name. “Can BERTopic discover themes?” is not an investment question. Neither is “Can an LLM classify earnings-call paragraphs?” Those are engineering questions. Useful, but incomplete.

An investment question names a decision.

“Can emerging cross-sector news themes predict future residual co-movement before ETF products appear?” is an investment question. “Can earnings-call discussion of data-center power raise exposure scores for industrial suppliers before estimate revisions?” is an investment

question. “Can prediction-market probabilities reduce risk around export-control events after the value-chain map identifies affected companies?” is an investment question.

A good research question specifies seven objects:

- the asset universe;
- the decision horizon;
- the text source and its availability lag;
- the signal mechanism;
- the portfolio expression;
- the benchmark and factor controls;
- the expected failure modes.

Without those objects, model selection becomes taste. The newest model wins because it is newest. A high-dimensional embedding replaces a phrase dictionary because it feels more sophisticated. A language model is added because the system can call one. The research may still produce insight, but the link to capital is accidental.

The investment question is the first guardrail against that drift. It forces the researcher to say what decision the text will change: monitor a theme, approve a theme, score a company, size a basket, hedge an event, reduce exposure, or kill a trade. Text is not the object of investment. The decision is.

30.3 Write the Mechanism Before the Model

A narrative signal should not enter a backtest until the researcher has written down why it could earn a return. The reason cannot be “because the model found it.” A model finds patterns. Markets pay for constraints.

The usual mechanisms are familiar from the rest of the book:

- slow information diffusion across sources;
- limited investor attention;
- complexity of value-chain mapping;
- delayed analyst-estimate revisions;
- short-sale or balance-sheet constraints;
- liquidity and capacity constraints;
- regulatory uncertainty;
- institutional mandate rigidity;
- behavioral extrapolation and crowding.

Each mechanism implies a different test. If the claim is slow diffusion, the signal should appear first in faster sources and later in slower ones. If the claim is value-chain complexity, the second-order beneficiaries should respond after the obvious pure plays. If the claim is crowding,

extreme attention and ETF flows should predict weaker forward residual returns. If the claim is institutional mandate rigidity, the opportunity should concentrate in names or geographies that large benchmark-aware portfolios cannot easily hold.

Writing the mechanism early does not make the thesis true. It makes the thesis falsifiable. A narrative that cannot name who is underreacting, why they are underreacting, and over what horizon the correction should occur belongs in the exploration file, not the production queue.

30.4 Separate Three Kinds of Evidence

The same text signal can answer three different questions, and confusing them is one of the book's central methodological traps.

Descriptive evidence measures what the market is talking about. Topic prevalence, attention acceleration, source breadth, sentiment, and narrative labels are descriptive. They tell us that a story exists, how it is changing, and which companies appear near it. Descriptive evidence is useful for monitoring and discovery, but it does not by itself prove forecast value.

Forecasting evidence asks whether a signal improves predictions out of sample. A topic score that predicts monthly residual returns, an earnings-call novelty measure that predicts estimate revisions, or an event-probability update that improves scenario performance belongs here. Forecasting evidence should be evaluated with walk-forward tests, fixed rules, pre-defined loss functions, and comparisons against fair alternatives. Diebold and Mariano's forecast-comparison framework is the canonical reference for testing whether two forecasts differ in expected accuracy [32]; Giacomini and White extend the logic to conditional predictive ability, which matters when one model works only in some states of the world [49].

Causal evidence asks whether one event or information shock caused a change in returns, spreads, estimates, flows, or exposures. That requires a research design: an event study with a credible window, a natural experiment, an instrumental variable, a difference-in-differences design, or another identification strategy. The potential-outcomes framework of Imbens and Rubin and the applied econometric discipline emphasized by Angrist and Pischke are useful anchors because they force the researcher to define the counterfactual: what would have happened to the treated companies without the event? [61, 6]

The three claims should not be ranked as moral categories. Descriptive evidence is enough for a monitoring dashboard. Forecasting evidence is enough for many trading signals if the evaluation is honest. Causal evidence is powerful when the question is whether a policy change, regulatory action, product launch, or contract announcement moved a set of exposed firms. The problem is using one kind of evidence while speaking in the language of another.

An AI infrastructure study can say, descriptively, that data-center power became a distinct topic in transcripts and news. It can say, predictively, that exposure to that topic forecast residual returns in a walk-forward basket. It can say, causally, that a Microsoft–Constellation power purchase agreement changed the market value of firms with comparable nuclear and baseload assets only if the event design supports that claim. Each statement has a different burden of proof.

30.5 Build Point-in-Time or Call It Descriptive

Text systems leak the future in more ways than price systems. A topic label can use future documents. A company mapping can use a later segment disclosure. ETF holdings can be

downloaded after publication and treated as if they were known earlier. A language model can classify a 2019 filing using knowledge it absorbed from 2024 commentary. A current ticker map can attach old documents to surviving firms and quietly erase bankrupt or acquired names.

The standard is the as-of invariant: every input to every simulated decision must have been available, in its recorded form, before the simulated decision time.

For narrative-aware research, that means storing at least:

- event time, publication time, vendor ingestion time, and availability time for each document;
- the document text as it was available then, not as revised later;
- point-in-time entity links, ticker mappings, and corporate-action histories;
- the theme specification and approval date;
- the exposure model, embedding model, prompt, schema, and parameters used at the time;
- factor files, prices, fundamentals, ETF holdings, and benchmark constituents as they were available then;
- analyst review decisions and overrides with timestamps.

If this record cannot be reconstructed, the result may still be useful. It can describe history, explain a past episode, or help design a future pipeline. It should not be presented as a tradable backtest. That distinction protects the reader and the researcher. A descriptive reconstruction of clean energy 2020–2022 can be excellent history. It becomes bad research only when it pretends the 2022 diagnosis was available to a 2020 portfolio.

30.6 Match Source, Signal, and Horizon

Every source has a natural tempo. Headlines move first and decay quickly. Earnings-call Q&A often carries monthly or quarterly information. SEC filings confirm slower-moving material exposure. ETF holdings are useful for benchmarking and crowding analysis but are usually late. Prediction markets can update event probabilities in real time, but only for events with contracts, liquidity, and clear resolution rules.

The signal horizon must match the source horizon and the portfolio rule. A monthly basket should not rely on a signal that decays in minutes. A quarterly filing signal should not be judged by next-day returns unless the filing change is itself the event. A dynamic-topic model fit on weekly windows should not be compared with an intraday news strategy. A prediction-market overlay should change event-sensitive exposures, not the whole thematic book.

Horizon mismatch creates a cruel kind of false negative. The signal can be real and still look useless because the test samples the wrong window. It also creates false positives. A signal can look predictive at a monthly horizon because it is really loading on momentum, attention, or delayed price reaction rather than on the narrative mechanism the researcher intended to test.

The research memo should state the source tempo and the return horizon in the same paragraph. If those two sentences do not fit together, the methodology is already warning you.

30.7 Choose the Simplest Adequate Model

The text-as-data literature emphasizes that text is high-dimensional and method-sensitive [48]. Finance adds a second complication: the cost of being wrong is not only classification error, but capital allocated to the wrong exposure. Complexity should therefore be earned.

Use keyword and phrase rules when the evidence is narrow, high precision, and auditable. Use embeddings when semantic similarity and retrieval matter. Use topic models when the goal is discovery, monitoring, or measuring attention across a corpus. Use supervised classifiers when the label is stable and training data exists. Use LLMs when the task requires contextual reading, structured extraction, or evidence-grounded reasoning. Use graph models when the exposure travels through customers, suppliers, contracts, or bottlenecks.

The model hierarchy is not a ladder of prestige. It is a set of tools matched to failure modes. Loughran and McDonald showed why finance-specific dictionaries can outperform generic sentiment dictionaries in filings [78]. Tetlock’s media-sentiment work showed that simple language measures can matter when tied to a clear market mechanism [108]. Ke, Kelly, and Xiu showed that supervised text methods can be adapted directly to return prediction rather than generic sentiment labels [72]. Bybee, Kelly, Manela, and Xiu showed that topic attention over a large news corpus can map macro narratives and forecast aggregate market returns [22]. None of these results says “use the biggest model.” They say: match method to question, and evaluate downstream.

A useful default for this book is hybrid but restrained: interpretable baselines first, embeddings for retrieval and semantic proximity, LLMs only where cited evidence and schema discipline make the output auditable, and graph models only when the economic mechanism truly travels through relationships.

30.8 Separate Discovery From Validation

Discovery rewards openness. Validation rewards suspicion. A research process that uses the same attitude for both phases either misses themes or believes too many of them.

The separation should be procedural:

1. explore a corpus and identify candidate topics;
2. write a point-in-time theme specification before testing returns;
3. map companies and score exposure using pre-specified evidence rules;
4. validate the mapping against independent evidence such as revenue, events, or analyst review;
5. run walk-forward portfolio tests with fixed construction rules;
6. monitor live performance, drift, costs, and falsification conditions.

The analyst who discovers a theme can be enthusiastic. The validation process cannot. This is not an insult to the analyst; it is a defense against narrative capture. A theme that survives a hostile validation process is more valuable than one that is merely well described.

30.9 Define Benchmarks Before Results

The benchmark is part of the hypothesis. Changing it after seeing performance is a quiet form of overfitting.

A narrative-aware strategy often needs several comparisons:

- a broad equity benchmark;
- a sector or industry benchmark;
- a style-factor benchmark;
- a thematic ETF or index benchmark;
- a custom eligible-universe benchmark;
- a factor-neutral long-short benchmark;
- a simple interpretable model that the complex model must beat.

The last benchmark is easy to neglect and often the most revealing. If a complex LLM workflow cannot outperform a revenue filter plus a handful of curated phrases, the complex workflow has not earned its operating risk. If a neural topic model cannot beat an NMF baseline on downstream portfolio utility, its prettier embeddings are not enough. If a thematic basket loses its alpha after FF5 plus momentum controls, the theme may have been a factor trade with better vocabulary.

Benchmarks are not bureaucratic. They are how the research process refuses to be impressed too easily.

30.10 Neutralize in Layers

Raw returns are the first line of a performance report, not the conclusion. A thematic strategy should be read in layers:

1. raw basket return;
2. benchmark-relative return;
3. sector- or industry-adjusted return;
4. factor-adjusted return;
5. residual return after turnover and transaction costs;
6. capacity-adjusted return;
7. regime- and crowding-conditioned return.

The pattern across layers tells the story. If raw performance is strong and residual performance vanishes after sector controls, the model found an industry rotation. If residual performance survives factor adjustment but disappears after costs, the model found an untradable signal. If performance exists only in one liquidity bucket or one regime, the strategy may still be useful, but the claim must be narrower.

Pre-orthogonalization belongs here as a research standard, not just a portfolio trick. A raw exposure score should be tested both before and after residualization against sectors and style factors. If the residualized score has no signal, the theme may still describe the market, but it does not yet justify a factor-aware portfolio.

30.11 Treat Human Review as Data

Human review is often described as a soft control. In a serious thematic system it is structured data.

Analysts approve labels, reject false positives, classify value-chain roles, judge materiality, read evidence packages, and identify economic mechanisms. Those decisions should be logged with the same care as model outputs:

- reviewer;
- timestamp;
- documents reviewed;
- decision and confidence;
- override reason;
- validity window;
- downstream authority of the decision.

This logging serves two purposes. It makes the process auditable, which matters for governance. It also creates a training set for the next version of the system. Overrides show where the model is wrong, where the rubric is unclear, and where analysts disagree with each other. Without that record, the human layer improves the current decision while teaching the system nothing.

30.12 Estimate Uncertainty, Not Just Scores

A company with sparse evidence should not receive the same treatment as a company with audited segment revenue, named customers, and repeated transcript confirmation. A theme with one week of source acceleration should not carry the same risk budget as a validated theme with residual co-movement and independent revenue evidence.

Exposure scores should therefore carry uncertainty. The simplest implementation uses confidence bands: high, medium, low. A better implementation stores a distribution or interval around the score. The portfolio can then use uncertainty to set position size, review priority, turnover limits, evidence thresholds, and monitoring frequency.

Uncertainty also changes the research conversation. A low-confidence high-exposure name is not a buy. It is a review item. A medium-exposure high-confidence name may be a better portfolio constituent because the evidence is stable and implementable. A method that emits only one score forces the optimizer to pretend the research team knows more than it does.

30.13 Respect Language, Jurisdiction, and Coverage

Most examples in this book use U.S. equities because the data are deep, the factor libraries are familiar, and the research literature is richest there. A real institutional system often has a wider footprint. European, Italian, Japanese, Korean, and emerging-market corpora introduce different disclosure rules, languages, market microstructures, ticker conventions, and source latencies.

International coverage is not a translation problem only. A multilingual encoder may map words across languages, but the economic meaning of a filing section, a regulator's phrase, or a labor-policy narrative depends on jurisdiction. Social-media data may be usable under one set of terms and restricted under another. Corporate-action histories, delisting data, and analyst-coverage depth differ by market. A system designed around SEC filings and U.S. newswires should not be assumed to work unchanged on European bank disclosures or Italian industrial filings.

The methodological rule is simple: define the supported universe honestly. If the system claims global thematic coverage, it needs multilingual corpora, jurisdiction-aware entity linking, local trading calendars, local factor models, and governance for data licensing and privacy. If it does not have those, it should say so. Scope discipline is a form of honesty.

30.14 The Research Memo

Before a narrative signal becomes a production candidate, the research memo should answer the following:

- What investment decision does the signal change?
- Is the claim descriptive, predictive, causal, or some combination?
- Why should the signal earn a return?
- What data were available point-in-time?
- What source horizon matches the portfolio horizon?
- What is the simplest adequate model, and what baseline does it beat?
- What benchmark stack is fair?
- What factor exposure explains the raw return?
- What costs, turnover, and capacity limits apply?
- What human review is required?
- What uncertainty attaches to the score?
- What would falsify the signal?

The checklist is demanding because the market is demanding. Narrative-aware investing is powerful because it can connect information before conventional maps do. It is dangerous for exactly the same reason: it can also connect stories that should never have met inside a portfolio.

The standards now become practical tests: where did text change an investment decision, what failure did the playbook prevent, and what evidence made the result worth believing?

Chapter 31

Use Cases, Anti-Patterns, and Playbooks

31.1 The Desk That Asked the Right Question

In late 2023, an AI infrastructure portfolio could have looked obvious if it started with the largest semiconductor names. The interesting question was not whether the market cared about GPUs. It did. The interesting question was whether the same narrative was beginning to reprice less obvious companies: memory suppliers, electrical-equipment manufacturers, cooling vendors, nuclear operators, grid contractors, and industrial landlords.

That is the practitioner's problem. A narrative-aware system does not win by naming a fashionable theme after everyone has named it. It wins, if it wins at all, by turning text into a decision that would otherwise be delayed: a company enters the watchlist earlier, a supplier receives a higher exposure score, a crowded proxy is capped, an event hedge is added, or a theme is killed before the story becomes an excuse.

The methodology of Chapter 30 becomes useful only when it changes a workflow. Each use case below follows the same pattern. First comes the investment decision. Then comes the failure mode that usually corrupts it. The playbook is the disciplined version of the workflow: data, signal, model, portfolio expression, validation, and production controls.

The cases are not equal in maturity. News sentiment and filing-change signals have long academic literatures. Earnings-call exposure measurement has strong published examples in political risk and climate risk. Alternative-data and event-intelligence workflows have strong industry adoption but often disclose less about realized alpha. Dynamic macro narratives and practitioner portfolio studies are closer to the frontier. Prediction-market overlays are useful, but only under strict liquidity and mapping constraints. The purpose here is not to pretend that every use case has the same evidentiary status. The purpose is to show how a serious desk should treat each one.

31.2 What Industry Success Actually Looks Like

Academic evidence usually asks whether a signal predicts returns under a defined test. Industry success often looks different. It appears as a product category that survives procurement, becomes part of analyst workflow, or is acquired by a larger financial-data platform because clients repeatedly need the capability. That is weaker than a clean out-of-sample alpha claim, but

it is not trivial. It tells us which parts of narrative-aware investing have become operationally valuable.

The strongest industry cases fall into six patterns.

News and event analytics turn unstructured news, social, and public data into alerts, sentiment, novelty, relevance, and event tags. RavenPack is the canonical investment example: it productizes NLP over news and other text for quant funds, asset managers, banks, economists, and risk teams [89]. Dataminr is the event-intelligence example: it productizes real-time detection of breaking events and risks from public data, more often for security, public-sector, and operational-risk users than for pure equity alpha [30]. The lesson is that speed, entity resolution, event taxonomy, and alert routing are themselves valuable, even before a portfolio rule exists.

Research workbenches make unstructured financial content searchable and explainable. AlphaSense and Tegus-like expert-call workflows belong here: they help investors and corporate strategy teams search filings, transcripts, broker research, news, expert calls, and internal material faster than manual research allows [4]. Kensho's role inside S&P Global shows the same institutional pattern from the data-owner side: AI is used to extract, organize, and retrieve meaning from trusted financial datasets rather than from the open web alone [73]. These platforms are not trading systems. They are evidence machines.

Alternative-data platforms connect narrative claims to observable behavior. YipitData, Similarweb, Exabel, and Bloomberg Alternative Data all reflect the same market need: investment teams want web, app, transaction, labor, location, pricing, and other nontraditional datasets inside usable research workflows [125, 100, 37, 17]. In this book, their role is not to replace text. It is to test whether text is pointing at a real economic change.

Geospatial and physical-world intelligence measures activity that companies may discuss only later. Kayrros, SpaceKnow, RS Metrics, and the Orbital Insight lineage belong here. Kayrros is especially relevant because it combines satellite data, AI, energy intelligence, emissions monitoring, and asset-level physical-risk work [71]. The investment lesson is simple: if a narrative says supply chains, energy flows, industrial activity, or emissions are changing, physical data can validate or contradict the story.

Exposure and index infrastructure converts themes into security-level maps. MSCI's Thematic Exposure Standard, FactSet RBICS with revenue, Solactive's ARTIS, and FactSet's alternative-data and Truvalue lines show that thematic exposure, revenue mapping, AI-driven ESG signals, and NLP-based index construction are now institutional data products [82, 40, 105, 39, 38]. The lesson is not that vendor scores are ground truth. It is that the industry has converged on the need for explicit exposure mapping.

Market, consumer, and technology intelligence platforms such as Quid, CB Insights, PatSnap, FiscalNote, Contify, and Similarweb are most relevant when the investment process needs early market maps, policy monitoring, startup ecosystems, patents, consumer demand, or competitive signals. They are often built for corporate strategy rather than portfolio construction. Used carefully, they expand the discovery layer. Used carelessly, they create attractive stories with no tradable expression.

This distinction matters for the use cases that follow. A successful use case is not only a paper with a coefficient. It can also be a repeatable institutional workflow that turns messy external information into earlier research, better exposure maps, better risk controls, or better questions for the portfolio team.

31.3 News Sentiment Drift

The first institutional use case for text in equities is still one of the most durable: convert news into firm-level information faster and more consistently than a human reader can. Tetlock's work on media pessimism showed that news tone can forecast market behavior when the signal is tied to investor attention and underreaction [108]. Loughran and McDonald later made the practical point that generic sentiment dictionaries misread financial language; in finance, words such as "liability" and "tax" are not automatically negative in the ordinary-language sense [78]. Ke, Kelly, and Xiu push the same lesson into supervised return prediction: the objective should be the investment task, not generic sentiment classification [72].

The industry version is RavenPack. Its importance is not that every client trades the same sentiment factor. It is that news analytics became a durable institutional product category: timestamped entity recognition, event classification, novelty, relevance, sentiment, and structured feeds that can enter quant models, economist dashboards, and risk workflows [89]. That is the right commercial analogue for this use case. The signal is useful only when it becomes structured enough for a decision system.

31.3.1 The Decision

The desk asks whether firm-specific news changes the next decision horizon: daily risk, weekly residual return, event follow-up, or review priority. A sentiment score that cannot change at least one of those decisions is a dashboard statistic, not a trading signal.

The natural sources are timestamped headlines, article bodies, newswire alerts, social-media streams when licensed and appropriate, and vendor event tags. The signal should separate tone from coverage. A company with ten neutral stories is not the same as a company with one negative litigation story, and neither is the same as a company that appears in sector-wide macro coverage.

31.3.2 The Anti-Pattern

The common failure is the universal sentiment bucket. Every article receives a positive, negative, or neutral label, the labels are averaged by ticker, and the highest-score stocks are bought. This looks scientific because the output is numeric. It fails because relevance, novelty, source duplication, event type, and horizon all matter.

The more subtle failure is treating mega-cap news as if it were underread. For the most liquid companies, short-horizon sentiment may decay before a realistic portfolio can trade it. The signal may still be useful for risk monitoring, but the return claim must survive latency, costs, and crowding.

31.3.3 The Playbook

The playbook begins with event-aware aggregation. Deduplicate related articles, keep the first credible timestamp, identify the company and event type, and compute sentiment conditional on relevance and novelty. Then test horizons separately. Intraday reaction, one-day drift, one-week drift, and one-month reversal are different claims.

The portfolio expression should start within sectors or industries so that a negative oil-price headline does not masquerade as negative firm-specific language. The benchmark stack should include a simple news-volume signal, a dictionary baseline, and a supervised model. If the

complex model cannot beat the simple baselines after costs, it belongs in monitoring rather than allocation.

Production controls are mechanical but important: source health, duplicate rates, entity-linking error rates, stale ticker mappings, publication-lag distributions, and performance by liquidity bucket. The signal is only as good as its timestamp discipline.

31.4 Earnings-Call Narrative Change

Earnings calls are not just disclosure events. They are structured conversations under pressure. Management chooses prepared remarks; analysts choose questions; both sides reveal what they believe investors need to model. That makes calls unusually useful for firm-level narrative measurement.

Published work gives this use case a strong foundation. Hassan, Hollander, van Lent, and Tahoun use earnings-call language to measure firm-level political risk and show that the measure relates to corporate actions and stock-market volatility [55]. Sautner, van Lent, Vilkov, and Zhang adapt the idea to climate-change exposure, separating opportunity, physical, and regulatory channels in calls across countries [93]. These are not generic text-mining exercises. They demonstrate that conference-call language can measure a firm-specific economic exposure that is not captured by a sector label.

31.4.1 The Decision

The investment question is whether a call changes the company's exposure, risk, or confidence score. In AI infrastructure, management may begin to quantify data-center revenue, discuss power constraints, or name memory bottlenecks. In clean energy, the same transcript machinery can reveal whether a company is speaking about subsidy eligibility, input-cost pressure, permitting delays, or margin compression.

Prepared remarks and Q&A should be modeled separately. Prepared remarks are controlled disclosure. Q&A is where analysts test the story. A repeated question about capex, backlog, pricing, or regulation is itself a signal of market uncertainty.

31.4.2 The Anti-Pattern

The weak version of the workflow reads the whole call as one bag of words. It then discovers that companies in the same industry use similar language. That is not a call signal; it is an industry classifier wearing a transcript costume.

The second failure is treating management optimism as truth. Managers learn which narratives investors reward. In a fashionable theme, the transcript can become part disclosure and part investor-relations theater.

31.4.3 The Playbook

Segment the transcript by speaker, section, and topic. Score prepared remarks, analyst questions, and management answers separately. Track novelty relative to the firm's own history and to its peer group. Store the passages that support each score, because the analyst should be able to inspect why the model changed the exposure.

The project dossier’s Microsoft narrative study is a useful template. It extracted firm-level narratives from Microsoft-related text between February 2021 and March 2025, then used ARIMA-X specifications with a U.S. economic-policy-uncertainty control to test whether narrative factors improved forecasts. The media-and-public-perception factor was positive in the decay-weighted specification, while currency/macro and technology/semiconductor ecosystem factors were negative; likelihood-ratio tests favored the narrative-augmented model over the baseline [46]. The important lesson is not that every coefficient should be copied. It is that the study connected a named narrative, a forecast object, a control variable, and a test against a baseline.

In production, call signals should flow into quarterly exposure updates, review queues, and risk flags. They should not automatically produce high-turnover trading unless the desk has shown that the post-call drift survives costs and factor controls.

31.5 Alternative Data as Narrative Validation

Many of the strongest industry cases are not pure text systems. They are alternative-data systems that test whether a narrative is visible in behavior. That matters for this book because text often arrives before hard fundamentals, but text is also easy to hype. Web traffic, app usage, transaction data, job postings, location data, satellite imagery, pricing data, and asset-level observations can tell the research team whether a narrative is becoming economic reality.

31.5.1 The Decision

The desk asks whether external activity data confirms, contradicts, or refines a text-derived exposure. If earnings calls say cloud demand is accelerating, cloud-spend datasets may test the claim. If a retailer describes resilient demand, transaction and web data may confirm it before the next quarter. If an energy narrative depends on shipping, production, or emissions activity, geospatial data may provide an independent view.

YipitData, Similarweb, Kayrros, Exabel, and Bloomberg’s alternative-data workflows are credible industry anchors for this use case. They represent different points in the workflow: YipitData and Similarweb provide company and digital-behavior indicators; Kayrros maps physical-world and environmental activity; Exabel helps investors combine and model alternative datasets; Bloomberg brings selected datasets into the terminal environment where many investment teams already work [125, 100, 71, 37, 17].

31.5.2 The Anti-Pattern

The failure is treating alternative data as magic confirmation. A rising web-traffic series may reflect curiosity rather than revenue. A transaction panel may be biased toward one customer segment. A satellite signal may capture activity but not margin. A job-posting spike may reflect replacement hiring, not growth. A vendor dashboard can make the data look cleaner than the economic inference really is.

The second failure is mixing frequencies without thought. A daily web series, monthly transaction panel, quarterly earnings call, and annual filing do not answer the same question. If the portfolio horizon is monthly, the signal construction must explain how high-frequency observations are aggregated, lagged, revised, and tested.

31.5.3 The Playbook

Use alternative data as an independent validation layer. Begin with a text-derived claim: demand acceleration, pricing pressure, hiring expansion, supply-chain bottleneck, traffic recovery, capacity shortage, or physical-risk exposure. Then choose the dataset that can observe the mechanism most directly. The model should store panel coverage, sampling bias, data revisions, vendor delivery time, missingness, and known breaks in methodology.

Portfolio use should be conservative at first. Activity data can raise or lower confidence, adjust exposure scores, prioritize analyst review, or trigger a scenario. It should become a direct alpha signal only after the research team tests it against fundamentals, returns, costs, and factor exposures.

This use case is especially useful for thematic investing because it prevents the naive narrative basket. A company that talks about AI infrastructure but shows no supporting capex, hiring, web demand, revenue exposure, or physical asset linkage receives lower confidence. A less promotional company whose activity data confirms the theme may deserve more attention.

31.6 SEC Filing Change Detection

Filings are slow, but they are legally meaningful. A company can promote a theme in an investor presentation with little consequence; changing a 10-K risk factor, business description, or MD&A section carries more weight.

Cohen, Malloy, and Nguyen’s “Lazy Prices” is the core reference for this use case. They show that active changes in regular filings predict future returns and operations, including future earnings, profitability, news, and bankruptcy risk [26]. For narrative-aware investing, the lesson is direct: a changed disclosure can be more informative than a high-frequency mention count. The industry counterpart is the rise of research workbenches that make filings, transcripts, expert calls, broker research, and company documents searchable at scale. AlphaSense is the clearest example in the report’s company map, and Kensho shows the same pattern inside a major data provider [4, 73].

31.6.1 The Decision

The desk asks whether a filing validates, weakens, or modifies a company’s theme exposure. If a cooling equipment company adds data-center demand to its business discussion, that may increase materiality confidence. If a clean-energy supplier expands risk-factor language around input costs and policy uncertainty, that may reduce position size or trigger analyst review.

31.6.2 The Anti-Pattern

The naive playbook treats all filing changes as equal. It compares whole documents, computes a distance score, and ranks companies by change. That can be useful for exploration, but it is too blunt for a portfolio. Boilerplate, formatting, legal cleanup, accounting-rule changes, and issuer-wide template revisions can all create false signals.

31.6.3 The Playbook

Parse filings by section and compare each section to the previous filing of the same type. Give special attention to business description, risk factors, MD&A, segment disclosure, liquidity,

customer concentration, and legal proceedings. Use semantic similarity to find changed passages, then classify whether the change relates to the theme, to general risk, or to unrelated disclosure hygiene.

The portfolio horizon should be slow. Filing-change signals are suited to quarterly rebalancing, exposure validation, and risk monitoring. The validation should test future residual returns, estimate revisions, volatility, bankruptcy risk, and event reactions. The as-of date is the filing acceptance and dissemination time, not the fiscal-period end date.

In production, the system should store the old passage, the new passage, the section label, the model's classification, and the reviewer decision. A filing signal without the actual changed text is not auditable.

31.7 Emerging Theme Discovery

Emerging-theme discovery is the case that attracts the most imagination and the most overfitting. The goal is to find a cross-sector narrative before it is fully named, packaged, and crowded. The danger is that every noisy cluster looks profound after the winners are known.

The literature offers both promise and caution. Bybee, Kelly, Manela, and Xiu show that news-topic attention can summarize macroeconomic states and forecast aggregate market returns [22]. Calomiris and Mamaysky show that topic, sentiment, frequency, and entropy in Reuters news can forecast risk and returns across countries [23]. These studies support the idea that topic attention is economically meaningful. They do not excuse retrospective theme naming.

31.7.1 The Decision

The decision is whether to promote a candidate cluster from monitoring to theme research. A promotion does not mean buying a basket. It means writing a theme specification, identifying candidate companies, and testing whether the narrative has economic structure.

The AI infrastructure case is the good version. "AI" alone is not a theme specification. The investable research question is whether specific sub-narratives—data-center power, high-bandwidth memory, networking, cooling, grid load, and baseload electricity—are diffusing across sources and linking to identifiable companies.

31.7.2 The Anti-Pattern

The failure is retrospective theme definition. The researcher observes that a set of stocks performed well, searches their historical text, and then names the common story. The backtest discovers the future because the human did.

Another failure is mistaking company clusters for themes. If every document in a topic mentions one famous semiconductor company, the model may have discovered a stock, not a cross-sector narrative.

31.7.3 The Playbook

Use a rolling or walk-forward discovery process. At each date, assign incoming documents to existing topic threads, buffer outliers, and measure novelty, attention acceleration, source breadth, entity breadth, and cross-sector diffusion. Mask entities during discovery when company names dominate, then reconstruct entity links after the topic is labeled.

The ForecastOS Hivemind macro-narrative study illustrates the portfolio version of this discipline. It tested 14 macro narratives from January 2020 to April 2026 on a top-500 U.S. equity universe, pre-orthogonalized narrative exposures against momentum, value, quality, leverage, and beta, and used a dual-gate activation rule. The industry-neutral narrative portfolio reported an information ratio of 1.49 with a 4.26% maximum drawdown, while a comparable momentum benchmark had an information ratio of 0.58 and much larger drawdowns; Fama-MacBeth tests were also positive for the narrative exposures [47]. As an unpublished practitioner study, it should be treated as evidence to inspect rather than authority to accept. Its methodological contribution is the design pattern: pre-orthogonalize, gate activation, compare against momentum, and report drawdown as well as return. The commercial analogue is the market-map and intelligence layer represented by Quid, CB Insights, PatSnap, FiscalNote, and Contify: these products help teams discover and organize emerging markets, technologies, companies, policy changes, and competitive movements, but the investment process still has to convert discovery into exposure, validation, and portfolio rules.

Production promotion requires a frozen theme card: name, economic mechanism, inclusion rules, exclusion rules, expected beneficiaries, expected losers, source evidence, first valid date, and kill conditions. Without that card, the theme is still a research note.

31.8 Supply-Chain and Value-Chain Mapping

Many thematic returns come from second-order exposure. The obvious winners are visible. The harder question is which suppliers, customers, infrastructure providers, and bottleneck assets participate in the economics of the theme. This is why value-chain mapping belongs in the practitioner track rather than in a footnote.

Hoberg and Phillips show that product-market similarity can be measured from firm text and used to form dynamic industry classifications [56]. Industry data products such as FactSet RBICS organize companies by granular business activities and revenue links rather than by broad sector labels [40]. For thematic investing, these tools are not substitutes for evidence. They are scaffolding.

31.8.1 The Decision

The desk asks whether a company should be treated as a pure play, supplier, enabler, adopter, beneficiary, disrupted firm, or hedge. In AI infrastructure, the distinction between a GPU designer, memory supplier, power provider, electrical-equipment manufacturer, cloud customer, and software adopter is not cosmetic. It determines the signal horizon, valuation anchor, risk control, and likely event sensitivity.

31.8.2 The Anti-Pattern

The common failure is graph centrality as thesis. A company appears central in a supplier graph, so the portfolio buys it. Centrality is not pricing power. It is not margin sensitivity. It is not revenue materiality. It may simply indicate that the company is large, disclosed often, or easy to link.

The LLM version of the same failure is worse: a model labels a company as an “AI infrastructure beneficiary” without retrieved evidence. The prose is fluent, but the evidence is missing.

31.8.3 The Playbook

Build value-chain roles from evidence, not labels. Retrieve passages from filings, calls, news, customer announcements, patent descriptions, segment data, and verified business descriptions. Extract relationships only when the source supports the edge. Store the passage, source, date, relation type, direction, confidence, and validity window.

Portfolio construction should cap exposures by role. A basket made only of high-beta pure plays will behave differently from a basket that includes bottleneck suppliers and infrastructure enablers. Role diversification is not a substitute for factor control, but it prevents the theme from collapsing into one obvious proxy.

Validation should use event reactions, revenue surprises, margin changes, customer announcements, and analyst-estimate revisions. If a named customer expands AI capex and the supposed supplier does not react, that is evidence against the edge or against its materiality.

31.9 Thematic ETF Replication and Improvement

Thematic ETFs are useful because they reveal how the industry packages narratives. They are dangerous because they can arrive late, hold diluted proxies, charge high fees, and attract flows after attention has already peaked.

Ben-David, Franzoni, Kim, and Moussawi show that specialized ETFs compete for investor attention and that newly launched specialized products tend to underperform after launch [9]. Index providers and data vendors nevertheless offer valuable methodological clues: MSCI publishes thematic exposure standards, Solactive describes NLP-based thematic index tools, and RBICS-style revenue classifications help separate broad proxies from economically exposed firms [82, 105, 40]. FactSet's acquisition of Truvalue Labs is another industry signal: large data platforms have absorbed AI-driven unstructured-text scoring because clients want external evidence, not only company self-reporting [38].

31.9.1 The Decision

The desk asks whether an existing ETF is a benchmark, a source of noisy labels, a crowding indicator, or an investable competitor. It should rarely be treated as truth.

31.9.2 The Anti-Pattern

The failure is training the model to reproduce ETF holdings and calling the result intelligence. ETF holdings reflect liquidity screens, index committee choices, marketing constraints, reconstitution schedules, and product design. A model that copies them inherits their timing and dilution.

31.9.3 The Playbook

Use ETF holdings as a benchmark and diagnostic. Compare model exposure scores with constituents and weights. Identify omissions, weak proxies, high-purity alternatives, crowded names, and factor concentrations. Ask whether the model improves at least one dimension: purity, timeliness, factor neutrality, capacity, turnover, or risk attribution.

The clean-energy 2020–2022 episode is the transition-investing case. The theme had real policy and technology support, but many public vehicles became concentrated in crowded growth

exposures at a moment when rates, valuation, and investor attention were turning. A good playbook would not have declared the theme false. It would have separated policy narrative from portfolio entry point, reduced crowded proxies, tested factor-adjusted performance, and required lifecycle gates before adding exposure. For a cleaner narrative-excess case, the metaverse ETF cycle or concept-stock SPACs do the job better: there the theme language often arrived before the cash-flow exposure was clear.

Production monitoring should track holdings overlap, ETF flows, valuation percentiles, attention percentiles, liquidity, short interest, and factor exposures. The ETF is not the strategy. It is the market's public packaging of the story.

31.10 Prediction-Market Event Overlays

Prediction markets are not a replacement for thematic equity research. They are event sensors. When a contract has clear resolution rules and enough liquidity, its price can summarize dispersed beliefs about a discrete event. The classic prediction-market literature explains why such markets can aggregate information, while also warning that prices are not frictionless truth [119, 104].

Modern venues make the use case more practical and more complicated. Kalshi operates as a CFTC-regulated event-contract exchange, while ICE's strategic investment in Polymarket shows that prediction-market data are moving toward institutional distribution [111, 62]. Dataminr represents the adjacent event-intelligence pattern: the market value comes from detecting and routing external events quickly, even when the output is an alert rather than a probability contract [30]. Those facts do not make every event feed tradable for equities. They make the data worth treating carefully.

31.10.1 The Decision

The desk asks whether a probability change should alter a theme's risk budget, hedge, or review priority. Export controls, elections, subsidy decisions, litigation outcomes, clinical approvals, and energy-policy events can all matter. The event only becomes investable after the value-chain map translates it into companies.

31.10.2 The Anti-Pattern

The failure is prediction-market overreach. A thin contract moves, the system treats the price as a clean probability, and the equity book trades as if the event exposure were obvious. That ignores liquidity, spreads, contract ambiguity, manipulation risk, jurisdiction, and mapping error.

31.10.3 The Playbook

Transform price changes into probability or logit changes, then filter by liquidity, spread, persistence, and contract clarity. Confirm with independent news or official sources when possible. Map the event to themes, roles, and companies through the exposure graph. Decide in advance whether the overlay changes gross exposure, hedges, review queues, or scenario weights.

Validation should use event windows and scenario analysis rather than ordinary monthly factor regressions. The post-event rule matters too. Once the event resolves, the probability signal disappears; the portfolio must decide whether the narrative consequence remains.

31.11 Attention, Crowding, and Contrarian Controls

Investor attention is both fuel and warning. Barber and Odean show that attention affects buying behavior by individual investors [8]. Da, Engelberg, and Gao use search activity as a direct attention proxy [29]. Shiller's narrative-economics framing reminds us that contagious stories can shape economic behavior, not merely describe it [99]. The trading problem is that attention has two regimes: early diffusion can be useful; late saturation can be dangerous.

31.11.1 The Decision

The desk asks whether attention should increase exposure, cap exposure, or trigger contrarian risk controls. Attention acceleration during early source diffusion may help discover a theme. Extreme attention after ETF launches, retail flows, valuation expansion, and media saturation may indicate crowding.

31.11.2 The Anti-Pattern

The failure is buying peak attention. The system identifies a theme when every source is talking about it, every issuer has added it to presentations, and every ETF provider has launched a product. The signal looks strong because the vocabulary is everywhere. The opportunity may already have moved into the price.

The Long Blockchain episode is the extreme version of the warning: a corporate name change connected a beverage company to a fashionable technology narrative and triggered enforcement attention after insider trading around the announcement [113]. Most thematic crowding is subtler, but the principle is the same. Language can be promotional.

31.11.3 The Playbook

Separate attention acceleration from attention saturation. Combine news volume, source breadth, search activity, social activity where appropriate, ETF flows, valuation, holdings overlap, short interest, and factor crowding. Classify the theme lifecycle before sizing the basket.

Early-stage attention can promote a research candidate. Mid-stage attention can support a portfolio if exposure scores and fundamentals confirm the mechanism. Late-stage attention should tighten caps, raise required evidence, reduce crowded proxies, or add hedges. In some cases it can support contrarian shorts, but only with borrow, timing, and squeeze risk explicitly modeled.

Clean energy belongs here as the constructive version of the same lesson. The error was not believing in decarbonization. It was failing to distinguish a durable transition narrative from a crowded public-equity expression at a fragile entry point. When the lesson is pure promotional excess, use Long Blockchain, metaverse, Web3 rebrands, or SPAC concept stocks instead.

31.12 The Operating Playbook

Across the cases, the workflow is the same even when the data differ.

1. Name the investment decision before selecting the model.
2. Identify the anti-pattern most likely to fool the research team.

3. Build the smallest credible signal that addresses that failure.
4. Store supporting evidence, timestamps, model versions, and human review.
5. Test against simple baselines and the right benchmark stack.
6. Neutralize sector and factor exposure before calling the result alpha.
7. Include costs, turnover, liquidity, capacity, and borrow.
8. Define lifecycle gates and kill rules before production.

The central lesson is practical: text does not become investable because it is interesting. It becomes investable when it changes a decision, survives a benchmark, fits a portfolio, and remains auditable after the story has stopped being exciting.

The final practitioner step is architecture: a machine that can produce these signals repeatedly without losing evidence, timing, or control.

Chapter 32

Architectural Patterns and System Design

32.1 The Weight That Had to Explain Itself

A portfolio manager looking at an AI infrastructure basket in 2024 might ask a simple question: why does this electrical-equipment company have a larger weight than a better-known semiconductor stock? A weak system answers with a score. A useful system answers with a chain of evidence: the documents that mentioned data-center power demand, the entities and relationships extracted from those documents, the sub-theme the company entered, the exposure score before and after factor neutralization, the liquidity cap, the analyst review, and the portfolio constraint that finally set the weight.

Architecture is where narrative research becomes investment control. It is not a back-office diagram drawn after the model is built. It decides whether the system can reproduce decisions, respect point-in-time constraints, monitor failures, and explain portfolio changes. In a narrative-aware thematic platform, every portfolio weight should be traceable backward to data, evidence, models, constraints, and human decisions.

The use cases become durable only when the machinery preserves timing, evidence, and accountability. Otherwise the system may produce a good answer once and lose the path that made the answer defensible.

32.2 Three Architectural Principles

The first principle is **separation of concerns**. Topic discovery, theme approval, exposure scoring, portfolio construction, and performance attribution are related, but they should not be the same service. A topic model should not silently choose portfolio weights. An optimizer should not silently rewrite exposure scores. A human override should not disappear into the next model run.

The second principle is **point-in-time persistence**. Every stage should produce durable artifacts: raw documents, cleaned sections, entity links, embeddings, topic assignments, theme versions, retrieved evidence, exposure scores, residualized scores, target weights, orders, and attribution reports. Temporary transformations that cannot be inspected later are acceptable in a notebook. They are not acceptable in production investing.

The third principle is **failure visibility**. A production system should reveal where it is breaking. Source latency, duplicate news, entity-linking drift, topic instability, stale evidence, review backlog, factor breaches, slippage, and alpha decay should trigger named alerts with owners and response rules. Monitoring is not a dashboard decoration. It is the system's memory of its own fragility.

32.3 Batch, Streaming, and Hybrid Workflows

The first design choice is tempo. Some tasks want batch research; others need live monitoring.

A **batch research pipeline** is appropriate when the desk is exploring a new corpus, validating a theme, running historical exposure studies, or preparing a periodic rebalance. It ingests a point-in-time corpus, cleans and sections documents, runs entity linking, generates embeddings, fits topic or classification models, scores exposure, constructs baskets, and evaluates performance against factors and costs. Its advantage is auditability. Its weakness is speed.

A **streaming monitoring pipeline** is appropriate when the desk monitors live news, social media, prediction markets, policy events, or abrupt narrative changes. It normalizes timestamps and entities as documents arrive, embeds incoming items, assigns them to topic threads or outlier buffers, updates attention and sentiment metrics, and triggers alerts for emergence, event probability shifts, or crowding. Its advantage is responsiveness. Its weakness is noise.

Most institutional systems need a **hybrid loop**. Batch research builds the theme library, validates models, and sets portfolio rules. Streaming monitoring updates attention, events, and review queues between scheduled rebalances. The two loops should share identifiers, theme versions, evidence stores, and scoring schemas. If the batch and streaming systems disagree because they use different entity maps or different theme definitions, the architecture has created a research problem.

32.4 Topic-First and Theme-First Patterns

A **topic-first** workflow starts from the corpus. It discovers clusters, tracks them through time, measures novelty and source breadth, and then asks which clusters deserve economic interpretation. This is the right pattern for emerging-theme discovery. It is also the dangerous pattern, because it can generate many elegant but uninvestable clusters.

A topic-first system should therefore include four gates. The first gate removes duplicate or source-driven clusters. The second checks whether the topic is broader than one company. The third asks for an economic mechanism. The fourth promotes only approved candidates into a point-in-time theme specification.

A **theme-first** workflow starts from an approved theme card. It retrieves evidence, scores company exposure, classifies value-chain roles, validates materiality, and builds baskets. This is the right pattern when the theme is already known: AI infrastructure, clean-energy equipment, obesity drugs, cybersecurity, reshoring, or defense modernization.

The failure of theme-first design is narrowness. If the theme card says “AI chips” and never allows the system to discover power, cooling, memory, or grid sub-themes, the architecture will miss the next ring of exposure. The hybrid solution is a discovery-to-scoring loop: topic-first systems propose new sub-themes, analysts approve or reject them, and theme-first systems score the companies that belong to the approved theme.

32.5 The Eleven-Stage Pipeline

The production pipeline can be described in eleven stages. The stages are not a rigid vendor architecture; they are control points. Each stage should have inputs, outputs, validation checks, and failure monitors.

1. **Ingest:** collect documents, prices, fundamentals, factors, ETF holdings, relationship data, and event data.
2. **Clean:** deduplicate, timestamp, section, normalize, translate where appropriate, and preserve raw-text references.
3. **Entity:** run financial named-entity recognition, entity linking, alias resolution, relationship extraction, and optional entity masking.
4. **Embed:** generate domain-aware document, sentence, section, or passage embeddings with versioned models.
5. **Discover:** identify topics, clusters, outliers, and candidate narrative threads.
6. **Track:** align topics through time, detect drift, split and merge threads, and maintain hierarchy.
7. **Reconstruct:** attach topics back to companies, roles, sources, and evidence after discovery.
8. **Score:** estimate company exposure, materiality, confidence, lifecycle stage, and value-chain role.
9. **Residualize:** pre-orthogonalize exposure scores against sectors, industries, and style factors.
10. **Optimize:** convert residual exposure into implementable weights under liquidity, cost, turnover, concentration, and risk constraints.
11. **Attribute:** explain returns, risk, costs, factor exposures, drift, overrides, and failures.

The survey material behind this book recommends the same broad flow: multi-source text streams enter entity-aware processing, semantic modeling, emerging-theme detection, hierarchical refinement, company exposure scoring, value-chain mapping, pre-orthogonalization, portfolio optimization, factor immunization, and event overlays. The important design choice is not the exact model brand at each step. It is preserving the chain from source document to traded weight.

32.6 The Ingestion Layer

The ingestion layer should treat every input as a point-in-time record. For documents, minimum fields include source identifier, source document identifier, source timestamp, vendor receipt timestamp, system ingestion timestamp, language, document type, revision identifier, license category, and raw-text checksum.

The same discipline applies to prices, fundamentals, factors, ETF holdings, benchmark constituents, relationship data, analyst labels, and prediction-market contracts. The system

should preserve historical versions rather than overwrite records silently. A current ticker map, current ETF holding file, or current factor model cannot be allowed to rewrite the past.

The most common ingestion failures are mundane: delayed feeds, duplicate syndication, missing body text, revised filings, stale ticker mappings, unlicensed redistribution, and broken contract metadata. They are mundane in engineering terms and lethal in backtests. If the system cannot say when it received an item, it cannot say whether a simulated decision could have used it.

32.7 The Document Processing Layer

Document processing includes deduplication, boilerplate removal, section parsing, language handling, transcript segmentation, and normalization. This layer should never destroy the link to the raw source. A cleaned paragraph should know where it came from.

The processing logic should differ by source. Filings need section parsing and amendment handling. Earnings calls need speaker labels, prepared-remarks versus Q&A separation, and analyst-management role identification. News needs headline-body linkage, duplicate clustering, source priority, and event-time handling. Prediction-market data need contract definition, resolution criteria, venue, price, spread, depth, and volume.

Minimum tests include duplicate-detection samples, section-parser validation, timestamp-ordering checks, language-detection checks, empty-document alerts, truncation alerts, and source-volume anomaly alerts. Bad processing produces bad themes. A topic model cannot rescue a corpus dominated by duplicate headlines, stale company descriptions, or boilerplate risk factors.

32.8 Entity and Relationship Services

Entity linking connects language to tradable securities. It should be a shared service, not an afterthought inside each model. The service should resolve company names, aliases, tickers, share classes, subsidiaries, products, executives, geographies, customers, suppliers, and regulators where they matter to the use case.

It should also support two opposite operations. During discovery, entity masking can prevent a topic model from clustering by famous company names rather than by economic narrative. After discovery, entity reconstruction attaches the topic back to companies and relationships so the portfolio can act.

Relationship extraction belongs beside entity linking because many themes travel through edges rather than nodes. AI infrastructure exposure may travel from cloud capex to chip designers, memory suppliers, power providers, cooling vendors, grid equipment firms, and construction contractors. A company should not receive value-chain exposure merely because it appears near a theme. The system should store the passage, relation type, direction, source, date, confidence, and validity interval.

The failure modes are predictable: aliases attach to the wrong issuer, ticker changes rewrite history, subsidiaries are mapped to the wrong parent, entity masking removes economically meaningful product names, and relation extraction turns co-mention into causality. Each of those failures needs sampling and review.

32.9 Embeddings, Topics, and Theme Objects

Embeddings are model outputs, not neutral data. The system should record the embedding model, version, tokenizer, fine-tuning corpus, dimensionality, normalization method, inference date, code version, and random seed where relevant. Topic models, classifiers, prompts, retrieval settings, scoring models, and optimizers belong in the same model registry.

For long-form corporate documents, contextual topic models, BERTopic-style workflows, and section-level embeddings can work well because filings and transcripts have enough semantic mass. For short-text feeds, headline sparsity often favors models designed for short text or workflows that aggregate related items before topic modeling. For dynamic narratives, chain-free tracking and hierarchical refinement are useful because themes can drift gradually, split into sub-themes, or appear suddenly.

The architecture should separate a **topic** from a **theme**. A topic is a statistical object. A theme is an approved investment object. A theme object should contain a name, version, narrative description, inclusion and exclusion rules, lifecycle stage, representative documents, source breadth, entity breadth, supporting and contradictory evidence, expected horizon, approval status, and review history.

This distinction prevents one of the central mistakes of thematic research: treating an interpretable cluster as an investable idea. The promotion from topic to theme is an investment decision, and the system should record who made it, when, and why.

32.10 Retrieval-Backed Exposure Scoring

Exposure scoring transforms a theme into company-level evidence. The service should output exposure, materiality, confidence, value-chain role, supporting passages, model components, analyst overrides, and validity intervals. A final score without a decomposition is not adequate.

A practical exposure score may combine several inputs:

- revenue or business-segment evidence;
- keyword and phrase evidence;
- embedding similarity to the theme specification and representative documents;
- filing-change and transcript-change evidence;
- relation-extraction evidence from value-chain documents;
- LLM classification grounded in retrieved passages;
- analyst review and override history.

The service should distinguish exposure from confidence. A company with one exciting article and no revenue evidence may have high possible exposure but low confidence. A slower company with repeated filings, segment discussion, and named customers may have less glamorous exposure but higher confidence. The optimizer should not be forced to pretend these cases are equally known.

LLMs belong here only under evidence discipline. The prompt should receive retrieved passages, a schema, allowed labels, uncertainty options, and instructions to cite the evidence used. The output should store the prompt, model version, retrieved context, structured response,

and reviewer outcome. A language model that classifies from memory is not a production exposure-scoring service.

32.11 The Residualization and Portfolio Layer

The portfolio layer should be separate from the text layer. It accepts exposure scores and produces target weights under explicit constraints. This separation prevents the NLP system from silently optimizing for raw returns and prevents the optimizer from hiding a weak theme behind factor exposure.

The residualization step should occur before basket construction. Cross-sectional exposure scores can be regressed on sector, industry, beta, size, value, profitability, investment, quality, momentum, leverage, and any other risk dimensions used by the desk. The residual becomes the cleaner thematic signal. This does not guarantee alpha. It prevents the strategy from calling a sector or style bet by a narrative name.

The optimizer should then apply universe filters, liquidity constraints, borrow constraints, single-name caps, sector and region bounds, value-chain role caps, sub-theme caps, turnover limits, transaction-cost estimates, rebalance calendars, and kill rules. It should output target weights, expected trades, factor exposures, active risk, turnover, cost estimates, binding constraints, and benchmark comparisons.

A production portfolio service should fail closed. If an input feed is stale, a factor exposure breaches bounds, a cost estimate is missing, or a required review is incomplete, the system should not produce trades as if nothing happened. Silence is not resilience.

32.12 Human Review as a Service

Human review should be routed by priority. The system should not ask analysts to review everything, and it should not let high-impact uncertainty pass unreviewed. Priority can be based on novelty, acceleration, expected portfolio impact, disagreement among models, low confidence, high turnover effect, or proximity to a risk limit.

A review item should include the model output, evidence package, suggested decision, uncertainty, affected companies, affected portfolios, required action, and deadline. The analyst's decision should store reviewer, timestamp, reviewed documents, approval or rejection, confidence, override reason, validity window, and downstream authority.

This design has a second benefit: review decisions become training data. Overrides reveal where the model is wrong, where the rubric is unclear, and where the theme specification needs to change. Without that record, human judgment improves one decision and teaches the system nothing.

32.13 Monitoring and Incident Response

Monitoring should cover five layers: data, models, themes, portfolios, and operations.

Data monitors track source latency, missing data, duplicate rates, volume anomalies, licensing flags, revised documents, and ingestion failures. Model monitors track embedding drift, classifier drift, prompt output changes, retrieval quality, topic stability, split-merge events, and exposure-score drift. Theme monitors track lifecycle stage, source breadth, entity breadth, attention saturation, contradictory evidence, and analyst-review backlog. Portfolio monitors track factor

exposure, sector concentration, turnover, cost, slippage, liquidity, borrow, drawdown, crowding, and residual alpha decay. Operational monitors track job failures, stale artifacts, approval bottlenecks, and broken downstream dependencies.

Each alert should have severity, owner, response time, escalation path, and allowed actions. A source-latency alert may pause new signal generation. An entity-linking failure may quarantine affected documents. A factor breach may block trades. A review backlog may reduce position sizes for low-confidence names. An alpha-decay alert may trigger a research review rather than an immediate liquidation.

Incident response should also preserve evidence. When a stage breaks, the system should record affected themes, companies, portfolios, artifacts, and decisions. The goal is not only to restore service. It is to know which investment decisions were made under degraded information.

32.14 Choosing the Architecture

Different use cases require different patterns:

Use Case	Primary Pattern	Critical Control
Emerging theme discovery	Topic-first hybrid	Promotion gate and point-in-time theme card
Exposure mapping	Theme-first scoring	Retrieved evidence and materiality confidence
Earnings-call change	Batch plus event update	Speaker segmentation and call availability time
Filing-change detection	Batch quarterly	Section-level diffs and filing dissemination time
Value-chain mapping	Graph-backed scoring	Evidence-backed edges and role caps
ETF improvement	Theme-first benchmark loop	Holdings as noisy labels, not truth
Prediction-market overlay	Streaming event monitor	Liquidity, contract clarity, and event-to-company map
Crowding control	Monitoring loop	Lifecycle stage, flows, valuation, and factor exposure

This table is a design guide, not a taxonomy for its own sake. Architecture should express the investment process. If the design is unclear, the investment thesis is probably unclear too.

32.15 AI Infrastructure as a System Test

AI infrastructure is a good system test because it forces the architecture to handle discovery, scoring, graph exposure, factor control, and crowding at the same time.

The topic service may first detect general AI language. That is not enough. The tracking layer should split the parent narrative into sub-themes: data-center power, cooling, networking, high-bandwidth memory, advanced packaging, cloud capex, electrical equipment, grid constraints, and baseload generation. The theme service should decide which sub-themes belong under the same parent and which deserve separate baskets.

The exposure service should then score companies by role: pure play, supplier, enabler, adopter, beneficiary, disrupted firm, or hedge. A semiconductor designer, memory supplier, power producer, cooling vendor, and electrical contractor should not receive the same explanation merely because all are near the AI narrative. The value-chain graph should show how each company connects to the theme.

The portfolio service should cap concentration in mega-cap technology, semiconductors, single suppliers, and crowded proxies. It should report performance before and after factor neutralization. The monitoring layer should watch whether the basket becomes a generic momentum trade, whether source breadth narrows, whether attention reaches saturation, and whether the second-order beneficiaries still respond to relevant events.

The architecture succeeds only if a portfolio manager can inspect a current weight and see the full path: source documents, extracted entities, sub-theme membership, exposure score, uncertainty, residualization, optimization constraints, human review, and governance approval.

32.16 Clean Energy as a Transition-Theme Test

Clean energy in 2020–2022 tests the architecture without asking the reader to dismiss the transition. The narrative was real, but many public-equity expressions became crowded, expensive, and factor-contaminated. A weak architecture would show rising attention and rising exposure scores, then keep buying. A stronger architecture would show the conflict: policy support and technology adoption on one side, valuation pressure, ETF flows, rate sensitivity, crowded ownership, and weakening residual performance on the other.

That conflict should trigger design behavior. The theme remains valid, but the portfolio expression changes. The optimizer reduces crowded proxies, raises evidence thresholds, shifts from broad enthusiasm to material exposure, tightens turnover and liquidity controls, and requires new validation before adding risk. The system does not need to be cynical about the narrative. It needs to be honest about the trade.

32.17 The Architectural Standard

The standard for this book is simple to state and demanding to implement: every investment decision should be reproducible, explainable, and falsifiable.

Reproducible means the system can reconstruct what data, models, parameters, prompts, factor files, constraints, and human decisions were available at the time. Explainable means the system can show why a theme, company, score, and weight existed. Falsifiable means the system has monitoring and kill rules that can show when the thesis is no longer behaving as expected.

The architecture is not successful because it uses fashionable models. It is successful when it turns narrative evidence into controlled investment decisions and leaves a trail strong enough for research, risk, operations, and governance to inspect.

Part VIII

Testing, Validation, and Production

Chapter 33

Backtesting Without Fooling Yourself

33.1 The Backtest That Knew the Ending

Imagine a research team in 2026 asking an LLM to classify 2019 filings for exposure to “generative AI infrastructure.” The model returns polished labels. It recognizes that data-center power, high-bandwidth memory, advanced packaging, and liquid cooling all belong to the same investment story. The backtest looks excellent. The portfolio buys the future winners early, avoids weak proxies, and reports a clean residual alpha.

The problem is not that the model is foolish. The problem is that it is too well educated. A current LLM may have absorbed the vocabulary, winners, regulatory debates, and market outcomes that made the theme obvious years later. If the 2019 decision did not have access to that knowledge, the backtest has not discovered foresight. It has measured hindsight with better grammar.

This is the central danger in narrative-aware investing. Price backtests can leak the future through returns, survivorship, and fundamentals. Text backtests can leak the future through documents, topic labels, embedding spaces, company mappings, analyst reviews, LLM training data, ETF holdings, and narrative definitions. A backtest is credible only when it reconstructs decisions that could have been made at the time.

The organizing rule is the *as-of invariant*: every input to every simulated decision must have been available, in its recorded form, before the simulated decision time.

33.2 What a Backtest Is For

A backtest is not a machine for producing attractive charts. It is a controlled reconstruction of a decision process. For this book, the question is not “would today’s best explanation have worked historically?” The question is “could a point-in-time system have identified, scored, sized, traded, and governed the theme using only what was then available?”

That standard changes the object of testing. The backtest does not test a stock list. It tests a workflow:

- which documents were available;
- how entities were linked;
- how topics were discovered and labeled;

- when a theme was approved;
- how companies were scored;
- which human reviews occurred;
- how scores were neutralized;
- how weights were optimized;
- how trades were simulated;
- how performance was attributed.

If those objects cannot be reconstructed, the result may still be useful as historical analysis. It should not be presented as evidence of a tradable strategy.

33.3 The Decision Object

At each rebalance date, the backtest should construct a complete decision object. The object should include the eligible universe, the point-in-time corpus, the entity map, the theme library, exposure scores, supporting evidence, factor files, risk estimates, portfolio constraints, transaction-cost assumptions, target weights, simulated trades, fills, and realized returns.

This object is the unit of reproducibility. It lets the researcher ask: why did the portfolio own this company on this date? A serious answer should trace the weight backward through evidence and forward through performance.

For narrative-aware systems, the decision object should also store theme versions. A theme is not timeless. “Clean energy” in 2020, “clean energy after the Inflation Reduction Act” in 2022, and “profitable grid-equipment suppliers under higher rates” in 2024 are not the same specification. “AI” before the generative-AI boom is not the same object as AI infrastructure after the market has learned to separate GPUs, memory, cooling, power, and cloud capex.

33.4 Timestamps Are Data

Financial text has more than one date. The event date, document creation date, publication date, vendor ingestion date, revision date, system-processing time, analyst-review time, and portfolio-decision time can all differ.

The backtest must use availability time, not narrative time. A filing accepted after the market close cannot influence a trade simulated before the close. A news article revised at noon cannot appear in its revised form at the open. ETF holdings published with a delay cannot be used on the holdings date. A transcript corrected by the vendor after publication cannot be treated as if the corrected text was available immediately.

The minimum timestamp record is:

- source timestamp;
- vendor receipt timestamp;
- system ingestion timestamp;
- processing timestamp;

- correction or revision timestamp;
- analyst-review timestamp;
- decision timestamp.

When the timestamps conflict, use the latest plausible availability time. This conservative rule will sometimes discard usable information. That is preferable to admitting future information quietly.

33.5 Universe, Corporate Actions, and ETF Holdings

The eligible universe must be point-in-time. A backtest that uses today's surviving companies overstates performance if failed, delisted, acquired, or bankrupt firms are missing. Themes are especially vulnerable because weak companies often disappear during theme busts.

The universe record should include listings, delistings, delisting returns where available, ticker changes, share-class mappings, mergers, acquisitions, spin-offs, historical sectors, historical industries, liquidity, borrow availability, and trading halts. Corporate identity should not be rewritten backward with today's ticker or business description.

ETF holdings create a separate look-ahead problem. A thematic ETF can be useful as a benchmark or noisy label, but holdings are published with timing conventions and delays. The backtest may use only the holdings file that was actually available as of the simulated decision. It may not use a later constituent list to define an earlier theme.

The ETF problem is also conceptual. Specialized ETFs often launch after investor attention has already risen, and their holdings can reflect liquidity, marketing, and index-design constraints. Ben-David, Franzoni, Kim, and Moussawi's evidence on specialized ETF underperformance after launch is a useful warning: an ETF can be a market artifact of late attention, not an early map of opportunity [9].

33.6 Leakage in Topic Models

Dynamic topic models can leak the future even when returns are handled correctly. A model fit on the full corpus can define a historical topic using later vocabulary, later documents, later cluster boundaries, and later sub-theme splits. A 2020 topic labeled after reading 2024 documents is not a 2020 signal.

A walk-forward topic backtest should use one of four designs:

- a model trained only on documents available before the decision time;
- a frozen model trained before the test period;
- a rolling model whose window ends before the decision time;
- a streaming model updated only with available documents.

Topic labels need the same discipline. If analysts rename or split a topic after future documents reveal its meaning, the system should store the new label with its approval date. The backtest may use the old label before that date and the new label after it. It may not pretend that the final taxonomy existed from the beginning.

This is where entity masking can also mislead. If names are masked for discovery and reconstructed later, the reconstruction must use point-in-time entity maps. Today's knowledge that a company became central to AI infrastructure cannot be used to attach old documents to a future theme without evidence available then.

33.7 A Leakage Framework for LLMs

LLMs introduce five leakage channels.

Model-memory leakage occurs when the model has learned facts, terminology, or outcomes from after the simulated date. A 2026 model may know which firms later became AI winners, even if the prompt contains only old documents.

Prompt leakage occurs when the researcher gives the model future labels, future examples, future performance, or future theme definitions. Asking a model to classify 2019 filings using a 2024 theme description can contaminate the test even if the retrieved documents are old.

Retrieval leakage occurs when the retrieval system searches a current corpus or current company profile instead of a point-in-time corpus. The model then grounds its answer in evidence that was not available.

Label leakage occurs when current analyst judgments are applied backward. A 2026 analyst may correctly know that a utility was an AI power beneficiary, but that judgment cannot become a 2021 label unless the evidence and approval existed in 2021.

Evaluation leakage occurs when thresholds, prompts, schemas, or model choices are selected after observing the test-period returns.

The controls are practical:

- use retrieval-constrained prompts with point-in-time evidence;
- forbid future performance, future labels, and future theme descriptions in prompts;
- store prompt, model identifier, retrieved passages, schema, output, and review decision;
- compare LLM outputs with simpler baselines;
- evaluate on post-cutoff or live periods whenever possible;
- separate exploratory historical labeling from tradable historical validation.

Lopez-Lira and Tang's GPT-4 headline study is a useful benchmark for the spirit of this control because it evaluates the model on headlines after the model's training cutoff [76]. Not every investment problem can use that exact design, but the principle is important: test the model where memory is least able to masquerade as forecasting.

Leakage Channel	Typical Symptom	Control
Model memory	Fluent labels using future vocabulary	Post-cutoff tests and retrieval-only claims
Prompt leakage	Current theme definition applied backward	Versioned theme cards and dated prompts
Retrieval leakage	Current documents retrieved for old dates	Point-in-time corpus filters
Label leakage	Current analyst labels used historically	Label creation timestamps
Evaluation leakage	Best prompt chosen after returns	Frozen test design and walk-forward validation

33.8 Discovery, Calibration, and Test

Narrative systems need a strict separation between discovery, calibration, and test.

Discovery is where researchers explore corpora, candidate themes, models, and mechanisms. It is allowed to be creative. Calibration is where thresholds, horizons, rebalance frequency, constraints, prompts, and review gates are chosen. Testing is where the frozen design is evaluated.

A walk-forward process repeats these steps through time. Parameters may update, but only by a rule that would have been known in advance. For example, a model may recalibrate exposure thresholds annually using the prior three years. It may not choose the threshold that would have worked best next year.

The production record should store the research state at each promotion gate: exploratory hypothesis, validation design, frozen specification, paper-trading configuration, live configuration, and retirement decision. Without those records, the team cannot later tell whether the system improved honestly or overfit quietly.

33.9 Multiple Testing and Researcher Degrees of Freedom

Thematic research offers many choices: corpora, source filters, topic models, embedding models, prompts, thresholds, horizons, universes, weighting schemes, factor controls, rebalance schedules, capacity assumptions, and benchmark definitions. If the researcher tries enough combinations, one will look good by chance.

This is the data-snooping problem. White's reality check and Hansen's superior predictive ability test are classic tools for asking whether the best-performing rule is genuinely superior after considering the search across alternatives [118, 53]. Harvey, Liu, and Zhu make the same point for factor discovery: statistical thresholds should rise when many candidates are tested [54]. Bailey, Borwein, Lopez de Prado, and Zhu frame the problem as the probability of backtest overfitting after searching a large strategy space [7].

The practical rule is simple: record the search. A research memo should list the alternatives tried, not only the winner. If the strategy survives only because the team tried twenty horizons and reported one, the result is weaker than it looks.

33.10 Benchmarks, Placebos, and Falsification

A credible backtest should try to disprove itself.

Useful falsification tests include:

- delayed signals that should arrive too late;
- shuffled theme labels;
- placebo themes with similar document volume but no economic mechanism;
- random baskets matched by sector, size, liquidity, and factor exposure;
- alternative rebalance dates;
- exclusion of the largest winners;
- tests by subperiod, region, sector, liquidity bucket, and regime;

- comparison with simple baselines such as keyword counts, news volume, or revenue screens.

The benchmark stack should be defined before results are reviewed. A long-only AI infrastructure basket should be compared with broad equity, technology, semiconductors, growth, and relevant thematic ETFs. A factor-neutral long-short strategy should be compared with cash, sector-neutral random baskets, and factor-neutral baselines. A prediction-market overlay should be evaluated around events and scenarios, not only by monthly return regression.

If the real strategy cannot beat matched placebos after costs and factor controls, the theme may be a label attached to ordinary risk exposure.

33.11 Costs, Borrow, Slippage, and Capacity

Theme strategies can look excellent before implementation costs because they trade names with changing attention, rising volatility, widening spreads, and crowded flows. Costs should be modeled before conclusions are drawn.

At minimum, the backtest should include commissions, fees, bid–ask spreads, market impact, borrow costs, locate availability, financing, turnover, delayed execution, and trade-size limits relative to average daily value traded.

Capacity should be reported as a curve, not a single number. A strategy may be attractive at \$50 million, weaker at \$500 million, and impossible at \$5 billion. The curve should reflect liquidity, participation rate, concentration, borrow, spread, volatility, and crowding. A realistic backtest should also measure overlap with thematic ETFs, popular baskets, and high-attention names.

The rule is blunt: if expected alpha disappears at realistic size, the backtest has discovered a research signal, not an investable product.

33.12 The Backtest Acceptance Checklist

A thematic backtest should not advance to paper trading unless the following are true:

- all documents, labels, scores, model versions, prompts, reviews, and weights are point-in-time;
- the eligible universe includes delisted names where relevant;
- dynamic topic models are trained or updated without future documents;
- LLM outputs are evidence-grounded and stored with retrieved context;
- ETF holdings, benchmark constituents, and factor files use availability dates;
- transaction costs, borrow, slippage, financing, and capacity are modeled;
- performance is shown raw, benchmark-relative, factor-adjusted, and cost-adjusted;
- placebo tests and matched random baskets are reported;
- sensitivity to thresholds, horizons, prompts, and rebalance frequency is documented;
- the economic mechanism and falsification conditions were written before promotion.

The checklist is not a bureaucratic ritual. It is the difference between a strategy that could have lived in time and a story written after time has done the hard work.

33.13 AI Infrastructure as a Backtest

For AI infrastructure, the backtest must reconstruct when each sub-theme became knowable. High-bandwidth memory, data-center power, liquid cooling, networking, advanced packaging, cloud capex, and baseload electricity should not be treated as one static theme throughout history unless that definition existed at the time.

The system should record when each sub-theme was detected, when analysts approved it, which companies had supporting evidence, which revenue or value-chain data were available, and how the basket would have been traded. Performance should then be reported against broad equity, technology, semiconductor, growth, and thematic ETF benchmarks.

The decisive question is not whether today's obvious AI infrastructure winners performed well in the past. The question is whether a point-in-time system could have identified a tradable set of beneficiaries before the market fully incorporated the narrative.

33.14 Clean Energy as a Point-in-Time Test

Clean energy is a point-in-time discipline test. A retrospective model can explain the 2020 surge and the 2021–2022 disappointment neatly: policy enthusiasm, falling technology costs, ETF flows, crowded growth exposure, rising rates, margin pressure, and valuation compression. That explanation is useful history. It becomes a valid backtest only if the system could have observed those conditions as they developed.

The clean-energy backtest should therefore ask when attention became extreme, when ETF flows accelerated, when valuation moved beyond historical ranges, when rate sensitivity rose, when input costs changed, and when residual performance weakened after factor controls. The trade was not simply “clean energy good” or “clean energy bad.” It was a lifecycle and portfolio-expression problem. A credible backtest has to show that the lifecycle signals were available before the drawdown explained them.

33.15 The Standard

The standard is severe because the subject demands it. Narrative-aware systems can connect information before conventional maps do. They can also connect future explanations to past decisions and call the result alpha.

A backtest is credible when it can answer four questions:

1. What exactly did the system know at the decision time?
2. What rule transformed that information into weights?
3. What benchmarks, costs, and falsification tests did the rule survive?
4. What evidence would have stopped the strategy from advancing?

If those questions have clear answers, the backtest can become evidence. If they do not, it remains an interesting story.

Chapter 34

Performance Measurement, Attribution, and Governance

34.1 The Strategy That Made Money for the Wrong Reason

A thematic portfolio can have a very good quarter for an embarrassing reason. Imagine an AI infrastructure basket in 2024 that beats its benchmark by several hundred basis points. The meeting begins well. The headline return is strong, the story is fashionable, and the largest positions have familiar names: semiconductors, cloud platforms, power equipment, data-center real estate.

Then the attribution arrives. Most of the active return came from one mega-cap semiconductor position, a long growth tilt, and a short exposure to utilities that happened to work before the market rediscovered electricity demand. The HBM suppliers contributed less than expected. The cooling and power-grid names barely mattered. The model had called the portfolio “AI infrastructure,” but the realized return looked closer to “large-cap growth plus one stock.”

That does not mean the strategy failed. It means the first performance number was not an explanation. In narrative-aware investing, measurement has to answer two questions at once: what did the portfolio earn, and did it earn that return through the mechanism the strategy claimed to exploit?

Governance begins at exactly that point. The same evidence trail that explains performance also defines accountability. Who approved the theme? Which signal changed the exposure score? Which analyst overrode the model? Which benchmark made the active return meaningful? Which risk limit allowed the position? A performance report without governance is a scorecard without custody of the decision. A governance process without attribution is theater.

34.2 Performance Is Evidence, Not Verdict

The clean-energy trade after 2020 shows why raw performance cannot carry the burden alone. A portfolio could have earned impressive returns during the policy-and-liquidity surge, then lost much of the advantage when rates rose and long-duration growth stocks sold off. The same theme name covered two different investment environments.

Performance measurement should therefore decompose the result into layers:

- market and benchmark exposure;

- sector, industry, and factor exposure;
- theme, sub-theme, and company-role contribution;
- source contribution from filings, calls, news, specialist research, and alternative data;
- implementation cost, turnover, liquidity, borrow, and capacity;
- residual return after known risks have been explained.

The purpose is not to force every thematic strategy into factor neutrality or to punish every exposure that can be named. Investors may choose to own growth, semiconductors, utilities, or small caps. The point is consent. A client who buys an AI infrastructure strategy should know whether the return came from the AI infrastructure mechanism, from a sector bet, from a factor bet, or from a few stocks that happened to sit inside the theme.

34.3 Core Return Metrics

The first line of any report is still return. If the portfolio did not make money after costs, the elegance of the topic model does not rescue it. But return needs a denominator, a horizon, and a cash alternative.

Let $R_{p,t}$ be the portfolio return in period t and $R_{f,t}$ the risk-free rate. Excess return is

$$r_{p,t} = R_{p,t} - R_{f,t}. \quad (34.1)$$

This is the return earned above cash. A strategy that returns 6 percent when cash yields 5 percent has not done the same job as a strategy that returns 6 percent when cash yields zero. The formula is simple because the question is simple: how much did the investor earn for accepting risk?

The core report should include annualized return, annualized volatility, maximum drawdown, hit rate, skewness, tail losses, turnover, transaction costs, borrow costs where relevant, and capacity-adjusted return. The Sharpe ratio then compresses one part of that picture:

$$\text{Sharpe} = \frac{\mathbb{E}[r_p]}{\sigma(r_p)}. \quad (34.2)$$

The numerator is average excess return. The denominator is the volatility of excess return. Sharpe made this trade-off central to performance evaluation, and it remains useful because it forces return to be read against risk [96]. It is also blunt. It can hide crash exposure, liquidity risk, stale pricing, and a strategy that earns quietly for many months before giving the gains back in one crowded unwind.

For thematic portfolios, the distribution matters. A clean-energy basket that rises steadily during ETF inflows and then falls sharply when discount rates move higher does not have the same risk profile as a slow-moving value portfolio with the same annualized volatility. The path tells the risk story that the average cannot.

34.4 Benchmark-Relative Metrics

A benchmark is a confession about the decision being evaluated. A long-only AI infrastructure strategy compared only with cash will almost always look better in an equity bull market than

it deserves. The same strategy compared only with the Nasdaq 100 may look worse than it deserves if it deliberately owns power equipment, utilities, industrials, and real estate alongside technology.

For a benchmark return $R_{b,t}$, active return is

$$a_t = R_{p,t} - R_{b,t}. \quad (34.3)$$

Tracking error is the volatility of that active return:

$$\text{TE} = \sigma(a_t). \quad (34.4)$$

The information ratio is

$$\text{IR} = \frac{\mathbb{E}[a_t]}{\text{TE}}. \quad (34.5)$$

The information ratio asks how much active return the strategy earned per unit of active risk. Grinold and Kahn place this benchmark-relative view at the center of active management because it evaluates skill against the opportunity set the manager actually chose [50].

The benchmark stack should match the mandate. A long-only AI infrastructure basket should be compared with broad equity, technology, semiconductors, growth, infrastructure, and relevant thematic ETFs. A factor-neutral long-short strategy should be compared with cash, sector-neutral random baskets, and factor-neutral baselines. A prediction-market overlay should be judged around the event risks it was designed to hedge or exploit, not only by a monthly equity benchmark.

34.5 Alpha and Factor Attribution

The most dangerous phrase in a performance meeting is “the theme worked” when the factor report says something else. During the clean-energy boom, many portfolios looked like climate alpha until rate sensitivity, high valuation, small-cap exposure, and growth exposure were separated from the environmental story.

A basic factor regression starts by explaining portfolio excess return with known risk factors:

$$R_{p,t} - R_{f,t} = \alpha + \beta_{\text{MKT}}\text{MKT}_t + \beta_{\text{SMB}}\text{SMB}_t + \beta_{\text{HML}}\text{HML}_t + \beta_{\text{MOM}}\text{MOM}_t + \epsilon_t. \quad (34.6)$$

Here β_{MKT} measures market exposure, β_{SMB} size exposure, β_{HML} value exposure, and β_{MOM} momentum exposure. The residual ϵ_t is the part not explained by those factors, and α is the average residual return in the regression. The Fama-French and Carhart traditions remain useful because they turn vague claims about skill into explicit claims about exposure [41, 24, 42].

Production reporting should show full-period and rolling alpha, rolling factor loadings, sector and industry contributions, residual return after factor hedging, and alpha statistics with caution about multiple testing. A theme that earns return only when its growth beta is high has not necessarily failed; it may be a mandate that intentionally accepts growth exposure. But the report should make that choice visible.

34.6 Theme, Role, and Source Attribution

AI infrastructure is not one economic object. It contains GPU designers, foundries, advanced-packaging suppliers, high-bandwidth-memory producers, networking firms, cooling specialists, cloud platforms, data-center landlords, power generators, electrical equipment manufacturers, and software adopters. If a portfolio owns all of them, the theme-level return can hide the real source of profit.

Theme attribution decomposes performance by theme and sub-theme. For each group, the report should show contribution to return, contribution to risk, average exposure score, confidence-weighted exposure, lifecycle stage, source breadth, attention level, valuation, and crowding. A strategy that appears diversified across AI infrastructure may still depend almost entirely on the semiconductor branch of the tree.

Company-role attribution adds the economic map built in Part V. Suppliers, enablers, core pure plays, adopters, downstream beneficiaries, and disrupted firms should be reported separately. For each role, the report should show gross exposure, net exposure, return contribution, risk contribution, turnover, average materiality, average confidence, and evidence quality. If most of the return comes from adopters rather than suppliers, the strategy's economic story may need revision. If shorts in disrupted firms dominate, the portfolio is not just a long theme basket; it is a disruption trade.

Source attribution asks which information channel actually added value. Earnings calls may identify management emphasis early. Filings may confirm durable exposure later. Specialist media may reveal supply-chain bottlenecks before broad financial news. Alternative data may validate demand, traffic, power use, hiring, or pricing. A costly data feed deserves renewal only if it contributes unique, timely, licensed, and implementable information after overlapping sources are removed.

34.7 Risk, Regime, and Scenario Attribution

The position that contributes most to return is not always the position that contributes most to risk. A small weight in a volatile supplier can dominate portfolio variance, while a large weight in a stable platform company can contribute less risk than its notional size suggests.

Given portfolio weights w and covariance matrix Σ , portfolio volatility is

$$\sigma_p = \sqrt{w^\top \Sigma w}. \quad (34.7)$$

The marginal contribution to risk of asset i is

$$\frac{(\Sigma w)_i}{\sigma_p}, \quad (34.8)$$

and the total contribution to risk is

$$w_i \frac{(\Sigma w)_i}{\sigma_p}. \quad (34.9)$$

The same calculation can be aggregated by sector, theme, sub-theme, source, company role, liquidity bucket, or analyst-approved sleeve. This matters because governance decisions rarely happen at the individual-stock level alone. A committee may accept a large total AI

infrastructure exposure but reject a concentration in one bottleneck, one country, one regulatory dependency, or one funding condition.

Regime attribution then asks when the strategy worked. The report should segment returns by rising-rate and falling-rate periods, growth-led and value-led markets, high-volatility and low-volatility regimes, liquidity expansion and contraction, risk-on and risk-off episodes, commodity shocks, and geopolitical shocks where relevant.

Scenario analysis should include both historical and hypothetical shocks. For AI infrastructure, plausible scenarios include semiconductor export restrictions, cloud capex cuts, memory oversupply, data-center permitting delays, power-grid constraints, water scarcity for cooling, and a rate shock that compresses long-duration growth multiples. For clean energy, the scenario set should include subsidy disappointment, higher financing costs, input inflation, grid bottlenecks, and ETF-flow reversal.

34.8 Governance Starts Where Attribution Ends

Attribution produces the uncomfortable facts that governance must act on. If the portfolio's alpha disappears after factor adjustment, someone must decide whether the mandate allows that exposure. If a source stops contributing unique information, someone must decide whether to keep paying for it. If analyst overrides improve outcomes in one sleeve and degrade them in another, the review process needs redesign.

Model risk guidance in banking makes this connection explicit. SR 11-7 defines model risk around adverse consequences from incorrect or misused model outputs and emphasizes development, validation, governance, policies, controls, inventories, and effective challenge [18]. The ECB's internal-model guidance similarly treats governance, validation, monitoring, and model-change control as part of the model's life cycle, not as paperwork after the mathematics is finished [35].

Narrative-aware investing is not bank capital modeling, but the lesson travels well. A topic model, exposure scorer, LLM classifier, portfolio optimizer, and risk monitor can all be correct in isolation and still produce a bad investment process if their outputs are misused, stale, undocumented, or unchallenged.

34.9 Decision Rights and Review Queues

An operating model should define what the system may do automatically and what requires human approval. The boundary can vary by mandate, but it cannot remain implicit.

The model may usually ingest and classify documents, assign them to existing topic threads, calculate preliminary novelty and attention metrics, generate draft labels, retrieve evidence, calculate provisional exposure scores, and produce pre-trade risk reports. These actions create candidates and measurements. They do not, by themselves, create authority.

Human approval should be required for promoting a topic to an investable theme, changing inclusion criteria, approving high-impact exposure changes, overriding materiality or value-chain role classifications, changing portfolio constraints, changing kill rules, promoting a model from research to paper trading or production, and continuing a strategy after a material breach or drawdown review.

Review queues should be prioritized by investment consequence. A low-confidence label on a tiny position can wait. A model-driven exposure increase in a large, liquid, high-volatility

name near a rebalance should move to the front. Queue priority should combine position size, proposed trade size, model confidence, disagreement among sources, novelty, liquidity, compliance sensitivity, and recent performance contribution.

34.10 Evidence Packages and Override Logs

An unsupported LLM label is not a production artifact. It is a suggestion with good formatting.

A sufficient evidence package includes representative documents, retrieved passages from filings or calls, entity links, value-chain relationships, revenue or segment evidence where available, contradictory evidence, model confidence, uncertainty, analyst decision, and rationale. For LLM-based components, the package should also store prompt version, model identifier, retrieval constraints, output schema, cited passages, refusal or uncertainty behavior, and examples of acceptable and unacceptable outputs.

Human overrides deserve the same discipline. They are necessary because models miss context, misread entities, overreact to language, and underweight business knowledge. They are dangerous because humans also introduce hindsight, inconsistency, incentives, and narrative attachment. Each override should record analyst identifier, timestamp, object changed, original model output, revised value, evidence reviewed, reason code, confidence, and validity interval.

Reason codes should be standardized: entity-linking error, stale evidence, missing revenue data, false positive keyword match, value-chain role correction, compliance exclusion, unsupported LLM output, or investment judgment. Standardization turns overrides into training data and governance evidence instead of private memory.

34.11 Feedback Without Future Leakage

Analyst feedback improves the system only if the time record remains honest. A 2026 analyst may correctly recognize that a 2021 utility had latent AI-power exposure, but that knowledge cannot be used to relabel the 2021 backtest unless the label creation date is treated as 2026.

Two datasets should coexist. The historical research dataset may contain retrospective labels, cleaned histories, improved taxonomies, and lessons learned. It helps the team understand what happened. The point-in-time decision dataset contains only what the system knew, in the form it knew it, at each decision date. It supports validation, simulation, and production audit.

This separation is especially important for LLM workflows. A current model may summarize old documents using current vocabulary. The output can be useful for research and writing, but it is not automatically valid as a historical trading signal. The evidence package must show whether the model merely organized point-in-time evidence or imported knowledge from outside the simulated date.

34.12 Responsible AI, Licensing, and Compliance

Narrative systems ingest messy information: news, social media, transcripts, filings, web documents, alternative data, prediction-market signals, and generated summaries. The governance process must address source licensing, permitted use, privacy, personal data, social-media terms of service, material nonpublic information risk, market-abuse concerns, explainability, record retention, fair allocation, and mandate constraints.

The responsible-AI vocabulary is broader than investment performance. MAS's FEAT principles emphasize fairness, ethics, accountability, and transparency in the use of AI and data analytics in financial services [80]. NIST's AI Risk Management Framework organizes AI risk around governance, mapping, measurement, and management of trustworthy systems [84]. Those frameworks do not tell a portfolio manager which AI infrastructure stock to own. They do tell the organization to make the system's purpose, limits, controls, accountability, and failure modes explicit.

Prediction-market signals require special care because liquidity, manipulation risk, insider-information concerns, and regulatory status vary by venue and contract. They should be treated as event-risk overlays or scenario inputs, not automatic trading instructions. A signal that looks precise because it trades as a probability may still be shallow, manipulable, or unusable at institutional scale.

34.13 Release Gates and Incident Response

A model or strategy should pass through release gates: research prototype, reproducible backtest, independent validation, paper trading or shadow mode, limited production, full production, periodic review, and retirement or redesign. Promotion requires evidence, not enthusiasm.

At minimum, the promotion package should include the investment thesis, theme card, data lineage, source permissions, model documentation, validation report, leakage analysis, performance attribution, benchmark stack, factor report, risk limits, monitoring plan, incident plan, rollback procedure, and named owners. A theme that cannot identify its owner is not ready for production.

The incident plan should define severity levels, owners, response times, and postmortem requirements. Incidents include missing source data, delayed ingestion, entity-linking failure, topic drift beyond threshold, unsupported LLM output, factor exposure breach, transaction-cost breach, unexpected drawdown, compliance restriction breach, analyst review backlog, and portfolio weight outside approved limits.

Postmortems should change something. They may update tests, monitors, prompts, documentation, constraints, review queues, source contracts, or kill rules. A postmortem that only explains why the problem was understandable has not controlled the risk.

34.14 AI Infrastructure as a Production Review

In an AI infrastructure strategy, the monthly production packet should show whether the portfolio still expresses the intended economic chain. The packet should separate semiconductors, memory, advanced packaging, cloud platforms, networking, cooling, data-center real estate, power generation, grid equipment, and software adopters. It should show which sources changed exposures, which analysts approved overrides, where the portfolio overlaps with popular ETFs and baskets, and how much active return came from mega-cap technology exposure.

Suppose the system adds a utility after detecting repeated discussion of data-center load growth. The evidence package should include the source passages, service territories, generation mix, grid constraints, contracted load, capex plan, regulatory context, and an analyst judgment about materiality. A generic electricity mention is not enough. The governance question is whether the company has investable exposure to the AI-power bottleneck or merely appears in a story about electricity.

The same packet should test the downside. Export controls, cloud capex reductions, memory oversupply, power-price regulation, permitting delays, and rate shocks should all appear as scenario lines. A strategy that cannot explain how it would behave if the theme's favorite assumptions fail is not fully governed.

34.15 Clean Energy as a Governance Review

Clean energy after 2020 offers the transition-investing lesson. The story had policy support, public enthusiasm, ETF flows, technological progress, and moral force. It also had valuation risk, rate sensitivity, unprofitable companies, supply-chain inflation, project-finance exposure, and crowded ownership.

A strong governance process would not have declared the theme invalid as soon as returns turned. It would have forced a sharper diagnosis. Did residual return weaken after controlling for growth and rates? Did ETF flows reverse? Did the exposure score still correspond to revenue materiality? Did input costs damage margins? Did the portfolio own durable grid-equipment beneficiaries or speculative concept stocks? Did the original thesis specify what evidence would reduce exposure?

That distinction matters. Some clean-energy companies remained economically important even as the 2020 public-equity expression unwound. Governance should protect the process from two opposite errors: staying loyal to a crowded trade because the story still sounds right, and abandoning a structural transition because one crowded version of the trade lost support.

34.16 The Production Review Packet

Every live strategy should produce a standard packet at each review cycle:

- performance after costs, financing, borrow, and slippage;
- benchmark-relative performance and information ratio;
- factor attribution and residual alpha;
- sector, industry, theme, sub-theme, role, and source attribution;
- risk contribution, concentration, liquidity, and capacity;
- regime and scenario attribution;
- turnover and implementation-cost diagnostics;
- evidence packages for material exposure changes;
- analyst overrides, reason codes, and unresolved disagreements;
- model confidence, drift, incidents, and validation status;
- compliance, licensing, privacy, and responsible-AI checks;
- decisions requested from the approval committee.

A strategy should not remain live simply because cumulative return is positive. It should remain live because current evidence still supports the mechanism, the portfolio still expresses

the approved theme, the risks are still within mandate, and the people responsible for the system can explain what it knew, why it acted, how it was protected, and who approved it.

Chapter 35

Research Frontiers and Open Problems

35.1 The Edge That Becomes Infrastructure

By late 2024, the AI infrastructure trade no longer looked like a secret. Nvidia was a household stock-market name. High-bandwidth memory had moved from specialist semiconductor vocabulary into investor presentations. Data-center power demand had turned utilities into technology-adjacent assets. Cooling, networking, cloud capex, nuclear restarts, and grid equipment had all been pulled into the same market conversation.

That is the strange fate of a good thematic signal. If the signal works, the world learns it. Data vendors package it. ETF issuers index it. Sell-side desks name it. Portfolio managers crowd it. The edge does not vanish in one clean moment; it becomes infrastructure, and then the next edge moves somewhere less comfortable.

This book has treated narrative-aware thematic trading as an investable process: point-in-time corpora, topic discovery, dynamic tracking, exposure scoring, value-chain evidence, factor control, backtesting, attribution, and governance. The machinery is usable. It is also incomplete. The hardest questions now sit between machine learning, behavioral finance, market microstructure, causal inference, and organizational design.

The research frontier is not “better AI” in the abstract. It is a set of sharper questions: which stories are real before the market agrees, which companies are economically exposed before the stock chart proves it, which model outputs survive time, and which investment process remains honest after success gives everyone a reason to believe.

35.2 Evaluation Has Not Caught Up With Models

BERTopic, Top2Vec, CTM, ETM, and LLM-assisted labeling can produce topics that look plausible on a slide. That is progress. It is also a trap. A cluster called “AI power bottlenecks” may sound coherent because the label is good, the top words are familiar, and the analyst already knows the 2024 ending.

The evaluation problem remains open because topic models serve different masters. A news product wants interpretable clusters. A research system wants stable semantic structure. A trading system wants information that changes decisions before prices adjust. A governance system wants evidence that can be audited. No single coherence score can satisfy all four.

The critique by Hoyle and coauthors still matters: automated coherence metrics do not reliably track human judgments for modern neural topic models [60]. For investment use, the problem is even harder because a topic can be semantically beautiful and financially useless. A clean cluster of documents about data-center energy may help an analyst understand the market, but it earns its place in a portfolio only if it improves exposure mapping, risk monitoring, timing, attribution, or return forecasts after costs.

Future work needs benchmarks that combine intrinsic quality, human review, and extrinsic investment value. The test set should include short texts, long filings, multilingual documents, topic splits, re-emerging themes, company-name contamination, and false positives that look convincing. The question is not which model wins on average. The question is which model fails in the way the investment process can tolerate.

35.3 Themes Do Not Evolve Like Tickers

A ticker keeps its identifier through most of its life. A theme does not. “AI” became generative AI, AI infrastructure, AI software, AI safety, AI regulation, AI power demand, AI cooling, and AI capex. Some branches became tradable. Some became marketing language. Some may later merge into broader productivity, energy, or automation narratives.

Dynamic topic models were built to handle change, but theme evolution is more complicated than smooth drift. Blei and Lafferty’s dynamic topic model gave the field a formal way to let topic-word distributions evolve through time [14]. Later systems improved scale and flexibility, and recent work such as BERTrend focuses on emerging weak signals in evolving corpora [19]. Chain-free dynamic topic models attack another failure mode: chained topics can become repetitive or lose association with their own history, so contrastive evolution tracking can preserve diversity while linking related topics [121].

The investment problem adds a further burden. A split is not merely a modeling event. It can create a new basket, a new risk report, a new exposure score, a new benchmark, and a new approval record. If the model splits “clean energy” into solar, wind, hydrogen, batteries, EVs, and grid equipment after the 2021 peak, the backtest must not pretend those branches were known in 2019.

The open problem is point-in-time lineage. A production system needs a dated family tree of themes: when a branch appeared, which evidence justified it, who approved it, which companies moved into it, and whether earlier returns are still attributed to the parent or to the newly discovered child. Without lineage, dynamic modeling becomes a polite form of hindsight.

35.4 The Global Story Does Not Speak One Language

A European bank, an Asian memory supplier, a U.S. cloud platform, and a Brazilian power producer can all sit inside the same AI infrastructure thesis. Their first meaningful disclosures may appear in different languages, under different reporting rules, on different calendars, and with different market audiences.

English-language corpora are convenient because they are large, liquid, and heavily studied. They are not a neutral map of global narratives. Defense procurement, energy policy, pharmaceutical approvals, supply-chain bottlenecks, and industrial subsidies often appear first in local filings, local media, ministry documents, regional trade publications, and earnings calls that global investors read late.

Cross-lingual contextualized topic models offer part of the answer. ZeroShotTM learns topics in one language and predicts topic distributions for documents in other languages, including European languages relevant to a cross-market investment process [11]. But multilingual modeling is not translation with a better accent. Sentiment, legal obligation, promotional style, political framing, and investor convention all travel badly across languages.

The frontier here is not only model quality. It is market mapping. A multilingual system must link local entities to global securities, distinguish parent companies from subsidiaries, handle private suppliers that affect public customers, and measure which market receives the narrative first. A theme that is old news in Korean supply-chain coverage may still be new to U.S. portfolio flows.

35.5 Causality Is Still the Scarce Asset

Narrative measures often forecast returns. That does not prove narratives cause returns. Rising prices attract journalists, analysts, influencers, ETF flows, and management language. Fundamentals, prices, and stories move together, and a backtest can confuse the order.

The empirical literature has strong predictive evidence. Bybee, Kelly, Manela, and Xiu show that topic attention in business news tracks macroeconomic conditions and forecasts aggregate market returns [22]. Calomiris and Mamaysky show that news topic, sentiment, frequency, and entropy predict country-level returns, volatility, and drawdowns across 51 countries [23]. Those results matter. They do not eliminate the need to distinguish forecasting evidence from causal evidence.

For thematic trading, causality has a practical meaning. The portfolio does not need a philosophical proof that a narrative moved prices. It needs a mechanism strong enough to survive falsification. Event studies around narrative shocks, difference-in-differences designs across exposed and unexposed firms, credible instruments for media attention, supply-chain natural experiments, and causal graphs linking documents, flows, fundamentals, and returns can all help.

The scarce object is an investable mechanism. “Attention rose and stocks rose” is not enough. “A new constraint appeared in the supply chain, management disclosed it before broad investors understood it, exposed firms had measurable revenue sensitivity, and prices adjusted with delay” is closer to evidence.

35.6 LLMs Force a New Standard of Time

LLMs are especially useful where thematic investing hurts: summarizing documents, extracting evidence, classifying company exposure, comparing filings, explaining topic labels, and organizing analyst review. They are also especially dangerous in backtests because they know too much.

Lopez-Lira and Tang’s post-cutoff headline design remains a useful model because it tests LLM forecasts where model memory is less likely to explain the result [76]. The newer memorization evidence is more severe: Lopez-Lira, Tang, and Zhu argue that when an LLM has seen realized economic or financial data, counterfactual forecasting ability is not identified, because apparent skill may reflect memorization rather than forecasting [77]. Ludwig, Mullainathan, and Rambachan make a complementary econometric distinction: prediction uses of LLMs require no training leakage, while measurement uses require validation samples to control downstream inference [79].

That distinction should shape investment research. If the LLM is measuring a document—extracting a revenue segment, summarizing a risk factor, or organizing passages for an analyst—the system needs validation data, error estimates, and human review. If the LLM is forecasting returns or classifying historical exposure for a backtest, the system needs strict model cutoffs, retrieval constraints, post-cutoff evaluation, and records showing exactly what evidence entered the prompt.

The open problem is institutional, not just technical. Investment teams need workflows where LLMs can improve productivity without laundering future knowledge into historical validation. The model should be allowed to help. It should not be allowed to time travel.

35.7 Benchmarks for Machine-Built Baskets

A model-generated AI infrastructure basket can beat the S&P 500 and still fail the test. It may have owned semiconductors during a semiconductor bull market. It may have loaded on momentum and large-cap growth. It may have copied a thematic ETF after the ETF had already absorbed the obvious companies. Benchmarking is where many attractive claims become smaller, but more honest.

The industry still lacks standardized benchmarks for AI-generated thematic baskets. Broad indexes are too easy. Narrow sector indexes can be too strict for cross-sector themes. Thematic ETFs are useful but contaminated by launch timing, liquidity screens, index methodology, and marketing constraints. Specialized ETF evidence is a warning rather than a solution: Ben-David, Franzoni, Kim, and Moussawi show that specialized ETFs tend to launch near peaks of investor attention and underperform in risk-adjusted terms after launch [9].

Future benchmarks should include matched random baskets by sector, size, liquidity, valuation, momentum, and factor exposure; delayed ETF-holdings benchmarks that respect publication lags; human analyst baskets built from the same information set; and capacity-adjusted versions that penalize concentration in illiquid pure plays. The benchmark should not ask whether the model found a story. It should ask whether the model found a better investable expression of the story than simpler alternatives.

35.8 Prediction Markets Are Inputs, Not Oracles

Event contracts turn uncertainty into prices. A contract that pays one dollar if an election, approval, lawsuit, or policy decision occurs can look like the cleanest possible narrative signal: a probability with a ticker.

The reality is messier. Prediction markets can aggregate information well, but they carry liquidity limits, favorite-longshot bias, regulatory complexity, manipulation concerns, and cross-venue differences. The classic prediction-market literature treats them as information-aggregation mechanisms [119], and political-market event studies show that equity markets can react to changing event probabilities [104]. Recent market-structure evidence is more sobering: Kalshi data show informative prices but also favorite-longshot bias [21], and Polymarket transaction data suggest highly concentrated profits among a small group of sophisticated participants [2].

For thematic portfolios, the promising role is not direct prediction-market trading. It is overlay design. Event probabilities can change risk budgets, hedge ratios, scenario weights, analyst review priority, or exposure caps after the value-chain map translates the event into

companies. An export-control probability matters only after the system knows which firms depend on restricted chips, which suppliers can substitute, and which revenues are exposed.

The open problem is event-to-equity transmission. A probability is not yet a portfolio weight.

35.9 Capacity Is a Research Problem

Many thematic signals look best before capital arrives. Then spreads widen, short borrow disappears, ETF flows distort baskets, pure plays become expensive, and exits happen through the same door. Capacity is often treated as an implementation detail after the alpha model has been built. In thematic investing, it belongs inside the research model.

The reason is structural. Themes are stories that attract coordinated attention. When the story becomes legible, the capital does not arrive randomly; it concentrates in the names that are easiest to explain, easiest to buy, and easiest to package. The purest companies may be too small. The liquid companies may be too diluted. The largest beneficiaries may already carry several other factor and sector exposures.

Future work needs capacity curves linked to theme lifecycle. A strategy should estimate how expected alpha changes as assets under management rise, ETF overlap increases, attention accelerates, volatility shifts, borrow tightens, and market impact grows. Capacity should also be sub-theme specific. The GPU designer, the memory supplier, the cooling specialist, and the regulated utility do not absorb capital in the same way.

The cleanest research signal is not always the most scalable portfolio. That is not a failure of the signal. It is a fact the signal must disclose.

35.10 Durable Themes and Narrative Bubbles

Metaverse in late 2021 is a cleaner example of narrative excess than clean energy. The word appeared everywhere: gaming, virtual real estate, augmented reality, enterprise collaboration, advertising, creator economies. Some of the technologies were real. The public-market package arrived before the cash-flow map was mature. That is the version of a theme where the label itself can run far ahead of the economics.

Clean energy raises a different and more important problem. The transition is durable, capital-intensive, policy-relevant, and morally consequential. A public-equity basket can still become crowded, expensive, and rate-sensitive. That is not a bubble verdict on the transition. It is a reminder that theme validity and trade validity are different objects.

The hardest unresolved question is how to distinguish a structural theme from a narrative bubble while the trade is still alive. The evidence will never be perfect. Useful clues include revenue conversion, order books, capex commitments, pricing power, customer concentration, subsidy durability, regulatory opposition, supply-chain constraints, valuation relative to realized fundamentals, and residual performance after attention peaks.

The danger runs in both directions. A process can stay loyal to a promotional narrative because the story remains exciting. It can also abandon a durable theme because the crowded expression of the trade unwinds. A better system would separate theme validity from trade validity: the economic transformation may continue even when the first public-market basket has become too expensive.

That distinction may be the most important frontier in the book. Markets do not only misprice stories. They also overpay for correct stories at the wrong time.

35.11 A Research Roadmap

The next generation of narrative-aware trading should reward decision quality more than model novelty. A research program worth funding would prioritize:

1. point-in-time benchmarks for topic discovery, exposure scoring, and theme lineage;
2. audited datasets of theme births, splits, merges, institutionalization, crowding, decay, and recycling;
3. evaluation protocols that compare simple exposure models, domain encoders, LLM workflows, and human analysts on the same dated evidence;
4. multilingual corpora and entity maps for European, Asian, and emerging-market narratives;
5. causal designs that separate attention feedback from information diffusion;
6. LLM validation standards for measurement tasks and strict leakage controls for prediction tasks;
7. value-chain graphs with timestamped relationships, private-company links, revenue evidence, and uncertainty scores;
8. capacity, crowding, ETF overlap, and transaction-cost models embedded directly into portfolio construction;
9. governance workflows that preserve analyst insight without allowing hindsight to enter the historical record.

The list is long because the problem is not one model. It is a market map, a data system, a research protocol, a portfolio engine, and an accountability structure.

35.12 The Four Questions

The book ends where the production review should begin. A narrative-aware thematic strategy is credible only when it can answer four questions:

1. What did the system know at the time?
2. Why should that information earn a return?
3. How was the portfolio protected against known risks and implementation costs?
4. Who approved the decision, and what evidence did they rely on?

The answers must be visible in the documents, the timestamps, the topic lineage, the exposure scores, the portfolio weights, the backtest, the attribution report, and the governance record. If the answers are missing, the strategy may still be an interesting story. It is not yet an investment process.

The promise of this field is real. Text can reveal changes before accounting data catches up. Topics can organize scattered evidence. Value-chain maps can connect stories to companies. Factor control can separate theme exposure from ordinary risk. Governance can keep the machine honest when the story becomes profitable.

But the standard should stay severe. Markets are full of narratives, and most of them are not alpha. The work is to find the few that become evidence before they become consensus, and to build portfolios that can survive the moment when the consensus finally arrives.

Part IX

Appendices

Appendix A

Mathematical Reference

A.1 How to Use This Appendix

The formulas in this appendix are the book's common notation. They are collected here so the main chapters can stay readable while still giving the practitioner a precise reference for returns, factor control, text similarity, topic evaluation, and network measures.

A.2 Returns

For price P_t and dividend or distribution D_t , the simple return is

$$R_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}}. \quad (\text{A.1})$$

The log return is

$$r_t = \log(P_t + D_t) - \log(P_{t-1}). \quad (\text{A.2})$$

Simple returns are natural for portfolio accounting because weighted simple returns aggregate across assets. Log returns are convenient for time aggregation and some statistical models. For small returns, the two are close.

A.3 Excess and Active Return

Excess return subtracts the risk-free rate:

$$R_{i,t}^e = R_{i,t} - R_{f,t}. \quad (\text{A.3})$$

Active return compares a portfolio with a benchmark:

$$A_t = R_{p,t} - R_{b,t}. \quad (\text{A.4})$$

In the book, alpha claims should usually be read through both lenses. A strategy can beat cash while failing to beat the benchmark that matches its mandate.

A.4 Annualization

For daily returns, annualized arithmetic return is often approximated as

$$\bar{R}_{\text{ann}} \approx 252 \bar{R}_{\text{daily}}. \quad (\text{A.5})$$

For compounded growth, use

$$R_{\text{ann}} = \left(\prod_{t=1}^T (1 + R_t) \right)^{252/T} - 1. \quad (\text{A.6})$$

Daily volatility is commonly annualized as

$$\hat{\sigma}_{\text{ann}} = \sqrt{252} \hat{\sigma}_{\text{daily}}. \quad (\text{A.7})$$

The square-root rule assumes approximately independent daily returns. It becomes less reliable when returns are autocorrelated, stale, smoothed, or dominated by rare jumps.

A.5 Mean, Variance, Covariance, and Correlation

The sample mean return is

$$\bar{R} = \frac{1}{T} \sum_{t=1}^T R_t. \quad (\text{A.8})$$

The sample variance is

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2. \quad (\text{A.9})$$

Covariance between returns R_i and R_j is

$$\text{Cov}(R_i, R_j) = \mathbb{E}[(R_i - \mathbb{E}[R_i])(R_j - \mathbb{E}[R_j])]. \quad (\text{A.10})$$

Correlation normalizes covariance:

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\sigma_i \sigma_j}. \quad (\text{A.11})$$

Correlation matters for diversification, residual co-movement, theme coherence, and risk contribution. In this book, the most important correlation is often the residual correlation left after sectors and factors have been removed.

A.6 Portfolio Return and Risk

For portfolio weights w_t and asset returns R_t , portfolio return is

$$R_{p,t} = w_t^\top R_t. \quad (\text{A.12})$$

With covariance matrix Σ , portfolio variance is

$$\sigma_p^2 = w^\top \Sigma w, \quad (\text{A.13})$$

and portfolio volatility is

$$\sigma_p = \sqrt{w^\top \Sigma w}. \quad (\text{A.14})$$

The marginal contribution to risk of asset i is

$$\frac{(\Sigma w)_i}{\sigma_p}, \quad (\text{A.15})$$

and the total contribution to risk is

$$w_i \frac{(\Sigma w)_i}{\sigma_p}. \quad (\text{A.16})$$

These quantities can be aggregated by sector, sub-theme, company role, source, or analyst-approved sleeve.

A.7 CAPM, Beta, and Jensen's Alpha

The CAPM relates expected excess return to market exposure:

$$\mathbb{E}[R_i - R_f] = \beta_i \mathbb{E}[R_m - R_f]. \quad (\text{A.17})$$

Market beta is

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}. \quad (\text{A.18})$$

In regression form:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t}. \quad (\text{A.19})$$

Jensen's alpha is the intercept α_i . It is the average excess return not explained by market beta in that regression.

A.8 Multifactor Regressions

A general multifactor model is

$$R_{p,t} - R_{f,t} = \alpha_p + \sum_{k=1}^K \beta_{p,k} F_{k,t} + \epsilon_{p,t}, \quad (\text{A.20})$$

where $F_{k,t}$ are factor returns.

The Fama-French three-factor model is

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \epsilon_t. \quad (\text{A.21})$$

The Carhart extension adds momentum:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \beta_{\text{MOM}} \text{MOM}_t + \epsilon_t. \quad (\text{A.22})$$

The five-factor model adds profitability and investment:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \beta_{\text{RMW}} \text{RMW}_t + \beta_{\text{CMA}} \text{CMA}_t + \epsilon_t. \quad (\text{A.23})$$

The book's working baseline is the five-factor model plus momentum:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \beta_{\text{RMW}} \text{RMW}_t + \beta_{\text{CMA}} \text{CMA}_t + \beta_{\text{MOM}} \text{MOM}_t + \epsilon_t. \quad (\text{A.24})$$

Thematic strategies often add sector, industry, country, currency, rates, commodity, liquidity, and volatility controls depending on the mandate.

A.9 Sharpe Ratio and Information Ratio

The Sharpe ratio is

$$\text{Sharpe} = \frac{\mathbb{E}[R_p - R_f]}{\sigma(R_p - R_f)}. \quad (\text{A.25})$$

Tracking error is the volatility of active return:

$$\text{TE} = \sigma(R_{p,t} - R_{b,t}). \quad (\text{A.26})$$

The information ratio is

$$\text{IR} = \frac{\mathbb{E}[R_{p,t} - R_{b,t}]}{\text{TE}}. \quad (\text{A.27})$$

Sharpe measures total risk efficiency. Information ratio measures benchmark-relative risk efficiency.

A.10 Drawdown

Let V_t be portfolio value and $M_t = \max_{s \leq t} V_s$ be the running maximum. Drawdown is

$$\text{DD}_t = \frac{V_t - M_t}{M_t}. \quad (\text{A.28})$$

Maximum drawdown is the most negative drawdown over the evaluation period.

A.11 Cross-Sectional Residualization

To remove known factor and sector exposure from a raw theme score $Y_{\text{raw},i,t}$, estimate

$$Y_{\text{raw},i,t} = \gamma_t + F_{i,t}^\top \theta_t + S_{i,t}^\top \delta_t + \eta_{i,t}. \quad (\text{A.29})$$

Here $F_{i,t}$ are firm characteristics or factor exposures, $S_{i,t}$ are sector or industry indicators, and $\eta_{i,t}$ is the residual score. The residual is the part of the theme score not explained by known characteristics.

A.12 Factor Hedge

For portfolio factor exposure vector b_p and hedge instrument exposure matrix B_h , a simple hedge solves

$$B_h h \approx b_p, \quad (\text{A.30})$$

where h are hedge weights. The hedged portfolio weights are then the original portfolio weights plus the hedge positions. In practice, hedge design must respect liquidity, borrow, financing, leverage, mandate, and basis risk.

A.13 Turnover and Transaction Costs

One-way turnover from weights w_{t-} before rebalancing to target weights w_t is

$$\text{Turnover}_t = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-}|. \quad (\text{A.31})$$

A simple cost model is

$$\text{Cost}_t = \sum_i \left(c_i |q_{i,t}| + s_i |q_{i,t}| + \kappa_i \left(\frac{|q_{i,t}|}{\text{ADV}_{i,t}} \right)^\phi |q_{i,t}| \right), \quad (\text{A.32})$$

where $q_{i,t}$ is trade value, c_i is commission and fees, s_i approximates spread cost, $\text{ADV}_{i,t}$ is average daily value traded, and the last term approximates market impact. The exponent ϕ is often between 0.5 and 1 in simple impact models, but should be calibrated to the asset class and execution style.

A.14 Capacity Curve

Let $\alpha(A)$ be expected alpha at assets under management A after implementation costs. A capacity curve reports

$$\alpha_{\text{net}}(A) = \alpha_{\text{gross}}(A) - \text{Costs}(A) - \text{CrowdingPenalty}(A). \quad (\text{A.33})$$

The capacity of a strategy is not a single number. It is the range of A over which expected net alpha remains attractive under realistic liquidity, turnover, borrow, and crowding assumptions.

A.15 Cosine Similarity

For embeddings x and y , cosine similarity is

$$\cos(x, y) = \frac{x^\top y}{\|x\| \|y\|}. \quad (\text{A.34})$$

It is commonly used for semantic retrieval, document clustering, and theme-company similarity.

A.16 c-TF-IDF

Class-based TF-IDF treats all documents in a cluster or topic as one class document. A simplified form is

$$\text{c-TF-IDF}_{w,c} = \text{tf}_{w,c} \log \left(\frac{C}{1 + \text{df}_w} \right), \quad (\text{A.35})$$

where $\text{tf}_{w,c}$ is the frequency of word w in class c , C is the number of classes, and df_w is the number of classes containing w .

A.17 NPMI

Normalized pointwise mutual information for words w_i and w_j is

$$\text{NPMI}(w_i, w_j) = \frac{\log \left(\frac{p(w_i, w_j)}{p(w_i)p(w_j)} \right)}{-\log p(w_i, w_j)}. \quad (\text{A.36})$$

Topic coherence metrics often average NPMI over top topic words. In financial applications, NPMI should be treated as one diagnostic, not as an investment-quality score.

A.18 KL Divergence

For discrete distributions P and Q , Kullback-Leibler divergence is

$$D_{\text{KL}}(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}. \quad (\text{A.37})$$

It can measure how much a topic distribution, attention distribution, or source distribution differs from a historical baseline.

A.19 Graph and Network Measures

For a graph with adjacency matrix A , degree centrality of node i is

$$d_i = \sum_j A_{ij}. \quad (\text{A.38})$$

Eigenvector centrality defines centrality c_i recursively:

$$c_i = \frac{1}{\lambda} \sum_j A_{ij} c_j. \quad (\text{A.39})$$

Betweenness centrality measures how often node i lies on shortest paths between other nodes:

$$BC(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}, \quad (\text{A.40})$$

where σ_{st} is the number of shortest paths from s to t , and $\sigma_{st}(i)$ is the number of those paths that pass through i .

Centrality can describe value-chain position, bottlenecks, or dependency concentration. It is not by itself an investment thesis.

Appendix B

Glossary

B.1 Market and Portfolio Terms

Alpha Return unexplained by the benchmark or risk model being used. Alpha is not the same thing as outperformance: a clean-energy basket that beats the market because it is long high-beta growth stocks has not necessarily produced thematic alpha. In this book, alpha is always residual to an explicit benchmark and factor specification.

Beta Sensitivity to a market, sector, or factor return. A beta of 1.2 to the equity market means the asset tends to move 1.2 percent when the market moves 1 percent, before residual noise. A thematic basket can have several betas at once: market beta, growth beta, semiconductor beta, utility beta, or rate sensitivity.

Benchmark The reference portfolio used to judge performance. Benchmark choice determines what counts as active return. A long-only AI infrastructure basket needs a different benchmark stack from a factor-neutral long-short strategy.

Capacity The amount of capital a strategy can plausibly manage before costs, market impact, crowding, borrow constraints, or liquidity limits erode the edge. Capacity is best reported as a curve, not a single number.

Drawdown A decline from a previous portfolio peak. Maximum drawdown is the worst peak-to-trough loss in the evaluation period. Drawdown matters because a strategy can have a respectable Sharpe ratio and still be impossible for clients to hold through a sharp thematic unwind.

Factor A systematic driver of returns, such as market, size, value, momentum, profitability, investment, quality, low volatility, sector, country, currency, rates, or commodities. Factors are useful because they reveal when a theme's apparent alpha is actually compensation for known risks.

Factor Neutralization Signal processing, hedging, or portfolio construction that reduces exposure to specified factors. Neutralization can happen before portfolio construction through residualized scores, or after construction through constraints and hedges.

Information Ratio Average active return divided by tracking error. It measures how efficiently a strategy uses benchmark-relative risk.

Liquidity The ability to trade without materially moving price. Liquidity depends on volume, spread, depth, volatility, ownership concentration, market stress, and the size of the order relative to normal trading activity.

Residual Return Return left after removing market, sector, industry, and factor effects. Residual return is where the theme-specific claim has to survive.

Sharpe Ratio Average excess return divided by excess-return volatility. It is useful, but it can hide crash risk, crowding, stale prices, and strategies that earn small gains for a long time before one large loss.

Short Selling Selling a borrowed security in expectation of repurchasing it later. Short books require borrow availability, locate processes, margin, financing, and drawdown controls because losses can grow when prices rise.

Tracking Error Volatility of active return relative to a benchmark. A strategy with high tracking error may still be attractive, but it should be judged as an active bet rather than a benchmark-like allocation.

Transaction Costs The full cost of trading: commissions, fees, bid-ask spreads, market impact, borrow, financing, taxes, and operational costs. For thematic strategies, transaction costs often rise when attention and volatility rise.

Turnover The amount of trading required to move from current weights to target weights. High turnover is not automatically bad, but it must be justified by signal decay, expected alpha, and realistic trading costs.

B.2 Narrative and Theme Terms

Acceleration The rate at which attention, source volume, entity breadth, search interest, or document frequency rises relative to a prior baseline. Acceleration is often more useful than level because it can identify a narrative before the level becomes obvious.

Attention The degree to which investors, media, analysts, companies, or the public focus on an asset, event, or theme. Attention can create delayed reaction when information is underprocessed, but it can also mark crowding when everyone is already looking.

Crowding A condition in which many investors hold similar exposures. Crowding can reduce future expected return, increase correlation during exits, and make liquidity disappear precisely when the theme starts to disappoint.

Lifecycle Stage A theme's state in its market life: weak signal, emerging, validated, crowded, institutionalized, decaying, or recycled. Lifecycle language prevents the mistake of treating a theme as the same object before discovery, during enthusiasm, and after ETF packaging.

Narrative A story that organizes facts into an interpretation investors can transmit, remember, and trade. Narratives matter because markets do not process all information as isolated facts; they process it through frames that determine attention and valuation.

Pure Play A company whose economics are highly exposed to a theme. A pure play is not just a company that uses the right words. It should have revenue, assets, costs, customers, or strategic dependency materially tied to the theme.

Sentiment Positive, negative, or neutral tone in text, measured at article, document, company, theme, or source level. In finance, generic sentiment dictionaries often fail because words such as “liability,” “tax,” or “capital” have domain-specific meanings.

Theme A transient, cross-sector, narrative-linked source of common residual return behavior. A theme is not a sector, factor, or product category, although it may overlap with all three.

Thematic Illusion A false connection that looks like a theme but lacks economic materiality, timing, residual return, or implementation feasibility. Examples include mention without exposure, greenwashing, AI-washing, and ETF holdings used as ground truth without checking launch timing.

Value Chain The network of suppliers, enablers, producers, adopters, downstream beneficiaries, and disrupted firms linked to a theme. Value-chain mapping turns a broad story into testable company exposure.

B.3 Text and Topic Modeling Terms

BERTopic A topic modeling pipeline that commonly combines sentence embeddings, dimensionality reduction, density clustering, and c-TF-IDF labels. It is powerful for exploration and labeling, but cluster stability and outlier handling need careful review.

BTM Biterm Topic Model, designed for short texts by modeling word co-occurrence pairs across the corpus. It is useful when individual documents are too short for document-level co-occurrence to be informative.

c-TF-IDF Class-based term-frequency inverse-document-frequency, often used to identify representative words for document clusters. It treats all documents in a cluster as one class document and highlights words that distinguish that cluster from others.

CTM Contextual Topic Model, a neural topic model that incorporates contextual embeddings while retaining topic-mixture structure. CTMs are useful when the system needs soft document-topic distributions rather than only hard clusters.

CWTM Contextualized Word Topic Model, a neural topic model family that uses contextualized word embeddings to improve topic representation. It is relevant when word meaning shifts by context, which happens often in financial text.

DTM Dynamic Topic Model, a model that represents topic evolution through time. DTM-style models are useful when the research question is how a topic’s vocabulary, attention, or meaning changes across periods.

Embedding A numeric vector representation of text, entities, securities, or concepts. Embeddings make semantic similarity computable, but they can also hide errors because nearby vectors are not automatically economically related.

Entity Linking Mapping a text mention to a canonical entity such as a company, security, product, person, geography, or regulator. Entity linking is essential because financial topics can otherwise become clusters of famous company names rather than economic concepts.

ETM Embedded Topic Model, which represents words and topics in a shared embedding space. It connects probabilistic topic modeling with semantic geometry.

GSDMM Gibbs Sampling Dirichlet Multinomial Mixture Model, often used for short texts under a one-document-one-topic assumption. It can work well for headlines and brief posts when the assumption is reasonable.

HDBSCAN A density-based clustering method used in many embedding topic pipelines. It can discover clusters without preselecting a fixed number of topics, but its parameters and treatment of noise points affect results materially.

LDA Latent Dirichlet Allocation, a classical probabilistic topic model that represents documents as mixtures of topics. It remains a useful baseline because it is interpretable and well understood, even when neural models outperform it on some corpora.

NER Named Entity Recognition, the task of identifying entities such as companies, people, places, products, and organizations in text. NER detects the span; entity linking resolves the span to a canonical identifier.

NMF Non-negative Matrix Factorization, a matrix decomposition method used for interpretable topic extraction. It is often simpler and more stable than more complex models, making it a strong baseline.

Outlier Document A document that does not fit any discovered topic or cluster under the current model. In emerging-theme detection, outliers can be noise, but they can also be first evidence of a new cluster.

RAG Retrieval-Augmented Generation, an architecture that grounds generated answers in retrieved evidence. In this book, RAG is useful only when retrieved passages, prompt versions, and output schemas are stored for audit.

Topic A recurring pattern in text represented by words, embeddings, documents, or probability distributions. A topic is not automatically an investment theme; it becomes useful only after interpretation, validation, and mapping to companies.

Topic Coherence A measure of how semantically related a topic's top words or documents are. Coherence helps diagnose topic quality, but investment usefulness requires extrinsic testing.

Topic Drift Change in a topic's vocabulary, documents, entities, or meaning over time. Drift can be genuine evolution, model instability, or contamination from later information.

Topic Merge A case where previously distinct topics become one broader theme. Merges require point-in-time lineage so old performance is not reinterpreted with a later taxonomy.

Topic Split A case where one topic evolves into multiple distinct sub-topics. Splits matter because each branch may have different companies, risks, benchmarks, and lifecycle stages.

Top2Vec An embedding-based topic model that learns topic vectors from document embeddings. It is useful for semantic discovery, but like other embedding methods it needs validation against financial meaning.

UMAP A dimensionality reduction method often used before clustering embeddings. UMAP can make clusters easier to detect, but it introduces parameter choices and stochasticity that must be controlled in production.

B.4 Production and Governance Terms

As-Of Invariant The rule that every simulated or live decision must use only information available before the decision time, in the form it was available then. This is the central control against future leakage.

Audit Log A durable record of data, model, analyst, portfolio, and operational decisions. An audit log should explain not only what the system did, but what evidence and authority supported the decision.

Evidence Package Supporting documents, passages, entity links, model outputs, confidence scores, contradictory evidence, and analyst notes behind a label or exposure score. Evidence packages turn model output into reviewable production artifacts.

Kill Rule A predefined condition that reduces, pauses, or retires a strategy or theme. Good kill rules are written before the drawdown that makes them emotionally difficult to apply.

Model Card A concise document describing a model's purpose, inputs, outputs, validation, limitations, owner, monitoring plan, and retirement criteria. For LLM components, it should include prompt versions, retrieval constraints, and unsupported-output tests.

Override A human change to a model output or system decision. Overrides are necessary, but they must be timestamped, reason-coded, and linked to evidence so they do not become unrecorded hindsight.

Point-in-Time Data Data stored and accessed as it was known at a historical decision date. Point-in-time data preserves historical availability, identifiers, revisions, and vendor delivery behavior.

Shadow Mode A production-like test in which signals, reviews, portfolios, and monitoring reports are generated but not traded. Shadow mode is where a strategy proves that the live workflow behaves like the backtest expected.

Source Latency Delay between an event, source publication, vendor delivery, system ingestion, processing, analyst review, and portfolio decision. Source latency determines which horizons a signal can plausibly trade.

Appendix C

Model and Method Comparison Tables

C.1 How to Read the Tables

These tables are decision aids, not rankings. A method is good when its assumptions match the corpus, investment question, horizon, review burden, and production constraints. The best production systems often combine a simple baseline, a richer semantic model, and a human review process.

C.2 Core Topic Discovery Methods

Method	Input Type	Assignment	Strength	Main Caution	Best Financial Use
LDA	Medium and long documents	Soft mixtures	Transparent probabilistic baseline	Weak on sparse headlines and entity-heavy text	Filings and transcripts with stable vocabulary
NMF	Document-term matrix	Soft components	Stable, fast, interpretable	Sensitive to vocabulary and preprocessing	Baseline topic extraction and audit-friendly prototypes
HDP	Long documents	Soft, variable topic count	Avoids fixed topic count	Harder to tune and explain	Exploratory corpora where topic count is unknown
hLDA	Long documents	Tree-structured topics	Explicit hierarchy	Tree can be brittle and hard to govern	Macro-theme and sub-theme taxonomies
BERTopic	Embedded text	Usually hard clusters	Strong discovery and labeling workflow	UMAP/HDBSCAN choices can be unstable	News, calls, filings, and mixed corpora exploration
Top2Vec	Embedded documents	Topic vectors	Natural semantic discovery	Less transparent for strict production audit	Exploratory semantic mapping
CTM	Text plus contextual embeddings	Soft mixtures	Combines mixtures with embeddings	More complex validation	Document-level exposure mixtures
ETM	Bag-of-words plus embeddings	Soft mixtures	Links topics to embedding geometry	Needs careful vocabulary design	Probabilistic topic modeling with semantic structure
CWTM	Contextualized word embeddings	Soft or neural topics	Captures contextual word meaning	Higher complexity and newer validation burden	Domains where word sense changes by context

C.3 Short-Text Methods

Method	Best Use	Key Assumption	Financial Use Case
BTM	Very short texts	Corpus-level biterns reveal topics	Headlines, alerts, social posts
GSDMM	Short texts	One document usually has one topic	Single-topic headlines and brief posts
Gibbs-BERTopic	Short embedded texts	Embeddings plus topic refinement improve grouping	Headline streams with semantic variants
Aggregation	Sparse text with metadata	Grouping creates richer pseudo-documents	Company-day or theme-week news bundles
Embedding clustering	Short semantic texts	Vector similarity captures meaning despite sparsity	Early detection of emerging narratives
LLM-assisted labeling	Sparse clusters needing explanation	Retrieved evidence constrains the label	Analyst-facing descriptions of short-text clusters

C.4 Temporal and Streaming Methods

Method	Temporal Support	Strength	Main Caution	Best Use
DTM	Explicit time evolution	Principled topic trajectories	Less convenient for fast sparse streams	Historical topic drift
D-ETM	Dynamic neural evolution	Embedding-aware topic drift	Complex implementation and validation	Research-grade dynamic semantics
RollingLDA	Windowed retraining	Simple walk-forward design	Requires topic alignment	Point-in-time baseline
BERTopic	Post-hoc time summaries	Easy diagnostics	Not a full temporal generative model	Dashboards and exploration
Windowed clustering	Independent time slices	Detects bursts and new clusters	Split-merge lineage can be fragile	Emerging theme detection
BERTrend	Online emerging-trend monitoring	Separates weak and strong signals	Still needs investment validation	Live corpus surveillance
CFDTM	Chain-free evolution tracking	Handles repetitive and unassociated-topic failures	More complex and newer	Theme lineage research
Streaming models	Incremental updates	Production-friendly	Drift and forgetting must be monitored	High-volume news feeds

C.5 Exposure Scoring Methods

Method	Strength	Weakness	Best Role
Keyword rules	Transparent and high precision	Miss semantic variants and context	Initial evidence and exclusion guardrails
Revenue mapping	Economically grounded	Often delayed or coarse	Materiality validation
Embedding similarity	Captures semantic relatedness	Can confuse mentions with exposure	Retrieval and candidate ranking
LLM classification	Flexible reasoning over evidence	Needs grounding and audit controls	Role classification and summaries
Relation extraction	Maps value-chain links	Entity and relation errors compound	Supplier, customer, and beneficiary mapping
Graph centrality	Highlights bottlenecks and dependencies	Centrality is not materiality	Value-chain diagnostics
Analyst review	Context and judgment	Costly and inconsistent without logs	High-impact approvals and overrides

C.6 Model Selection by Investment Problem

Investment Problem	Preferred Starting Point	Escalate When
Find broad unknown themes	BERTopic or NMF baseline	Vocabulary is diverse, multilingual, or short-text heavy
Track an approved theme	Rolling model plus alignment layer	Split, merge, and re-emergence events become frequent
Score company exposure	Revenue mapping plus retrieval	Materiality requires synthesis across filings, calls, and relationships
Detect early weak signals	Windowed embeddings and novelty metrics	The system must monitor live streams continuously
Build an auditable basket	Simple scoring plus factor residualization	Semantic evidence materially improves coverage or purity
Review LLM outputs	Retrieval-constrained classification	Model disagreement affects large positions or governance decisions

C.7 Index and Data-Provider Methodologies

Provider / Method	What It Is Good For	Main Limitation	Book Use
MSCI thematic exposure frameworks	Broad security-level exposure analysis across themes	Methodology may be opaque and product-specific	External benchmark and sanity check
FactSet RBICS with revenue	Segment-level revenue mapping into granular industries	Revenue categories may lag new narratives	Materiality and pure-play validation
S&P thematic and Kensho indexes	Tradable index construction around named themes	Index rules may encode liquidity and investability compromises	Benchmarking AI-generated baskets
Solactive thematic indexes	Fast index construction for emerging product themes	Product design may follow attention cycles	ETF overlap and late-cycle launch diagnostics
ETF holdings	Actual market implementation by products	Holdings are delayed and not ground truth	Crowding, overlap, and product-design comparison

C.8 When to Prefer Simpler Methods

Prefer a simpler method when:

- the decision requires auditability more than semantic breadth;
- the corpus is small or noisy;
- labels are scarce;
- the expected edge cannot support complex infrastructure;
- a transparent rule already performs well in live or out-of-sample tests;
- human reviewers need to understand every inclusion decision.

Prefer richer neural or LLM-assisted methods when:

- themes appear through diverse vocabulary;
- entity names must be separated from conceptual content;
- evidence is distributed across filings, calls, news, and relationships;
- role classification requires synthesis;
- semantic retrieval materially improves exposure scoring;
- the system can store prompts, evidence, model versions, and outputs.

Appendix D

Source and Dataset Guide

D.1 Source Selection Principles

Data sources differ in latency, coverage, licensing, survivorship, revision behavior, and investment horizon. A source should enter the system because it improves a decision, not because it is available or fashionable.

For every source, record:

- provider, contract owner, and permitted use;
- coverage universe, geography, language, and historical start date;
- source timestamp, vendor timestamp, ingestion timestamp, and revision timestamp;
- identifier mapping and point-in-time security linkage;
- survivorship, backfill, and archive risks;
- expected latency and observed latency distribution;
- production failure modes and responsible owner;
- retention, privacy, and compliance constraints.

D.2 Source Qualification Matrix

Source Type	Best Use	Latency Profile	Main Controls
Regulatory filings	Materiality, risk factors, segment evidence	Slow but authoritative	Filing acceptance time, amendments, section parsing
Earnings calls	Management narrative and analyst pressure	Event-driven, vendor-dependent	Speaker labels, transcript release time, revisions
Newswires	Fast events and sentiment	Fast, often duplicated	Deduplication, correction handling, entity linking
Financial press	Narrative framing and synthesis	Medium latency	Source bias, republication, headline-body mismatch
GDELT / event databases	Global breadth and geopolitical monitoring	Near real-time, noisy	Event-code quality, translation artifacts, source bias
Social / web attention	Retail attention and diffusion	Fast but unstable	Bot filtering, terms of service, privacy, archive changes
ETF holdings	Product overlap and crowding	Often delayed	Publication lag, index rules, survivorship
Prices and corporate actions	Backtesting and implementation	Market-close or intraday	Delistings, splits, dividends, stale prices
Factor datasets	Attribution and neutralization	Daily or monthly	Definition stability, region fit, publication timing
Relationship data	Value-chain mapping	Sparse and vendor-specific	Timestamped relationships, direction, confidence
Prediction markets	Event probabilities	Venue-specific, liquidity-dependent	Contract terms, bid-ask, manipulation, regulation

D.3 SEC EDGAR and Regulatory Filings

Regulatory filings provide legally reviewed evidence about business lines, risks, segments, and management discussion. They are especially useful for exposure validation and slow-moving theme confirmation.

Use cases include:

- risk-factor change detection;
- business-description comparison;
- segment revenue mapping;
- materiality checks;
- point-in-time evidence retrieval for LLM classification;
- detecting whether narrative claims appear in legal disclosure or only in management commentary.

Cautions include filing lag, boilerplate, section parsing errors, amended filings, fiscal-calendar alignment, and issuer-level rather than security-level mapping. A filing accepted after market close should not influence a trade simulated before it was available.

D.4 Earnings-Call Transcripts

Earnings calls combine management narrative, analyst questions, guidance, and emerging concerns. They are useful for narrative change, uncertainty, and theme materiality. AI infrastructure often appeared in management discussion and analyst questions before it became clear in business segments.

Important metadata includes call date, call start and end time, transcript publication date, vendor delivery date, speaker labels, prepared remarks versus Q&A, company identifiers, fiscal period, and transcript version.

Cautions include transcript delay, speaker attribution errors, edited transcripts, missing smaller companies, survivorship in vendor archives, and inconsistent treatment of conference presentations versus earnings calls.

D.5 Newswires and Financial Press

News sources are central for sentiment, attention, emergence detection, and event monitoring. Newswires may be faster and more structured; financial press may provide deeper narrative framing.

Required controls include:

- deduplication of syndicated stories;
- correction, update, and deletion handling;
- source timestamp versus vendor timestamp;
- entity-linking validation;
- headline-body consistency;
- source-specific latency measurement;
- embargo and redistribution restrictions.

News is useful for short and medium horizons, but generic mega-cap news may be priced quickly. Its value often comes from cross-source diagnostics, neglected companies, supply-chain links, and early specialist coverage.

D.6 GDELT and Event Databases

Global event databases can provide broad coverage of geopolitical, regulatory, social, and macro narratives. They are useful for source breadth, geographic monitoring, event intensity, and multilingual signal discovery.

Cautions include automated extraction errors, noisy event coding, translation artifacts, media-source bias, duplicated local coverage, and entity ambiguity. Event databases should usually complement curated financial sources rather than replace them.

D.7 Social Media and Web Attention

Social platforms, forums, search trends, job postings, app traffic, web traffic, and product-review data can capture attention and early diffusion. They are most useful when attention itself is part of the mechanism, as in retail-driven bubbles, product adoption, consumer demand, or sentiment shocks.

Risks include spam, manipulation, bot activity, changing platform access rules, privacy restrictions, terms-of-service limitations, short historical archives, and severe selection bias. Social data should rarely drive a production position without corroboration from other sources.

D.8 ETF Holdings and Index Data

ETF and index holdings are useful as benchmarks, overlap diagnostics, crowding indicators, and noisy labels of industry practice. They are also useful because they reveal how product providers translate themes into investable rules.

They should be used with care:

- holdings are often published with a delay;
- index rules may change;
- launched products are survivorship-biased;
- holdings reflect product design, liquidity, and marketing constraints;
- thematic ETFs often launch after investor attention is already high.

ETF holdings can help evaluate a basket. They should not define ground truth for a backtest unless their publication timing and index methodology are handled point-in-time.

D.9 Prices, Corporate Actions, and Fundamentals

Backtests require point-in-time prices, returns, corporate actions, delisting returns, share classes, market capitalization, liquidity, and fundamentals. These datasets are not background plumbing; they define the investable universe and implementation reality.

Minimum controls include adjusted and unadjusted prices, delisting handling, currency, exchange, historical identifiers, split and dividend treatment, stale price checks, share-class mapping, trading halts, and corporate-action effective dates.

Fundamentals need their own timing record. Report date, filing date, vendor availability date, fiscal period, restatement status, and currency conversion can all change what a historical decision could have known.

D.10 Factor and Risk Model Data

Factor datasets support neutralization, attribution, benchmarking, and residual return analysis. Sources may include academic factors, commercial risk models, sector and industry classifications, and custom factor portfolios.

Use the factor set appropriate to the strategy. A global equity strategy may need country, currency, sector, industry, style, rates, commodities, and liquidity factors. A factor model selected after viewing performance is a research degree of freedom and should be treated as such.

D.11 Corporate Relationship Data

Supplier, customer, competitor, patent, product, ownership, and private-company relationship data can improve value-chain mapping. These datasets are often sparse, vendor-specific, and time-varying.

Important checks include relationship timestamp, direction, strength, source evidence, confidence, historical validity, and whether the relationship is operationally material. A current supplier relationship should not be applied backward unless it existed at the historical date.

D.12 Prediction Markets and Event Contracts

Prediction-market data can provide event probabilities. Required fields include contract definition, venue, price, bid, ask, volume, open interest, resolution criteria, resolution date, fee schedule, market status, and regulatory constraints.

Production use requires liquidity filters, manipulation checks, contract-definition review, and event-to-theme mapping. Prediction markets should usually affect scenario weights, risk budgets, hedges, or monitoring priority rather than directly replacing equity signals.

D.13 Minimum Source Acceptance Checklist

Before a source enters production, confirm:

- the license permits the intended research, backtesting, storage, and production use;
- historical and live schemas match or differences are documented;
- timestamps support point-in-time reconstruction;
- identifiers can be mapped to securities and entities;
- coverage gaps and survivorship risks are understood;
- revisions, corrections, and deletions are preserved;
- source latency is measured and monitored;
- quality checks are automated;
- the source adds value beyond existing sources;
- failure modes have owners, alerts, and fallback behavior.

Appendix E

Implementation Checklists

E.1 How to Use These Checklists

These checklists are minimum operating standards. They should be adapted to the mandate, asset universe, data providers, and regulatory setting. A checklist item is complete only when the evidence is stored and reviewable.

E.2 Corpus Ingestion

- Source license and permitted use recorded.
- Raw document stored with checksum.
- Source timestamp, vendor timestamp, ingestion timestamp, and processing timestamp stored.
- Revision, correction, and deletion identifiers preserved.
- Language, geography, source type, and document type detected.
- Duplicate and near-duplicate logic applied.
- Entity mentions extracted and mapped point-in-time.
- Missing, empty, malformed, or truncated documents flagged.
- Ingestion latency measured against source publication.
- Quality checks and failure alerts assigned to an owner.

E.3 Preprocessing and Entity Handling

- Boilerplate and template text handling specified.
- Tokenization, lemmatization, stopword, and phrase-detection settings versioned.
- Company names, tickers, products, people, geographies, and regulators identified.
- Entity-linking confidence stored.

- Ambiguous entity mentions routed to review or conservative handling.
- Entity masking or retention decision documented.
- Security identifiers linked to company identifiers point-in-time.
- Rebrands, mergers, spin-offs, ticker changes, and delistings handled.

E.4 Topic Model Training

- Investment question and corpus scope defined.
- Training window respects the as-of invariant.
- Baseline model selected before richer models are tuned.
- Preprocessing configuration versioned.
- Embedding model and tokenizer versioned.
- Random seeds and hyperparameters stored.
- Topic count or clustering parameters justified.
- Outlier handling documented.
- Intrinsic metrics computed.
- Representative documents stored.
- Human review sample completed.
- Baseline comparison performed and retained.

E.5 Temporal Alignment

- Rolling, streaming, or retraining update rule specified.
- Window start and end dates stored.
- Topic matching method documented.
- Split, merge, drift, and re-emergence events recorded.
- Label versions and validity intervals stored.
- Future documents excluded from training and labeling.
- Drift and stability metrics monitored.
- Old labels preserved rather than overwritten.

E.6 Emerging-Theme Review

- Novelty, burstiness, volume, acceleration, source breadth, and entity breadth computed.
- Candidate theme evidence package generated.
- Economic mechanism written in plain language.
- Lifecycle stage assigned.
- Inclusion and exclusion criteria drafted.
- Potential beneficiaries, suppliers, adopters, and disrupted firms listed.
- Factor, sector, liquidity, and capacity risks sketched.
- Analyst review decision stored.
- Monitoring-only and investable statuses distinguished.
- Falsification conditions and kill rules drafted.

E.7 Exposure Scoring

- Eligible company universe defined point-in-time.
- Evidence retrieval performed for each candidate company.
- Keyword, embedding, revenue, relationship, and LLM components separated.
- Materiality and confidence scored separately.
- Value-chain role assigned.
- Supporting and contradictory evidence stored.
- Sparse-evidence companies flagged.
- Analyst overrides logged with reason codes.
- Score validity interval defined.
- Comparison with external benchmarks or ETF holdings documented.

E.8 LLM Workflow

- Task classified as measurement, classification, extraction, summarization, or prediction.
- Model identifier, version, and cutoff information recorded where available.
- Prompt template and schema versioned.
- Retrieval corpus restricted to point-in-time evidence when used for backtesting.
- Retrieved passages stored with source timestamps.
- Output confidence, uncertainty, refusal, or unsupported-claim behavior tested.

- Validation sample reviewed by humans.
- Hallucination and future-knowledge tests documented.
- Production outputs linked to evidence packages.

E.9 Portfolio Construction

- Mandate and benchmark stack defined.
- Eligibility filters applied point-in-time.
- Liquidity, borrow, price, and market-cap constraints applied.
- Exposure threshold selected before test results.
- Weighting rule specified.
- Single-name, sector, industry, country, and sub-theme caps applied.
- Factor exposure bounds defined.
- Turnover limit applied.
- Transaction-cost and slippage models applied.
- Rebalance calendar fixed.
- Pre-trade risk report generated.

E.10 Backtesting

- Discovery, calibration, and test windows separated.
- Universe includes delisted names where relevant.
- All documents, labels, scores, and weights are point-in-time.
- Dynamic topic models avoid future information.
- LLM outputs are evidence-grounded and stored.
- Costs, slippage, borrow, financing, and capacity included.
- Raw, benchmark-relative, factor-adjusted, and cost-adjusted results reported.
- Placebo and matched random basket tests performed.
- Sensitivity to horizons, thresholds, prompts, and rebalancing tested.
- Results reviewed by regime, liquidity bucket, source, and sub-theme.

E.11 Factor Neutralization

- Factor model selected before performance review.
- Sector and industry exposures measured.
- Style factor loadings estimated.
- Raw exposure scores residualized where appropriate.
- Portfolio-level factor constraints applied where appropriate.
- Rolling betas monitored.
- Residual returns reported.
- Neutralization trade-off documented.

E.12 Attribution and Review

- Performance shown after costs.
- Benchmark-relative performance reported.
- Factor attribution and residual alpha reported.
- Theme, sub-theme, source, and company-role attribution reported.
- Risk contribution aggregated by relevant sleeves.
- Drawdowns linked to regime, factor, source, or theme explanations.
- Analyst overrides reviewed for value added and error patterns.
- Production committee decisions recorded.

E.13 Production Monitoring

- Data-source latency monitored.
- Document volume anomalies detected.
- Entity-linking drift monitored.
- Embedding drift monitored.
- Topic stability monitored.
- Exposure score changes explained.
- Analyst review backlog tracked.
- Factor exposure breaches alerted.
- Turnover and cost breaches alerted.
- Drawdown and residual alpha decay monitored.

- Capacity, crowding, ETF overlap, and liquidity monitored.
- Incident response procedure tested.

E.14 Governance Review

- Theme approval record complete.
- Model card complete.
- Evidence packages reviewable.
- Override log complete.
- Compliance restrictions checked.
- Licensing and retention requirements verified.
- Kill rules defined.
- Shadow-mode results reviewed.
- Promotion decision documented.
- Review cadence scheduled.

E.15 Release Gate Checklist

Gate	Required Evidence	Exit Decision
Research prototype	Hypothesis, corpus sample, model output, initial evidence	Continue exploration or stop
Validation candidate	Frozen design, point-in-time data, baseline comparison	Approve validation or redesign
Paper trading	Reproducible backtest, monitoring plan, review workflow	Enter shadow mode or reject
Limited production	Governance approval, risk limits, incident plan, rollback process	Allocate pilot capital or pause
Full production	Live evidence, attribution, capacity, compliance, stable operations	Scale, hold, redesign, or retire

Appendix F

Reproducible Research Standards

F.1 The Minimum Standard

A narrative-aware trading experiment is reproducible when another researcher can reconstruct the same corpus, features, labels, scores, portfolio weights, trades, and performance report from stored artifacts. Reproducibility is not only a scientific ideal. It is a model-risk control.

Each experiment should have a unique identifier, a frozen experiment record, and links to immutable artifacts.

F.2 Experiment Metadata

The experiment record should include:

- experiment identifier;
- owner and reviewer;
- creation timestamp and freeze timestamp;
- research question;
- hypothesis and economic mechanism;
- asset universe;
- time period;
- decision horizon;
- benchmark set;
- intended use: exploration, validation, paper trading, production, or monitoring;
- promotion status;
- links to reports and artifacts.

F.3 Data Snapshot

The data snapshot should identify:

- corpus sources and versions;
- source licenses and allowed uses;
- raw document checksums;
- source, vendor, ingestion, processing, and revision timestamps;
- vendor versions and schema versions;
- pricing dataset and adjustment method;
- corporate-action dataset;
- fundamental dataset;
- factor dataset;
- ETF holdings dataset;
- relationship dataset;
- prediction-market dataset where used.

If a source is mutable, the snapshot should preserve the exact version used in the experiment. Backfilled data should be identified as backfilled.

F.4 Universe and Identifier Metadata

The universe definition should include:

- security identifiers;
- company identifiers;
- ticker mapping;
- exchange and currency;
- share-class rules;
- index membership where relevant;
- delisting treatment;
- corporate-action treatment;
- liquidity filters;
- borrow filters;
- compliance exclusions.

Identifier mapping must be point-in-time. Today's ticker should not silently rewrite yesterday's security.

F.5 Model and Feature Metadata

For every model or feature generator, store:

- code version;
- model name and version;
- training data window;
- inference data window;
- tokenizer and embedding model;
- hyperparameters;
- random seed;
- preprocessing configuration;
- feature definitions;
- fitted model artifact;
- validation report.

For LLM components, also store:

- model identifier;
- known model cutoff or vendor statement, where available;
- prompt template;
- retrieved evidence;
- retrieval filters and corpus date limits;
- generation parameters;
- output schema;
- post-processing rules;
- human review decision;
- hallucination or unsupported-claim checks.

F.6 Theme and Label Metadata

Theme records should include:

- theme identifier;
- theme name and version;
- narrative description;
- inclusion and exclusion criteria;

- representative documents;
- lifecycle stage;
- parent and child themes;
- split and merge history;
- approval timestamp;
- approving analyst or committee;
- validity interval.

Labels should never be overwritten without version history. A historical backtest should use the label that existed at the simulated date.

F.7 Exposure Score Metadata

Each company-theme score should store:

- company identifier;
- security identifier where relevant;
- theme identifier;
- score timestamp;
- exposure score;
- materiality score;
- confidence score;
- value-chain role;
- component scores;
- supporting evidence;
- contradictory evidence;
- model version;
- analyst overrides;
- validity interval.

This makes inclusion decisions explainable and auditable.

F.8 Backtest Configuration

The backtest record should include:

- rebalance calendar;

- portfolio rule;
- optimization objective;
- constraints;
- factor model;
- transaction-cost model;
- slippage model;
- borrow-cost model;
- capacity assumptions;
- execution timing;
- cash and financing assumptions;
- benchmark definitions;
- evaluation metrics.

The record should also store every target portfolio, trade list, realized holding, return series, attribution report, and exception.

F.9 Analyst and Governance Metadata

Human decisions are data. Store:

- review queue item;
- analyst identifier;
- review timestamp;
- evidence reviewed;
- decision;
- confidence;
- override reason;
- approval or rejection status;
- committee notes;
- incident references where relevant.

This protects the process from unrecorded hindsight and makes future model improvement possible.

F.10 Reporting Artifacts

Each experiment should produce a durable report containing:

- hypothesis;
- data summary;
- model summary;
- backtest design;
- raw performance;
- benchmark-relative performance;
- factor-adjusted performance;
- cost-adjusted performance;
- attribution;
- robustness tests;
- placebo tests;
- limitations;
- production recommendation.

F.11 Structured Experiment Template

Field	Required Content
Question	What investment decision does this experiment support?
Mechanism	Why should the signal earn a return after costs and known risks?
Data	Which point-in-time sources, universes, prices, factors, and identifiers are used?
Model	Which preprocessing, embedding, topic, LLM, scoring, and portfolio models are used?
Time Discipline	What information is available at each simulated decision time?
Controls	Which benchmarks, factors, placebo tests, and robustness checks are required?
Costs	Which spread, impact, borrow, financing, turnover, and capacity assumptions apply?
Governance	Who approved the theme, model, overrides, and promotion decision?
Decision	Promote, monitor, redesign, pause, or retire.

F.12 Promotion Gates

A research artifact can advance only when the required records exist:

1. exploratory analysis may use flexible data, but it must be labeled exploratory;
2. validation candidate requires a frozen hypothesis and point-in-time design;
3. paper trading requires reproducible backtest artifacts and monitoring;

4. limited production requires risk limits, review queues, and incident procedures;
5. full production requires governance approval, auditability, capacity analysis, and rollback procedures;
6. retirement requires a record of why the strategy was stopped.

F.13 The Reproducibility Test

Before promotion, ask another researcher to reproduce:

- the corpus;
- the model outputs;
- the theme labels;
- the exposure scores;
- the portfolio weights;
- the performance report;
- the attribution and governance packet.

If they cannot, the system is not ready. In this field, irreproducibility is not a clerical flaw; it is often where hidden leakage, overfitting, and governance failures live.

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Draft Note

This working draft contains Parts I through VIII and Appendices A through F. Later revisions should continue tightening references, examples, implementation detail, and chapter-level consistency.